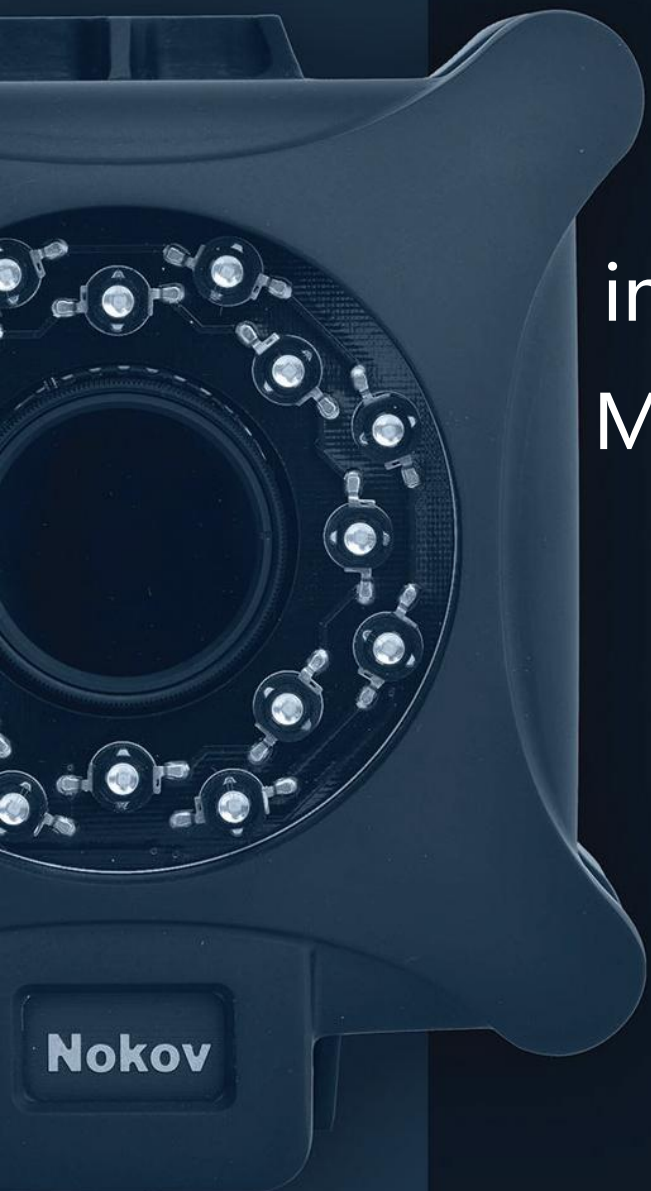




Published Papers in English Using NOKOV Motion Capture Systems

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Part I UAV, UGV

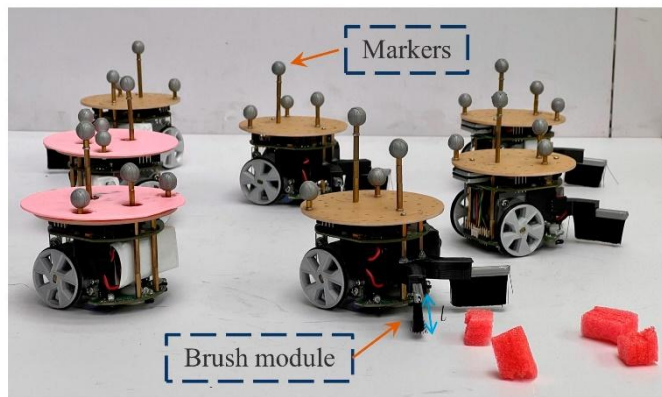
Self-Organized Patchy Target Searching and Collecting with Heterogeneous Swarm Robots Based on Density Interactions

Keywords

heterogeneous swarm robots; target search; target collection; targets with patchy distribution; shrinking strategy; density interaction

Abstract

The issue of searching and collecting targets with patchy distribution in an unknown environment is a challenging task for multiple or swarm robots because the targets are unevenly dispersed in space, which makes the traditional solutions based on the idea of path planning and full spatial coverage very inefficient and time consuming. In this paper, by employing a novel framework of spatial-density-field-based interactions, a collective searching and collecting algorithm for heterogeneous swarm robots is proposed to solve the challenging issue in a self-organized manner. In our robotic system, two types of swarm robots, i.e., the searching robots and the collecting robots, are included. To start with, the searching robots conduct an environment exploration by means of formation movement with Levy flights; when the targets are detected by the searching robots, they spontaneously form a ring-shaped envelope to estimate the spatial distribution of targets. Then, a single robot is selected from the group to enter the patch and locates at the patch's center to act as a guiding beacon. Subsequently, the collecting robots are recruited by the guiding beacon to gather the patch targets; they first form a ring-shaped envelope around the target patch and then push the scattered targets inward by using a spiral shrinking strategy; in this way, all targets eventually are stacked near the center of the target patch. With the cooperation of the searching robots and the collecting robots, our heterogeneous robotic system can operate autonomously as a coordinated group to complete the task of collecting targets in an unknown environment. Numerical simulations and real swarm robot experiments (up to 20 robots are used) show that the proposed algorithm is feasible and effective, and it can be extended to search and collect different types of targets with patchy distribution.



How NOKOV Contributes to the Research

In our experiments, the robot's position, speed, and heading are captured in real time by the NOKOV moving capture system. The real-time position of target units is collected by a top-view camera. Note that although this information is collected globally, the control of each robot only uses the local information within its sensing range.

References Xiang Y, Lei X, Duan Z, Dong F, Gao Y. Self-Organized Patchy Target Searching and Collecting with Heterogeneous Swarm Robots Based on Density Interactions. *Electronics*. 2023; 12(12):2588. <https://doi.org/10.3390/electronics12122588>
Link <https://www.mdpi.com/2079-9292/12/12/2588>

Robo-Centric ESDF: A Fast and Accurate Whole-Body Collision Evaluation Tool for Any-Shape Robotic Planning

Keywords

Shape, Trajectory planning, Real-time systems, Safety, Planning, Mobile robots, Reliability

Abstract

For letting mobile robots travel flexibly through complicated environments, increasing attention has been paid to the whole-body collision evaluation. Most existing works either opt for the conservative corridor-based methods that impose strict requirements on the corridor generation, or ESDF-based methods that suffer from high computational overhead. It is still a great challenge to achieve fast and accurate whole-body collision evaluation. In this paper, we propose a Robo-centric ESDF (RC-ESDF) that is pre-built in the robot body frame and is capable of seamlessly applied to any-shape mobile robots, even for those with non-convex shapes. RC-ESDF enjoys lazy collision evaluation, which retains only the minimum information sufficient for whole-body safety constraint and significantly speeds up trajectory optimization. Based on the analytical gradients provided by RC-ESDF, we optimize the position and rotation of robot jointly, with whole-body safety, smoothness, and dynamical feasibility taken into account. Extensive simulation and real-world experiments verified the reliability and generalizability of our method.

How NOKOV Contributes to the Research

In real-world experiments, to validate our method can be applied to any-shape robots, we especially implement experiments with rectangle-shape (convex) and L-shape (non-convex) McNamee-based wheeled robots. Environment information is stored by a pre-build precise point cloud map and real-time localization is obtained by a NOKOV motion capture system.

References S. Geng, Q. Wang, L. Xie, C. Xu, Y. Cao and F. Gao, "Robo-Centric ESDF: A Fast and Accurate Whole-Body Collision Evaluation Tool for Any-Shape Robotic Planning," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 290-297, doi: 10.1109/IROS55552.2023.10342074.
Link <https://ieeexplore.ieee.org/document/10342074>

Privacy-Preserving Distributed Iterative Localization for Wireless Sensor Networks

Keywords

Distributed localization, privacy preservation, wireless sensor network (WSN)

Abstract

Effectively determining the accurate locations of sensor nodes while preserving data privacy is an interesting and challenging problem in the research of wireless sensor networks. On the basis of the well-performing distributed iterative localization framework in wireless sensor networks, this article proposes a dual privacy-preserving scheme based on node-intermittent updates and noises injection. A comprehensive analysis of the localization algorithm in terms of convergence and privacy-preserving performance is presented. It is theoretically demonstrated that the localization algorithm can achieve global accurate localization without leaking the privacy information of sensor nodes. At last, the performance of the privacy-preserving localization algorithm is verified by building an experimental platform containing six robots.

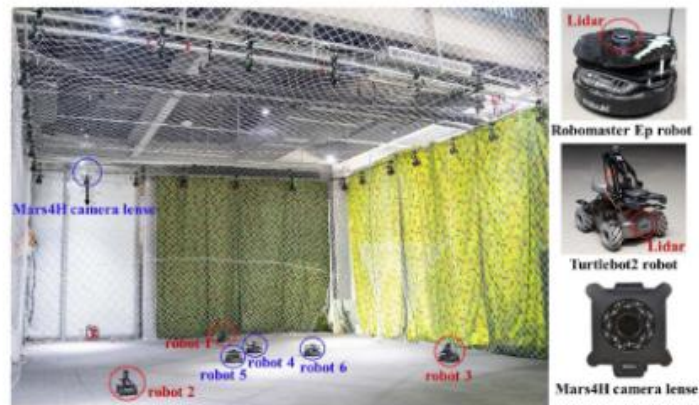


Fig. 3. Experimental platform composed of six robots and an NOKOV optical 3-D motion capture system, where robots 1,2,3 are the anchor nodes and robots 4,5,6 are the nonanchor nodes.

How NOKOV Contributes to the Research

The robots are randomly deployed over a 10×10 m² square area, which is covered by a NOKOV optical 3-D motion capture system composed of 24 Mars4H camera lenses. Fig. 4 shows the robots' deployment and the motion capture system's coverage. The motion capture system can provide submillimeter level location data. In the 2-D space, each robot's accurate location is decomposed into $x_i = (x_1, x_2)$. By the motion capture system, we can acquire all robots' accurate locations, i.e.

References L. Shi, W. X. Zheng, Q. Liu, Y. Liu and J. Shao, "Privacy-Preserving Distributed Iterative Localization for Wireless Sensor Networks," in IEEE Transactions on Industrial Electronics, vol. 70, no. 11, pp. 11628-11638, Nov. 2023, doi: 10.1109/TIE.2022.3231272.
Link <https://ieeexplore.ieee.org/document/10002381>

Multi-UAV online path planning algorithm based on improved Hybrid A*

Keywords

path planning, obstacle avoidance, anti-collision, velocity obstacle model, hybrid A* algorithm, potential field method

Abstract

Real-time path planning of multi-UAV includes obstacle avoidance and anti-collision, which is an important condition to ensure the coordinated operation of multi-UAV. This paper studies a multi-UAV online path planning algorithm based on improved Hybrid A*. Each UAV uses the Hybrid A* algorithm for distributed path planning. Through the collision risk assessment of the planned path, if there is a collision risk, the path adjustment is based on the defined UAV priority level and interactive information. In this paper, the existing two-dimensional velocity obstacle model is improved, and a three-dimensional velocity obstacle model suitable for 6-DOF UAV is constructed to calculate the safe speed range of UAV. As the optimization constraint condition, the path is adjusted based on the potential field method to achieve the effect of obstacle avoidance and collision avoidance. Finally, the simulation system and experimental platform are built to verify the real-time and stability of the proposed algorithm.



How NOKOV Contributes to the Research

The depth camera provides a depth map to sense the surrounding environment. NOKOV motion capture system provides accurate position estimation for each UAV. The on-board computer is responsible for the path information interaction between UAVs and running the path planning algorithm.

References Z. Wang, C. Wan, X. Lv, C. Ni, Z. Mao and Y. Li, "Multi-UAV online path planning algorithm based on improved Hybrid A*," 2023 6th International Symposium on Autonomous Systems (ISAS), Nanjing, China, 2023, pp. 1-6, doi: 10.1109/ISAS59543.2023.10164537.

Link <https://ieeexplore.ieee.org/abstract/document/10164537>

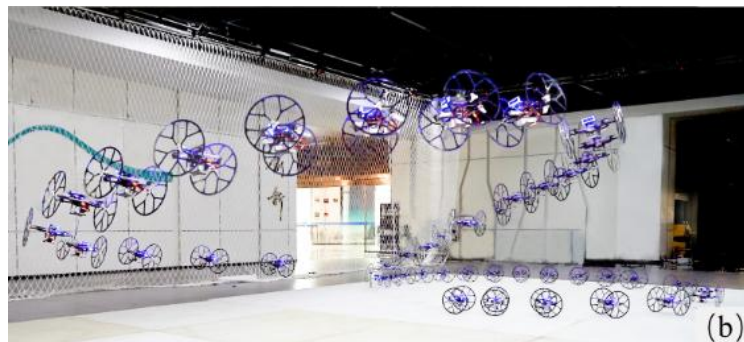
Model-Based Planning and Control for Terrestrial-Aerial Bimodal Vehicles with Passive Wheels

Keywords

Trajectory tracking, Computational modeling, Dynamics, Wheels, Benchmark testing, Predictive models, Planning

Abstract

Terrestrial and aerial bimodal vehicles have gained widespread attention due to their cross-domain maneuverability. Nevertheless, their bimodal dynamics significantly increase the complexity of motion planning and control, thus hindering robust and efficient autonomous navigation in unknown environments. To resolve this issue, we develop a model-based planning and control framework for terrestrial aerial bi-modal vehicles. This work begins by deriving a unified dynamic model and the corresponding differential flatness. Leveraging differential flatness, an optimization-based trajectory planner is proposed, which takes into account both solution quality and computational efficiency. Moreover, we design a tracking controller using nonlinear model predictive control based on the proposed unified dynamic model to achieve accurate trajectory tracking and smooth mode transition. We validate our framework through extensive benchmark comparisons and experiments, demonstrating its effectiveness in terms of planning quality and control performance.



How NOKOV Contributes to the Research

To demonstrate and verify the proposed approaches in real-world environments, we customize a TABV platform, as shown in Fig.4. The corresponding physical parameters are listed in Tab.I. The 6D localization of the TABV and the environmental information are obtained from a NOKOV motion capture system¹ and a Realsense D430 stereo camera, respectively.

References R. Zhang et al., "Model-Based Planning and Control for Terrestrial-Aerial Bimodal Vehicles with Passive Wheels," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 1070-1077, doi: 10.1109/IROS55552.2023.10342188.

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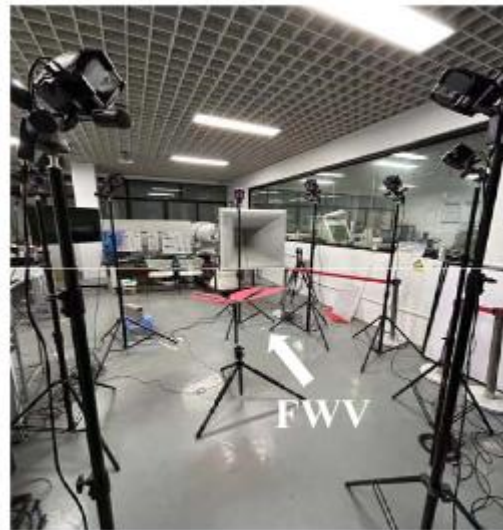
Lift enhancement of a butterfly-like flapping wing vehicle by reinforcement learning algorithm

Keywords

butterfly-like; flapping wing vehicle; lift peak in upstroke; reinforcement learning; take-off

Abstract

In order to enhance the take-off lift of a butterfly-like flapping wing vehicle (FWV), we implemented an integrated experimental platform and applied a reinforcement learning algorithm. The vehicle, which has a wingspan of 81 cm and is mounted on a stand with a force sensor, is driven by two servos that are powered and controlled wirelessly. To achieve the goal of enhancing take-off lift, we used a model-free, on-policy actor-critic proximal policy optimization algorithm. After 300 learning steps, the average aerodynamic lift force increased significantly from 0.044 N to 0.861 N. This enhanced lift force was sufficient to meet the take-off requirements of the vehicle without the need for any additional aids or airflow. Additionally, we observed a strong lift peak in the upstroke after analyzing the learning results. Further experiments showed that this lift peak is directly related to the elastic release of the wing twist and the opening and closing of the gap between the forewing and hindwing in the early stage of the upstroke. These findings were not easily predicted or discovered using traditional aerodynamic methods. This work provides valuable reinforcement learning experience for the future development of FWVs.



How NOKOV Contributes to the Research

We take a simple measurement of the opening distance between the forewing and hindwing during the flight. In figures 12(a) and (b), we place the FWV in NOKOV 3D motion capture system Mars2H to measure the motion of mark points in real time, and paste multiple infrared reflection markers on the forewing and the hindwing.

References Xiong M, Wei Z, Yang Y, Chen Q, Liu X. Lift enhancement of a butterfly-like flapping wing vehicle by reinforcement learning algorithm. Bioinspir Biomim. 2023 May 26;18(4). doi: 10.1088/1748-3190/acd3cc. PMID: 37160126.

Link <https://iopscience.iop.org/article/10.1088/1748-3190/acd3cc/meta>

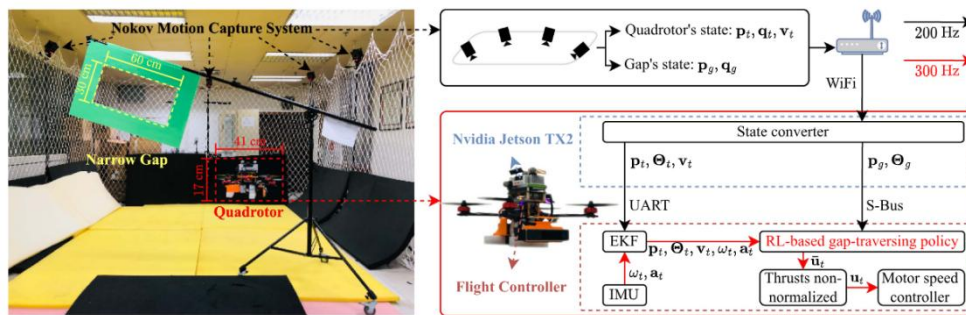
Learning Real-Time Dynamic Responsive Gap-Traversing Policy for Quadrotors With Safety-Aware Exploration

Keywords

Quadrotors, Trajectory, Task analysis, Vehicle dynamics, Trajectory planning, Safety, Real-time systems

Abstract

In this paper, we propose a dynamic responsive policy for a quadrotor to traverse through narrow gaps based on deep reinforcement learning (RL). The policy provides agile and dynamically adjusted gap-traversing actions for the quadrotor in real-time by directly mapping the observed states of the quadrotor and gaps to motor thrusts. In contrast to existing optimization-based methods, our RL-based policy couples trajectory planning and control modules, which are strictly independent and computationally complex in previous works. Moreover, a safety-aware exploration is presented to reduce the collision risk and improve the safety of the policy, which also facilitates the transfer of the policy to real-world environments. Specifically, we formulate a safety reward on the state spaces of both position and orientation to inspire the quadrotor to traverse through the narrow gap at an appropriate attitude closer to the center of the gap. With the learned gap-traversing policy, we implement extensive simulations and real-world experiments to evaluate its performance and compare it with the related approaches. Our implementation includes several gap-traversing tasks with random positions and orientations, even if never trained specially before. All the performances indicate that our RL-based gap-traversing policy is transferable and more efficient in terms of real-time dynamic response, agility, time-consuming, and generalization.



How NOKOV Contributes to the Research

The size of the narrow gap used in the experiments is $60 \times 30 \text{ cm}^2$, which indicates that the quadrotor can never safely traverse through the gap unless flying at a suitable tilted attitude. A Nokov motion capture system is used for the environmental perception, which outputs both the state information of the quadrotor and the gap at 200 Hz.

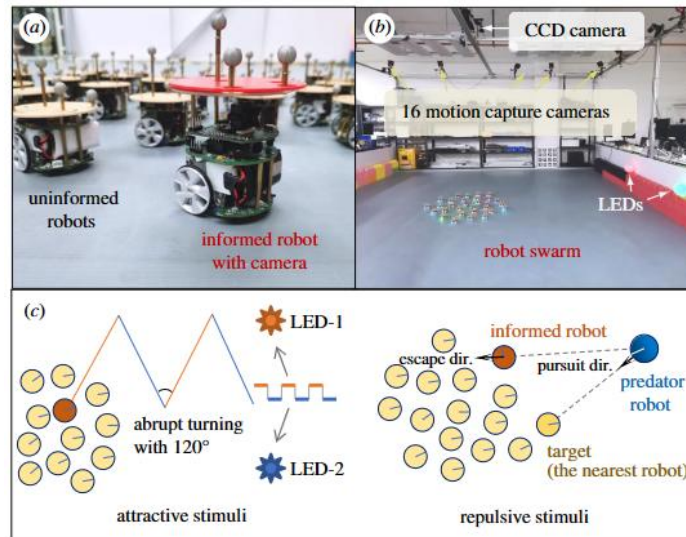
References S. Chen, Y. Li, Y. Lou, K. Lin and X. Wu, "Learning Real-Time Dynamic Responsive Gap-Traversing Policy for Quadrotors With Safety-Aware Exploration," in IEEE Transactions on Intelligent Vehicles, vol. 8, no. 3, pp. 2271-2284, March 2023, doi: 10.1109/TIV.2022.3229723.

Link <https://ieeexplore.ieee.org/abstract/document/9991071>

Exploring the criticality hypothesis using programmable swarm robots with Vicsek-like interactions

Abstract

A widely mentioned but not experimentally confirmed view (known as the ‘criticality hypothesis’) argues that biological swarm systems gain optimal responsiveness to perturbations and information processing capabilities by operating near the critical state where an ordered-to-disordered state transition occurs. However, various factors can induce the ordered–disordered transition, and the explicit relationship between these factors and the criticality is still unclear. Here, we present an experimental validation for the criticality hypothesis by employing real programmable swarm-robotic systems (up to 50 robots) governed by Vicsek-like interactions, subject to time-varying stimulus-response and hazard avoidance. We find that (i) not all ordered–disordered motion transitions correspond to the functional advantages for groups; (ii) collective response of groups is maximized near the critical state induced by alignment weight or scale rather than noise and other non-alignment factors; and (iii) those non-alignment factors act to highlight the functional advantages of alignment-induced criticality. These results suggest that the adjustability of velocity or directional coupling between individuals plays an essential role in the acquisition of maximizing collective response by criticality. Our results contribute to understanding the adjustment strategies of animal interactions from a perspective of criticality and provide insights into the design and control of swarm robotics.



How NOKOV Contributes to the Research

Our experiments are conducted in an approximately $6.0\text{ m} \times 6.0\text{ m}$ arena with boundary, see figure 1b, in which a CCD camera (Basler a2A4504-18ucPRO with Kowa LM6FC24M lens) with top-down view is used to record experimental videos. We employ a motion capture system (NOKOV) to track the location and orientation of each robot, from which the robot's real-time linear speed and angular speed are derived by differential calculation.

References X. Lei, Y. Xiang, M. Duan, and X. Peng, "Exploring the criticality hypothesis using programmable swarm robots with Vicsek-like interactions," *Journal of The Royal Society Interface*, vol. 20, no. 204, Jul. 2023.

Link <https://royalsocietypublishing.org/doi/10.1098/rsif.2023.0176>

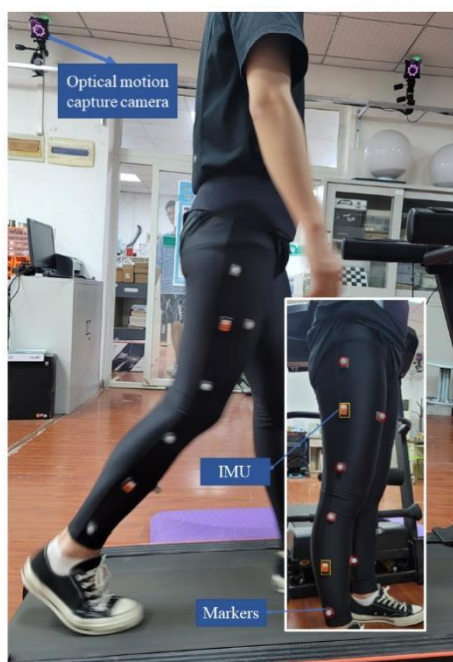
Effects of Dynamic IMU-to-Segment Misalignment Error on 3-DOF Knee Angle Estimation in Walking and Running

Keywords

IMU; DPSO; joint constraint; dynamic IMU-to-segment alignment

Abstract

The inertial measurement unit (IMU)-to-segment (I2S) alignment is an important part of IMU-based joint angle estimation, and the accurate estimation of the three degree of freedom (3-DOF) knee angle can provide practical support for the evaluation of motions. In this paper, we introduce a dynamic weight particle swarm optimization (DPSO) algorithm with crossover factor based on the joint constraint to obtain the dynamic alignment vectors of I2S, and use them to perform the quaternion-based 3-DOF knee angle estimation algorithm. The optimization algorithm and the joint angle estimation algorithm were evaluated by comparing with the optical motion capture system. The range of 3-DOF knee angle root mean square errors (RMSEs) is 1.6° – 5.9° during different motions. Furthermore, we also set up experiments of human walking (3 km/h), jogging (6 km/h) and ordinary running (9 km/h) to investigate the effects of dynamic I2S misalignment errors on 3-DOF knee angle estimation during different motions by artificially adding errors to I2S alignment parameters. The results showed differences in the effects of I2S misalignment errors on the estimation of knee abduction, internal rotation and flexion, which indicate the differences in knee joint kinematics among different motions. The IMU to thigh misalignment error has the greatest effect on the estimation of knee internal rotation. The effect of IMU to thigh misalignment error on the estimation of knee abduction angle becomes smaller and then larger during the two processes of switching from walking to jogging and then speeding up to ordinary running. The effect of IMU to shank misalignment error on the estimation of knee flexion angle is numerically the largest, while the standard deviation (SD) is the smallest. This study can provide support for future research on the accuracy of 3-DOF knee angle estimation during different motions.



How NOKOV Contributes to the Research

Twelve high-speed infrared cameras from Nokov (Beijing, China) form an optical motion capture system to capture the position of markers, and both systems operate at 60 Hz. The three non-coplanar markers are set on the thighs and shank of both legs to define the attached rigid bodies, and the rotational data of each segment is output through Nokov's software.

References Jiang C, Yang Y, Mao H, Yang D, Wang W. Effects of Dynamic IMU-to-Segment Misalignment Error on 3-DOF Knee Angle Estimation in Walking and Running. *Sensors*. 2022; 22(22):9009. <https://doi.org/10.3390/s22229009>
Link <https://www.mdpi.com/1424-8220/22/22/9009>

Demonstration Learning and Generalization of Robotic Motor Skills Based on Wearable Motion Tracking Sensors

Keywords

Robots, Trajectory, Task analysis, Motion capture, Robot sensing systems, Optimization, Heuristic algorithms

Abstract

Learning from demonstration (LfD) is an important way for robots to learn new skills. Its principle is to obtain an optimized trajectory through an instructor's action teaching or action coding regression. By learning the teaching trajectory, the robot can master the movement skills. However, adapting the learned motor skills to meet additional constraints that arise during the task can be challenging. This article proposed a noise exploration method policy improvement through black-box optimization with an adaptive covariance matrix (PI BB -CMA), which improves the PI BB algorithm by using an adaptive covariance matrix. In our method, the teaching trajectory is collected with our designed inertial motion capture system. After that, a more reasonable trajectory is synthesized as the input to dynamic movement primitives (DMPs) using the Gaussian mixture model (GMM) and Gaussian mixture regression (GMR). Finally, the improved algorithm PI BB -CMA is used to perform noise exploration to ensure that the robot arm can pass the designated points in the path planning. Furthermore, three baseline methods are chosen for comparison to validate the robustness of our proposed method. The experimental results demonstrated that our proposed method outperforms the baseline algorithms and is feasible for improving robot intelligence. We believe that our method has a certain potential in the aspect of skill learning for robots.

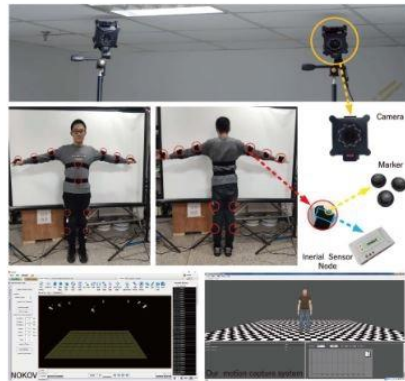


Fig. 4. Overview of the validation experiment for NOKOV and our system.

TABLE II
ESTIMATION CURVES COMPARISON BETWEEN
OUR METHOD AND NOKOV

Coordinates axis	Correlation coefficient
x	0.9876
y	0.9912
z	0.9991

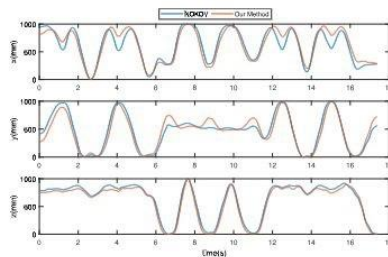


Fig. 5. Comparison of attitude data collected by NOKOV and our method.

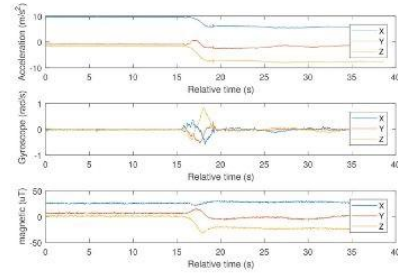


Fig. 6. Sensor data collected while writing the letter c.

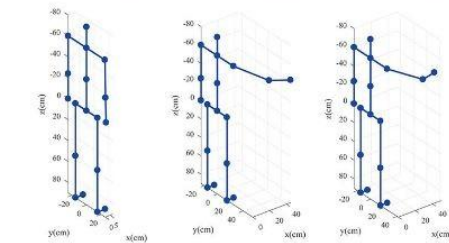


Fig. 7. Human posture corresponding to the upright, raised hand, and writing during the demonstration.

How NOKOV Contributes to the Research

In the experiment, the golden standard, optical system NOKOV, was chosen for validating the accuracy of trajectory tracking. Imitative learning and obstacle avoidance experiments are also carried out to assess the feasibility of the proposed algorithm.

References X. Liu, Z. Wang, J. Li, A. Cangelosi and C. Yang, "Demonstration Learning and Generalization of Robotic Motor Skills Based on Wearable Motion Tracking Sensors," in IEEE Transactions on Instrumentation and Measurement, vol. 72, pp. 1-15, 2023, Art no. 2519515, doi: 10.1109/TIM.2023.3288240.

Link <https://ieeexplore.ieee.org/abstract/document/10159577>

Decentralized Planning for Car-Like Robotic Swarm in Cluttered Environments

Keywords

Robot kinematics, Wireless networks, System recovery, Real-time systems, Robustness, Planning, Trajectory

Abstract

Robot swarm is a hot spot in robotic research community. In this paper, we propose a decentralized framework for car-like robotic swarm which is capable of real-time planning in cluttered environments. In this system, path finding is guided by environmental topology information to avoid frequent topological change, and search-based speed planning is leveraged to escape from infeasible initial value's local minima. Then spatial-temporal optimization is employed to generate a safe, smooth and dynamically feasible trajectory. During optimization, the trajectory is discretized by fixed time steps. Penalty is imposed on the signed distance between agents to realize collision avoidance, and differential flatness cooperated with limitation on front steer angle satisfies the non-holonomic constraints. With trajectories broadcast to the wireless network, agents are able to check and prevent potential collisions. We validate the robustness of our system in simulation and real-world experiments. Code will be released as open-source packages.

How NOKOV Contributes to the Research

We conduct real-world experiments in a $3\text{m} \times 8\text{m}$ environment, where three car-like robots move from one side to the other. Each robot is equipped with an NVIDIA Jetson Nano as the onboard processor and a single-line lidar with a 3m sensing range, which is equal to the horizon of locally planned trajectories. Localization is done by the NOKOV motion capture system.

References C. Ma et al., "Decentralized Planning for Car-Like Robotic Swarm in Cluttered Environments," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 9293-9300, doi: 10.1109/IROS55552.2023.10342360.

Link <https://ieeexplore.ieee.org/document/10342360/>

Cross-Domain Representation Learning for Clothes Unfolding in Robot-Assisted Dressing

Keywords

Clothes unfolding, Grasping point recognition, Robot-assisted dressing, Human-robot interaction

Abstract

Assistive robots can significantly reduce the burden of daily activities by providing services such as unfolding clothes and dressing assistance. For robotic clothes manipulation tasks, grasping point recognition is one of the core steps, which is usually achieved by supervised deep learning methods using large amount of labeled training data. Given that collecting real annotated data is extremely labor-intensive and time-consuming in this field, synthetic data generated by physics engines is typically adopted for data enrichment. However, there exists an inherent discrepancy between real and synthetic domains. Therefore, effectively leveraging synthetic data together with real data to jointly train models for grasping point recognition is desirable. In this paper, we propose a Cross-Domain Representation Learning (CDRL) framework that adaptively extracts domain-specific features from synthetic and real domains respectively, before further fusing these domain-specific features to produce more informative and robust cross-domain representations, thereby improving the prediction accuracy of grasping points. Experimental results show that our CDRL framework is capable of recognizing grasping points more precisely compared with five baseline methods. Based on our CDRL framework, we enable a Baxter humanoid robot to unfold a hanging white coat with a 92% success rate and assist 6 users to dress successfully.



How NOKOV Contributes to the Research

We gather real depth images by constantly changing the hanging positions of the white coat with the help of the Kinect v2 camera. While taking depth images, the spatial Cartesian coordinates of the grasping points are recorded with a NOKOV Motion Capture System by placing markers at the collar areas. After repeating the above steps, we obtain a total of 5000 pieces of real labeled data, which takes approximately 50 hours. The non-clothing segments are filtered from the real images by thresholding the depth between 80 cm and 110 cm.

References Qie, J., Gao, Y., Feng, R., Wang, X., Yang, J., Dasgupta, E., Chang, H.J., & Chang, Y. (2022). Cross-Domain Representation Learning for Clothes Unfolding in Robot-Assisted Dressing. *ECCV Workshops*.

Link https://link.springer.com/chapter/10.1007/978-3-031-25075-0_44

Continuous Implicit SDF Based Any-shape Robot Trajectory Optimization

Keywords

Geometry, Pipelines, Planning, Computational efficiency, Collision avoidance, Robots, Trajectory optimization

Abstract

Optimization-based trajectory generation methods are widely used in whole-body planning for robots. However, existing work either oversimplifies the robot's geometry and environment representation, resulting in a conservative trajectory or suffers from a huge overhead in maintaining additional information such as the Signed Distance Field (SDF). To bridge the gap, we consider the robot as an implicit function, with its surface boundary represented by the zero-level set of its SDF. We further employ another implicit function to lazily compute the signed distance to the swept volume generated by the robot and its trajectory. The computation is efficient by exploiting continuity in space-time, and the implicit function guarantees continuous collision evaluation even for nonconvex robots with complex surfaces. We also propose a trajectory optimization pipeline applicable to the implicit SDF. Simulation and real-world experiments validate the high performance of our approach for arbitrarily shaped robot trajectory optimization.



Fig. 6: The real-world indoor experiment: The quadrotor must traverse three consecutive circles while precisely avoiding nearby obstacles. We highlight the most critical frames when the robot is closest to the obstacles.

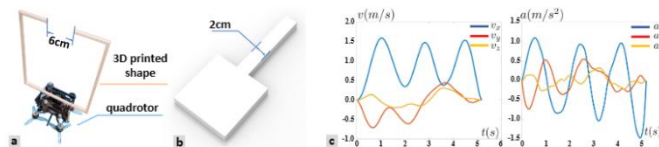


Fig. 7: The UAV platform and the obstacle in the experiment are shown in Fig.a and Fig.b. The velocity and acceleration of the drone in the real experiment are shown in Fig.c.

How NOKOV Contributes to the Research

We conduct a real-world experiment with the motion capture NOKOV system. Fig.7a shows the quadrotor robot in our experiment. We deliberately install a structure on the robot to give it a complex shape. We hang the obstacle shown in Fig.7b inside a ring so that the robot has to cross the ring from the correct position to ensure that it does not collide.

References T. Zhang, J. Wang, C. Xu, A. Gao and F. Gao, "Continuous Implicit SDF Based Any-Shape Robot Trajectory Optimization," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 282-289, doi: 10.1109/IROS55552.2023.10342104.

Link <https://ieeexplore.ieee.org/document/10342104>

Application of Dynamic Weight Particle Swarm Optimization with Cross Factor in Joint Calibration

Keywords

Heuristic algorithms, Motion segmentation, Dynamics, Particle measurements, Motion measurement, Particle swarm optimization, Optimization

Abstract

The alignment of inertial measurement units(IMUs) to segment is an important step in inertial motion capture, which directly affects whether the imu data can fully represent the motion of the segment. Inspired by the gene crossover and mutation of Genetic Algorithm(GA), we propose a dynamic inertial weighted particle swarm optimization algorithm with cross factor to solve the joint constraint problem, and compared our algorithm with Particle Swarm Optimization(PSO) and Dynamic Inertial Weighted Particle Swarm Optimization(DPSO) algorithms to show the superiority of our algorithm during human lower limb movements. The experiment shows that introduced the random cross mechanism between particles with larger fitness and only the effective cross retained, makes the new algorithm show better search ability and convergence effect in this project, the stability and effectiveness are also improved. Our current work provides a good support for accurate calculation of joint angles in the future.

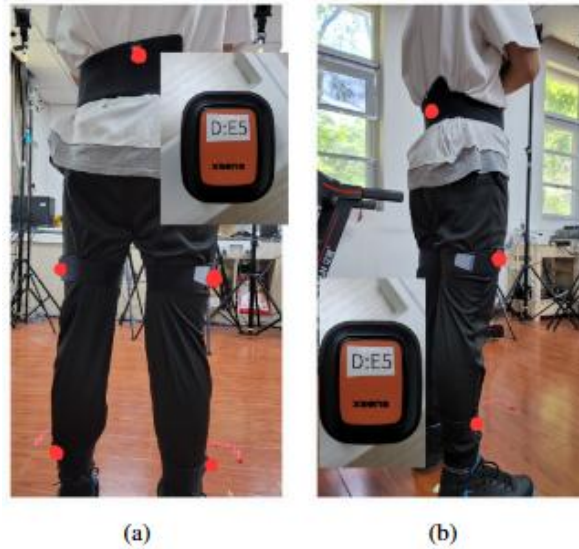


Fig. 3. Overview of the motion capture test equipment

How NOKOV Contributes to the Research

The Xsens Dot is set to collect data at a sampling frequency of 120 Hz. The Nokov infrared optical motion capture system (Beijing, China) was used to obtain high precision knee angles as reference values. Alignment between IMU and segment is accomplished through the joint constraint model to complete the IMU-based joint angle estimation. The obtained joint angle is used as the observed value and the joint angle obtained through Nokov is used as the reference value to calculate the RMSE, the results are shown in Fig.5. The RMSE ranges were from 1.62 to 2.94 for 3DOF joint angle.

References C. Jiang, W. Wang, D. Yang, Y. Yang and H. Mao, "Application of Dynamic Weight Particle Swarm Optimization with Cross Factor in Joint Calibration," 2022 6th International Conference on Communication and Information Systems (ICCIS), Chongqing, China, 2022, pp. 159-163, doi: 10.1109/ICCIS56375.2022.9998132.

Link <https://ieeexplore.ieee.org/document/9998132>

An Efficient Trajectory Planner for Car-Like Robots on Uneven Terrain

Keywords

Tracking, Heuristic algorithms, Dynamics, Stochastic processes, Approximation algorithms, Tires, Planning

Abstract

Autonomous navigation of ground robots on uneven terrain is being considered in more and more tasks. However, uneven terrain will bring two problems to motion planning: how to assess the traversability of the terrain and how to cope with the dynamics model of the robot associated with the terrain. The trajectories generated by existing methods are often too conservative or cannot be tracked well by the controller since the second problem is not well solved. In this paper, we propose terrain pose mapping to describe the impact of terrain on the robot. With this mapping, we can obtain the $SE(3)$ state of the robot on uneven terrain for a given state in $SE(2)$. Then, based on it, we present a trajectory optimization framework for car-like robots on uneven terrain that can consider both of the above problems. The trajectories generated by our method conform to the dynamics model of the system without being overly conservative and yet able to be tracked well by the controller. We perform simulations and real-world experiments to validate the efficiency and trajectory quality of our algorithm.

How NOKOV Contributes to the Research

In order to validate the performance of our method in realworld applications, we deploy it on a car-like robot, as shown in Fig. 3. All computations are performed by an onboard computer NVIDIA Jetson Nano. We utilize NOKOV Motion Capture System2 for localization. Furthermore, a MPC controller with position feedback is fitted to the robot for trajectory tracking.

References L. Xu et al., "An Efficient Trajectory Planner for Car-Like Robots on Uneven Terrain," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 2853-2860, doi: 10.1109/IROS55552.2023.10341558. Link <https://ieeexplore.ieee.org/document/10341558>

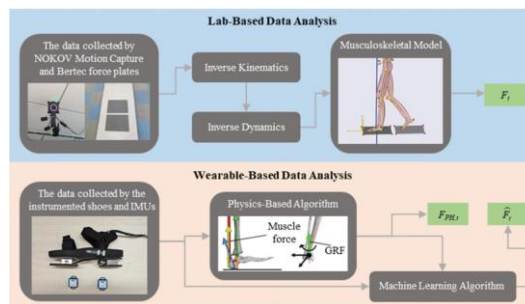
A Wearable Biofeedback Device for Monitoring

Keywords

Tibial force, rehabilitation, partial weight-bearing, machine learning, ambulatory monitoring

Abstract

Patients with tibial fractures are usually advised to follow a partial weight-bearing gait rehabilitation program after surgery to promote bone healing and lower limb functional recovery. Currently, the biofeedback devices used for gait rehabilitation training in fracture patients use ground reaction force (GRF) as the indicator of tibial load. However, an increasing body of research has shown that monitoring GRF alone cannot objectively reflect the load on the lower limb bones during human movement. In this study, a novel biofeedback system was developed utilizing inertial measurement units and custom instrumented insoles. Based on the data collected from experiments, a hybrid approach combining a physics-based model and neural network architectures was used to predict tibial force. Compared to the traditional physics-based algorithm, the physical guided neural networks method showed better predictive performance. The study also found that regardless of the type of weight-bearing walking, the peak tibial force was significantly higher than the peak tibial GRF, and the time at which the peak tibial compression force occurs may not be consistent with the time at which the peak vertical GRF occurs. This further supports the idea that during gait rehabilitation training for patients with tibial fractures, monitoring and providing feedback on the actual tibial force rather than just the GRF is necessary. The developed device is a non-invasive and reliable portable device that can provide audio feedback, providing a viable solution for gait rehabilitation training outside laboratory and helping to optimize patients' rehabilitation treatment strategies.



How NOKOV Contributes to the Research

On the day of data collection, 26 retro-reflective markers were placed on each subject's body. During walking, the marker trajectories were recorded with 8-camera motion capture system (Nokov, Beijing, China) at 200 Hz, and two force plates (Bertec, Ohio, USA) embedded in the walkway were used to collect 3D GRF during walking.

References T. Ma, X. Xu, Z. Chai, T. Wang, X. Shen and T. Sun, "A Wearable Biofeedback Device for Monitoring Tibial Load During Partial Weight-Bearing Walking," in IEEE Transactions on Neural Systems and Rehabilitation Engineering, vol. 31, pp. 3428-3436, 2023, doi: 10.1109/TNSRE.2023.3305205.

Link <https://ieeexplore.ieee.org/document/10217015>

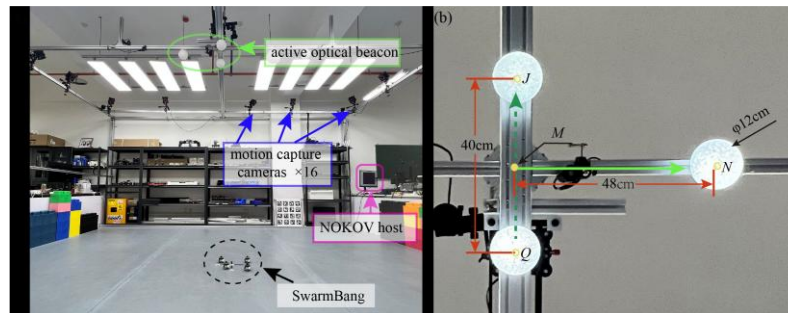
A Minimalist Self-Localization Approach for Swarm Robots Based on Active Beacon in Indoor Environments

Keywords

swarm robots; visual localization; optical beacon; self-localization; indoor environment

Abstract

When performing indoor tasks, miniature swarm robots are suffered from their small size, poor on-board computing power, and electromagnetic shielding of buildings, which means that some traditional localization methods, such as global positioning system (GPS), simultaneous localization and mapping (SLAM), and ultra-wideband (UWB), cannot be employed. In this paper, a minimalist indoor self-localization approach for swarm robots is proposed based on active optical beacons. A robotic navigator is introduced into a swarm of robots to provide locally localization services by actively projecting a customized optical beacon on the indoor ceiling, which contains the origin and the reference direction of localization coordinates. The swarm robots observe the optical beacon on the ceiling via a bottom-up-view monocular camera, and extract the beacon information on-board to localize their positions and headings. The uniqueness of this strategy is that it uses the flat, smooth, and well-reflective ceiling in the indoor environment as a ubiquitous plane for displaying the optical beacon; meanwhile, the bottom-up view of swarm robots is not easily blocked. Real robotic experiments are conducted to validate and analyze the localization performance of the proposed minimalist self-localization approach. The results show that our approach is feasible and effective, and can meet the needs of swarm robots to coordinate their motion. Specifically, for the stationary robots, the average position error and heading error are 2.41 cm and 1.44°; when the robots are moving, the average position error and heading error are less than 2.40 cm and 2.66°.



How NOKOV Contributes to the Research

In As depicted in Figure 6a, our experiments are conducted in a ~30 m² indoor arena with flat ground and no obstacles. Around the arena, 16 NOKOV motion capture cameras are mounted, which connect to a host. This motion capture system serves to monitor the real-time position and heading of each SwarmBang robot. The NOKOV system has a positioning and orientation accuracy of 0.2 mm and 0.28°, respectively, so these collected data can be treated as the true pose to check the robot's self-localization results.

References Duan, M.; Lei, X.; Duan, Z.; Zheng, Z. A Minimalist Self-Localization Approach for Swarm Robots Based on Active Beacon in Indoor Environments. *Sensors* 2023, 23, 4926. <https://doi.org/10.3390/s23104926>
 Link <https://www.mdpi.com/1424-8220/23/10/4926>

3D-RPP: a novel 3D vision-based Pose Perception Approach for Industrial Robots

Abstract

This paper considers a multi-target coverage problem where a robot team aims to efficiently cover multi-targets while maintaining connectivity in a distributed manner. A novel knowledge-incorporated policy framework is proposed to derive a distributed, efficient, and connectivity guaranteed coverage policy. In particular, a knowledge-guided policy network (KGPnet) is designed, which consists of observation attention representation, interaction attention representation, and knowledge-guided policy learning. Giving credit to the KGPnet, the connectivity guaranteed coverage policy can be applied to different number targets. Moreover, based on the knowledge of the algebraic connectivity and coverage rate, a comprehensive reward is designed to guide the training of the behavior of multi-target coverage with connectivity maintenance. Furthermore, since the policy learned through deep reinforcement learning (DRL) can not guarantee the connectivity of the robot team, a knowledge-nested policy filtering is designed to filter dis-connectivity policies to satisfy the connectivity constraint based on the knowledge model of connectivity maintenance. Various simulations are conducted to verify the effectiveness of the proposed method. Besides, numerous real-world experiments with three-wheel omnidirectional cars and a motion capture system are presented to demonstrate the practicability of the proposed method.

How NOKOV Contributes to the Research

In high-precision localisation, high-precision location tracking devices such as Nokov, Vicon are introduced as infrastructure . Such devices are capable of precise robot position sensing

References Li, Mo & Niu, Jianwei & Dou, Yuanbo & Tao, Ren & Li, Qingfeng & Yu, Xiaolong. (2022). 3D-RPP: a novel 3D vision-based Pose Perception Approach for Industrial Robots. Journal of Physics: Conference Series. 2188. 012013. 10.1088/1742-6596/2188/1/012013.

Link <https://iopscience.iop.org/article/10.1088/1742-6596/2188/1/012013/meta>

Multi-target Coverage with Connectivity Maintenance using Knowledge-incorporated Policy Framework

Abstract

This paper considers a multi-target coverage problem where a robot team aims to efficiently cover multi-targets while maintaining connectivity in a distributed manner. A novel knowledge-incorporated policy framework is proposed to derive a distributed, efficient, and connectivity guaranteed coverage policy. In particular, a knowledge-guided policy network (KGPnet) is designed, which consists of observation attention representation, interaction attention representation, and knowledge-guided policy learning. Giving credit to the KGPnet, the connectivity guaranteed coverage policy can be applied to different number targets. Moreover, based on the knowledge of the algebraic connectivity and coverage rate, a comprehensive reward is designed to guide the training of the behavior of multi-target coverage with connectivity maintenance. Furthermore, since the policy learned through deep reinforcement learning (DRL) can not guarantee the connectivity of the robot team, a knowledge-nested policy filtering is designed to filter dis-connectivity policies to satisfy the connectivity constraint based on the knowledge model of connectivity maintenance. Various simulations are conducted to verify the effectiveness of the proposed method. Besides, numerous real-world experiments with three-wheel omnidirectional cars and a motion capture system are presented to demonstrate the practicability of the proposed method.

How NOKOV Contributes to the Research

In order to verify the practicability of the proposed method in practical application, some real-world experiments are conducted. As shown in Fig.3 (b), a robot team consisting of three holonomic robots is used to test the derived policy. To improve the real-time of the policy generating, each robot is equipped with onboard computer (Nvidia Jetson TX2). In addition, the positions and velocities of the robots are provided by the NOKOV motion capture system.

References Wu S, Pu Z, Liu Z, et al. Multi-target Coverage with Connectivity Maintenance using Knowledge-incorporated Policy Framework[C]//2021 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2021: 8772-8778.

Link <https://ieeexplore.ieee.org/abstract/document/9562017>

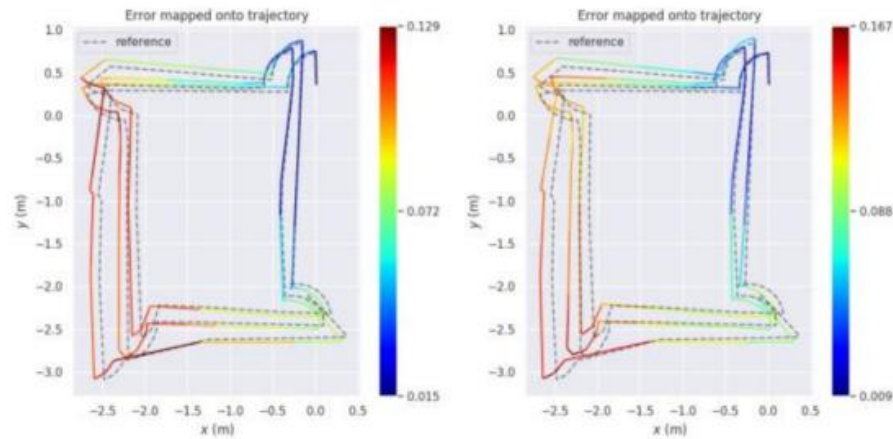
Stereo Visual Inertial Mapping Algorithm for Autonomous Mobile Robot

Keywords

Visual inertial mapping; Stereo camera; AMR; SLAM

Abstract

Simultaneous Localization and Mapping (SLAM) is a fundamental problem for autonomous mobile robots (AMRs). AMRs are widely used in automated warehousing, factory material transfer systems, flexible assembly systems, and other intelligent transportation sites. The visual Inertial Odometry (VIO) which consists of the camera and inertial-measurement-unit (IMU), is a popular approach to achieve accurate 6-DOF state estimation. However, such locally accurate VIO is prone to drift and cannot provide a global consistent map. The prerequisite for re-localizing the robot and ensuring precise autonomous navigation is an accurate and global consistent map of its environment. In this study, we propose a stereo visual-inertial mapping system. The front-end is a robust stereo VIO based on a tightly-coupled sliding window optimization. The core of the back-end is the global Bundle-Adjustment (BA) which is a nonlinear optimization, in which IMU is also added as a time-domain constraint. Meanwhile, stereo-camera-IMU extrinsic calibration is performed in BA to improve mapping accuracy. The selection principles of keyframes and map points are also designed according to the AMRs application characteristics. Further, the forward and backward Perspective-n-Point (PNP) method is also adopted to avoid the loop-detection mismatch. The performance of the system was validated and compared against other state-of-the-art algorithms. The findings revealed the effectiveness and robustness of this stereo visual-inertial mapping algorithm.



How NOKOV Contributes to the Research

The algorithm is verified by comparing two different scenes: one is a small-scale laboratory scene and the other is a large-scale scene (≥ 1000 square meters). A motion capture system (Nokov Mars1.3H) is a physically-based ground truth in laboratory scenes. In a large-scale scene, the surrounding environment is not friendly to visual SLAM, which contains pedestrians, a long corridor, low-light condition, textureless area, glass, and reflection. In this experiment, we compare our algorithm with another state-of-the-art algorithm VINS-Fusion [36]. For a fair comparison, most config parameters of both our algorithm and VINS-Fusion are similar.

References Zhang M, Han S, Wang S, et al. Stereo Visual Inertial Mapping Algorithm for Autonomous Mobile Robot[C]//2020 3rd International Conference on Intelligent Robotic and Control Engineering (IRCE). IEEE, 2020: 97-104.

Link <https://ieeexplore.ieee.org/abstract/document/9199252>

Continuous-time Gaussian Process Trajectory Generation for Multi-robot Formation via Probabilistic Inference

Abstract

In this paper, we extend a famous motion planning approach GPMP2 to multi-robot cases, yielding a novel centralized trajectory generation method for the multi-robot formation. A sparse Gaussian Process model is employed to represent the continuous-time trajectories of all robots as a limited number of states, which improves computational efficiency due to the sparsity. We add constraints to guarantee collision avoidance between individuals as well as formation maintenance, then all constraints and kinematics are formulated on a factor graph. By introducing a global planner, our proposed method can generate trajectories efficiently for a team of robots which have to get through a width-varying area by adaptive formation change. Finally, we provide the implementation of an incremental replanning algorithm to demonstrate the online operation potential of our proposed framework. The experiments in simulation and real world illustrate the feasibility, efficiency and scalability of our approach.

How NOKOV Contributes to the Research

We test the proposed framework with a team of quadrotors on three common scenarios: formation maintenance, replanning for a changed destination and adaptive formation change for moving through a width-varying area. Experiments in the real world are conducted with a group of Crazyflie nano-quadrotors flying under the supervision of a NOKOV motion capture system.

References Guo S, Liu B, Zhang S, et al. Continuous-time Gaussian Process Trajectory Generation for Multi-robot Formation via Probabilistic Inference[J]. arXiv preprint arXiv:2010.13148, 2020.

Link <https://arxiv.org/abs/2010.13148>

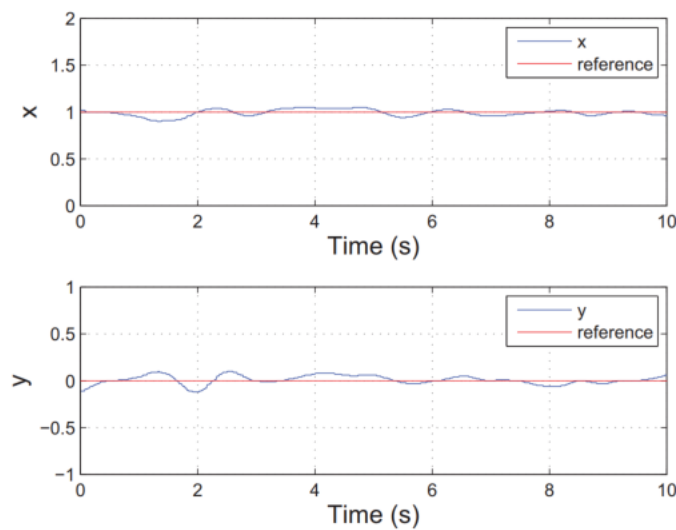
A Feedback Linearization and Saturated Control Structure for Quadrotor UAV

Keywords

UAV; Feedback linearization control; LESO

Abstract

In this paper, attitude controller and position controller are investigated for a small quadrotor unmanned aerial vehicle (UAV). A feedback linearization method is proposed to design attitude tracking control of UAV, with linear extended state observer (LESO) to estimate and compensate for the disturbances. Meanwhile, a saturated control law with arc tangent functions is adopted as the position controller. Then the convergence of LESO, stability of attitude system and position system are proved. Finally, experiment results are given to illustrate the effectiveness of the proposed method.



How NOKOV Contributes to the Research

The quadrotor position experimental platform is shown in Fig.3b, which exits three parts: Nokov motion capture system, quadrotors and communication system. Nokov motion capture system gets the position of the quadrotor by Nokov markers (reflective balls), the quadrotor can get the information of its own precise position via communication system (WiFi). Based on the quadrotor position experimental platform, a hovering test is done to verify the effectiveness of position system control. The parameters of position system are given in Table.2. In this case, the experimental results are shown in Fig.6 and Fig.7. Fig.6 shows the position tracking performance and Fig.5 shows the inputs of position system. From results, the errors of position from x and y axes are less than 0.1 meter and inputs are reasonable. In other word, it can accurately track the reference position.

References Jiang H, Xia Y, Hu R, et al. A feedback linearization and saturated control structure for quadrotor UAV[C]//2019 Chinese Control Conference (CCC). IEEE, 2019: 8277-8282.

Link <https://ieeexplore.ieee.org/abstract/document/8866172>

On-demand Precise Tracking for Energy-constrained UAVs in Underground Coal Mines

Keywords

Unmanned aerial vehicle; unscented Kalman; filter; energy-efficient; on-demand precise tracking; coal mines.

Abstract

The precise tracking of unmanned aerial vehicle (UAV) in coal mines remains a challenging task that faces strict energy constraints. The existing approaches mainly focus on achieving better accuracy but ignore energy consumption. This paper proposes an on-demand precise tracking (OPT) framework based on the fusion of the ultra-wideband (UWB) and inertial measurement unit (IMU). The fusion unscented Kalman filter (fusion UKF) is devised for energy-efficient tracking with customized performance. First, OPT provides an adaptive adjustment mechanism of UWB signal transmission to trade-off between accuracy and energy consumption for UAV local positioning in coal mines. Second, we propose an on-demand trigger algorithm for UAV remote tracking to determine whether the data is transmitted to the ground server. Especially, the on-demand trigger condition is ameliorated owing to the existence of packet drops in complicated coal mines. Finally, extensive simulations in the Gazebo Platform and field experiments utilizing P440 UWB nodes in the indoor and coal mine laboratories have been conducted to verify the feasibility. Results show that OPT is efficient, practical and balances the performance and energy consumption under appropriate parameters. Compared with the existing event-triggered extended Kalman filter (ET-EKF) scheme, the accuracy is improved by 10.3%, and the communication rate is contracted by 11.4% in the coal mine environment.



Fig. 11. The experiments in the indoor laboratory.

How NOKOV Contributes to the Research

Since the ground truth of position cannot be obtained directly in experiments, an optical motion capture system (NOKOV VICON) that provides almost minor positioning errors is utilized to provide the reference position. We arrange eight cameras forming a square to capture the position of the UAV in the indoor laboratory.

References Zhang K, Chen P, Ma T, et al. On-Demand Precise Tracking for Energy-Constrained UAVs in Underground Coal Mines[J]. IEEE Transactions on Instrumentation and Measurement, 2022, 71: 1-14.

Link <https://ieeexplore.ieee.org/abstract/document/9694638>

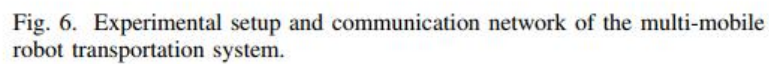
Motion Planning of Multi-Robots Object Transport with Deformable Sheet

Keywords

Multi-robot manipulation; Path planning for multiple mobile robots

Abstract

Using a deformable sheet to handle objects is convenient and found in many practical applications. Modeling of deformable sheet-object interactions and robot formation are among the main challenges in multi-robot object manipulation through deformable sheets. Motion planning of robotic team is critical for object manipulation quality. We present a computational robots-sheet-object model and the dual-capability robotic motion planner for optimal trajectory. A computational virtual variable cables model (VVCM) is presented to simplify the modeling of the deformable sheet with an arbitrary number of used robots. The motion planner is first formulated as a formation optimization problem. A path generation algorithm is then employed for obstacle avoidance and crossing features. The motion planner successfully produces an obstacle crossing path in complex scenarios where other benchmark planners fail. The effectiveness and versatility of the motion planner are verified and demonstrated by a robotic team of three robots. We further extend the results to a larger number of robots to illustrate the algorithms and analysis.



The position of robots and object were acquired at a rate of 60 Hz by the NOKOV motion capture system with 8 cameras.

Link <https://arxiv.org/abs/2111.09046>

Risk-aware Trajectory Sampling for Quadrotor Obstacle Avoidance in Dynamic Environments

Keywords

Collision Avoidance; Motion and path planning; Aerial Systems; Perception and Autonomy

Abstract

Obstacle avoidance of quadrotors in dynamic environments, with both static and dynamic obstacles, is still a very open problem. Current works commonly leverage traditional static maps to represent static obstacles and the detection and tracking of moving objects (DATMO) method to model dynamic obstacles separately. The dynamic obstacles are pretrained in the detector and can only be modeled with certain shapes, such as cylinders or ellipsoids. This work utilizes our dual-structure particle-based (DSP) dynamic occupancy map to represent the arbitrary-shaped static obstacles and dynamic obstacles simultaneously and proposes an efficient risk-aware sampling-based local trajectory planner to realize safe flights in this map. The trajectory is planned by sampling motion primitives generated in the state space. Each motion primitive is divided into two phases: short-term planning with a strict risk limitation and relatively long-term planning designed to avoid high-risk regions. The risk is evaluated with the predicted future occupancy status, represented by particles, considering the time dimension. With an approach to split from and merge to global trajectories, the planner can also be used with an arbitrary preplanned global trajectory. Comparison experiments show that the obstacle avoidance system composed of the DSP map and our planner performs the best in dynamic environments. In real-world tests, our quadrotor reaches a speed of 6 m/s with the motion capture system and 2 m/s with everything computed on a low-price single board computer.



Fig. 8. The hardware structure of the quadrotor.

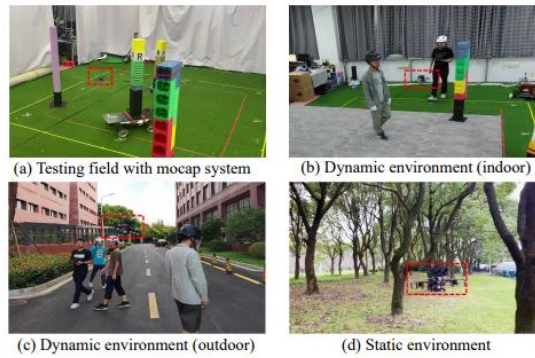


Fig. 9. Testing scenarios in the real world. In subplot (a), the position of the quadrotor and the position of obstacles are given by the motion capture system. The dynamic obstacles are two foam pillars mounted on mobile robots. Subplot (b) and (c) show two dynamic environments. Subplot (d) shows a static woods.

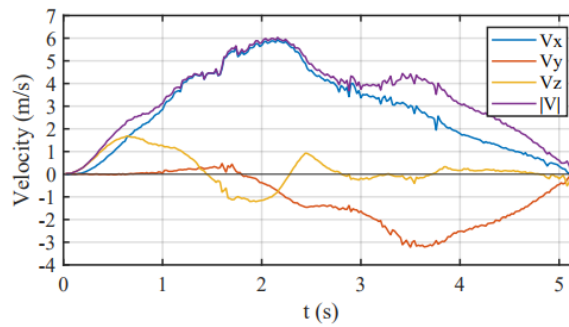


Fig. 10. The velocity curve of a test with the motion capture system.

How NOKOV Contributes to the Research

Tests with the motion capture system: As is shown in Fig. 9 (a), two static obstacles and two dynamic obstacles were in the testing field. The dynamic obstacles moved with a constant velocity, about 1m/s for the faster one and 0.5m/s for the slower one. The Nokov motion capture system was used to capture the position of the quadrotor and the obstacles. The future occupancy status of the dynamic obstacles was predicted with the constant velocity model. The quadrotor was able to fly rapidly and safely in the tests. The maximum speed reached 6m/s. The corresponding velocity curve is shown in Fig. 10.

References Chen G, Peng P, Zhang P, et al. Risk-aware Trajectory Sampling for Quadrotor Obstacle Avoidance in Dynamic Environments[J]. arXiv preprint arXiv:2201.06645, 2022. Link <https://arxiv.org/abs/2201.06645>

Obstacle Crossing by Multi-mobile Robots in Object Transportation with Deformable Sheet

Keywords

Multi-robot systems; Intelligent transportation systems; Path planning for multiple mobile robots;

Abstract

Multi-robot transportation (MRT) is to transport the object to the destination by the cooperation of multiple robots. In the process of object transportation, obstacle avoidance is an indispensable feature. In traditional local planners, obstacles are usually considered insurmountable, so the robot team bypasses the obstacles as a whole. However, many obstacles can be crossed in real situation. Studying the obstacle crossing ability of robot team can improve the efficiency of MRT and increase the planning success rate in complex environment. Inspired by the patient transfer through bed sheet, this paper focuses on the object transportation by multi-mobile robots with deformable sheet. A new local planner with obstacle crossing capability is proposed, which consists of three parts: deformable sheet modeling, formation optimization and local path generation. It can successfully find an obstacle crossing path in complex scenarios where other planners fail. The effectiveness and the versatility of the planner is verified by a case study with three mobile robots in the experiment and a simulation with four robots.

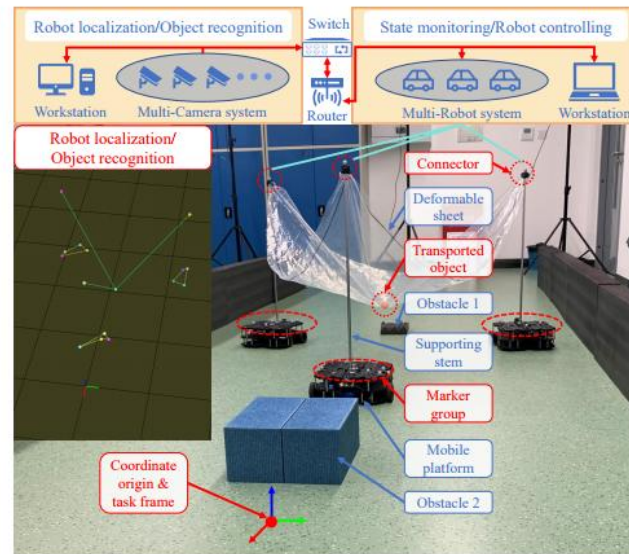


Fig. 7. Experimental setup and communication network of the multi-mobile robot transportation system.

How NOKOV Contributes to the Research

Three robots (turtlebot3 waffle-pi) are applied in this experiment. The transported object is a billiard ball. The position of robots and object can be acquired at a rate of 60 Hz by the NOKOV motion capture system with 8 cameras. The deformable sheet are fabric cloth and plastic cloth, the latter is convenient for the camera to capture the position of the robot in real time. Each side length of the original sheet is 1600 mm, and the height of each vertex is 790 mm. The experimental environment is a corridor (2 meters wide, 6 meters long) with two different obstacles in the middle. One has a radius of 100 mm and a height of 50 mm, and the other has a radius of 200 mm and a height of 200 mm. The simulation result is obtained by MATLAB 2019b, and the whole control architecture is implemented based on robot operating system (ROS). A remote laptop (Intel i7-4710HQ CPU with four cores and 8 GB RAM) is set up for state monitoring and robot controlling. Experimental setup and communication network of the system are shown in Fig. 7.

References Hu J, Liu W, Zhang H, et al. Obstacle Crossing by Multi-mobile Robots in Object Transportation with Deformable Sheet[J]. arXiv preprint arXiv:2111.09046, 2021.
Link <https://arxiv.org/abs/2111.09046>

Stabilizing Angle Rigid Formations With Prescribed Orientation and Scale

Keywords

Bearing measurement; formation control; multi-agent systems; prescribed orientation and scale;

Abstract

Angle rigid formations have the advantage of requiring only bearing measurements in their implementation. However, the capability of controlling the orientation and scale of these formations has not been explored. This undetermined orientation and scale can degrade the robustness of the formation against measurement noise. To maintain both advantages of requiring less sensor measurements and sustaining robustness against measurement noise, this paper aims to achieve a desired angle rigid formation while simultaneously controlling its orientation and scale. We first design a formation algorithm for the first three agents to achieve a desired triangular formation with prescribed orientation and scale. Using the control gain design technique, we then design formation control algorithms for the remaining agents such that the overall desired formation can be achieved under a vertex addition operation. We present the role of generic property from angle rigidity for the formation's stability analysis. We also highlight that with one additional relative position measurement or two additional communication channels, the local convergence to the desired formation can be improved to a global convergence. Experiments are conducted to validate the theoretical results and the advantages are highlighted in comparison with other two formation control laws.

How NOKOV Contributes to the Research

In this section, we validate the results of Theorems 1-4 using four wheeled robots to achieve a desired rectangular formation. The size of each robot is 60cm length, 46cm width, and 46cm height. Each robot has four wheels and is controlled by an on-board computer. Since wheeled robots are unicycles, we apply feedback linearization [20], [21] towards a reference point that is inside of the robot to obtain the single-integrator dynamics (1). The measurements of relative positions or bearings among robots are captured by the NOKOV Mocap system sampling at the rate of 120 Hz. Each robot will calculate the control input along its X-axis and Y-axis according to the designed controllers, then apply it to the robots.

References Chen L, Yang Q, Shi M, et al. Stabilizing Angle Rigid Formations With Prescribed Orientation and Scale[J]. IEEE Transactions on Industrial Electronics, 2021, 69(11): 11654-11664.

Link <https://ieeexplore.ieee.org/abstract/document/9583880>

Event-Triggered Control for Inter-Vehicle Communications of Cyber-Physical Transportation Systems

Keywords

Cyber-Physical Transportation Systems, reliability, event-triggered broadcast control

Abstract

In Cyber-Physical Transportation Systems (CPTSs), inter-vehicle communication is a basis of traffic safety and vehicle applications. Vehicles need to transmit motion data with each other via wireless networks. Despite many strategies that make the vehicle network more reliable and efficient under the noise, the separate design of estimation and communication makes them incapable of ensuring state received from multiple vehicles around reliability. To address the gap, this paper proposes an event-triggered broadcast control strategy that fuses communication scheduling with estimators. The estimator is designed to estimate the information transmitted by the strategy and the event-triggered condition is designed based on estimation states. To verify the performance of the estimator and the tracking accuracy of the broadcast strategy, an autonomous driving simulation platform is built. The communication rates of our strategy in the traffic flow are evaluated by SUMO and NS3. The experiment result shows that the strategy can dynamically adjust the event-triggered conditions according to the accuracy of the vehicle estimator and the network transmission rate improve significantly.



Fig. 5: Layout of the platform.

How NOKOV Contributes to the Research

In our platform, the position and yaw angle of the vehicle can be obtained by the infrared high-precision motion capture system (Nokov) and Robot state sensors include inertial measurement unit (IMU) and Odometer. The Nokov updates every 300ms. the odometer update at 10Hz and the IMU updates at 100 Hz. According to the structure of the vehicle and the data collected on our platform, we choose the secant model [9] as the time propagation equation.

References Jiang F, Liu X D, Zhang L. Event-triggered control for inter-vehicle communications of cyber-physical transportation systems[C]//2021 40th Chinese Control Conference (CCC). IEEE, 2021: 4709-4714.

Link <https://ieeexplore.ieee.org/abstract/document/9550400/>

Integrated Positioning System of Unmanned Automatic Vehicle in Coal Mines

Keywords

Extended Kalman filter (EKF), inertial navigation system (INS), odometer, ultra-wideband (UWB) positioning, underground coal mine.

Abstract

Automatic positioning technology of underground vehicles has become the key technology for the intelligent development of coal mines. To improve the comprehensive positioning accuracy of unmanned automatic vehicles in coal mines, an 18-D model of an odometer-aided inertial navigation system (INS) and a positioning model of an extended Kalman filter-based ultra-wideband (UWB) are established based on the vehicle kinematics equation. A tight integrated state estimation model is proposed to restrain the long-time drift of the INS and enhance the instability of the UWB system. A local positioning reference system based on an infrared motion capture system is established for the first time. Experimental tests show that the average positioning error of the proposed algorithm is 6.03 cm in the forward direction, and the root-mean-square error increased by 36.09% and 38.24% compared with those of the INS and UWB system, respectively. It is experimentally verified that the integrated positioning system achieves decimeter-level positioning accuracy in a coal mine roadway environment.

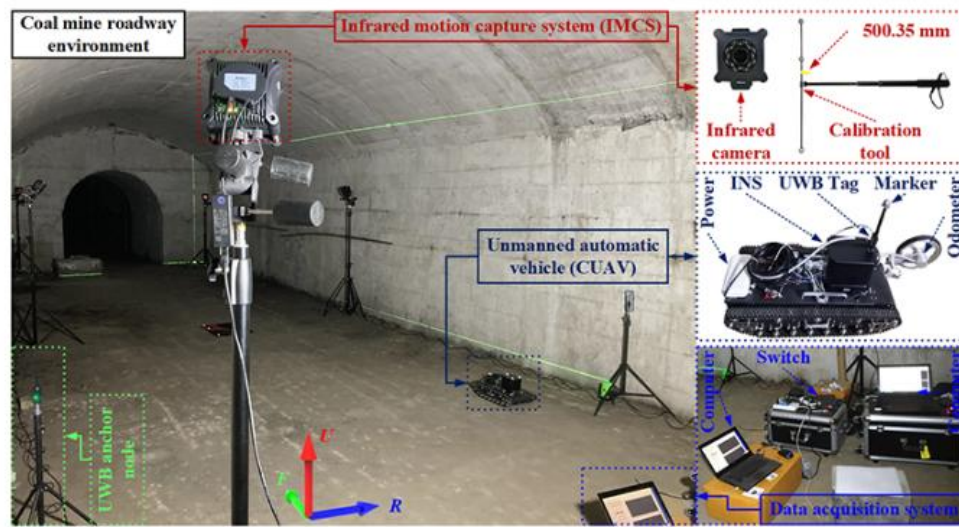


Fig. 10. Field experimental layout of CUAV positioning system.

How NOKOV Contributes to the Research

To verify the actual positioning performance of the proposed method in an underground roadway, a laser inertial sensor (FOG INS300), a UWB positioning module (LinkTrack P), and an absolute odometer (CAS38R12E06R4B) were used to build the underground positioning experimental system. Particularly, the bias and noise of the gyroscope were $0.3^{\circ}/h$ and $0.03^{\circ}/\sqrt{h}$, respectively, whereas those of the accelerometer were 0.07 mg and $0.013\text{ mg}/\sqrt{\text{Hz}}$, respectively. The experimental system coordinate frame was based on the UWB frame. The forward and right directions of the INS initial frame were aligned to the UWB frame using a laser range finder. The actual trajectory of the CUAV was captured by an IMCS comprising several infrared cameras (NOKOV Mars 2H). Time synchronization was realized through the time stamp information of the UWB system with the highest update frequency.

References Cui Y, Liu S, Yao J, et al. Integrated positioning system of unmanned automatic vehicle in coal mines[J]. IEEE Transactions on Instrumentation and Measurement, 2021, 70: 1-13.

Link <https://ieeexplore.ieee.org/abstract/document/9440987/>

Spatial-temporal constrained particle filter for cooperative target tracking

Keywords

Cooperative target tracking; Accumulative error; Particle filter; Error ellipse resampling ; Spatial temporal constraints

Abstract

Localization is one of the most important topics of cyber physical system. In the last decades, much attention has been paid to the precise localization and performance evaluation in wireless sensor networks. Inertial-measurement-unit and Time-of-Arrival fusion is a state-of-the-art method to solve the accumulative error and drifting problem faced by sole inertial-measurement-unit positioning and navigation. Network cooperative technology could effectively suppress the accumulative error. This paper presents a spatial-temporal constrained particle filter algorithm for cooperative target tracking, so as to solve the problem of multi-target high-precision position tracking in complex and highly dynamic environments. Firstly, we propose an error-ellipse-resampling particle filter method. In the resampling stage of the particle filter, error ellipses with different confidence probabilities would be established with the use of the known estimated center and confidence scale, to achieve hierarchical resampling optimization based on the geometrical position of particles. As for cooperative tracking, an optimization method of state estimation based on spatial distance constraint is proposed, so that the Bayesian filter can benefit from spatial information and achieve cooperative tracking of spatial-temporal information fusion. Numerical experimental results show that the proposed error-ellipse-resampling particle filter could decrease the growth rate of cumulative errors and reach a positioning accuracy of 1.05 m, multi-target cooperative error-ellipse-resampling particle filter can effectively eliminate the cumulative error and achieve a positioning accuracy of 0.24 m.

How NOKOV Contributes to the Research

*To verify the effectiveness, practical experiments are conducted to compare the proposed method with above-mentioned state-of-the-art methods. A total of 10 navigation robots are programmed to travel randomly by applying the collision-avoidance algorithm (Thrun, 2002), and each robot maintains a distance-ranging frequency of once a second with its neighboring nodes. A high precision optical motion tracking system (Mars 2H) (<https://www.nokov.com/en/>, 2020) is deployed so as to provide the ground truth measurements. Considering the size of the tracking target and the deployment requirements of optical devices, an open field covered 10 m*10 m is chosen as a testing scenario.*

References Xu C, Wang X, Duan S, et al. Spatial-temporal constrained particle filter for cooperative target tracking[J]. Journal of network and computer applications, 2021, 176: 102913.

Link <https://www.sciencedirect.com/science/article/abs/pii/S1084804520303726>

Distributed H_∞ -Constraint Robust Estimator for Multi-Sensor Networked Hybrid Uncertain Systems

Keywords

Consensus weight strategy, Distributed H_1 -constraint robust estimator, Multi-sensor networked hybrid uncertain systems, Network-induced uncertainties.

Abstract

Due to the complex communication channel environment, uncertainties, including random measurement delays, missing measurements and uncertain measurement noise variances, often exist in multi-sensor networked hybrid uncertain systems. These uncertainties will lead to poor accuracy and robustness for traditional information fusion methods. Hence, this paper presents a distributed robust estimator based on the H_∞ constraint to solve the mentioned issues in sensor networked systems. The nonlinear pseudomeasurement model is first constructed with Bernoulli distributed random variables to describe the mentioned uncertainties in the communication channels. Then, a sufficient condition with a local optimal upper bound on estimation error covariance is derived rigorously with respect to the robustness definition, which is based on the H_∞ constraint inequality to improve the robustness of state estimation. Furthermore, the proposed estimator is proven to be stable through boundedness analysis for the upper bound matrix. The numerical simulation and the target tracking experiment are finally applied to evaluate the effectiveness of the novel fusion estimator.



Fig. 4. Experimental platform for target tracking.

How NOKOV Contributes to the Research

To verify the efficiency of the proposed estimator, the experimental platform for multi-sensor target tracking displayed in Fig. 4 is established. As shown in Fig. 4, the devices of the experimental platform include the assembled test car, the NOKOV motion capture system (six Mars 2 cameras), the 24G millimeter-wave radar sensors, PCs, etc. The camera parameters are a detection range of 01m and precision of 0.3mm. Radar parameters are detection range 0.13.5m and precision 50mm. As the tracked target, the test car (system model can refer to [38]) moves in a circular motion under electromagnetic drive, and concurrently, the NOKOV system (submillimeter accuracy) captures the position as the target's real state vector.

References Xia J, Gao S, Li G, et al. Distributed H_{∞} -Constraint Robust Estimator for Multi-Sensor Networked Hybrid Uncertain Systems[J]. IEEE Transactions on Network Science and Engineering, 2021, 8(4): 3335-3348.

Link <https://ieeexplore.ieee.org/abstract/document/9540386>

Obstacle-Aware Topological Planning over Polyhedral Representation for Quadrotors

Abstract

In this paper, we propose a novel mapping-planning framework for autonomous quadrotor navigation. First, a polyhedron-based mapping algorithm is presented to fully exploit the information of the onboard sensor data. Polyhedra are generated to approximate the segmented clusters of occupied voxels. Then, customized data structures are designed to extract information for motion planning in real time. With complete knowledge of the shape, position, and number of the observed obstacles, we can conveniently generate smooth trajectories with sufficient obstacle clearance along the most desired direction. Before searching for the initial path, a local topological graph is constructed to keep the path expanding in the most favorable topology class. The following path search is segmented based on the graph vertices, which allows fast convergence. The refined trajectory is obtained after smoothing, and large deviations are penalized in the formulated optimization problem to preserve the original clearance. Finally, we analyze and validate the proposed framework through extensive simulations and real-world quadrotor flights.

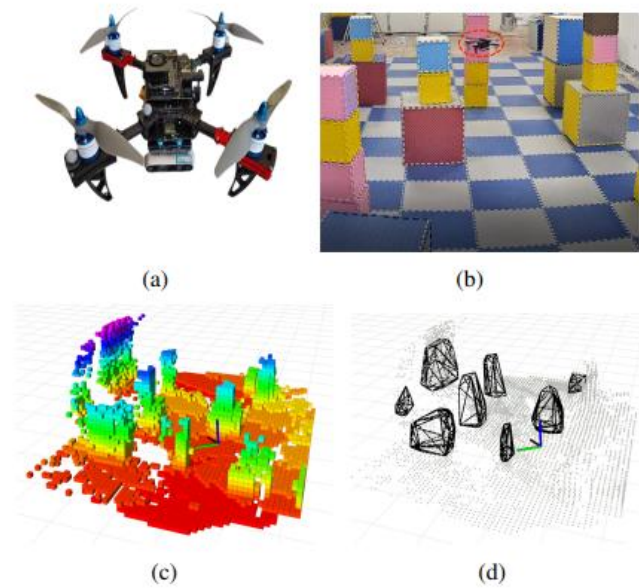


Fig. 8. (a) Hardware setting of our custom-built quadrotor. (b) A snapshot of the quadrotor crossing the unknown environment, and the average flight speed is $1.5m/s$ (c) The original occupancy grid map. (d) The polyhedral environment within the sensing range. The surface noise of the original map is covered by the generated planes, resulting in smoother trajectories.

How NOKOV Contributes to the Research

The custom-built quadrotor platform (shown in Fig. 8(a)) is equipped with a JETSON Xavier NX for onboard computing, a Pixhawk flight controller for stabilization, and a forward-facing RealSense D435i to provide depth images. The localization information is obtained from the NOKOV motion capture system. Real-world flight experiments are conducted in an unknown indoor environment as shown in Fig. 8(b). The original occupancy grid map and the corresponding polyhedron map are shown in Fig. 8(c) and Fig. 8(d), respectively. We abandon the occupied voxels below $0.15m$ and above $2.3m$ to segment the floor and the ceiling. The height constraint is posed during the path search. The collision check is implemented by regarding the quadrotor as a sphere with a radius of $0.2m$.

References G. H. Qin, Z. Meng, W. Meng, X. Chen, H. Sun, F. Lin, and M. H. Ang, "Autonomous exploration and mapping system using heterogeneous UAVs and UGVs in GPS-denied environments," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1339–1350, Feb. 2019.

Link <https://ieeexplore.ieee.org/document/10161295>

Self-organized multi-target trapping of swarm robots with density-based interaction

Abstract

The task of multi-target trapping in swarm robots can often be solved by global shape planning and target assignment, but it still remains a challenge to achieve fully self-organized multi-target trapping behavior based on local information. In this paper, inspired by the concept of spatial density in physics and biology, we proposed a novel density-based method to enable the swarm robots to entrap multiple targets with either single-ring, multi-ring or multi-subgroup formation in a distributed and self-organized way while neither communication among robots nor encirclement function is required. Each robot's local spatial density is considered as the main clue for the individual's motion decision-making and the enclosed configurations emerge from such individual-level interactions rather than being explicitly designed. Numerical simulations and real robotic experiments are conducted to validate the effectiveness of the proposed method. The results show that the proposed self-organized trapping method allows a swarm of robots to entrap multiple moving targets in a stable, flexible, noise-tolerate and size-scalable fashion.

How NOKOV Contributes to the Research

We employed a motion capture system manufactured by NOKOV to track the positions and headings of all robots. The global information is broadcasted to all robots via a customized communication protocol based on a 2.4 G RF module. Each working robot makes its own decision according to the local information extracted from the global message. That is, the setup of our experiment is centralized-perception and decentralized-decision-making. This pseudo-distributed structure is flexible in conducting a wide range of swarm robotic experiments., see Fig. 11. The experiments were conducted within an area of 5140mm × 4700mm with the origin located at the center of the area. We employed a motion capture system manufactured by NOKOV to track the positions and headings of all robots. The global information is broadcasted to all robots via a customized communication protocol based on a 2.4 G RF module. Each working robot makes its own decision according to the local information extracted from the global message. That is, the setup of our experiment is centralized-perception and decentralized-decision-making. This pseudo-distributed structure is flexible in conducting a wide range of swarm robotic experiments.

References Garnier S, Gautrais J, Theraulaz G (2007) The biological principles of swarm intelligence. *Swarm Intell* 1(1):3–31.

Link <https://link.springer.com/article/10.1007/s40747-023-01014-6>

Vision-Based Autonomous Landing of a Hybrid Robot on a Powerline

Abstract

In recent years, various types of inspection robots have been developed to automate powerline inspection. The hybrid robot combines the advantages of climbing and flying robots and has a promising prospect in powerline inspection. But landing a hybrid robot on the target powerline among multiple ones is challenging. Flights require robust detection of powerlines and stable tracking of the target powerline. We propose a complete solution for the autonomous landing of a hybrid robot on a powerline. First, a special feature extraction operator and the corresponding density-based feature recognition algorithm are designed to detect multiscale powerlines. Second, a binocular vision-based depth estimation method for the landing point in the powerline is described. Third, two spatio-temporal dictionaries are established to track the target one in multiple powerlines. Meanwhile, landing strategies and control methods are presented to achieve a stable landing task. Finally, a hybrid robot is designed to validate the proposed method. The experiment results demonstrate the accuracy of the powerline detection and depth estimation algorithm, as well as the effectiveness of the robot in tracking and landing tasks.

How NOKOV Contributes to the Research

Depth accuracy is crucial for the landing task. An optical NOKOV Mocap was used to evaluate the accuracy of the proposed ranging method. Fig. 11 shows the relative height between the powerline and camera measured by our method and Mocap, respectively. Since the measurement accuracy of the stereo camera decreases with distance, we divide the measurements into four intervals to calculate the errors, respectively. The results of the data analysis are shown in Table II. When the depth is less than 0.8 m, the mean absolute error (MAE), root mean squared error (RMSE) and max error are less than 5.09, 6.38, and 18.05 mm, respectively. Considering the margin brought by the guiding mechanism, the depth accuracy can meet the requirements of the actual landing task.

References Y. Zhai, X. Yang, Q. Wang, Z. Zhao, and W. Zhao, "Hybrid knowledge R-CNN for transmission line multifitting detection," IEEE Trans. Instrum. Meas., vol. 70, pp. 1–12, 2021.

Link https://ieeexplore.ieee.org/search/searchresult.jsp;jsessionid=fOppHzyKKtEoAZ-KPQFW90_zrb0DAdajisTsiX1FUb8hzSY4JdLP!-54960200?queryText=Vision-Based%20Autonomous%20Landing%20of%20a%20Hybrid%20Robot%20on%20a%20Power%20line&newsearch=true&type=alt2

Evolution of UAV landing structures in the bistable space of Kresling origami structures

Abstract

Landing gear plays an important role in enabling unmanned aerial vehicles to engage in complex missions and protect expensive airborne sensors. Most existing landing gears are designed to satisfy only specific maneuvers in predefined scenarios. Although effective in predefined scenarios, these landing gears cannot guarantee the adaptiveness in multiple scenarios. A promising solution is to build a model automatically generated based on the actual requirements, so that the expected model structure can be obtained through a constraint generation process. However, translating the actual requirements into reasonable design constraints is complex. For this purpose, we propose a landing structure design process in the bistable space of the Kresling tube. Furthermore, we use a combination of graph search and controller to optimize the landing structure. We validate the bistatic and graph search combined design structure in a realistic environment and test the mechanisms flexibility and assisted landing capability. Experiments show that the structure combining spatial bistability with a graph search approach can safely and successfully land a quadrotor under various conditions while enhancing the UAV's shock absorption during landing. A video of the experiments is available at <https://youtu.be/oLeERbCij0>.

TABLE III
EXPERIMENTAL EQUIPMENT DETAILS

Drone wheelbase	250(mm)
Onboard computer	Intel NUC
Analysis Tool	NOKOV optical motion capture system

How NOKOV Contributes to the Research

This section details the experimental procedure, experimental setup, and experimental results to validate the bistable design for the verification landing. To reduce the influence of the experimental subjects on the results, we adjusted the weight of all three sets of equipment to 1.44 (kg), where the weight of the bistable design and the non-bistable design structure is the same; both are 383 (g). Additional detailed experimental equipment is shown in Table III. Fig. 9. shows the velocity and acceleration changes of different landing systems within 2.5 seconds after contacting the ground in the indoor experiment, and Fig. 10. shows the testing process of varying landing systems in outdoor scenarios. The experimental results show that the bistable landing structure performs better in the landing process in both indoor and outdoor scenarios.

References J. Bass and A. L. Desbiens, "Improving multirotor landing performance on inclined surfaces using reverse thrust," IEEE Robotics and Automation Letters, vol. 5, no. 4, pp. 5850–5857, 2020.

Link <https://ieeexplore.ieee.org/document/10048537>

Mixed-Integer and Conditional Trajectory Planning for an Autonomous Mining Truck in Loading/Dumping Scenarios: A Global Optimization Approach

Abstract

Trajectory planning for a heavy-duty mining truck near the loading/dumping sites of an open-pit mine is difficult. As opposed to trajectory planning for a small-sized passenger car in a parking lot, trajectory planning for a heavy-duty mining truck involves complex factors in vehicle kinematics and environment. These factors make the concerned trajectory planning scheme a mixed-integer nonlinear program (MINLP) incorporated with conditional constraints (denoted as C-MINLP). MINLP solvers can neither deal with conditional constraints nor find global optima in real time. Instead of solving the C-MINLP directly, we build a from-coarse-to-fine framework so that the coupled difficulties (the mixed integral variables, conditional constraints, and the demand for global optimality) are divided and conquered. At the coarse search stage, a global-optimality-enhanced hybrid A* search algorithm is proposed to find a near-optimal coarse trajectory with the mixed integral variables, conditional kinematic constraints, and global optimality considered. The coarse trajectory is further polished at the refinement stage, wherein the nominal C-MINLP is simplified as a small-scale NLP. The solution to the NLP is an optimized trajectory, which does not violate the complex constraints in the nominal C-MINLP. This indicates that conversion from the C-MINLP to an NLP is efficient with the help of a high-quality coarse trajectory.



Fig. 7. An indoor workspace and a small-scale autonomous vehicle platform deployed for the field tests.

How NOKOV Contributes to the Research

Experiments were conducted on a small-size autonomous vehicle platform in a 1.75m 1.20m × indoor workspace (Fig. 7). Six infrared sensors from the NOKOV Motion Capture System were deployed for obstacle perception and ego vehicle localization. Concretely, reflective marking points were placed on the surface of each stone-like obstacle and the top of the autonomous vehicle (Fig. 8). The infrared light sources would emit infrared beams, which are captured by the infrared sensors after being reflected by the marking points. Through this, the location of each marking point is identified. The localization information was collected on a desktop PC, where trajectory planning was implemented. The trajectory planning module ran once to generate a trajectory before the ego vehicle began to move. The planned trajectory, together with the localization information would be sent from the PC to the ego vehicle platform via ZigBee communication technology. PID and pure pursuit controllers were used for longitudinal and lateral tracking under the frequency of 10 Hz, respectively.

References X. Zhang, A. Guo, Y. Ai, et al., "Real-time scheduling of autonomous mining trucks via flow allocation-accelerated Tabu search," IEEE Transactions on Intelligent Vehicles, doi: 10.1109/TIV.2022.3166564.

Link <https://ieeexplore.ieee.org/document/9919391>

Accurate integrated position measurement system for mobile applications in GPS-denied coal mine

Keywords

Inertial navigation system; Multi-sensor fusion; Roadheader positioning; Underground coal mine; Unscented Kalman filter

Abstract

The automatic positioning of underground mobile applications plays a crucial role in enabling intelligent coal mining. However, due to the diverse kinematics and dynamics of these applications, various positioning methods have been proposed to match different targets. Nonetheless, the accuracy and applicability of these methods still fall short of meeting the requirements for field applications. Based on the vibration characteristics of underground mobile devices, a multi-sensor fusion positioning system is developed to enhance the accuracy of positioning in long and narrow global positioning system denied (GPS-denied) underground coal mine roadways. The system combines inertial navigation (INS), odometer, and ultra wide band (UWB) technologies through extended Kalman filter (EKF) and unscented Kalman filter (UKF). This approach enables accurate positioning by recognizing target carrier vibrations and facilitating fast conversion between multi-sensor fusion modes. The proposed system is tested on both a small unmanned mine vehicle (UMV) and a large roadheader, demonstrating that UKF enhances stability for roadheaders with strong nonlinear vibrations while EKF is more suitable for flexible UMVs. Detailed results confirm that the proposed system achieves an accuracy level of 0.15 m, meeting most coal mine application requirements.

How NOKOV Contributes to the Research

To investigate the positioning characteristics of the two different scenarios, we arrange the proposed positioning system on UMV and roadheader separately. The inertial sensor is FOG IMU300 (Fogsins Technology Co., Ltd.), the UWB sensor is the LinkTrack P module (Nooploop Technology Co., Ltd.), and the absolute odometer is CAS38R12E06R4B (Shanghai Qiyi Electrical & Mechanical Equipment Co., Ltd.). The three-dimensional piezoelectric vibration sensor is 1A314E (DONGHUA Test Technology Co., Ltd.), and the infrared motion capture camera is Mars 2H (Beijing NOKOV Science & Technology Co., Ltd.). The sensors are arranged as Fig. 7 and the parameters are presented in Table 1.

References Wang G, Liu F, Meng X, Fan J, Qunying WU, Ren H, et al. Research and practice on intelligent coal mine construction (primary stage). Coal Sci Technol 2019;47(8):1–36.

Link <https://www.sciencedirect.com/science/article/abs/pii/S0019057823001830>

CSI-Based Calibration Free Localization with Rotating Antenna for Coal Mine

Keywords

Indoor localization; Channel state information; Smartphone

Abstract

Accurate mine localization has always been an urgent demand. With the widespread emergence of cheap wireless devices in our life, channel state information (CSI) based indoor localization has attracted people's attention. In the existing CSI-based localization methods, fingerprint-based localization needs to build a huge database with poor robustness, and the methods based on angle of arrival (AOA) need specific antenna arrays and complex phase calibration. Because mobile devices such as smartphones only have one or two antennas, and the antennas inside the devices cannot form a receiving array, the conventional AOA estimation system cannot be deployed on such mobile devices. Therefore, we propose an AOA estimation method using rotating AP antennas. This method only needs one antenna at the receiving end to work and does not need to calibrate the phase. Firstly, the CSI phase of the rotating antenna is collected and the phase difference is calculated. Then, empirical mode decomposition (EMD) algorithm based on mutual information is used to remove the noise, and subcarrier selection is used to reduce the multipath effect in the data processing part. Finally, the AOA is estimated according to the relative position of the antenna. We conduct extensive experiments in mine and laboratory scenarios, and the median localization errors in mine and laboratory are 0.4 m and 0.6 m, the median angle errors are 4.2° and 5.2° respectively.



Fig. 7. Experimental scenarios

How NOKOV Contributes to the Research

In the experiment, two TL-wdr 5610 are used as access points, and one nexus 6p with nexmon [7] patch and bcm4358 wifi chip is used as receiver object. TL-wdr 5610 works in 2.4 GHz band, 20 MHz bandwidth, and has two omnidirectional antennas. Eight Nokov optical 3D motion capture cameras were used to measure the ground truth. We choose the laboratory and mine for the experiment, and the experimental scene is shown in the Fig. 7.

References IEEE standard for a precision clock synchronization protocol for networked measurement and control systems. IEEE STD 1588-2019 (Revision of IEEE STD 1588-2008), pp. 1–499 (2020). <https://doi.org/10.1109/IEEESTD.2020.9120376>.

Link https://link.springer.com/chapter/10.1007/978-3-030-85928-2_21

Collective fission behavior in swarming systems with density-based interaction

Keywords

Swarming systems; Robotic swarm; Density field interaction; Collective fission behavior; Spontaneous split

Abstract

In this paper we revisit the self-organized collective behavior of swarms with density field interaction that is inspired from the model of smoothed particle hydrodynamics. For a homogeneous swarming system a novel collective fission behavior is seen to emerge where equal to or more than two sub-clusters spontaneously come from a single connected cluster. The focus of this study is on the analysis of collective fission behavior in trigger conditions, function manners and dynamic characteristics. For the first time, we are able to predict the stability of a single connected cluster, the trigger conditions of collective fission behavior, the number of sub-clusters and the group size of each sub-cluster. The analysis results also show that the collective fission behavior is irreversible by changing the control parameters and the group size. However, it is reversible when the sensing range is enlarged to encompass all sub-clusters. These characteristics are evaluated in simulations as well as on the real swarm robotic system. The study of collective fission behavior provides an elegant insight for how self-organized segregation emerges in biological swarms and artificial systems.

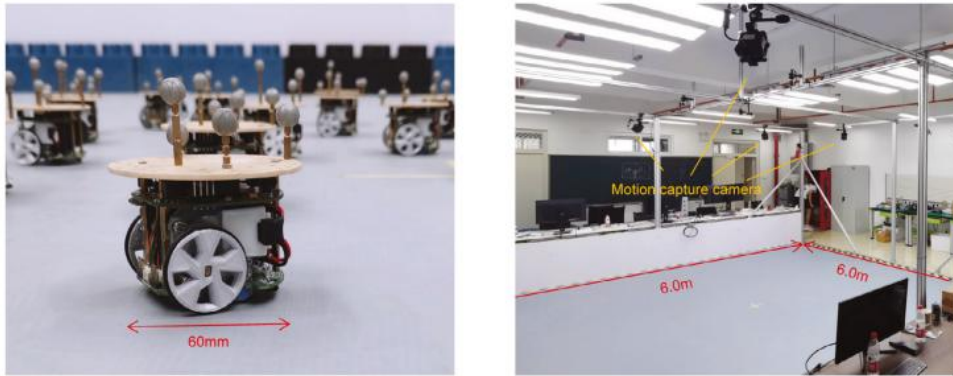


Fig. 8. SwarmBang robot and the overhead tracking system.

How NOKOV Contributes to the Research

Firstly, each robot needs to obtain the heading and position information of its local neighbors, which means the robot should have a localized sensing ability. To achieve this, we use the overhead tracking system, which can estimate the headings and positions of all robots in a global reference frame by employing a motion capture system manufactured by NOKOV. The information of headings and positions that contains all robots then is broadcasted to the working robots. For each working robot, it only decodes the information of positions and headings of its neighbors, such that the distributed computation is approximated.

References A. Ward, J.E. Herbert-Read, D. Sumpter, J. Krause, Fast and accurate decisions through collective vigilance in fish shoals, Proc. Natl Acad. Sci. USA 108 (6) (2011) 2312–2315.

Link <https://www.sciencedirect.com/science/article/abs/pii/S0378437122004794>

Trajectory Optimization for 3D Shape-Changing Robots with Differential Mobile Base

Abstract

Service robots have attracted extensive attention due to specially designed functions, such as mobile manipulators or robots with extra structures. For robots that have changing shapes, autonomous navigation in the real world presents new challenges. In this paper, we propose a trajectory optimization method for differential-drive mobile robots with controllable changing shapes in dense 3D environments. We model the whole-body trajectory as a polynomial trajectory that satisfies the nonholonomic dynamics of the base and dynamics of the extra joints. These constraints are converted into soft constraints, and an activation function for dense sampling is applied to avoid nonlinear mutations. In addition, we guarantee the safety of full shape by limiting the system's distance from obstacles. To comprehensively simulate a large extent of height and width changes, we designed a novel Shape-Changing Robot with a Differential Base (SCR-DB). Our global trajectory optimization gives a smooth and collision-free trajectory for SCR-DB at a low computational cost. We present vast simulations and real-world experiments to validate our performance, including coupled whole-body and independent differential-driven vehicle motion planning.

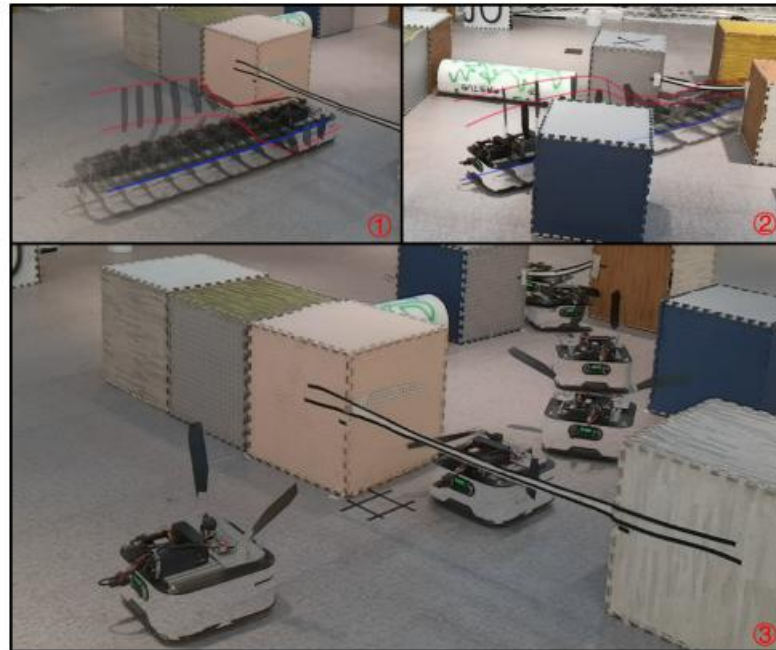


Fig. 8. Real-world experiment.

How NOKOV Contributes to the Research

We use the motion capture system for positioning and map the environment in advance. The robot's movement is shown in Fig. 8, which shows that the robot can avoid obstacles following the planned trajectory.

References "Hello robot: Simply useful robots," <https://hello-robot.com/>, [Accessed 12-Sep-2022].

Link <https://ieeexplore.ieee.org/abstract/document/10160911>

Multi-Robot Object Transport Motion Planning With a Deformable Sheet

Keywords

Multi-robot manipulation, collaborative manipulation, multi-robot motion planning.

Abstract

Using a deformable sheet to handle objects is convenient and found in many practical applications. For object manipulation through a deformable sheet that is held by multiple mobile robots, it is a challenging task to model the object-sheet interactions. We present a computational model and algorithm to capture the object position on the deformable sheet with changing robotic team formations. A virtual variable cables model (VVCM) is proposed to simplify the modeling of the robot-sheet-object system. With the VVCM, we further present a motion planner for the robotic team to transport the object in a three-dimensional (3D) cluttered environment. Simulation and experimental results with different robot team sizes show the effectiveness and versatility of the proposed VVCM. We also compare and demonstrate the planning results to avoid the obstacle in 3D space with the other benchmark planner.

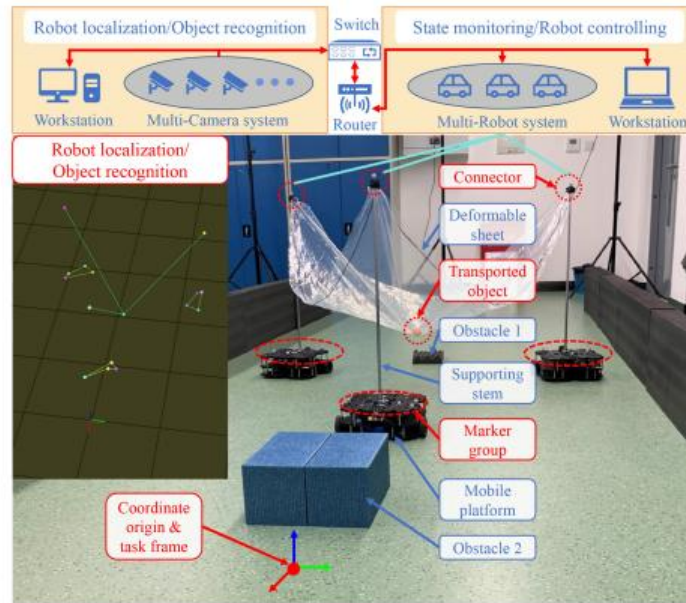


Fig. 6. Experimental setup and communication network of the multi-mobile robot transportation system.

How NOKOV Contributes to the Research

Fig. 6 shows the experimental setup and communication network of the system. Three robots (Turtlebot3 Waffle-Pi) were used in experiment. The transported object was a billiard ball. The position of the robots and the object were acquired by a motion capture system (8 cameras from NOKOV) at a rate of 60 Hz. Plastic cloth sheets were used in experiments. Each side length of the sheet was 1.6 m and the height of each holding point was $z_r = 0.79$ m.

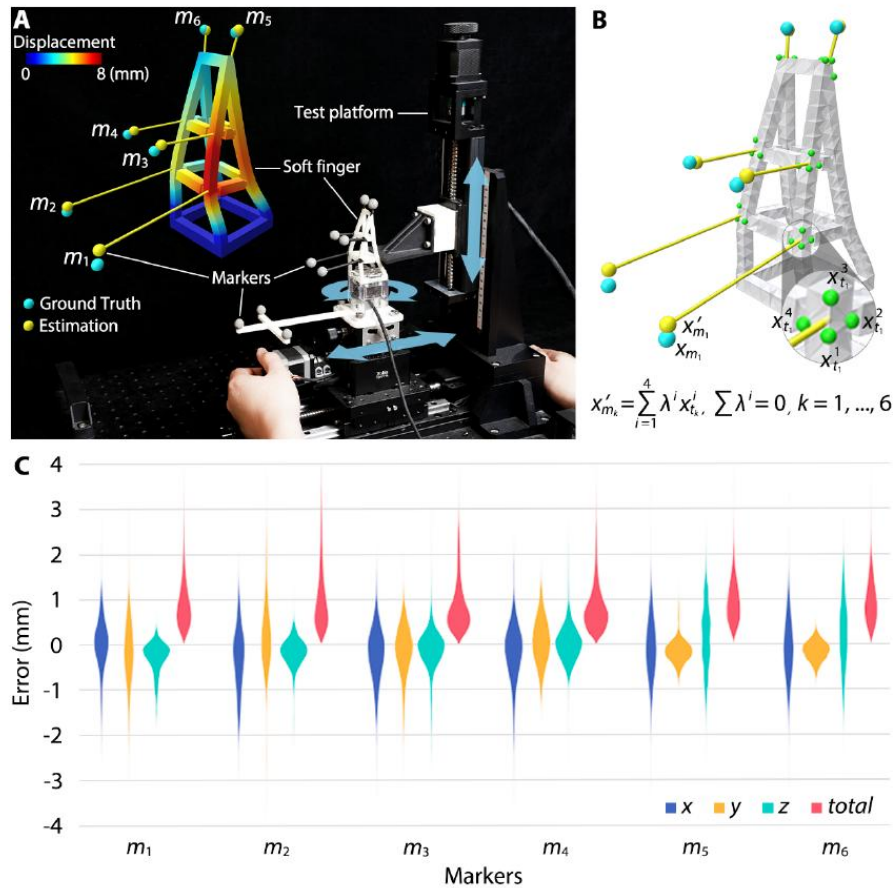
References B. Roy, A. Basnajian, and H. H. Asada, "Repositioning of a rigid body with a flexible sheet and its application to an automated rehabilitation bed," IEEE Trans. Automat. Sci. Eng., vol. 2, no. 3, pp. 300–307, Jul. 2005.

Link <https://ieeexplore.ieee.org/document/9830876>

Proprioceptive State Estimation for Amphibious Tactile Sensing

Abstract

This paper presents a novel vision-based proprioception approach for a soft robotic finger that can estimate and reconstruct tactile interactions in terrestrial and aquatic environments. The key to this system lies in the finger's unique metamaterial structure, which facilitates omni-directional passive adaptation during grasping, protecting delicate objects across diverse scenarios. A compact in-finger camera captures high framerate images of the finger's deformation during contact, extracting crucial tactile data in real time. We present a volumetric discretized model of the soft finger and use the geometry constraints captured by the camera to find the optimal estimation of the deformed shape. The approach is benchmarked using a motion capture system with sparse markers and a haptic device with dense measurements. Both results show state-of-the-art accuracies, with a median error of 1.96 mm for overall body deformation, corresponding to 2.1% of the finger's length. More importantly, the state estimation is robust in both on-land and underwater environments as we demonstrate its usage for underwater object shape sensing. This combination of passive adaptation and real-time tactile sensing paves the way for amphibious robotic grasping applications.



How NOKOV Contributes to the Research

A motion capture system (Mars2H by Nokov, Inc.) was used to track finger deformations through nine markers with an 8 mm radius. Among them, six markers were divided into three pairs, which were rigidly attached to the fingertip (m_5, m_6), the first layer (m_3, m_4), and the second layer (m_1, m_2) of the soft finger, respectively. The other three markers were attached to the platform and used as the reference reading to align the motion capture system's reference frame with the platform's coordinate frame.

References N. Guo et al., "Proprioceptive State Estimation for Amphibious Tactile Sensing," in IEEE Transactions on Robotics, vol. 40, pp. 4684-4698, 2024, doi: 10.1109/TRO.2024.3463509.

Link <https://ieeexplore.ieee.org/abstract/document/10684118>

MERSYS: A Collaborative Estimation and Dense Mapping System for Multi-Agent Generic SLAM

Abstract

Multi-agent collaborative Simultaneous Localization and Mapping (SLAM) is an effective way for large-scale mapping. However, this approach, which relies on Visual-Inertial Odometry(VIO) as input, suffers from limitations such as susceptibility to environmental influences and the difficulty in accurately constructing dense 3D maps. To address these challenges, this paper presents Multi-Estimation Robust SLAM System (MERSYS), a novel framework for three-dimensional dense mapping based on the fusion of Lidar-Inertial Odometry(LIO) and VIO. Benefiting from lower communication's costs and dense information acquisition capability, the proposed framework aims to achieve compatibility in processing both LIO and VIO inputs, establish joint loop closure detection to enable multi-map fusion, and then create a comprehensive global 3D dense point cloud map. Furthermore, an efficient communication strategy has been proposed to enable bidirectional transmission of dense and voluminous data. Experimental evaluations conducted on the publicly available HILTI SLAM 2021 dataset[10] as well as a real world dataset. Experimental results show that MERSYS achieves better results than state-of-the-art methods. The source code is available on the GitHub1.

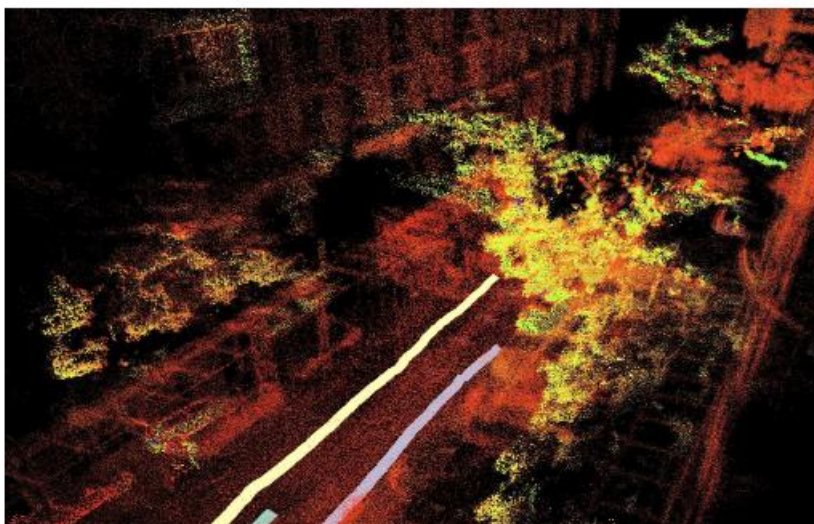


Fig. 1: We introduce MERSYS, a multi-agent SLAM framework compatible with both LIO and VIO.

How NOKOV Contributes to the Research

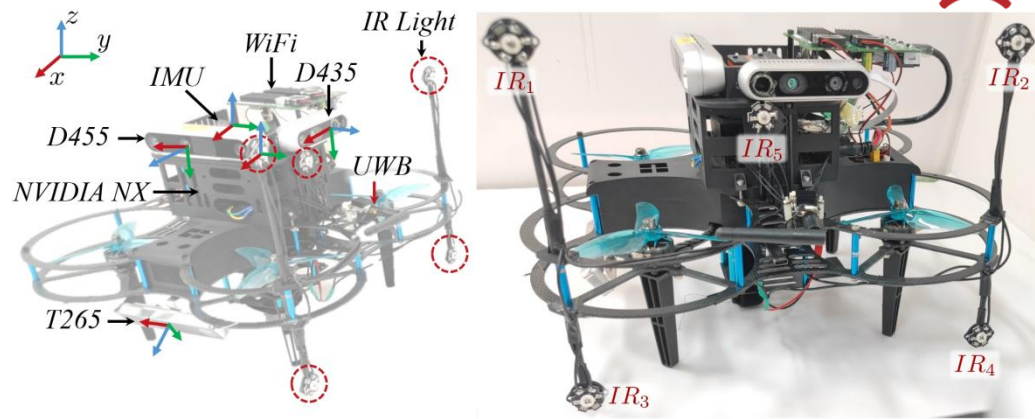
To capture ground truth data, we use RTK sensors for outdoor environments and the NOKOV motion capture system for indoor environments.

References Q. Lai, E. Zhao, S. Fan, and J. Zou, "MERSYS: A collaborative estimation and dense mapping system for multi-agent generic SLAM", IROS 2024

A Collaborative Stereo Camera with Two UAVs for Long-distance Mapping of Urban Buildings

Abstract

For a swarm of Unmanned Aerial Vehicle (UAV), long-distance visual mapping is advantageous for pre-planning navigation paths in unknown urban building environments. This work leverages two cameras on two UAVs to build a wide baseline collaborative stereo camera, which can construct a navigable mesh map for remote building obstacles. We present a complete framework of the collaborative stereo camera for long-distance mapping, including online extrinsic parameter estimation of the stereo camera, real-time cross-camera feature association, and semantic mesh map generation of remote buildings. Extensive simulations and real-world experiments verify the effectiveness of the collaborative stereo camera. With a 3m baseline, the collaborative stereo camera achieves long-distance mapping of buildings (20m ~ 50m) away with a relative error of approximately 10%. The constructed remote map enables UAVs to pre-detect large obstacles and pre-plan navigation paths in large-scale building environments. Hopefully, this work can provide a novel and practical approach for collaborative visual tasks of UAV swarm.



How NOKOV Contributes to the Research

The evaluation of the extrinsic parameter estimation of the stereo camera is performed under the NOKOV motion capture system.

References Z. Wang and W. Dong, "A Collaborative Stereo Camera with Two UAVs for Long-distance Mapping of Urban Buildings," 2023.

Link

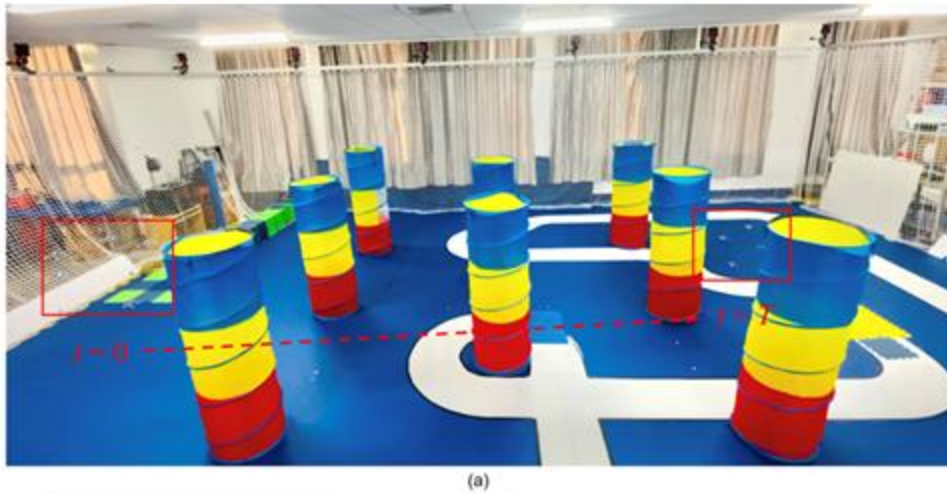
Self-Organized Reynolds Swarms of Unmanned Aerial Vehicles in Dense Environments

Keywords

self-organized swarm, UAV, collision avoidance

Abstract

The achievement of fully distributed self-organized swarm control of unmanned aerial vehicles (UAVs) in realistic environments is a current challenge in the field of UAV research. This paper investigates the problem of self-organized swarm control of UAVs in dense environments. Based on the behavioral characteristics of biological swarms, a distributed self-organizing Reynolds (SOR) swarm model is proposed to improve the environmental adaptability of UAV swarms. In order to evaluate the performance of the SOR swarm in actual deployment, the proposed SOR swarm method is validated in a real indoor environment based on the Nokov motion capture system using five quadrotor UAVs, and the autonomous navigation of the UAV swarm in an obstacle environment, similar to the flocking behavior of a bird, is successfully achieved



(a)

How NOKOV Contributes to the Research

This paper builds a self-organized UAV swarm flight experiment platform, which mainly consists of a Nokov 3D optical motion capture system, a ground control station equipped with a robot operating system (ROS), and a Crazyflie [23] UAV platform (see Fig.1).

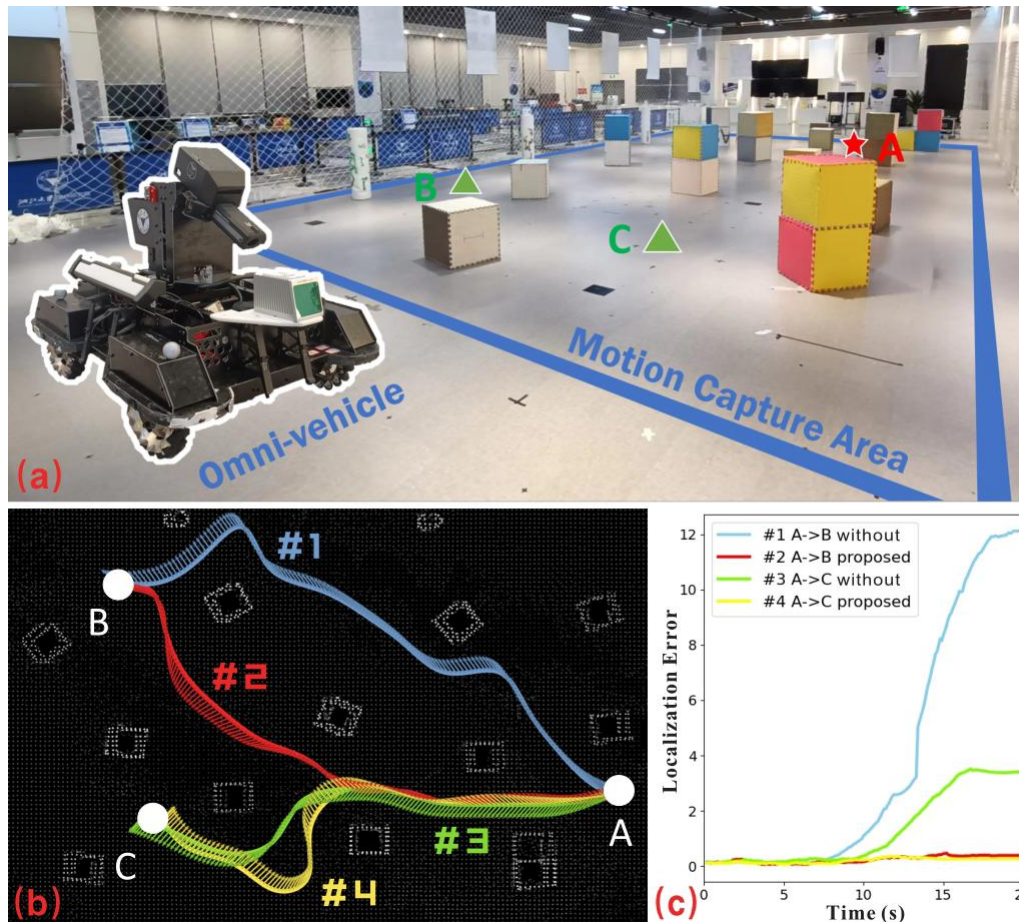
References Y. Ding, H. Xiong, X. Shi, J. Liu, Y. Chen, and J. Wang, "Self-Organized Reynolds Swarms of Unmanned Aerial Vehicles in Dense Environments," 2023.

Link

LF-3PM: a LiDAR-based Framework for Perception-aware Planning with Perturbation-induced Metric

Abstract

Just as humans can become disoriented in featureless deserts or thick fogs, not all environments are conducive to the Localization Accuracy and Stability (LAS) of autonomous robots. This paper introduces an efficient framework designed to enhance LiDAR-based LAS through strategic trajectory generation, known as Perception-aware Planning. Unlike vision-based frameworks, the LiDAR-based requires different considerations due to unique sensor attributes. Our approach focuses on two main aspects: firstly, assessing the impact of LiDAR observations on LAS. We introduce a perturbation-induced metric to provide a comprehensive and reliable evaluation of LiDAR observations. Secondly, we aim to improve motion planning efficiency. By creating a Static Observation Loss Map (SOLM) as an intermediary, we logically separate the time-intensive evaluation and motion planning phases, significantly boosting the planning process. In the experimental section, we demonstrate the effectiveness of the proposed metrics across various scenes and the feature of trajectories guided by different metrics. Ultimately, our framework is tested in a real-world scenario, enabling the robot to actively choose topologies and orientations preferable for localization. The source code is accessible at this [https URL](https://github.com/ZJU-FAST-Lab/LF-3PM).The source code is accessible at <https://github.com/ZJU-FAST-Lab/LF-3PM>.



How NOKOV Contributes to the Research

The ground truth of robot pose is provided by the motion tracking system (NOKOV2).

References K. Chai, L. Xu, Q. Wang, C. Xu, P. Yin, and F. Gao, "LF-3PM: a LiDAR-based Framework for Perception-aware Planning with Perturbation-induced Metric," 2023.
Link <https://arxiv.org/abs/2408.01649>

Preserving Relative Localization of FoV-Limited Drone Swarm via Active Mutual Observation

Abstract

Relative state estimation is crucial for vision-based swarms to estimate and compensate for the unavoidable drift of visual odometry. For autonomous drones equipped with the most compact sensor setting -- a stereo camera that provides a limited field of view (FoV), the demand for mutual observation for relative state estimation conflicts with the demand for environment observation. To balance the two demands for FoV limited swarms by acquiring mutual observations with a safety guarantee, this paper proposes an active localization correction system, which plans camera orientations via a yaw planner during the flight. The yaw planner manages the contradiction by calculating suitable timing and yaw angle commands based on the evaluation of localization uncertainty estimated by the Kalman Filter. Simulation validates the scalability of our algorithm. In real-world experiments, we reduce positioning drift by up to 65% and managed to maintain a given formation in both indoor and outdoor GPS-denied flight, from which the accuracy, efficiency, and robustness of the proposed system are verified.

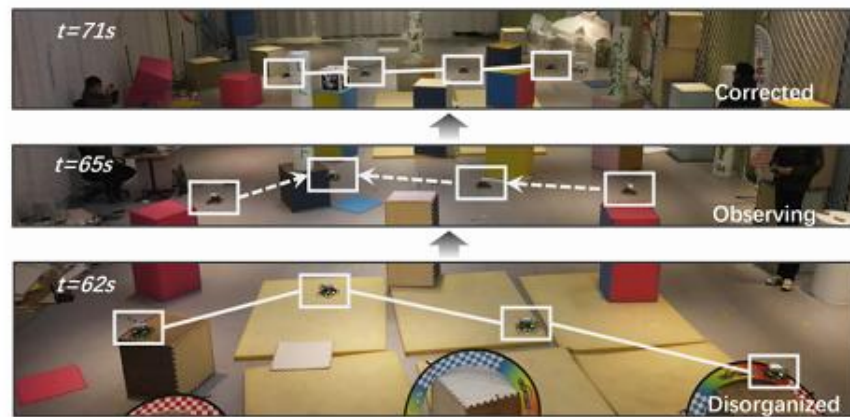


Fig. 9: Four drones conduct line formation flight in the indoor experiment.

How NOKOV Contributes to the Research

For the indoor experiment, the position information provided by the NOKOV motion capture system is introduced as the ground truth for evaluating the localization accuracy of our method.

References L. Guo, Z. Gongye, Z. Xu, Y. Wang, X. Zhou, J. Zhou, and F. Gao, "Preserving Relative Localization of FoV-Limited Drone Swarm via Active Mutual Observation," arXiv:2407.01292 [cs.RO], 2024. [Online]. Available: <https://doi.org/10.48550/arXiv.2407.01292>

Link <https://arxiv.org/abs/2407.01292>

A cross-platform deep reinforcement learning model for autonomous navigation without global information in different scenes

Abstract

This paper employs a deep reinforcement learning algorithm named Twin Delayed Deep Deterministic algorithm into autonomous navigation in intelligent transportation systems. It trains a fully connected neural network model in a simulation environment, which outputs the expected linear and angular velocity of the vehicle based on real-time data measured by embedded sensors. Through continuous epochs of training, the model gradually navigates the vehicle to reach a provided destination by making rational motion decisions at each discrete time instant without knowing global environment information. Especially, to improve the model's generalization ability across various scenes, an input preprocessing function is proposed to eliminate the singularity and uniformity of raw input data. A large number of simulation tests are carried out, where the proportion that the vehicle moves from a start position to a destination without collision within a specified limited time exceeds 90%. The remaining failures are mainly due to the vehicle's inability to approach the destination immediately adjacent to obstacles for its safety. Furthermore, traditional mapless navigation algorithms suffer from locally optimal solutions in the face of U-shaped obstacles. This paper introduces a virtual obstacle mechanism designed to prevent the vehicle from entering the U-shaped region, effectively addressing the aforementioned issue. Finally, the model trained from the simulation environment can be directly loaded onto a physical vehicle without considering the different processor architectures. Large quantities of experiments show that the model improves the autonomous navigation capability of vehicles when global environment information cannot be obtained by the system, which optimizes the functions of the navigation module in intelligent transportation systems.



(a) Platform panorama



(b) Navigation scene



(c) Robot

Fig. 9. A physical platform composed of obstacles, vehicles and positioning devices is built to test the actual navigation performance of the trained model.

How NOKOV Contributes to the Research

A 4 m×4 m navigation field, as shown in Fig. 9 is constructed to test the autonomous navigation capability of a vehicle loaded with our trained model. The field contains randomly placed cardboard boxes that serve as obstacles. To accurately obtain the real-time pose of the vehicle, a set of motion capture device developed by Nokov Science & Technology Co., Ltd is equipped in the field. The device provides sub millimeter positioning accuracy and its data can be read by various terminals.

References Chuanxin Cheng, Hao Zhang, Yuan Sun, Hongfeng Tao, Yiyang Chen, A cross-platform deep reinforcement learning model for autonomous navigation without global information in different scenes, Control Engineering Practice, Volume 150, 2024, 105991, ISSN 0967-0661, <https://doi.org/10.1016/j.conengprac.2024.105991>. Link <https://www.sciencedirect.com/science/article/abs/pii/S0967066124001515>

MTABot: An Efficient Morphable Terrestrial-Aerial Robot With Two Transformable Wheels

Keywords

Aerial systems: Mechanics and control, mechanism design, multimodal locomotion, terrestrial-aerial robot

Abstract

Terrestrial-aerial robots, capable of swift aerial navigation and enduring terrestrial operations, possess significant potential for utilization in exploration and rescue missions. However, achieving their capability to negotiate diverse terrains with a high-power-efficient structure remains a formidable challenge. This letter presents a morphable terrestrial-aerial robot, named MTABot, which achieves three modalities through the deployment of two multifunctional appendages. These include: 1) rolling mode, 2) climbing mode, both achieved with transformable two-wheeled configuration, and 3) flying mode, achieved with a bicopter configuration. Moreover, the radius and sector angle of transformable wheel have been optimized to enhance the obstacle-climbing capability; the position of the robot's body center of gravity has been optimized to balance ground gripping capacity and flight dynamic response speed. Finally, the robot's multi-terrain overcoming capability is validated through obstacle-climbing experiments and continuous terrestrial-aerial transformation experiments, and the high power efficiency of robot is affirmed, demonstrating feasibility of the design.

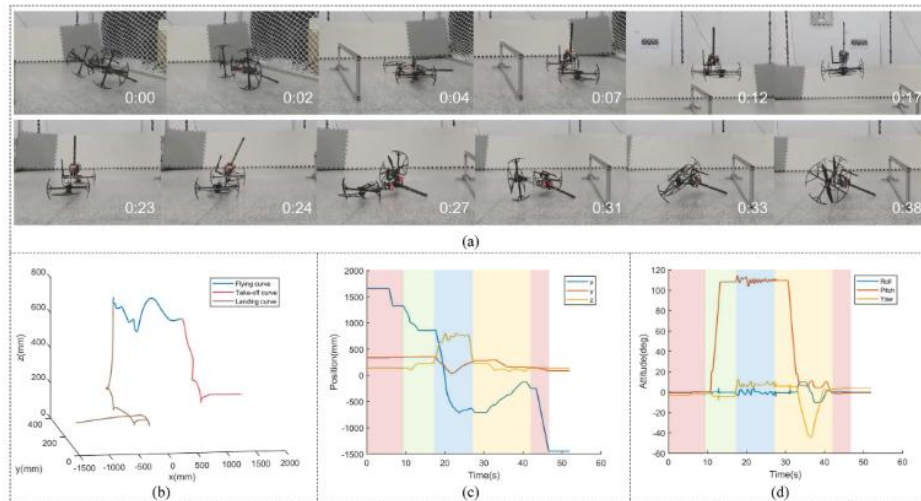


Fig. 10. Terrestrial-aerial transformation result. (a) Time sequence of terrestrial-aerial transformation. (b) 3D trajectory points corresponding to (a). (c) Position response corresponding to (a). (d) Attitude response corresponding to (a).

How NOKOV Contributes to the Research

The experiment took place indoors, using the NOKOV motion capture system at a 60Hz sampling frequency for data collection.

References K. Shi, Z. Jiang, L. Ma, L. Qi and M. Jin, "MTABot: An Efficient Morphable Terrestrial-Aerial Robot With Two Transformable Wheels," in IEEE Robotics and Automation Letters, vol. 9, no. 2, pp. 1875-1882, Feb. 2024, doi: 10.1109/LRA.2024.3349962.

Link <https://ieeexplore.ieee.org/abstract/document/10380664>

Autonomous Navigation of Quadrotors in Dynamic Complex Environments

Keywords

Aerial robotics, Motion planning, Velocity obstacle.

Abstract

This article introduces a novel framework utilizing velocity obstacles to enhance the autonomous navigation of quadrotors in dynamic complex environments. In this framework, quadrotors rely on onboard sensors to perceive the surrounding environment and construct an occupancy grid map for environmental representation. The density-based spatial clustering of applications with noise (DBSCAN) algorithm is employed to extract the positions and velocities of dynamic obstacles within the environment. Based on these results, we propose a velocity obstacle-based gradient field, called gradient velocity obstacle (GVO), for generating collision-free velocities ensuring safety. Compared with existing methods, GVO preserves the original feasible set while ensuring computational efficiency. Moreover, it exhibits excellent fault tolerance to environmental perception noise. Additionally, we design motion primitives based on B-spline parameterization. By optimizing within position and velocity state spaces, collision-free trajectories are dynamically constructed in real-time. Extensive simulations and experiments validate our framework's effectiveness, showcasing significant improvements in navigation efficiency and safety. The experimental section of the entire work can be found at the following link: <https://www.youtube.com/watch?v=TOEeoFO4OxY>.

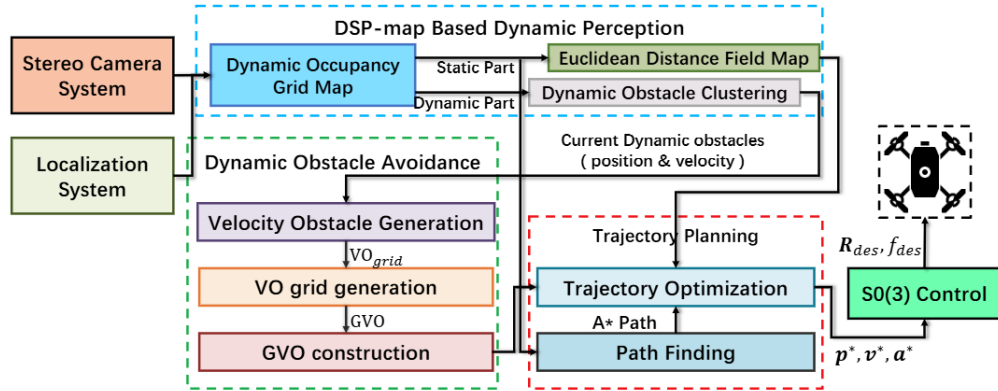


Fig. 1: A block diagram illustrating the full pipeline of the proposed planning framework.

How NOKOV Contributes to the Research

To validate the performance of the proposed approach in real-world scenarios, we conduct extensive field experiments in complex indoor environments. The quadrotor platform we used is shown in Fig. 9. We perform localization based on our previous work [3], use DSP-Map [33]+DBSCAN to update the occupancy grid map and use the NOKOV motion capture system as the ground truth.

References R. Li and B. Xin, "Autonomous Navigation of Quadrotors in Dynamic Complex Environments," in IEEE Transactions on Industrial Electronics, doi: 10.1109/TIE.2024.3433585.

Link <https://ieeexplore.ieee.org/abstract/document/10649014>

Design and Analysis of Hybrid Robots for Powerline Inspection

Abstract

Hybrid robots, which combine crawling and flying capabilities, have gained significant attention in recent years due to their potential in powerline inspection. This paper provides insights into the design of hybrid robots, offering a comprehensive analysis of the system's design requirements and its solutions. Firstly, we present a detailed overview of the hybrid robot's system architecture, encompassing the design aspects of its mechanical, hardware, and software components. Secondly, we analyze the dynamics of the robot in crawling mode, deducing the conditions necessary for maintaining balance on powerlines and traversing inclined powerlines. Thirdly, we investigated the kinematics and dynamics of the robot in flight mode, examining how the structural design of the hybrid robot specifically impacts its system dynamics and control responsiveness. Finally, we verify our analyses through a series of simulations and real-world tests, confirming the effectiveness and feasibility of the design strategies presented.

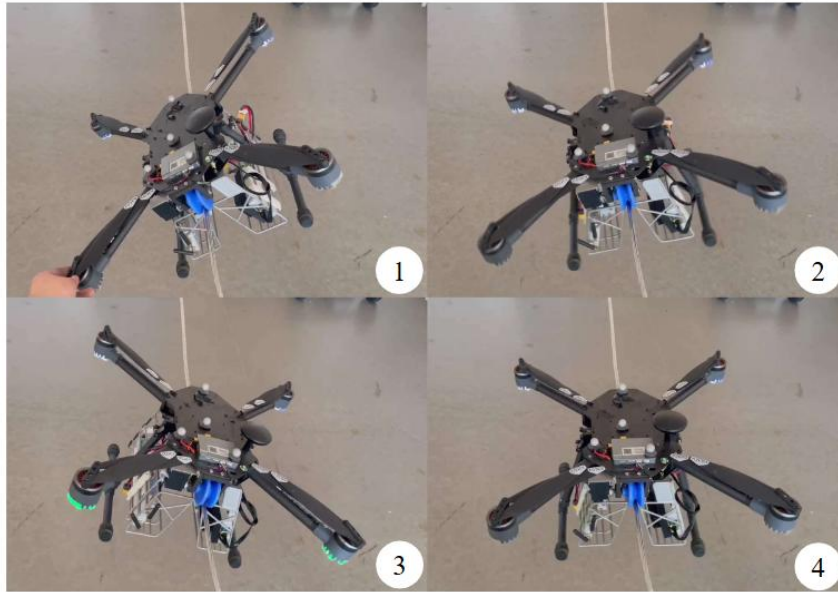


Fig. 8. Balance test of the hybrid robot on powerline. The numbers denote the evolution of the hybrid robot returning to the balance state.

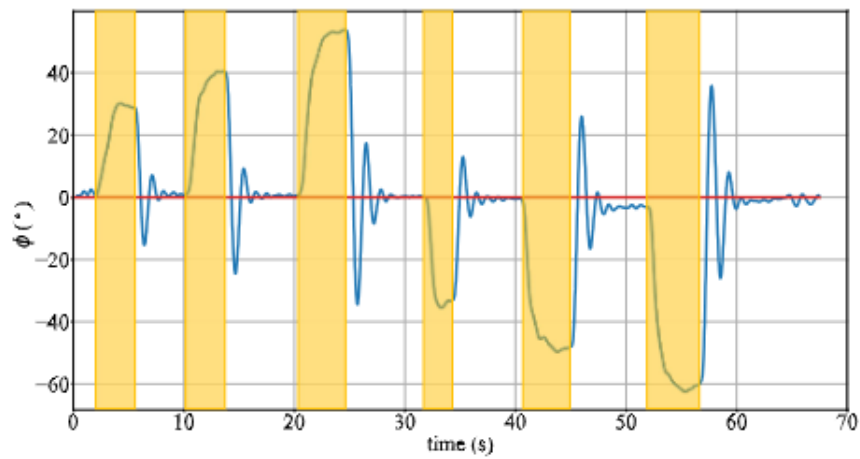


Fig. 9. The roll angle of the hybrid robot during the balance test. The yellow shaded area indicates the artificially applied disturbance.

How NOKOV Contributes to the Research

Fig. 9 presents the roll angle of the robot recorded by the NOKOV Motion Capture System.

References

Zhishuo Li et al., "Design and Analysis of Hybrid Robots for Powerline Inspection", 2024 IEEE 20th International Conference on Automation Science and Engineering (IEEE CASE 2024)

Link

<http://ir.ia.ac.cn/handle/173211/57160>

Real-Time Multi-Modal Active Vision for Object Detection on UAVs Equipped With Limited Field of View LiDAR and Camera

Abstract

This letter aims to solve the challenging problems in multi-modal active vision for object detection on unmanned aerial vehicles (UAVs) with a monocular camera and a limited Field of View (FoV) LiDAR. The point cloud acquired from the low-cost LiDAR is firstly converted into a 3-channel tensor via motion compensation, accumulation, projection, and up-sampling processes. The generated 3-channel point cloud tensor and RGB image are fused into a 6-channel tensor using an early fusion strategy for object detection based on a Gaussian YOLO network structure. To solve the low computational resource problem and improve the real-time performance, the velocity information of the UAV is further fused with the detection results based on an extended Kalman Filter (EKF). A perception-aware model predictive control (MPC) is designed to achieve active vision on our UAV. According to our performance evaluation, our pre-processing step improves other literature methods running time by a factor of 10 while maintaining acceptable detection performance. Furthermore, our fusion architecture reaches 94.6 mAP on the test set, outperforming the individual sensor networks by roughly 5%. We also described an implementation of the overall algorithm on a UAV platform and validated it in real-world experiments.

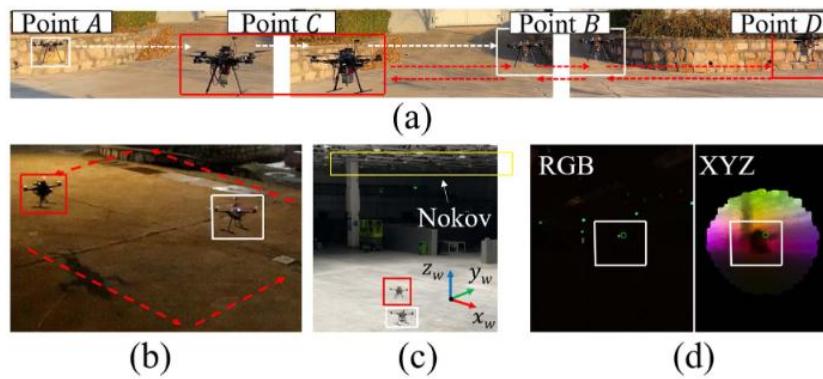


Fig. 5. Real-world experiments. Red box: the detector UAV and white box: the target UAV. (a) LF. (b) SF in LL conditions. (c) Indoor experimental environments. (d) The first-person view of the detector UAV of the circular flight in CD conditions.

How NOKOV Contributes to the Research

The NOKOV motion capture system was utilized to localize UAVs during indoor flight.

References

C. Shi, G. Lai, Y. Yu, M. Bellone and V. Lippiello, "Real-Time Multi-Modal Active Vision for Object Detection on UAVs Equipped With Limited Field of View LiDAR and Camera," in *IEEE Robotics and Automation Letters*, vol. 8, no. 10, pp. 6571-6578, Oct. 2023, doi: 10.1109/LRA.2023.3309575.

Link

<https://ieeexplore.ieee.org/document/10233103>

Multi-Agent Visual-Inertial Localization for Integrated Aerial Systems With Loose Fusion of Odometry and Kinematics

Abstract

Reliably and efficiently estimating the relative pose and global localization of robots in a common reference for Integrated Aerial Platforms (IAPs) is a challenging problem. Unlike unmanned aerial vehicle (UAV) swarms, where the agent individual is able to move freely, IAPs connect UAV agents with mechanical joints, such as spherical joints, and form a rigid central platform, limiting the degree of freedom (DOF) of agents. Traditional methods, which rely on forming loop closures, object detection, or range sensors, suffer from degeneration or inefficiency due to the restricted relative motion between agents. In this paper, we present a centralized multi-agent localization system that fuses the internal kinematic constraints of IAPs and odometry measurements, using only visual-inertial suits for ego-motion estimation for agents and an additional 9-DOF Inertial Measurement Unit (IMU) attached to the central platform for posture estimation. A general formulation for kinematic constraints is derived without requiring knowledge about detailed kinematic parameters. A sliding-window optimization-based state estimator is constructed to estimate the relative transformation between agents. Our proposed approach is validated in our collected dataset. The results show that the proposed method reduces the global localization drift by 27.15% and relative localization error by 53.4% in the translation part and 36.99% in the rotation part compared to the baseline.

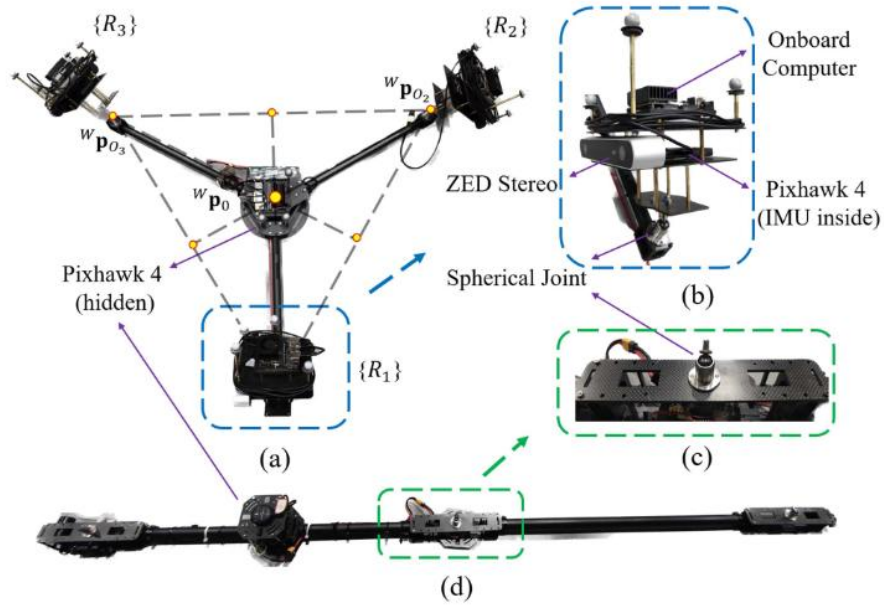


Fig. 4. Experiment platforms.

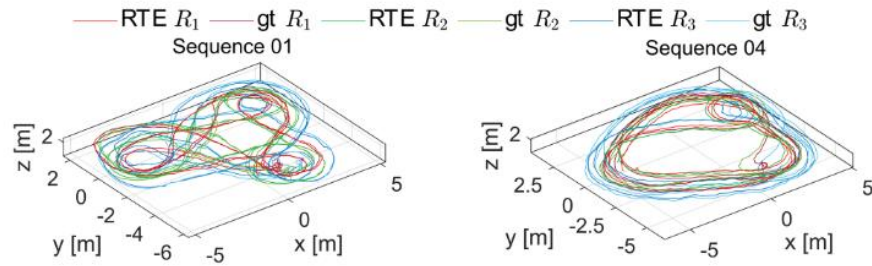


Fig. 5. Overview of some trajectories in our dataset.

How NOKOV Contributes to the Research

The NOKOV motion capture system¹ is utilized to provide the ground truth pose.

References

G. Lai, C. Shi, K. Wang, Y. Yu, Y. Dong and A. Franchi, "Multi-Agent Visual-Inertial Localization for Integrated Aerial Systems With Loose Fusion of Odometry and Kinematics," in IEEE Robotics and Automation Letters, vol. 9, no. 7, pp. 6504-6511, July 2024, doi: 10.1109/LRA.2024.3407579.

Link

<https://ieeexplore.ieee.org/abstract/document/10542321>

17-Point Algorithm Revisited: Toward a More Accurate Way

Abstract

17-point algorithm is a popular method in relative pose estimation of multi-cameras. However, the role of overlap in 17-point algorithm remains unexplored. And the relaxed way in solving constrained normal equation leads to sub-optimal results. Both of them influence accuracy of the estimated pose. In this paper, we theoretically analyze the influence of overlap and the solvability of 17-point algorithm. In addition, we show that the abuse of overlap can harm accuracy in practice. In light of these findings, we propose an improved 17-point algorithm, which avoids using overlaps and derives a simple way to solve normal equation on manifold. Both simulations and real world data experiments demonstrate the proposed one outperforms the traditional 17-point algorithm in term of accuracy.

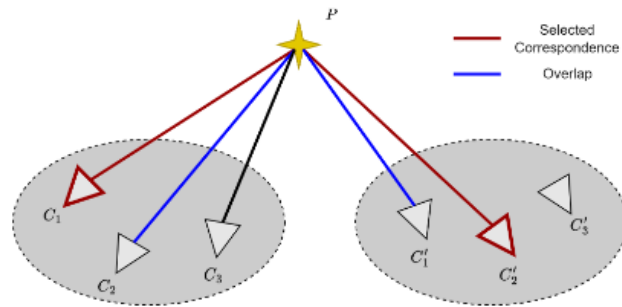


Fig. 1: An example to illustrate the concept of overlap, which denotes for the extra correspondence.

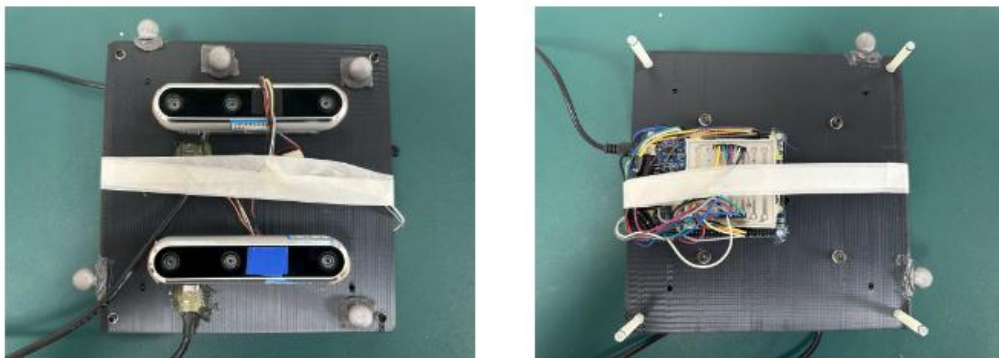


Fig. 8: Customized, synchronized multi-camera rig.

How NOKOV Contributes to the Research

We employ a frame-to-frame matching within a NOKOV motion capture-enabled environment which can provide the ground truth pose.

References

C. Xie, R. Xing, N. Hao and F. He, "17-Point Algorithm Revisited: Toward a More Accurate Way," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 10208-10214, doi: 10.1109/ICRA57147.2024.10611149.

Link

<https://ieeexplore.ieee.org/document/10611149>

CoNi-MPC: Cooperative Non-inertial Frame Based Model Predictive Control

Abstract

This letter presents a novel solution for UAV control in cooperative multi-robot systems, which can be used in various scenarios such as leader-following, landing on a moving base, or specific relative motion with a target. Unlike classical methods that tackle UAV control in the world frame, we directly control the UAV in the target coordinate frame, without making motion assumptions about the target. In detail, we formulate a non-linear model predictive controller of a UAV, referred to as the agent, within a non-inertial frame (i.e., the target frame). The system requires the relative states (pose and velocity), the angular velocity and the accelerations of the target, which can be obtained by relative localization methods and ubiquitous MEMS IMU sensors, respectively. This framework eliminates dependencies that are vital in classical solutions, such as accurate state estimation for both the agent and target, prior knowledge of the target motion model, and continuous trajectory re-planning for some complex tasks. We have performed extensive simulations to investigate the control performance with varying motion characteristics of the target. Furthermore, we conducted real robot experiments, employing either simulated relative pose estimation from motion capture systems indoors or directly from our previous relative pose estimation devices outdoors, to validate the applicability and feasibility of the proposed approach.

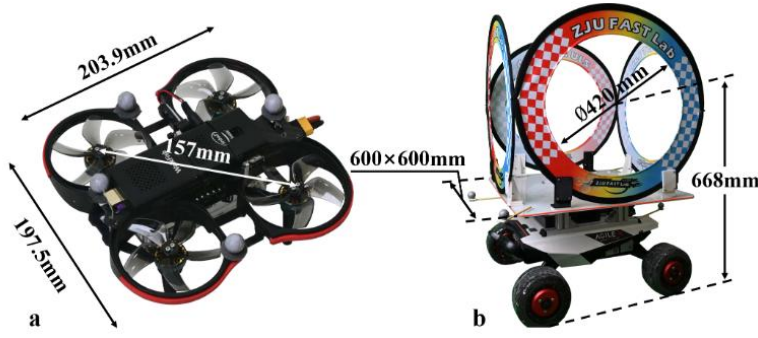


Fig. 8: The quadrotor and UGV in our real experiments.

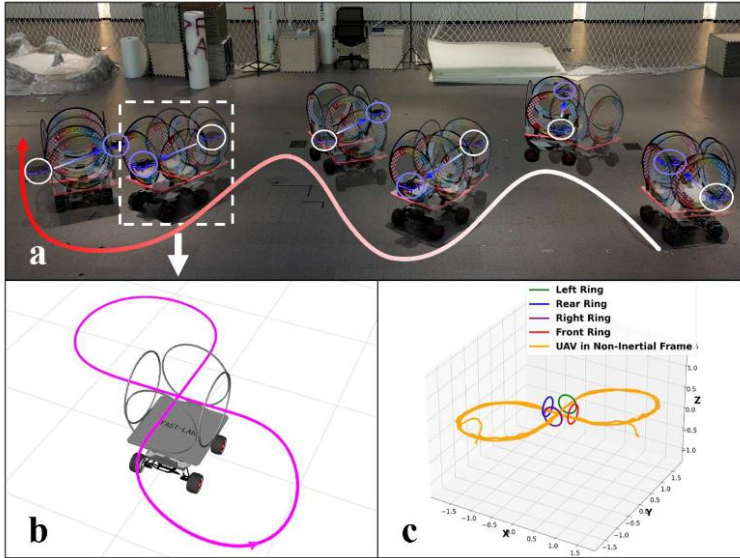


Fig. 11: Rings crossing experiment. The UGV moves with an average speed of 0.15 m/s and a maximal speed of 0.38 m/s while the UAV can cross the ring attached to it.

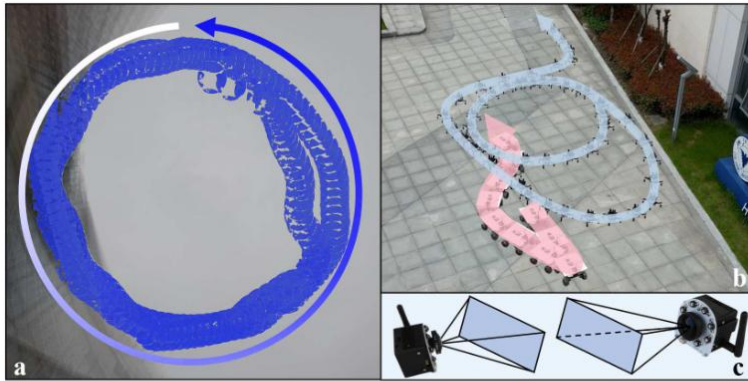


Fig. 12: Outdoor experiment. (a) is shot from the camera on the UGV, and (b) is shot from third-person view. The UAV follows a circular trajectory while the UGV arbitrarily moves. (c) shows CREPES devices we use in the outdoor experiment.

How NOKOV Contributes to the Research

The relative estimation can be generated directly from CREPES or computed from NOKOV motion capture system for outdoor or indoor experiments respectively.

References

B. Zhang et al., "CoNi-MPC: Cooperative Non-inertial Frame Based Model Predictive Control," in IEEE Robotics and Automation Letters, vol. 8, no. 12, pp. 8082-8089, Dec. 2023, doi: 10.1109/LRA.2023.3325776.

Link

<https://ieeexplore.ieee.org/document/10287548>

Active Collision-Based Navigation for Wheeled Robots

Abstract

Collision is typically avoided in robot navigation for safety guarantee. However, when a robot's exteroceptive sensors fail, which means it becomes "blind", collision can actually be leveraged to improve localization performance. Our research demonstrates the informative nature of collisions in this context. Moreover, we show that a robot is able to navigate in a known environment with only proprioceptive sensors by actively colliding with its surroundings for more reliable localization. Firstly, we design a collision-based observation model, which is differentiable and can be easily applied to various estimators. Secondly, we integrate this model into a collision-aided localization framework and implement it in two widely used estimators, the Kalman filter and the particle filter. Thirdly, we propose an active collision path planning method, which effectively reduces localization uncertainty.

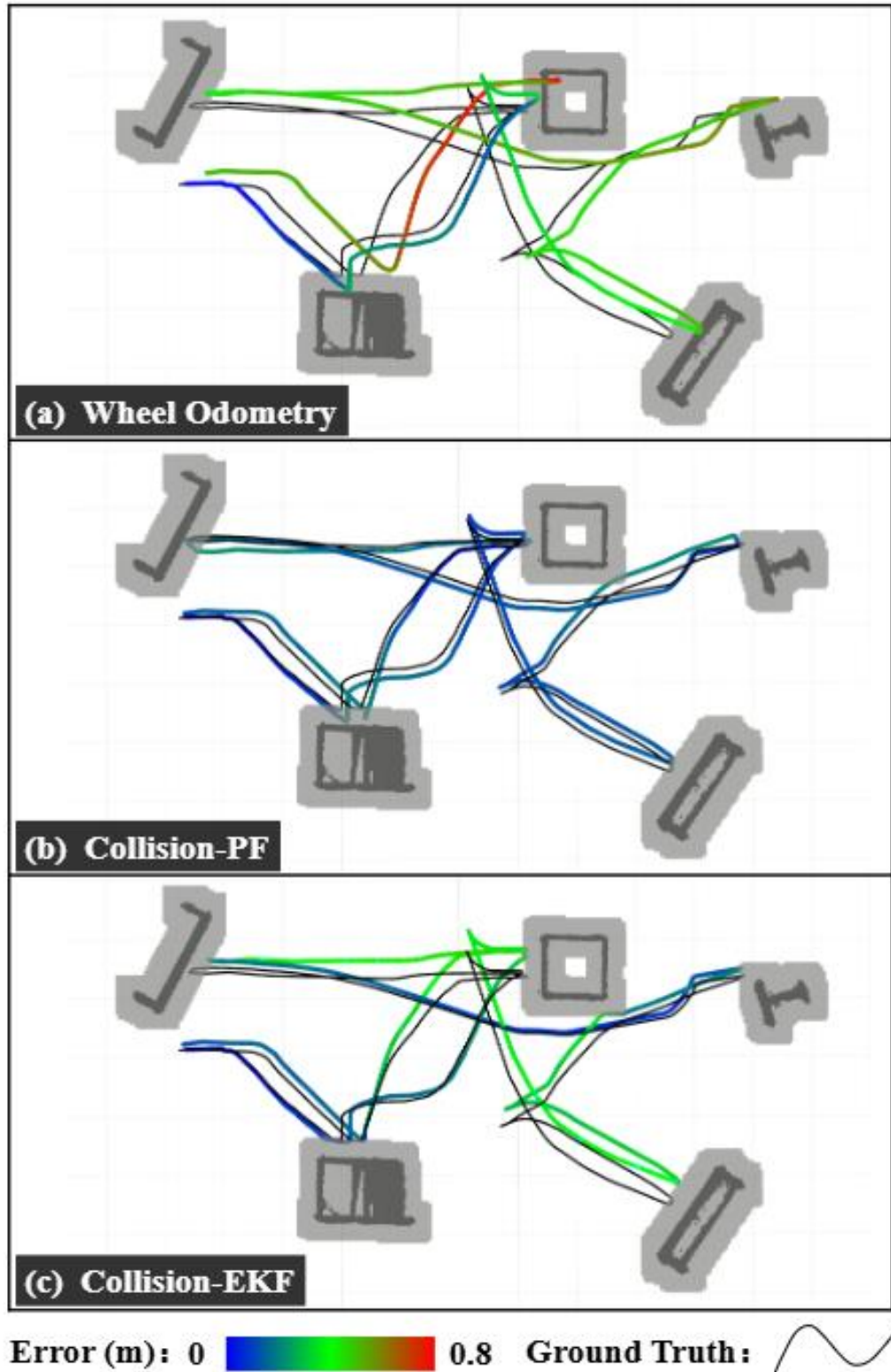


Fig. 6. Snapshots of trajectories in localization experiments. (a) The trajectory of wheel odometry without exploiting collision information. (b) The trajectory of collision-aided particle filter. (c) The trajectory of collision-aided extended Kalman filter.

How NOKOV Contributes to the Research

We use the NOKOV motion capture system to provide the ground truth of localization.

References

J. Li, J. Ji, Q. Wang, H. Yu, Y. Pan and F. Gao, "Active Collision-Based Navigation for Wheeled Robots," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 10932-10938, doi: 10.1109/ICRA57147.2024.10610726.

Link

<https://ieeexplore.ieee.org/document/10610726>

A Trajectory-based Flight Assistive System for Novice Pilots in Drone Racing Scenario

Abstract

Drone racing has become a popular international competition and has attained wide attention in recent years. However, the requirements of high-level operation keep the novice pilots away from participating in it. This paper presents a trajectory-based flight assistive system that enables various operators to fly the drone in a racing scene at a high speed. The whole system is structured hierarchically, consisting of both offline and online components. In the offline part, a global time-optimal trajectory is generated as the expert reference, and a dense flight corridor is constructed to provide sufficiently large safe region. In the online part, a remote control-mapped primitive is designed to fast encapsulate pilots' inputs, and the time mapping based trajectory progress is customized to further capture intention. Then, a trajectory planner is proposed to generate intention-aligned, smooth, feasible, and safe trajectories periodically. Additionally, a yaw planning that provides the pilot with the best suitable view angle is employed to further alleviate the operation difficulty. Simulations and real world experiments are implemented to verify the performance of our system. The maximum flight speed can reach 6.0 m/s for a novice drone pilot in a real racing scene. Our code is released as an open-source package.

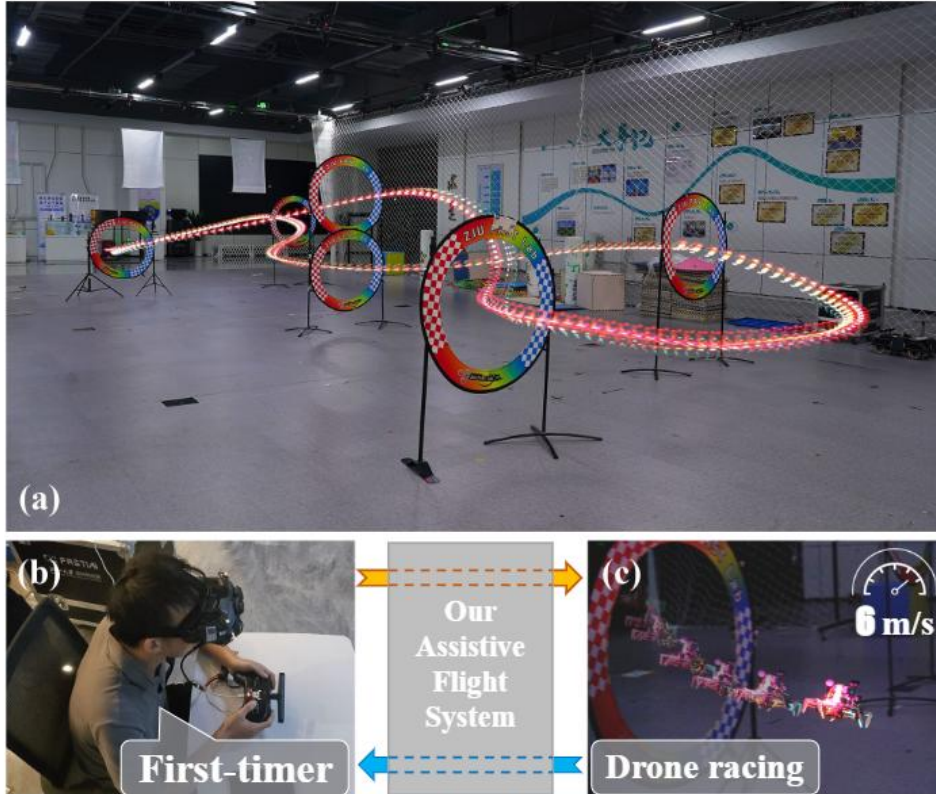


Fig. 1. Illustration of the real world experiments of our flight assistive system in a racing scene. (a) The demonstration of the real drone racing performance. (b) The first-time flying pilot wears goggles to access the FPV and operates the joystick to control the drone. (c) The snapshot of the racing drone at a high flight speed.

How NOKOV Contributes to the Research

The localization of drones is achieved by motion capture system.

References

Y. Zhong, G. Zhao, Q. Wang, G. Xu, C. Xu and F. Gao, "A Trajectory-based Flight Assistive System for Novice Pilots in Drone Racing Scenario," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 8714-8720, doi: 10.1109/ICRA57147.2024.10610179.

Link

<https://ieeexplore.ieee.org/document/10610179>

RMSC-VIO: Robust Multi-Stereoscopic Visual-Inertial Odometry for Local Visually Challenging Scenarios

Abstract

We present a Multi-Stereoscopic Visual-Inertial Odometry (VIO) system capable of integrating an arbitrary number of stereo cameras, exhibiting excellent robustness in the face of visually challenging scenarios. During system initialization, we introduce multi-view keyframes for simultaneous processing of multiple image inputs and propose an adaptive feature selection method to alleviate the computational burden of multi-camera systems. This method iteratively updates the state information of visual features, filtering out high-quality image feature points and effectively reducing unnecessary redundancy consumption. In the backend phase, we propose an adaptive tightly coupled optimization method, assigning corresponding optimization weights based on the quality of different image feature points, effectively enhancing localization precision. We validate the effectiveness and robustness of our system through a series of datasets, encompassing various visually challenging scenarios and practical flight experiments. Our approach achieves up to a 90% reduction in Absolute Trajectory Error (ATE) compared to state-of-the-art multi-camera VIO methods.

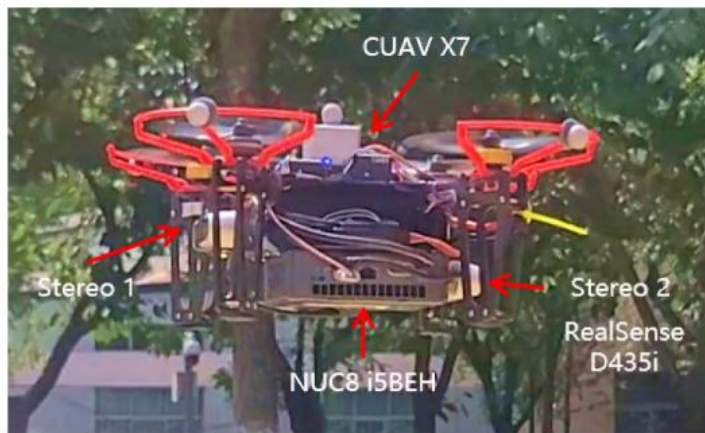


Fig. 4. Quadrotor drone employed in our experiment, equipped with two cameras featuring distinct orientations.

How NOKOV Contributes to the Research

For indoor sequences, we utilized NOKOV Motion Capture System to acquire ground truth poses with millimeter-level accuracy.

References

T. Zhang, J. Xu, H. Shen, R. Yang and T. Yang, "RMS-VIO: Robust Multi-Stereoscopic Visual-Inertial Odometry for Local Visually Challenging Scenarios," in *IEEE Robotics and Automation Letters*, vol. 9, no. 5, pp. 4130-4137, May 2024, doi: 10.1109/LRA.2024.3377008.

Link

<https://ieeexplore.ieee.org/document/10472018>

Simultaneous Time Synchronization and Mutual Localization for Multi-robot System

Abstract

Mutual localization stands as a foundational component within various domains of multi-robot systems. Nevertheless, in relative pose estimation, time synchronization is usually underappreciated and rarely addressed, although it significantly influences estimation accuracy. In this paper, we introduce time synchronization into mutual localization to recover the time offset and relative poses between robots simultaneously. Under a constant velocity assumption in a short time, we fuse time offset estimation with our previous bearing-based mutual localization by a novel error representation. Based on the error model, we formulate a joint optimization problem and utilize semi-definite relaxation (SDR) to furnish a lossless relaxation. By solving the relaxed problem, time synchronization and relative pose estimation can be achieved when time drift between robots is limited. To enhance the application range of time offset estimation, we further propose an iterative method to recover the time offset from coarse to fine. Comparisons between the proposed method and the existing ones through extensive simulation tests present prominent benefits of time synchronization on mutual localization. Moreover, real-world experiments are conducted to show the practicality and robustness.

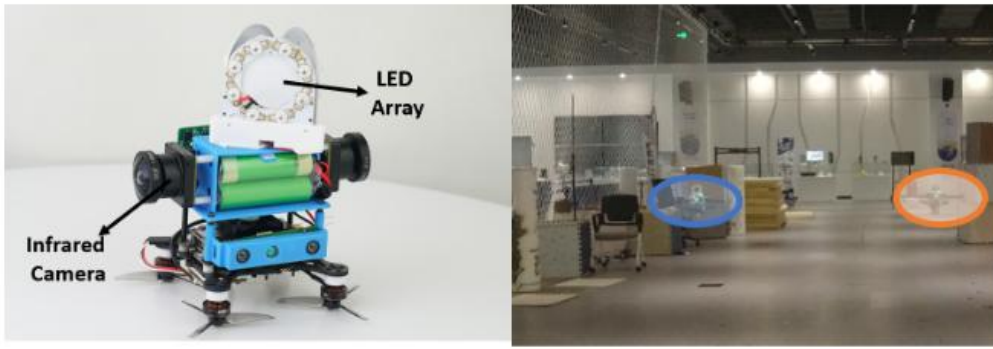


Fig. 6. Platform and snapshot in real-world experiment.

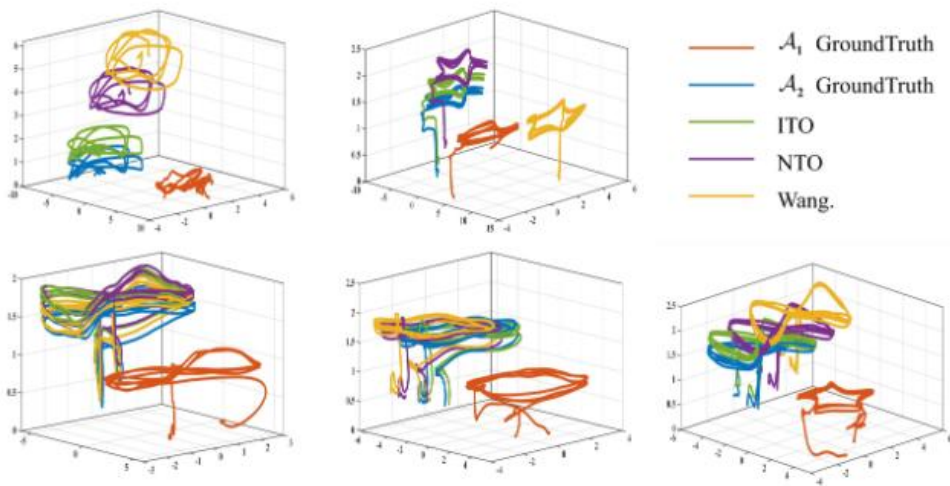


Fig. 7. Comparisons of estimated trajectories in real-world experiments. Two robots' trajectories are shown in a common frame.

How NOKOV Contributes to the Research

We utilize NOKOV motion capture system collect the ground truth.

References

X. Wen, Y. Wang, X. Zheng, K. Wang, C. Xu and F. Gao, "Simultaneous Time Synchronization and Mutual Localization for Multi-robot System," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 2603-2609, doi: 10.1109/ICRA57147.2024.10610915.

Link

<https://ieeexplore.ieee.org/document/10610915>

Mean-Shift Shape Formation of Multi-Robot Systems Without Target Assignment

Abstract

The methods of shape formation in robot swarms are usually classified into two categories by whether assignment is used or not. The first is to use target assignment to assemble precise formation. However, the additional algorithm for re-assignment is required to handle unreasonable situations, which results in lower efficiency. The second, also called assignment-free method, is to use local behaviors to assemble formation, however, existing methods can rarely achieve the precise formation. In this letter, we present a distributed assignment-free algorithm to achieve the precise shape formation based on the mean-shift algorithm. Specifically, each target location in robot's perception range is equally regarded as a point of the mean-shift vector. Then, the weight value of each point is computed according to the density of the target location. Here, each robot obtains the density of the target location according to the distribution of its neighbors. Moreover, this density calculation also considers the states of non-neighboring robots via the hop-count algorithm, thus avoiding conflicts among robots. Subsequently, each robot can regard the calculated mean-shift vector as its control command. Finally, simulation results show that our algorithm can form precise shapes at least 8 times more efficient than the assignment-based approach and physical experiment results confirm that the proposed algorithm exhibits promising potential for practical applications.

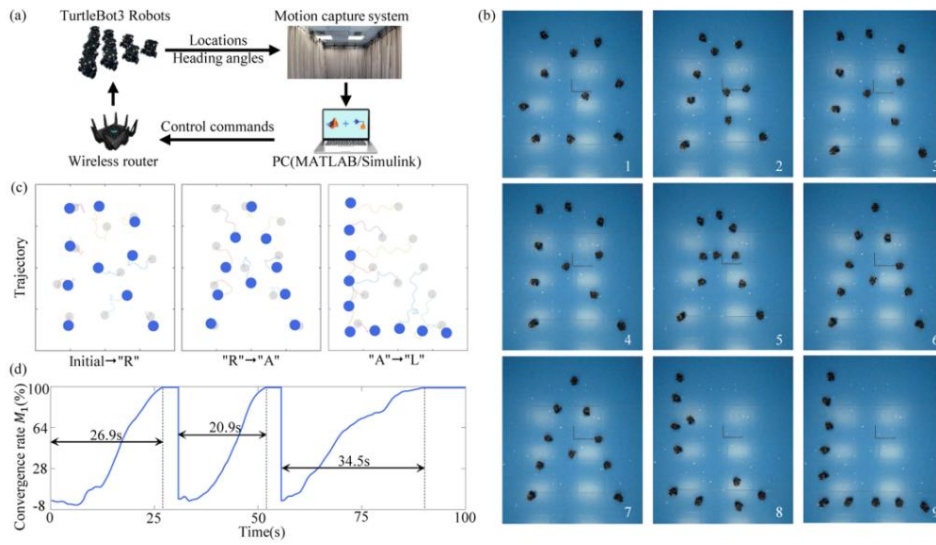


Fig. 7. System configuration and experiment results. (a) Experiment system configuration. (b) Snapshots of the experiment in forming "R," "A," and "L." The snapshots 1–3, 4–6, and 7–9 show the robots from initial state to the letter "R," the letter "R" to "A," and the letter "A" to "L," respectively. (c) Trajectories of the experiment process. (d) The convergence rate of the experiment over time.

How NOKOV Contributes to the Research

The Nokov Mocap system is used to obtain real-time positions of robots

References

Y. Zhang, R. Zhou, X. Li and G. Sun, "Mean-Shift Shape Formation of Multi-Robot Systems Without Target Assignment," in IEEE Robotics and Automation Letters, vol. 9, no. 2, pp. 1772-1779, Feb. 2024, doi: 10.1109/LRA.2024.3349926.

Link

<https://ieeexplore.ieee.org/document/10380669>

Time-Varying Formation Planning and Distributed Control for Multiple UAVs in Clutter Environment

Abstract

In this article, we present a novel method to achieve optimization based obstacle avoidance and distributed regular polygon time-varying formation control for multiple unmanned aerial vehicle systems (UAVs) in clutter environment. Under a leader-following structure, directed communication graph is considered. The formation size is time-varying and the trajectory of the leader UAV is planned in real time. Different from most of the existing trajectory planning algorithms for single UAV, an optimization-based safe trajectory and formation size online planning algorithm are proposed. To deal with information loss caused by directed communication topology, distributed smooth adaptive filters are designed for each UAV to asymptotically estimate the safe trajectory and formation size. Besides, a geometric tracking controller is adopted to track the desired trajectory for each UAV. Experimental results are provided to demonstrate the effectiveness of the proposed method.

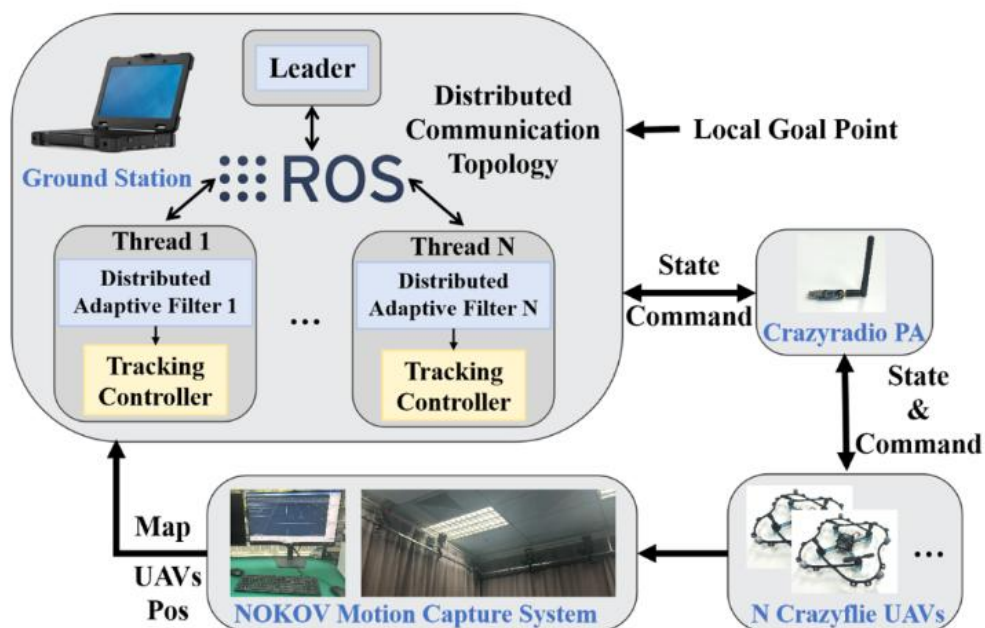


Fig. 3. Diagram of the experimental system.

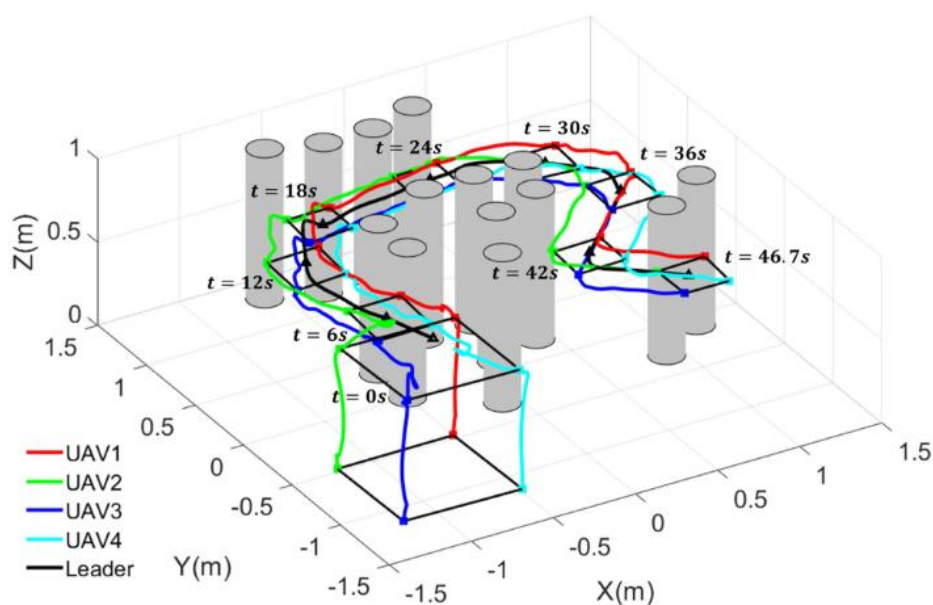


Fig. 9. Trajectory of follower UAVs in real world.

How NOKOV Contributes to the Research

The NOKOV motion capture system obtains the markers' information on a PC and provides localization and obstacle sensing.

References

K. Ze, W. Wang, K. Liu and J. Lü, "Time-Varying Formation Planning and Distributed Control for Multiple UAVs in Clutter Environment," in IEEE Transactions on Industrial Electronics, vol. 71, no. 9, pp. 11305-11315, Sept. 2024, doi: 10.1109/TIE.2023.3335448.

Link

<https://ieeexplore.ieee.org/abstract/document/10350055>

Air-M: A Visual Reality Many-Agent Reinforcement Learning Platform for Large-Scale Aerial Unmanned System

Abstract

Reinforcement learning for swarms of flying robots is a challenging task that requires a large number of data samples. Moreover, the problem of sim-to-real transfer has long been a challenge in robotics algorithm deployment. To address these issues, we propose Air-M, a platform that facilitates large-scale drone swarm learning in a distributed docker container environment and deployment in a virtual reality setting. Air-M trains the policy network using physics engines and creates replicas of agents in docker containers, which helps amortize the computational cost. In addition, Air-M establishes an intermediate link between the simulation and the real world, allowing real drones to interact with virtual objects via virtual sensors. This enables the policy network to be trained using virtual agents and seamlessly transferred to real drones. Air-M is highly scalable, accommodating hundreds of agents with dynamic models and virtual sensors. We evaluate the effectiveness of our approach by conducting experiments in three representative virtual scenarios with an increasing number of agents. Our results demonstrate that our method outperforms the state-of-the-art in terms of training efficiency and transferability, making it a promising platform for swarm robotics applications.

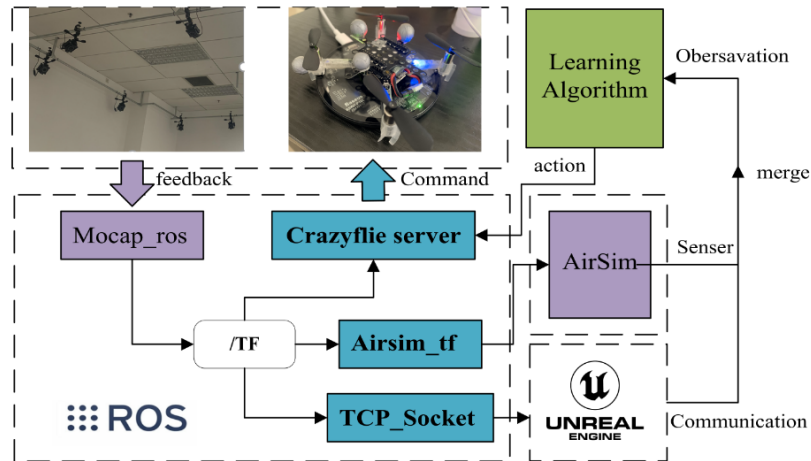


Fig. 4: Physical Experiment Testbed, it contains the external positioning system, the UAV control system, and the UAV ontology

How NOKOV Contributes to the Research

As shown in Figure. 4, we used the NOKOV Motion Capture System and modified the ROS driver of the crazyflie to enable the Mocap positioning.

References

J. Lou, W. Wu, S. Liao and R. Shi, "Air-M: A Visual Reality Many-Agent Reinforcement Learning Platform for Large-Scale Aerial Unmanned System," 2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS), Detroit, MI, USA, 2023, pp. 5598-5605, doi: 10.1109/IROS55552.2023.10341405.

Link

<https://ieeexplore.ieee.org/document/10341405>

Embodied Footprints: A Safety-Guaranteed Collision-Avoidance Model for Numerical Optimization-Based Trajectory Planning

Abstract

Optimization-based methods are commonly applied in autonomous driving trajectory planners, which transform the continuous-time trajectory planning problem into a finite nonlinear program with constraints imposed at finite collocation points. However, potential violations between adjacent collocation points can occur. To address this issue thoroughly, we propose a safety-guaranteed collision-avoidance model to mitigate collision risks within optimization-based trajectory planners. This model introduces an “embodied footprint”, an enlarged representation of the vehicle’s nominal footprint. If the embodied footprints do not collide with obstacles at finite collocation points, then the ego vehicle’s nominal footprint is guaranteed to be collision-free at any of the infinite moments between adjacent collocation points. According to our theoretical analysis, we define the geometric size of an embodied footprint as a simple function of vehicle velocity and curvature. Particularly, we propose a trajectory optimizer with the embodied footprints that can theoretically set an appropriate number of collocation points prior to the optimization process. We conduct this research to enhance the foundation of optimization-based planners in robotics. Comparative simulations and field tests validate the completeness, solution speed, and solution quality of our proposal.

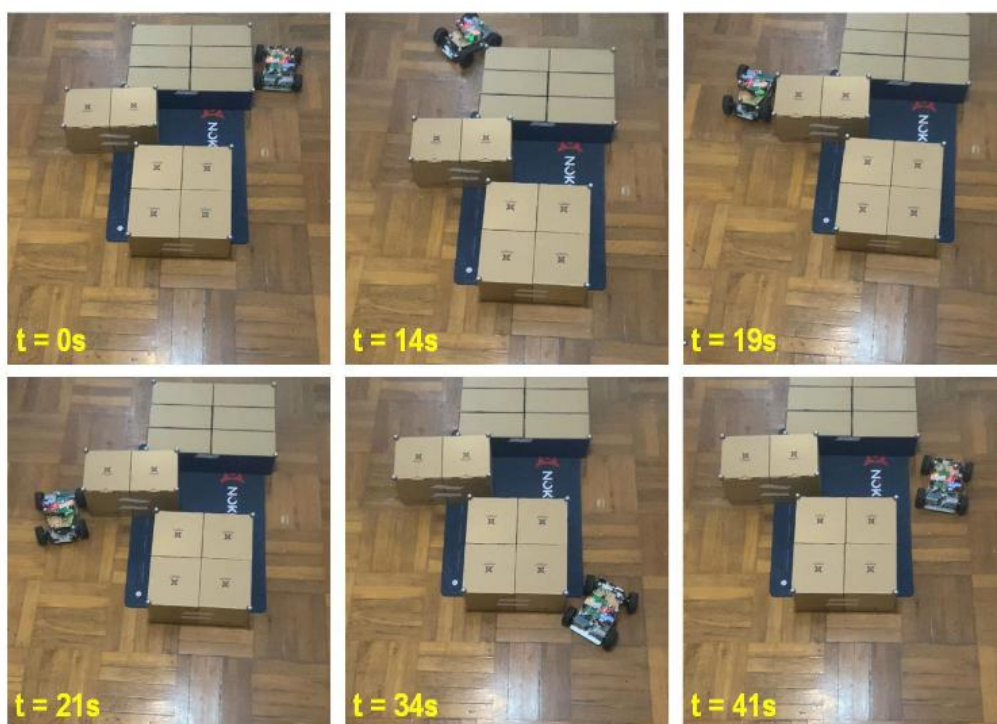


Fig. 15. Snapshots of closed-loop control performance when the ego vehicle is tracking a planned open-loop trajectory, which is derived by a trajectory planner with safe collision-avoidance constraints.

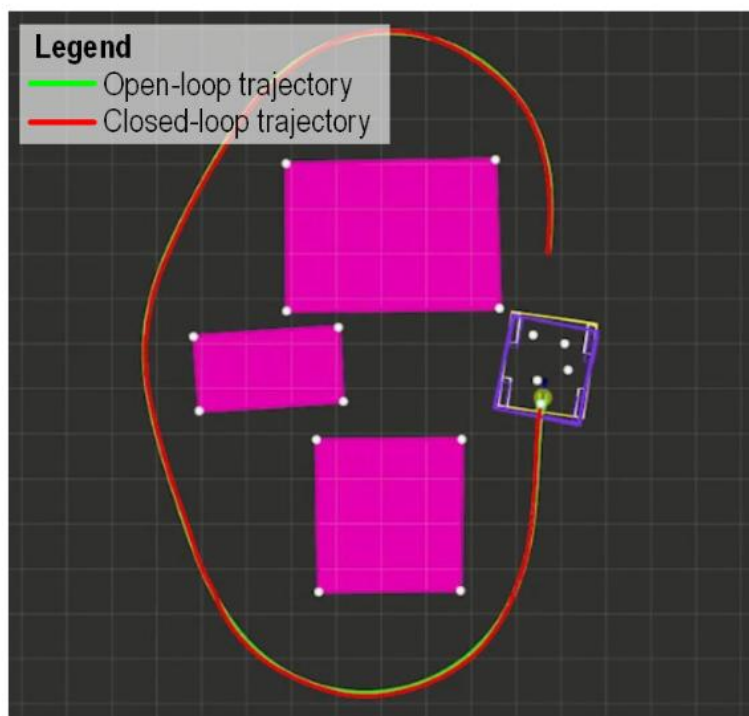


Fig. 16. Comparison between open-loop and closed-loop trajectories in field test.

How NOKOV Contributes to the Research

The indoor infrared solution, provided by NOKOV®, was implemented in this study. The infrared sensors from this solution operated at a frequency of 60Hz. The data collected for perception and localization was processed on a desktop computer.

References

B. Li et al., "Embodied Footprints: A Safety-Guaranteed Collision-Avoidance Model for Numerical Optimization-Based Trajectory Planning," in IEEE Transactions on Intelligent Transportation Systems, vol. 25, no. 2, pp. 2046-2060, Feb. 2024, doi: 10.1109/TITS.2023.3316175.

Link

<https://ieeexplore.ieee.org/abstract/document/10296483>

Sharing Traffic Priorities via Cyber–Physical–Social Intelligence: A Lane-Free Autonomous Intersection Management Method in Metaverse

Keywords

Autonomous intersection management (AIM), cyber–physical–social intelligence, metaverse, numerical optimal control.

Abstract

Replacing traffic signals with roadside vehicle-to-infrastructure systems in the era of connected and autonomous vehicles (CAVs) is promising. Managing CAVs in a signal-free intersection, known as autonomous intersection management (AIM), controls the driving behavior of each intersection-traverse CAV to maximize the throughput. Although AIM improves the gross throughput, the fairness of each individual vehicle in its right of way is not seriously considered. This study sets up an AIM system in the cyber–physical–social space to trade traverse priorities quantitatively and fairly. To that end, one needs an AIM method that is optimal and stable, otherwise no convincing trades of traverse priorities could be made. This study proposes a near-optimal lane-free AIM method based on numerical optimal control, wherein log-exp functions are deployed to convexify nondifferentiable collision-avoidance constraints. Besides that, a parameterized social force model (SFM) is proposed to provide a tunable initial guess for numerical optimal control. By tuning the urgency weights in SFM, one may get cooperative trajectories in different homotopy classes, which are further utilized to decide the amount of virtual currency to reward those CAVs who tend to share their traverse priorities. The overall method improves the traverse throughput with individual fairness respected. In experiencing this system, passengers learn how to behave with politeness when they drive manually. Experiments show the efficiency and robustness of the AIM method and also show the efficacy of the overall priority-sharing system.

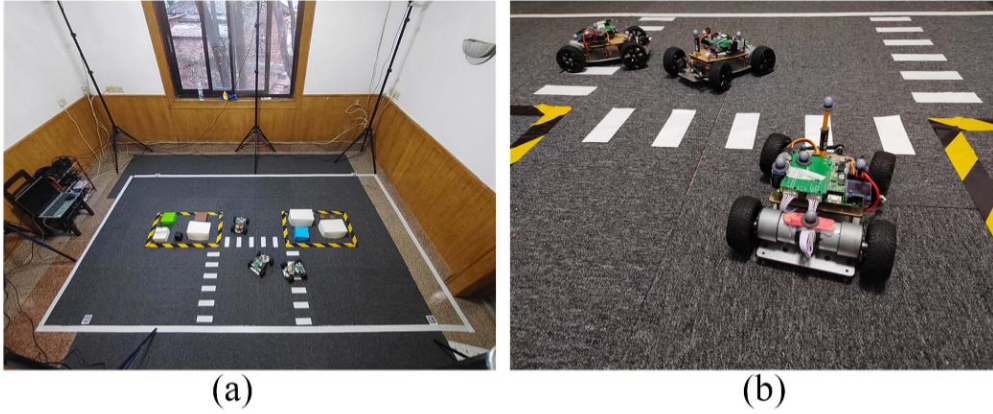


Fig. 8. Set up of indoor field experiments: (a) field layout of a *T*-junction scenario and (b) reflective marking points on tops of each robot for localization.

How NOKOV Contributes to the Research

Six infrared sensors of NOKOV Motion Capture System were set up for CAV localization. Specifically, reflective marking points were uniquely placed on the top of each car-like robot [Fig. 8(b)]. The reflective marking points reflected the beams emitted from the six light sources before the reflected beams got captured by the infrared sensors. Through this, the position of a robot can be estimated.

References

B. Li et al., "Sharing Traffic Priorities via Cyber-Physical-Social Intelligence: A Lane-Free Autonomous Intersection Management Method in Metaverse," in *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 53, no. 4, pp. 2025-2036, April 2023, doi: 10.1109/TSMC.2022.3225250.

Link

<https://ieeexplore.ieee.org/document/9982626>

Adaptive Pure Pursuit: A Real-Time Path Planner Using Tracking Controllers to Plan Safe and Kinematically Feasible Paths

Abstract

Path planning is an essential function in an intelligent vehicle, especially when driving in scenarios cluttered by large-scale static obstacles. Traditional path planners often struggle to find a balance among speed, accuracy, and optimality in their solutions. In this article, we introduce an Adaptive Pure Pursuit (APP) planner, which is designed to be fast and near-optimal for autonomous driving in cluttered environments. The APP planner generates feasible paths through a simulated closed-loop tracking control process of a virtual vehicle. If a derived path encounters obstacles, an adaptive refinement step is taken to locally reduce these collisions. Unlike search-based planners that suffer from the “curse of dimensionality” and optimization-based methods that often run slowly, the APP planner operates extremely fast. The high speed stems from the fact that both the virtual controller simulation and the refinement step involve computations with zero degrees of freedom. The proposed APP planner outperforms the prevalent optimization-based and search-based path planners, as shown by comparative simulations.



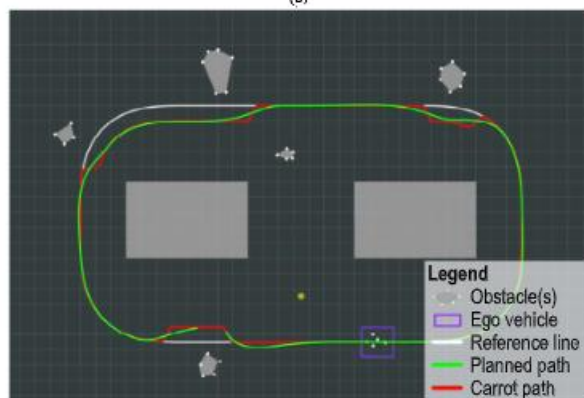
Fig. 8. Field test scenario layout and device setups.



(a)



(b)



(c)

Fig. 9. Typical experimental operations and results: (a) Relocation of an obstacle during the real-time tracking process; (b) evasive maneuvers of ego vehicle captured via time-lapse photography; (c) environmental layout and technical details of APP planner visualized in ROS rviz interface.

How NOKOV Contributes to the Research

A car-like robot of dimensions $0.211\text{m} \times 0.191\text{m}$ was utilized as the ego vehicle. Six infrared sensors from the NOKOV Motion Capture System were deployed for precise vehicle and obstacle localization.

References

B. Li et al., "Adaptive Pure Pursuit: A Real-Time Path Planner Using Tracking Controllers to Plan Safe and Kinematically Feasible Paths," in IEEE Transactions on Intelligent Vehicles, vol. 8, no. 9, pp. 4155-4168, Sept. 2023, doi: 10.1109/TIV.2023.3296435.

Link

<https://ieeexplore.ieee.org/document/10185581>

Enhancing the terrain adaptability of a multirobot cooperative transportation system via novel connectors and optimized cooperative strategies

Keywords

Mobile Robot, Motion planning

Abstract

Given limited terrain adaptability, most existing multirobot cooperative transportation systems (MRCTSs) mainly work on flat pavements, restricting their outdoor applications. The connectors' finite deformation capability and the control strategies' limitations are primarily responsible for this phenomenon. This study proposes a novel MRCTS based on tracked mobile robots (TMRs) to improve terrain adaptability and expand the application scenarios of MRCTSs. In structure design, we develop a novel 6-degree-of-freedom passive adaptive connector to link multiple TMRs and the transported object (the communal payload). In addition, the connector is set with sensors to measure the position and orientation of the robot with respect to the object for feedback control. In the control strategy, we present a virtual leader–physical follower collaborative paradigm. The leader robot is imaginary to describe the movement of the entire system and manage the follower robots. All the TMRs in the system act as follower robots to transport the object cooperatively. Having divided the whole control structure into the leader robot level and the follower robot level, we convert the motion control of the two kinds of robots to trajectory tracking control problems and propose a novel double closed-loop kinematics control framework. Furthermore, a control law satisfying saturation constraints is derived to ensure transportation stability. An adaptive control algorithm processes the wheelbase uncertainty of the TMR. Finally, we develop a prototype of the TMR-based MRCTS for experiments. In the trajectory tracking experiment, the developed MRCTS with the proposed control scheme can converge to the reference trajectory in the presence of initial tracking errors in a finite time. In the outdoor experiment, the proposed MRCTS consisting of four TMRs can successfully transport a payload weighing 60 kg on an uneven road with the single TMR's maximum load limited to 15 kg. The experimental results demonstrate the effectiveness of the structural design and control strategies of

the TMR-based MRCTS.

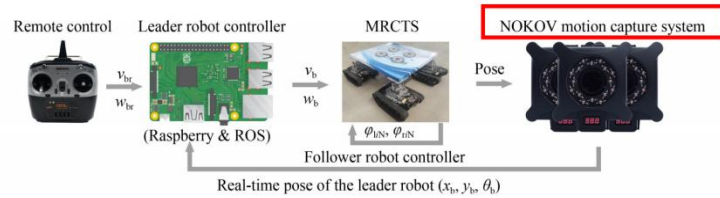


Fig. 10 Experimental setup and information flow in the trajectory tracking experiment. MRCTS: multirobot cooperative transportation system, ROS: robot operating system.

How NOKOV Contributes to the Research

The follower-robot controller1..deployed on the MCU board with a Wi-Fi module. Forthe NOKOV motion capture system isfeedback control.adopted to collect the leader robot's real-time poseinformation.

References Liu Q, Gong Z, Nie Z, et al. Enhancing the terrain adaptability of a multirobot cooperative transportation system via novel connectors and optimized cooperative strategies[J]. Frontiers of Mechanical Engineering, 2023, 18(3): 38.
Link <https://link.springer.com/article/10.1007/s11465-023-0754-2>

Dynamic Obstacle Avoidance for Cable-Driven Parallel Robots With Mobile Bases via Sim-to-Real Reinforcement Learning

Keywords

collision avoidance, machine learning for robot control, parallel robots, wire mechanism

Abstract

A Cable-Driven Parallel Robot (CDPR) with Mobile Bases (MBs) can modify its geometric architecture and is suitable for manipulation tasks in constrained environments. In manipulation tasks, a CDPR with MBs inevitably encounters obstacles, including dynamic obstacles. However, the high dimensional state space and a considerable number of constraints caused by multiple cables and MBs make the real-time dynamic obstacle avoidance of a CDPR with MBs challenging. This letter proposes a Reinforcement Learning (RL)-based dynamic obstacle avoidance method for a CDPR with MBs to deal with dynamic obstacles in real time. To explain the RL-based dynamic obstacle avoidance method, this letter focuses on a CDPR with four fixed-length cables connected to four MBs. An RL-based Obstacle Avoidance Controller (OAC) is developed and integrated into a trajectory tracking controller to address the dynamic obstacle avoidance problem of a CDPR with MBs tracking a target trajectory. To explain and evaluate the RL-based dynamic obstacle avoidance method further, an RL-based OAC is trained in a Mujoco simulator and transferred to a CDPR with four fixed-length cables connected to four MBs in the real world.

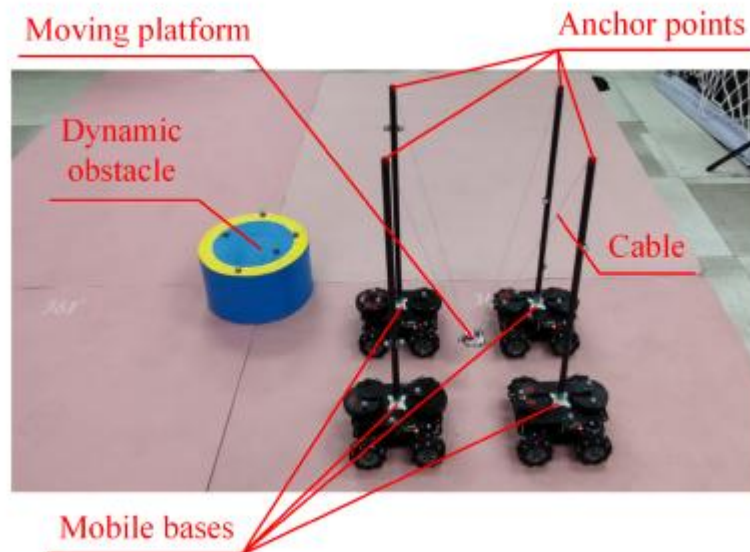


Fig. 1. A CDPR with four mobile bases.

How NOKOV Contributes to the Research

Unit (CPU) and 16 gigabits memory is used in simulation, and in experiments. The Graphics Processing Unit (GPU) of the personal computer is not used in simulation or in experiments. In experiments, the position of the MP, the position of MBs, and the position and shape of a dynamic obstacle are observed based on a NOKOV motion capture system.

References Liu Y, Cao Z, Xiong H, et al. Dynamic obstacle avoidance for cable-driven parallel robots with mobile bases via sim-to-real reinforcement learning[J]. IEEE Robotics and Automation Letters, 2023, 8(3): 1683-1690.

Link <https://ieeexplore.ieee.org/document/10035491>

Modeling and Robust Adaptive Practical Predefined Time and Precision Tracking Control of Unmanned Fire Fighting Robot

Keywords

Modeling, predefined precision tracking, predefined time tracking, robust adaptive control, unmanned fire fighting robots (UFFRs)

Abstract

This article studies the modeling and tracking control problems for a class of towed unmanned fire fighting robots. Considering that no similar modeling results exist, we take the lead in building a novel system model that takes into consideration both system uncertainties and external disturbances, including unknown friction factors and drag force. Then, to compensate for the adverse effects of system uncertainties and external disturbances, a novel robust control algorithm is proposed, which utilizes adaptive control and scaling techniques. Moreover, innovative predefined performance functions are designed to ensure that tracking processes meet predefined transient and steady-state requirements. Unlike most of the existing works, our predefined time performance function has the advantage that the convergence time and convergence accuracy can be arbitrarily changed. In the end, a novel robust adaptive control scheme with predefined time and precision tracking is designed using the backstepping recursive method. Based on Lyapunov stability theory, it is demonstrated that all signals in the closed-loop system are ultimately bounded, and both predefined transient and steady-state processes are never violated. To validate the effectiveness of this proposed control scheme, numerical simulations and practical platform experiments have been conducted. To ensure repeatability, our codes are open sourced on Github: <https://github.com/JiannanChen/Modeling-and-Robust-Control-UFFR.git>.

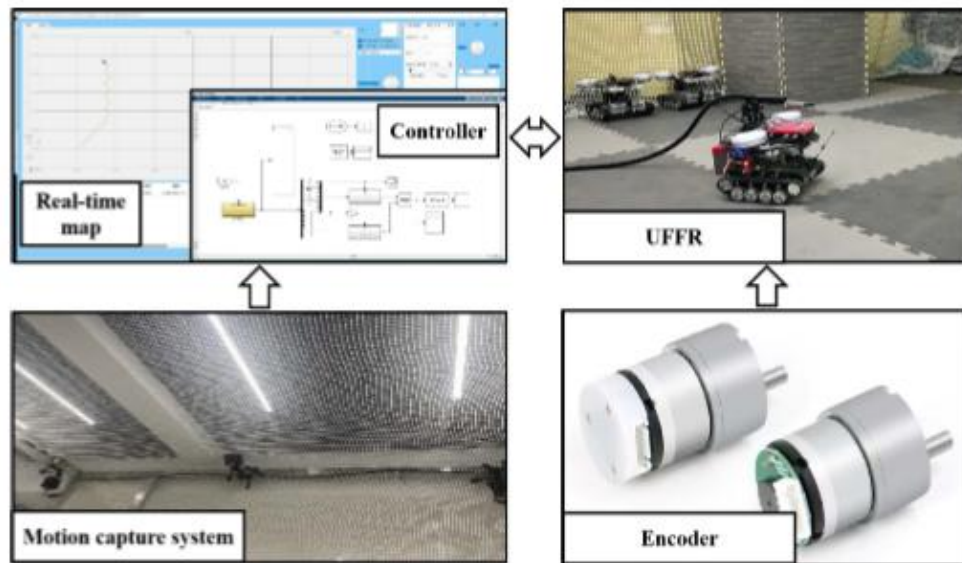


Fig. 5. System structure block diagram of our self-made unmanned fire fighting system platform.

How NOKOV Contributes to the Research

The operation process of the entire platform is shown in Fig.5:1) UFR onboard STM32 uses PWM input capture mode to acquire the pulse number of the encoder unit, then the velocity of real-time calculating is transmitted to onboard Raspberry through serial communication, and further transmitted to MATLAB control unit through the network; 2) the position and attitude of the UFR measured by the NOKOV motion capture system are transmitted to the MATLAB control unit through the network cable; 3) the MATLAB control unit receives these information to calculate the control signal according to the designed controller, and further sends it to the UFR through the network with UDP protocol; and 4) in the end, the upper computer displays the UFR trajectory in real time.

References Guo H, Ma Y, Ma Z, et al. Modeling and Robust Adaptive Practical Predefined Time and Precision Tracking Control of Unmanned Fire Fighting Robot.....[J]
Link <https://ieeexplore.ieee.org/abstract/document/10495058>

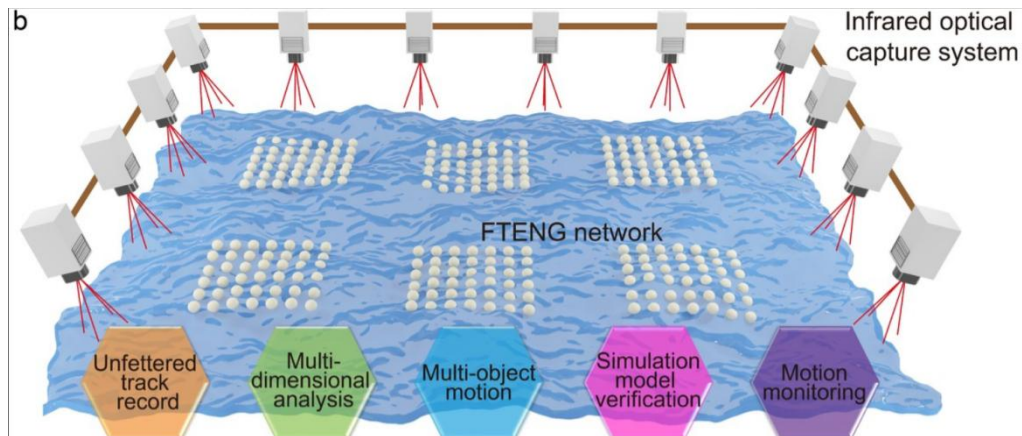
Dynamic behavior and energy flow of floating triboelectric nanogenerators

Keywords

Blue energy, Triboelectric nanogenerator, Six degrees of freedom test, Kinetics analysis, Optical motion capture system

Abstract

The comprehension of the dynamic behavior and energy flow characteristics of triboelectric nanogenerators on water is a prerequisite for enhancing wave energy conversion efficiency and output performance, yet these data have always been lacking, impeding their progress in harnessing blue energy. Herein, a series of meticulously conducted experiments aim to unveil the interaction between water waves and the geometrical structures of floating triboelectric nanogenerators (F-TENGs). The six degrees of freedom (6DoF) information of various F-TENGs under wave excitation is received for the first time by employing the new high-tech Infrared Optical Capture System and precision numeric analysis. Based on statistical data from 118 research papers, the most universally applicable modes/parameters are systematically explored, as well as the non-rigid body models. The six-dimensional kinematic radar matrices and energy gradient curves, peeled from the calculation and statistics of the 6DoF data, have comprehensively illuminated the dynamic behavior and energy flow of F-TENGs while interacting with water waves. These results can serve as an enlightening framework for structural design and facilitate the optimization of F-TENG in harnessing blue energy.



How NOKOV Contributes to the Research

The IOCS system used for testing was constructed with the assistance of NOKOV Corporation, utilizing their NOKOV Motion Capture System, which comprises four high-definition cameras, a splitter, a three-axis gimbal, markers, and other components. Among these components, the lenses play a central role in the entire IOCS system, as their resolution and field of view determine the effectiveness and precision of F- TENG motion tracking. Data decoding for the marker points is performed using NOKOV's specialized software Suixing and subsequent kinematic calculations are carried out in collaboration between Matlab and Origin.

References Xu S, Zhang J, Su E, et al. Dynamic behavior and energy flow of floating triboelectric nanogenerators[J]. Applied Energy, 2024, 367: 123468.

Link <https://www.sciencedirect.com/science/article/abs/pii/S0306261924008511>

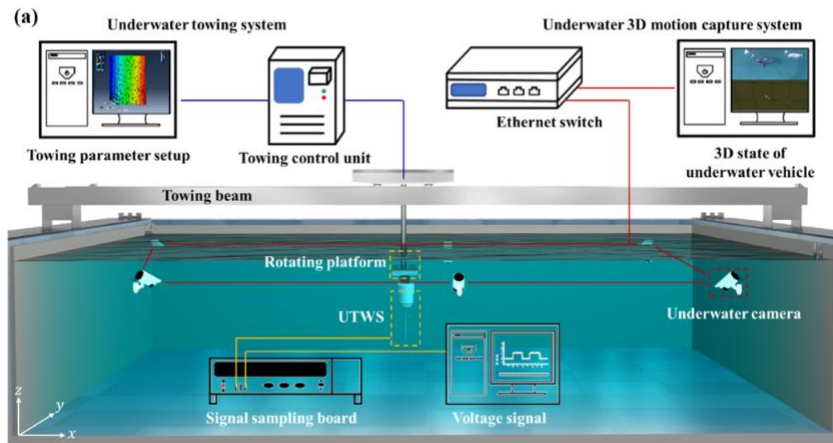
Deep-learning-assisted triboelectric whisker for near field perception and online state estimation of underwater vehicle

Keywords

Underwater perception, Triboelectric nanogenerators, Whisker sensor, Self-powered, Deep learning

Abstract

Agile near field perception remains a challenge for underwater vehicles, which could significantly enhance their capability of online state estimation. Harbor seals have evolved a whisker array that can accurately measure and identify environmental information in their surroundings. Inspired by the "smart" whiskers of harbor seals, the study has designed a deep learning-assisted bionic underwater triboelectric whisker sensor (UTWS) to passively perceive diverse hydrodynamic flow fields, including omnidirectional steady flow and wakes induced by bluff body upstream. The device mainly comprised an elliptical whisker shaft with a high-aspect-ratio of 0.403, four flexible triboelectric sensing units mimicking the nerve in the follicular sinus complex, and a flexible corrugated joint imitating the surface skin of marine organisms' cheeks. The UTWS demonstrated impressive advantages of rapid response times of 21 ms, high sensitivity of 1.16 V/m.s^{-1} , high signal-to-noise-ratio of 61.66 dB. By implementing deep learning analytics to process the multi-channel signals, the underwater vehicle equipped with the UTWS can accomplish online velocity estimation proficiently, with an approximate root mean square error of approximately 0.093 in the verification case. Thus, this UTWS-based, deep-learning-assisted perception could become a promising tool for integration with underwater vehicles in the local navigation tasks.



How NOKOV Contributes to the Research

The 3D position and speed of the ROV were acquired using a commercial motion capture system called Nokov, as depicted in Fig. 3a and Supplementary Fig. 5.

Fig. 7. Verification of the UTWS in online state estimation of the ROV. a, Data of randomly turning motion captured by Nokov. b, Real-time velocity estimation of the ROV. c, Angular velocity of the ROV the during the turning motion trajectory. d, Error band for the predicted trajectory. e, Angular velocity of the ROV the during the circle trajectory. f, Acceleration based on inertial measurement unit (IMU). g, Comparison among the truth trajectory, the

References Xu P, Liu J, Liu B, et al. Deep-learning-assisted triboelectric whisker for near field perception and online state estimation of underwater vehicle[J]. Nano Energy, 2024, 129: 11001

Link <https://www.sciencedirect.com/science/article/abs/pii/S2211285524007614>

Visual-Inertial Fusion-Based Five-Degree-of-Freedom Motion Measurement System for Vessel-Mounted Cranes

Keywords

stereo camera, object detection, IMU, Kalman filtering

Abstract

Vessel-mounted cranes operate in complex marine environments, where precise measurement of cargo positions and attitudes is a key technological challenge to ensure operational stability and safety. This study introduces an integrated measurement system that combines vision and inertial sensing technologies, utilizing a stereo camera and two inertial measurement units (IMUs) to capture cargo motion in five degrees of freedom (DOF). By merging data from the stereo camera and IMUs, the system accurately determines the cargo's position and attitude relative to the camera. The specific methodology is introduced as follows: First, the YOLO model is adopted to identify targets in the image and generate bounding boxes. Then, using the principle of binocular disparity, the depth within the bounding box is calculated to determine the target's three-dimensional position in the camera coordinate system. Simultaneously, the IMU measures the attitude of the cargo, and a Kalman filter is applied to fuse the data from the two sensors. Experimental results indicate that the system's measurement errors in the x , y , and z directions are less than 2.58%, 3.35%, and 3.37%, respectively, while errors in the roll and pitch directions are 3.87% and 5.02%. These results demonstrate that the designed measurement system effectively provides the necessary motion information in 5-DOF for vessel-mounted crane control, offering new approaches for pose detection of marine cranes and cargoes.

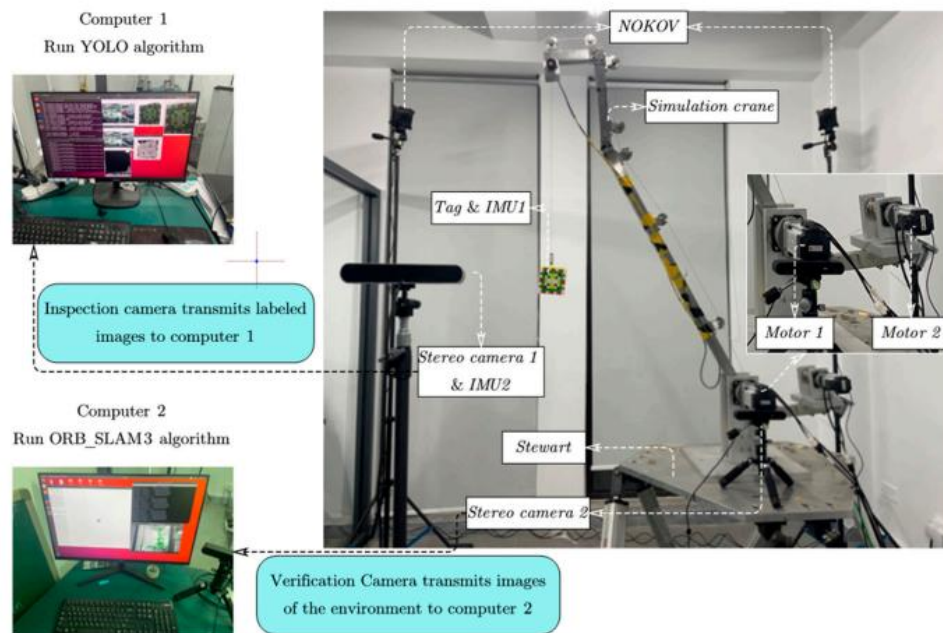


Figure 6. Schematic diagram of the experimental scene.

How NOKOV Contributes to the Research

To assess the accuracy of measurements during the experiment, measured by the NOKOV motion capture system composed of eight Mars2H were used as the ground truth. The experimentally generated data were benchmarked against the standards set by the International Maritime Organization (IMO) resolution MSC.363(92).

References Yu B, Cheng Y, Xia X, et al. Visual-Inertial Fusion-Based Five-Degree-of-Freedom Motion Measurement System for Vessel-Mounted Cranes[J]. Machines, 2024, 12(11): 748

Link <https://www.mdpi.com/2075-1702/12/11/748>

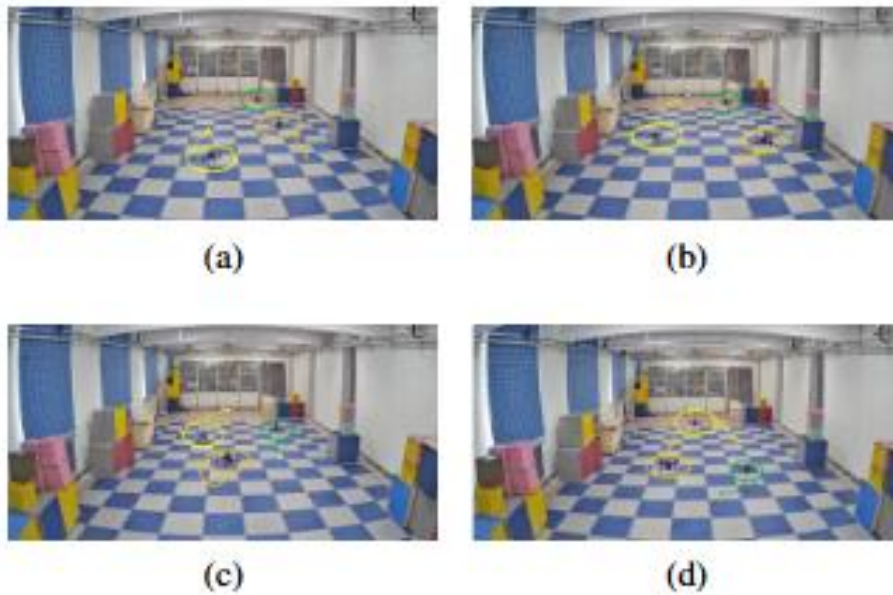
KD-EKF: A Consistent Cooperative Localization Estimator Based on Kalman Decomposition

Keywords

Cooperative localization, Kalman decomposition, nonlinear estimation, extended Kalman filter, consistency

Abstract

In this paper, we revisit the inconsistency problem of EKF-based cooperative localization (CL) from the perspective of system decomposition. By transforming the linearized system used by the standard EKF into its Kalman observable canonical form, the observable and unobservable components of the system are separated. Consequently, the factors causing the dimension reduction of the unobservable subspace are explicitly isolated in the state propagation and measurement Jacobians of the Kalman observable canonical form. Motivated by these insights, we propose a new CL algorithm called KD-EKF which aims to enhance consistency. The key idea behind the KD-EKF algorithm involves perform state estimation in the transformed coordinates so as to eliminate the influencing factors of observability in the Kalman observable canonical form. As a result, the KD-EKF algorithm ensures correct observability properties and consistency. We extensively verify the effectiveness of the KD-EKF algorithm through both Monte Carlo simulations and real-world experiments. The results demonstrate that the KD-EKF outperforms state-of-the-art algorithms in terms of accuracy and consistency.



How NOKOV Contributes to the Research

In addition, we adopt visual-inertial odometry to sense the linear and angular velocities. To evaluate the localization results, we use the Nokov motion capture system to track the poses of these robots as ground truth.

References Hao N, He F, Tian C, et al. Kd-ekf: A consistent cooperative localization estimator based on kalman decomposition[C]//2023 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2023: 11064-1107
Link <https://ieeexplore.ieee.org/document/10341604>

Synthesis of a single-degree-of-freedom noncircular pulley-five-bar finger rehabilitation mechanism considering force transfer performance

Keywords

Noncircular pulley-five-bar mechanism, finger rehabilitation, joint force, solution-region synthesis method, optimization

Abstract

Currently, most of the finger rehabilitation mechanisms achieve the desired motion by scale synthesis or using multi-degree-of-freedom (DOF) mechanisms, which cannot simultaneously achieve the requirements of posture and trajectory for finger rehabilitation. Aiming to solve this problem, this study proposes a single-DOF noncircular pulley-five-bar finger rehabilitation mechanism, obtains the pose requirements of finger movement and extracts four important poses. The solution domain of the five-bar mechanism with four poses was solved using the solution-region synthesis method, and the noncircular pulley was used to constrain two rods, making it a single-DOF mechanism. According to the requirements of the finger rehabilitation mechanism and considering the force transmission characteristics of the mechanism, the mechanism parameter optimization model is established, and the optimal parameters are optimized. The calculation model of noncircular pulley transmission is established, and the profile of noncircular pulley is solved according to the obtained rod rotation angle relationship. A finger rehabilitation device was developed, and an experimental study was performed. It was found that the device passed four poses, and the actual trajectory was consistent with the target trajectory. It is confirmed that the single-DOF noncircular pulley-five-bar finger rehabilitation mechanism proposed is effective, and solution method are correct.



Figure 3. Motion capture experiment environment.

How NOKOV Contributes to the Research

In this study, the motion capture experiment is performed by the NOKOV Motion Capture System, which calculates the coordinates, angles, speeds and other kinematic data of the marker points through calculation and analysis.

References Chen J, Yao K, Yu C, et al. Synthesis of a single-degree-of-freedom noncircular pulley-five-bar finger rehabilitation mechanism considering force transfer performance[J]. Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science, 2025, 239(1): 55-70.

Link <https://journals.sagepub.com/doi/10.1177/09544062241271746>

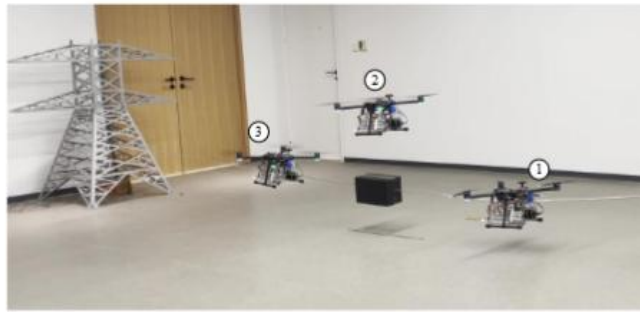
Model-Based Trajectory Planning of a Hybrid Robot for Powerline Inspection

Keywords

Aerial systems: Applications, constrained motion planning, motion and path planning

Abstract

This letter presents the first trajectory planning method for hybrid robot to perform powerline inspection involving obstacle navigation and landing. We develop a geometric model that incorporates constraints for landing the hybrid robot on a powerline, obstacle avoidance, and objectives that maximize the visibility of the powerline during flight. The trajectory generation is achieved via solving a multiple shooting nonlinear programming problem with respect to system dynamics and geometric constraints. The formulation of the problem accommodates both powerline-to-powerline and air-to-powerline trajectory planning scenarios. It runs onboard and is capable of generating trajectories within 50 ms, regardless of whether the hybrid robot's initial state is positioned on the powerline or hovering above it. Through simulation experiments, we illustrate the impact of our proposed geometric model on trajectory planning. Furthermore, real-world experimental results validate the efficacy of the proposed planning method. Compared with the existing feedback-control-based work, the landing and obstacle navigation time are significantly reduced.



(a)



(b)

How NOKOV Contributes to the Research

The state estimates of the hybrid robot are derived from an Extended Kalman Filter (EKF) that fuses data from external positioning systems like the NOKOV Motion Capture (Mocap) System or GPS, with IMU data from the PX4 Autopilot.

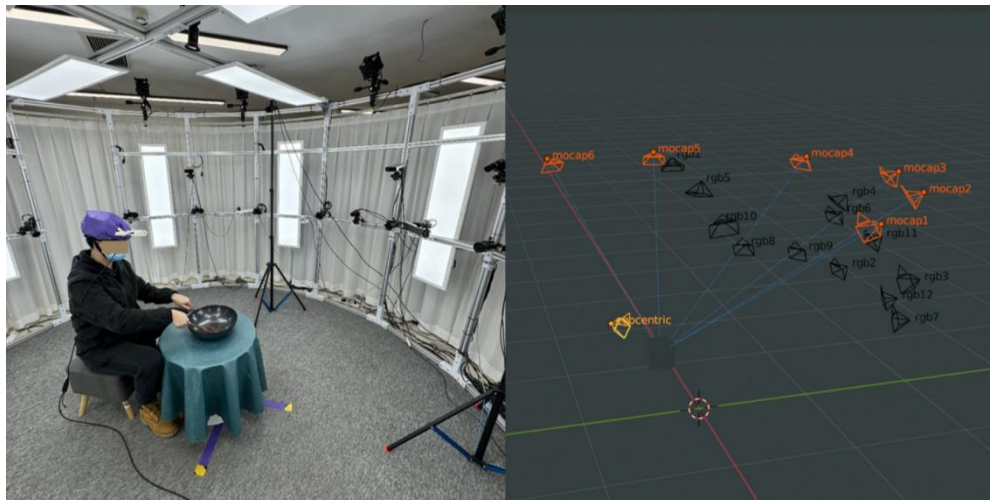
References Li Z, Tian Y, Yang G, et al. Model-Based Trajectory Planning of a Hybrid Robot for Powerline Inspection[J]. IEEE Robotics and Automation Letters, 2024, 9(4): 3443-3450.

Link <https://ieeexplore.ieee.org/abstract/document/10423842>

TACO: Benchmarking Generalizable Bimanual Tool- ACtion-Object Understanding

Abstract

Humans commonly work with multiple objects in daily life and can intuitively transfer manipulation skills to novel objects by understanding object functional regularities. However existing technical approaches for analyzing and synthesizing hand-object manipulation are mostly limited to handling a single hand and object due to the lack of data support. To address this we construct TACO an extensive bimanual hand-object-interaction dataset spanning a large variety of tool-action-object compositions for daily human activities. TACO contains 2.5K motion sequences paired with third-person and egocentric views precise hand-object 3D meshes and action labels. To rapidly expand the data scale we present a fully automatic data acquisition pipeline combining multi-view sensing with an optical motion capture system. With the vast research fields provided by TACO we benchmark three generalizable hand-object-interaction tasks: compositional action recognition generalizable hand-object motion forecasting and cooperative grasp synthesis. Extensive experiments reveal new insights challenges and opportunities for advancing the studies of generalizable hand-object motion analysis and synthesis. Our data and code are available at <https://taco2024.github.io>.



How NOKOV Contributes to the Research

To this end, as shown in Figure 2, our data capturing system incorporates 12 synchronized industrial FLIR cameras and a NOKOV optical motion capture suite with 6 infrared Mars4H cameras.

References Liu Y, Yang H, Si X, et al. Taco: Benchmarking generalizable bimanual tool-action-object understanding[C]//Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2024: 21740-21751.

Link

https://openaccess.thecvf.com/content/CVPR2024/html/Liu_TACO_Benchmarking_Generalizable_Bimanual_Tool-Action-Object_Understanding_CVPR_2024_paper.html

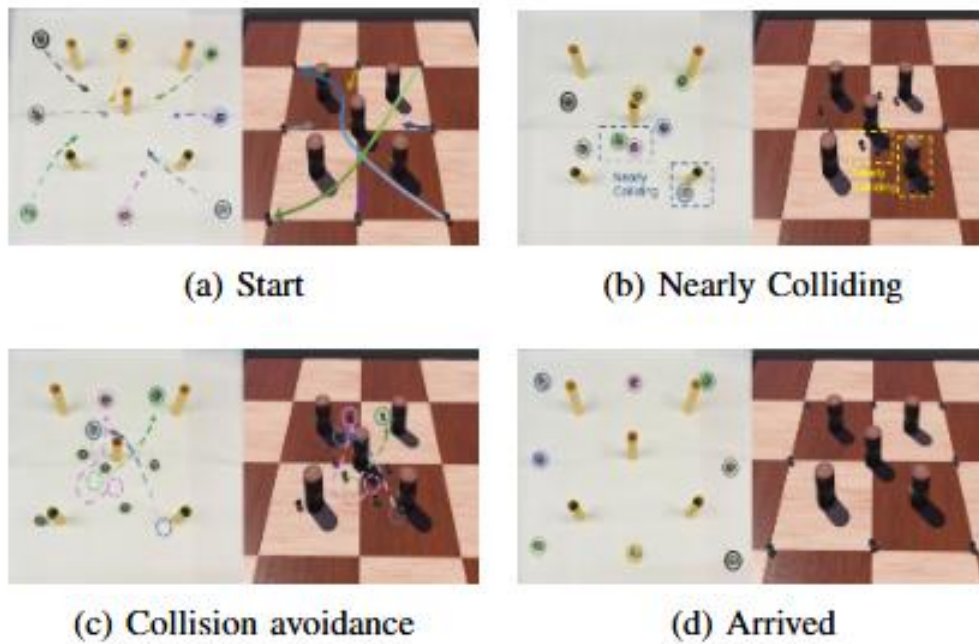
Hierarchical Consensus-Based Multi-Agent Reinforcement Learning for Multi-Robot Cooperation Tasks

Keywords

Training, Hands, Collaboration, Reinforcement learning, Contrastive learning, Strategic planning, Consensus protocol, Multi-robot systems, Standards, Intelligent robots

Abstract

In multi-agent reinforcement learning (MARL), the Centralized Training with Decentralized Execution (CTDE) framework is pivotal but struggles due to a gap: global state guidance in training versus reliance on local observations in execution, lacking global signals. Inspired by human societal consensus mechanisms, we introduce the Hierarchical Consensus-based Multi-Agent Reinforcement Learning (HC-MARL) framework to address this limitation. HC-MARL employs contrastive learning to foster a global consensus among agents, enabling cooperative behavior without direct communication. This approach enables agents to form a global consensus from local observations, using it as an additional piece of information to guide collaborative actions during execution. To cater to the dynamic requirements of various tasks, consensus is divided into multiple layers, encompassing both short-term and long-term considerations. Short-term observations prompt the creation of an immediate, low-layer consensus, while long-term observations contribute to the formation of a strategic, high-layer consensus. This process is further refined through an adaptive attention mechanism that dynamically adjusts the influence of each consensus layer. This mechanism optimizes the balance between immediate reactions and strategic planning, tailoring it to the specific demands of the task at hand. Extensive experiments and real-world applications in multi-robot systems showcase our framework's superior performance, marking significant advancements over baselines.



How NOKOV Contributes to the Research

We validated the real-world applicability of HC-MARL by conducting experiments on E-puck swarm. We utilized the NOKOV motion capture system for indoor positioning.

References Feng P, Liang J, Wang S, et al. Hierarchical Consensus-Based Multi-Agent Reinforcement Learning for Multi-Robot Cooperation Tasks[C]//2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2024: 642-649.

Link <https://ieeexplore.ieee.org/abstract/document/10802212>

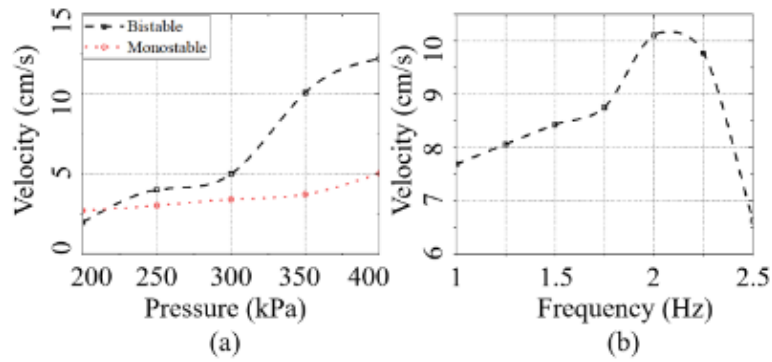
Manta Ray-Inspired Soft Robotic Swimmer for High-speed and Multi-modal Swimming

Keywords

Artificial muscles, Parametric study, Force, Soft robotics, Propulsion, Turning, Energy efficiency, Robots, Intelligent robots, Sports

Abstract

Manta rays exhibit complex motion behavior through their flexible fins. This study proposes a novel manta ray-inspired soft robotic swimmer with bistable flapping wings for high-speed and multi-modal swimming. The wings are created with prestressed bistable composite and actuated by small McKibben artificial muscles. Pressurizing and depressurizing the McKibben actuators integrated into the flapping wings generates alternating snap-throughs between two stable states, yielding swimming. Experiments are set up and conducted to analyze how the robot responses vary as a function of input pressures and actuation frequencies for both bistable and monostable modes. Experimental results show that the highest swimming velocity is 0.58 body lengths (BL) per second (equivalent to 12.23 cm/s), and the maximum turning angle speed is 22.5° per second with a smaller turning radius by holding the fins in asymmetric positions for bistable modes. Multimodal swimming motions are achieved including forward and backward translating, turning, and flip-turning.



How NOKOV Contributes to the Research

To quantify swimming speed, the robot is directed to perform ten cycles of PWM square wave motions under different pressures. Velocity is subsequently recorded by a NOKOV motion capture system. The number of measurements is three times for each pressure, taking the average values as the final results as shown in Fig. 5(a).

References Xu Z, Liang J, Zhou Y. Manta Ray-Inspired Soft Robotic Swimmer for High-speed and Multi-modal Swimming[C]//2024 IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS). IEEE, 2024: 235-240.

Link <https://ieeexplore.ieee.org/document/10802678>

A Swarm Model with Constraint Coordination Mechanism for Unmanned Aerial Vehicle Swarm Formation Maintenance in Dense Environments

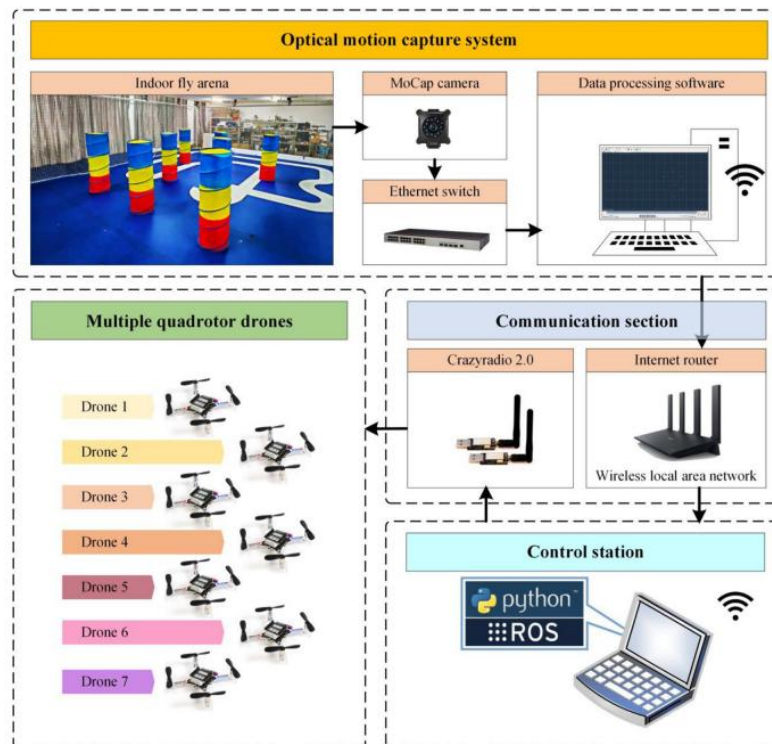
Keywords

Formation maintenance, Constraint coordination, Self-organized behavior, UAV swarm

Abstract

The formation of unmanned aerial vehicle (UAV) swarm plays a critical role in numerous applications, such as unmanned agriculture, environmental monitoring and cooperative fencing. Meanwhile, the self-organized swarm model exhibits excellent performance in amorphous formation flight, and its collective motion pattern displays great potential in dense obstacle avoidance. The paper aims to realize the formation maintenance of UAVs while combining the advantage of the self-organized swarm model in avoiding dense obstacles. Thereby enhancing the flexibility, adaptability and safety of UAV swarms in dense and unpredictable scenarios.

Figure 4 Self-organized UAV swarm flight experimental platform



How NOKOV Contributes to the Research

The self-organized UAV swarm flight experiment platform is shown in Figure 4, which includes the NOKOV Motion Capture System, the Robot Operating System, a Wireless Local Area Network, a pair of UAV communication modules and a set of quadcopters.

References Xiong H, Shi X, Liu J Z, et al. A swarm model with constraint coordination mechanism for unmanned aerial vehicle swarm formation maintenance in dense environments[J]. Industrial Robot: the international journal of robotics research and application, 2024.

Link <https://www.emerald.com/insight/content/doi/10.1108/ir-07-2024-0316/full/html>

A Dual-Mode Robot-Assisted Plate Implantation Method for Femoral Shaft Fracture

Keywords

digitalization of surgical operation, dual-mode surgery, femoral shaft fracture, fracture plate implantation, robot-assisted surgery

Abstract

Background: Minimally invasive internal fixation is the preferred treatment option for femoral shaft fractures. However, there are problems such as invisibility, inaccuracy and instability in the process of plate implantation. Methods: In this paper, a dual-mode robot-assisted plate implantation method was proposed by combining a starting point determination algorithm, motion capture, deep learning and robotics. Results: The neural network model planned the plate implantation trajectory according to patient's condition. Then, the advantages of high stability and high precision of the robotic arm were used for plate implantation. Conclusion: The trend and fluctuation of the plate implantation trajectories obtained using this method met clinical requirements. Furthermore, the robotic arm implantation process was safe.

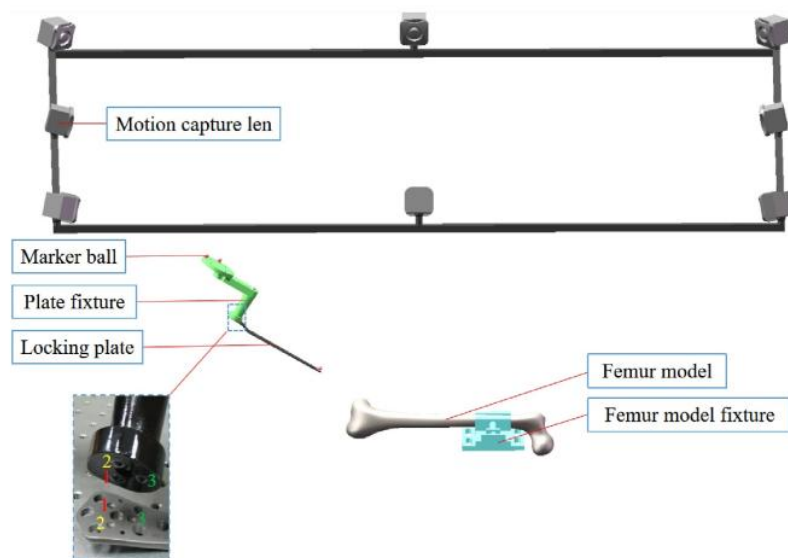


FIGURE 5 | Trajectory acquisition device.

How NOKOV Contributes to the Research

The plate fixture can be either hand-held or robot arm connected. The motion capture system is an eight-lens NOKOV optical three-dimensional motion capture system. After the plate is fastened to the plate fixture, three marker points are attached to it.

References Guo Y, Chen Y, Zhou X, et al. A Dual-Mode Robot-Assisted Plate Implantation Method for Femoral Shaft Fracture[J]. The International Journal of Medical Robotics and Computer Assisted Surgery, 2024, 20(6): e70008.
Link <https://onlinelibrary.wiley.com/doi/10.1002/rcs.70008>

Development of a Load-Bearing, Terrain-Adaptive Hexapod Robot With Chebyshev-Linkage Legs

Keywords

Chebyshev linkage, dynamic simulation, hexapod robots, load-bearing capacity, terrain adaptability

Abstract

This study introduces an exploratory design for a hexapod load-carrying robot, aiming to address challenges commonly associated with quadruped robots, including limited load capacity and high control complexity. By leveraging a Chebyshev linkage-based overconstrained leg-foot architecture that combines high rigidity and low friction, along with a multi-drive system for lateral movement, we propose a robot with improved adaptability to diverse terrains, such as snow, sand, puddles, ice, and deserts. This versatility suggests potential applications in areas such as factory inspection, field reconnaissance and transport, and desert exploration. Using static and dynamic analyses, we evaluated the leg structure's stress distribution and motion requirements, employing simulations in Workbench and Adams to cross-validate our theoretical calculations. The development of a prototype and its subsequent testing under various environmental conditions aimed to demonstrate the design's practicality and the robot's operational capabilities in real-world settings. Although the findings indicate progress toward enhancing load-bearing capabilities and terrain adaptability with reduced control complexity, further research and development are necessary to fully realize the potential of such robotic systems in practical applications.

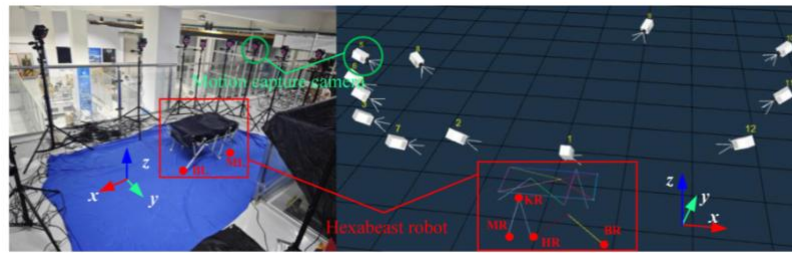


FIGURE 25 | Motion data validation of the Hexabeast robot. [Color figure can be viewed at [wileyonlinelibrary.com](https://onlinelibrary.wiley.com)]

How NOKOV Contributes to the Research

To verify the differences between the simulated and actual movements of the robot, we used the NOKOV Motion Capture System to capture the motion data of the Hexabeast robot. The specific testing environment is shown in Figure 25.

References Ju Z, Wu R, Guo D, et al. Development of a Load-Bearing, Terrain-Adaptive Hexapod Robot With Chebyshev-Linkage Legs[J]. Journal of Field Robotics, 2024.

Link <https://onlinelibrary.wiley.com/doi/abs/10.1002/rob.22463>

Controlling Soft Robotic Arms Using Hybrid Modelling and Reinforcement Learning

Keywords

Robot control, soft robotics, robot learning, reinforcement learning, hybrid modeling

Abstract

Soft robotic arms exhibit high deformability and degrees of freedom, which brings challenges in modeling accuracy and susceptibility to gravitational effects, resulting in imprecise control. This study proposes a reinforcement learning control strategy based on a hybrid model for precise control of soft robotic arms. The hybrid model is formulated by gathering a small dataset of analytical modeling errors of coordinates under various loading (0 g~300 g) conditions, followed by employing a Multilayer Perceptron (MLP) to fit these errors, thereby reducing the forward kinematics MAEs (mean absolute errors) from a range of 75.2 mm~95.3 mm to 5.9 mm~9.1 mm. The reinforcement learning virtual environment is built upon the hybrid model and the Proximal Policy Optimization (PPO) algorithm is used to train control policies. The efficiency of the control policy is validated for different trajectories and loading conditions both in simulation and on the soft robotic arm prototype, revealing distance errors of 3.9 mm~6.7 mm and 12.1 mm~15.4 mm, respectively, representing 1.1%~1.9% and 3.5%~4.4% of the total arm length.

How NOKOV Contributes to the Research

T Finally, the motion capture system with eight Mars2H (NOKOV) cameras collected endpoint coordinates, which are transmitted to and stored on the computer. Initially, the soft robotic arm was suspended, and four reflective markers were affixed at the front endpoint of the arm.

References Lou G, Wang C, Xu Z, et al. Controlling soft robotic arms using hybrid modelling and reinforcement learning[J]. IEEE Robotics and Automation Letters, 2024. Link <https://ieeexplore.ieee.org/document/10569050>

Soft Pneumatic Helical Actuators With Programmable Variable Curvatures

Keywords

Analytical modeling, helical actuators, pneumatic actuators, soft gripper, soft robotics

Abstract

This letter proposes novel variable-curvature soft helical actuators (VC-SHAs) using fluidic prestressed composites (FPCs) for enhanced grasping performance. A third-order three-dimensional chained composite model called 3D-CCM is developed to pre-program variable curvatures of SHAs and predict their actuation responses. The proposed actuator design increases shape-matching areas between the objects and the soft actuators, highly improving grasping performance. Various VC-SHAs are fabricated and tested to validate the model and parametric studies are conducted to identify the effects of key parameters on actuator responses. A bioinspired passive dynamic gripper is developed with three VC-SHAs. Loading tests are conducted to evaluate its grasping success rates and maximum payload for various objects and object sizes, achieving a maximum payload of 26.56 N. The gripper is integrated to a drone to demonstrate its capability to perch on pipelines of different sizes and angles, and various objects.

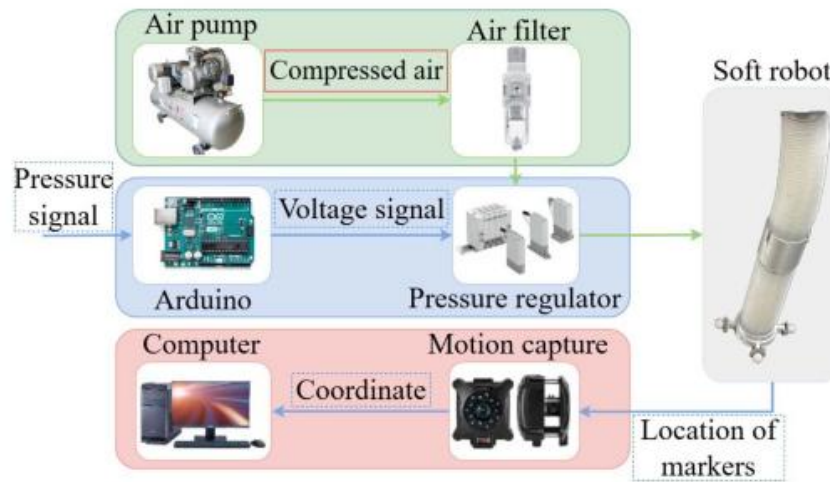


Fig. 6. Experimental setup for soft robotic arm tip coordinates collection.

How NOKOV Contributes to the Research

Finally, the motion capture system with eight Mars2H (NOKOV) cameras collects endpoint coordinates, which are transmitted to and stored on the

References Xu Z, Liang J, Zhou Y. Soft pneumatic helical actuators with programmable variable curvatures[J]. IEEE Robotics and Automation Letters, 2024.

Link <https://ieeexplore.ieee.org/document/10549798>

Upper-Limb Rehabilitation with a Dual-Mode Individualized Exoskeleton Robot: A Generative-Model-Based Solution

Keywords

Upper-limb rehabilitation, diffusion-based trajectory generation, individualized assistance, compliantly actuated and cable-driven exoskeleton

Abstract

Several upper-limb exoskeleton robots have been developed for stroke rehabilitation, but their rather low level of individualized assistance typically limits their effectiveness and practicability. Individualized assistance involves an upper-limb exoskeleton robot continuously assessing feedback from a stroke patient and then meticulously adjusting interaction forces to suit specific conditions and online changes. This paper describes the development of a new upperlimb exoskeleton robot with a novel online generative capability that allows it to provide individualized assistance to support the rehabilitation training of stroke patients. Specifically, the upper-limb exoskeleton robot exploits generative models to customize the fine and fit trajectory for the patient, as medical conditions, responses, and comfort feedback during training generally differ between patients. This generative capability is integrated into the two working modes of the upper-limb exoskeleton robot: an active mirroring mode for patients who retain motor abilities on one side of the body and a passive following mode for patients who lack motor ability on both sides of the body. In addition, the upperlimb exoskeleton robot has three other attractive features. First, it has six degrees of freedom (DoFs), namely five active DoFs and one passive DoF, to assist the shoulder and the elbow joints and cover the full range of upper-limb movement. Second, most of its active joints are driven by series elastic actuators (SEAs) and a cable mechanism, which absorb energy and have low inertia. These compliantly driven high DoFs provide substantial flexibility and ensure hardware safety but require an effective controller. Thus, based on the singular perturbation approach, a model-based impedance controller is proposed to fully exploit the advantages of the hardware. Third, the safety of the upper-limb exoskeleton robot is guaranteed by its hardware and software. Regarding hardware, its SEAs are tolerant to impacts and have high backdrivability. Regarding

software, online trajectory refinement is performed to regulate the assistance provided by the upper-limb exoskeleton robot, and an anomaly detection network is constructed to detect and relax physical conflicts between the upper-limb exoskeleton robot and the patient. The performance of the upper-limb exoskeleton robot was illustrated in experiments involving healthy subjects and stroke patient.

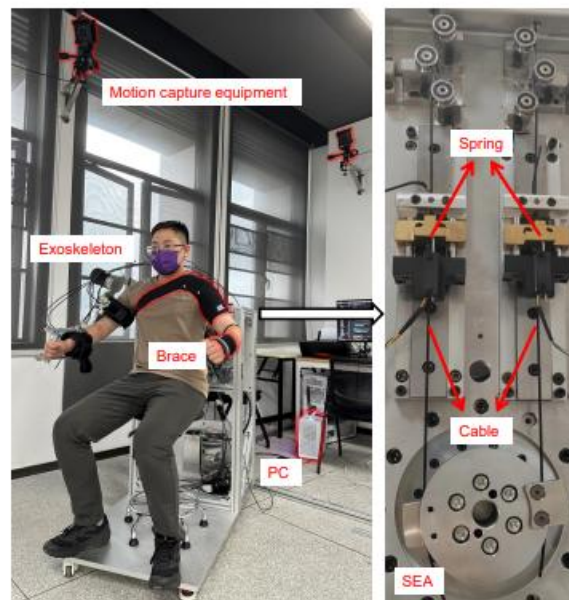


Figure 8. The experimental setup in active mirroring mode used motion capture equipment to obtain feedback from the unaffected side of the body. The SEA was constructed by connecting the motor output and the joint end with a spring and cable.

How NOKOV Contributes to the Research

Figure 8 depicts the experimental configuration, in which the main board controlled the impedance of the upper-limb exoskeleton robot. This board was interfaced via a serial port connection with a PC, which was outfitted with an Intel i5-13490F CPU and an RTX 4060Ti graphics card. The PC executed the anomaly detection module, calculated the reference trajectory q_r , and dynamically updated the desired trajectory q_d online. Subjects performed rehabilitation training in passive following mode or active mirroring mode. In the latter mode, the subject wore a brace equipped with markers on the unaffected side of the body to facilitate engagement in rehabilitation exercises. Optical motion capture equipment (Nokov) was used to accurately capture motion intentions.

References Chen Y, Miao S, Ye J, et al. Upper-Limb Rehabilitation with a Dual-Mode Individualized Exoskeleton Robot: A Generative-Model-Based Solution[J]. arXiv preprint arXiv:2409.03193, 2024.
Link <https://arxiv.org/abs/2409.03193>

Transformer-based segmented learning for kinematics modelling of a cable driven parallel continuum robot

Keywords

Parallel continuum robot, kinematics modelling, segmented learning, neural network

Abstract

Continuum robots with structural compliance exhibit significant operational potential in unstructured environments. However, theoretical kinematics modelling for continuum robots is complex due to their uncertain nonlinear characteristics. Although learning-based algorithms can effectively address the inherent challenges, unfortunately they frequently fail to capture internal information in robotic systems involving multiple spatial mappings, which reduces the accuracy of data-driven approaches in such applications. In this regard, we propose a segmented learning approach applied to kinematic modelling for a cable-driven parallel continuum robot (PCR), using Transformer networks to segmentally learn the mapping from task space to configuration space, and from configuration space to actuation space. Additionally, in order to evaluate the performance of the proposed approach, we employ multiple sets of neuralnetwork models and various segmented learning configurations in the experiments. The experimental results across different trajectories show that, the Transformer network model under the segmented learning approach achieves superior trajectory tracking accuracy, compared to state-of-the-art modelling approaches.

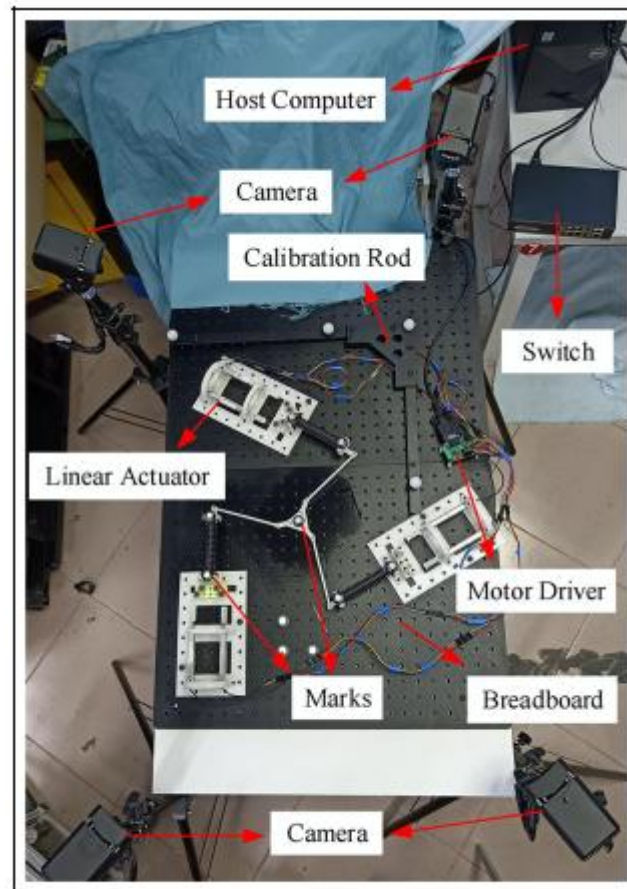


Figure 5. Experimental platform, including the PCR, motion capture cameras, motor driver and host computer.

How NOKOV Contributes to the Research

Calibration rods establish the camera's base coordinate system, and a switch facilitates communication between the host computer and the camera system. The motion capture camera, Pluto 1.3C from Beijing NOKOV Science and Technology, has a 3D accuracy of 1 mm. Four sets of cameras are arranged, achieving sub-millimetre calibration accuracy.

References Liu X, Wang Z, Li Z, et al. Transformer-based segmented learning for kinematics modelling of a cable-driven parallel continuum robot[J]. Proceedings of the Institution of Mechanical Engineers, Part C: Journal of Mechanical Engineering Science, 2024: 09544062241306683.

Link <https://journals.sagepub.com/doi/10.1177/09544062241306683>

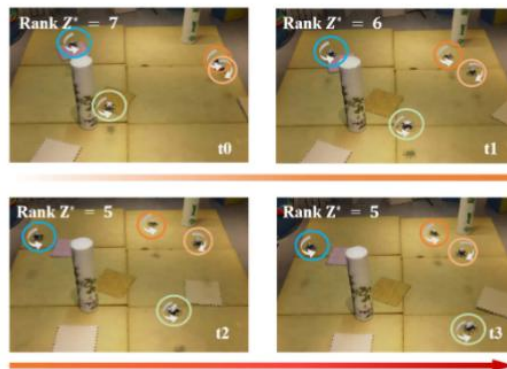
FACT: Fast and Active Coordinate Initialization for Vision-Based Drone Swarms

Keywords

Aerial systems, perception and autonomy, swarm robotics, vision-based navigation

Abstract

Coordinate initialization is the first step in accomplishing collaborative tasks within robot swarms, determining the quality of tasks. However, fast and robust coordinate initialization in vision-based drone swarms remains elusive. To this end, our letter proposes a complete system for initial relative pose estimation, including both relative state estimation and active planning. Particularly, our work fuses onboard visual-inertial odometry with vision-based observations generating bearing and range measurements, which are anonymous, partially mutual, and noisy. It is the first method based on convex optimization to initialize coordinates with vision-based observations. Additionally, we designed a lightweight module to actively control the movement of robots for observation acquisition and collision avoidance. With only stereo cameras and inertial measurement units as sensors, we validate the practicability of our system in simulations and real-world areas with obstacles and without signals from the Global Navigation Satellite System. Compared to methods based on local optimization and filters, our system can achieve the global optimum for coordinate initialization more stably and quickly, which is suitable for robots with size, weight, and power constraints. The source code is released for reference.



(a) Real-world experiments with 4 drones.



(b) Real-world experiments with 3 drones.

How NOKOV Contributes to the Research

To analyze errors in real-world experiments, we use the NOKOV Motion Capture System to get the ground truth of rotation estimation, including poses of all robots in the motion capture system coordinate.

References Li Y, Zhao A, Wang Y, et al. Fact: Fast and active coordinate initialization for vision-based drone swarms[J]. IEEE Robotics and Automation Letters, 2024.

Link <https://ieeexplore.ieee.org/document/10803028>

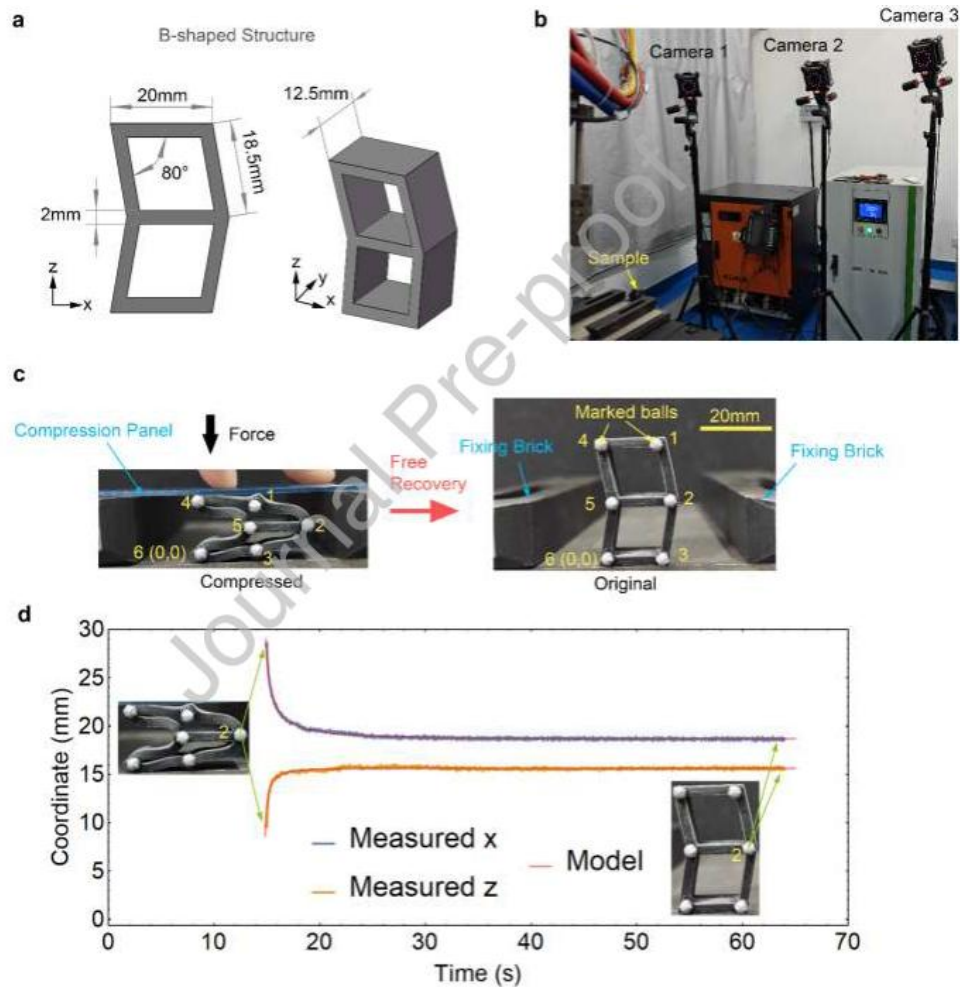
Enabling sequential logic leveraging time delays of thin-walled soft matter

Keywords

Mechanical computing; Sequential logic; Information processing; Origami structures; Soft materials; Property programming;

Abstract

Combinational logic is a common feature of modern engineered materials for general computation, but incorporating sequential logic is comparatively less advanced, which impedes further uses in environmental responses and adaptations. Sequential logic needs the history of the system outputs concomitantly with the present inputs, and we expect only one iteration of outputs and inputs for each operation. Thus, the main thrust of our research is to propose an original and general solution that utilizes the time delay effect of thin-walled soft matter inspired by origami structures to coordinate the work pace of the overall system. We create four basic sequential systems for ordered performance leveraging this device: a time delayer, a clock generator, a self-locking system (memory), and a pulse generator. Furthermore, we build five complex sequential systems based on the basic systems (a value assigner, a flip-flop, a shift register, a cycle shift register, and a counter). The novel contribution is making mechanical integrated circuit materials suitable for complex sequential control. Our approach is innovative, scalable to different sizes, and robust because it works only once after each input pulse. The inputs and outputs of our sequential systems can be force, light, or electricity for communication with other devices and general applicability.



How NOKOV Contributes to the Research

Six balls were marked on the B-shaped structure to measure its deformation (Fig. 1). Three cameras of the motion capture system (NOKOV MARS2H, Beijing) emitted infrared radiation that was reflected by the marked balls and then received by the cameras.

A different high-speed camera system (FASTCAM Mini AX200 with interval between consecutive frames of 3.81 ms) was used in Figs. 2-8 for monitoring purposes since the laser or LED spots in these experiments may disturb the recognition of the NOKOV motion capture system.

References Yang N, Huang K, Qian Z, et al. Enabling sequential logic leveraging time delays of thin-walled soft matter[J]. Thin-Walled Structures, 2025, 209: 112968.

Link <https://www.sciencedirect.com/science/article/abs/pii/S026382312500062X>

Part II Exoskeleton



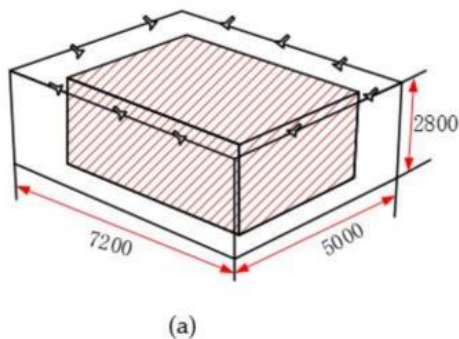
Design and Experimental Research of Knee Joint Prosthesis Based on Gait Acquisition Technology

Keywords

Gait acquisition; Lower limb prosthesis; Instantaneous movement; Bionic performance

Abstract

Whether the lower limb prosthesis can better meet the needs of amputees, the biomimetic performance of the knee joint is particularly important. In this paper, Nokov(metric) optical 3D motion capture system was used to collect motion data of normal human lower limbs, and the motion instantaneous center of multi-gait knee joint was obtained. Taking the error of knee joint motion instantaneous center line as the objective function, a set of six-bar mechanism prosthetic knee joint was designed based on a genetic algorithm. The experimental results show that the movement trajectory of the instantaneous center of the knee joint is basically similar to that of the human knee joint, so it can help amputees complete a variety of gaits and has good biomimetic performance. Gait acquisition technology can provide important data for prosthetic designers and it will be widely used in prosthetic design and other fields.



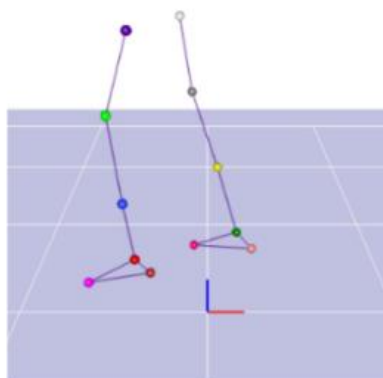


Figure 4. Gait acquisition interface.



(a)



(b)



(c)

How NOKOV Contributes to the Research

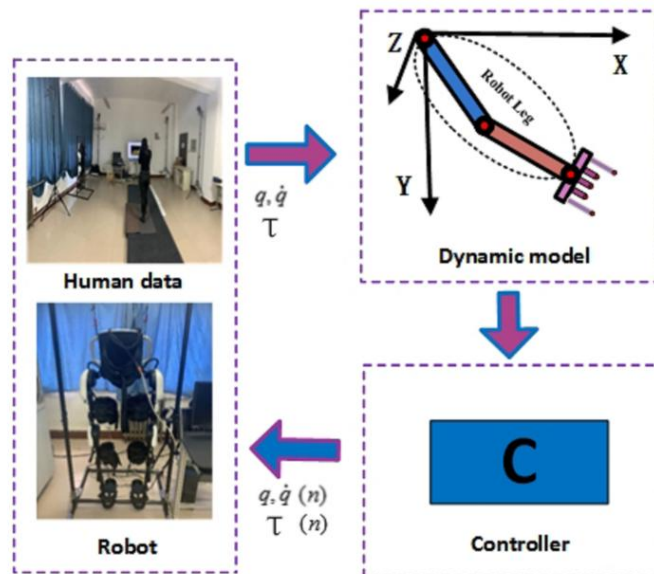
A Nokov motion information collection system was established, and 30 bone healthy and normal gaiters were selected to collect motion data on their knee and ankle joints. This paper mainly studies the motion status of the lower limbs, namely the motion data of the knee joint and the ankle joint, and then marks the positions shown in Figure 3 in combination with the human body structure.

References Zhang Y, Wang E, Wang M, et al. Design and Experimental Research of Knee Joint Prosthesis Based on Gait Acquisition Technology[J]. Biomimetics, 2021, 6(2): 28.
Link <https://www.mdpi.com/2313-7673/6/2/28>

Human-Like Robust Adaptive PD Based Human Gait Tracking for Exoskeleton Robot

Abstract

For patients with dyskinesias caused by central nervous system diseases such as stroke, in the early stage of rehabilitation training, lower limb rehabilitation robots are used to provide passive rehabilitation training. This paper proposed a human-like robust adaptive PD control strategy of the exoskeleton robot based on healthy human gait data. When the error disturbance is bounded, a human-like robust adaptive PD control strategy is designed, which not only enables the rehabilitation exoskeleton robot to quickly track the human gait trajectory obtained through the 3D NOKOV motion capture system, but also can well identify the structural parameters of the system and avoid excessively initial output torque for the robot. MATLAB simulation verifies that the proposed method has a better performance to realize tracking the experimental trajectory of human movement and anti-interference ability under the condition of ensuring global stability for a lower limb rehabilitation exoskeleton robot.





How NOKOV Contributes to the Research

For patients with dyskinesias caused by central nervous system diseases such as stroke, in the early stage of rehabilitation training, lower limb rehabilitation robots are used to provide passive rehabilitation training. This paper proposed a human-like robust adaptive PD control strategy of the exoskeleton robot based on healthy human gait data. When the error disturbance is bounded, a human-like robust adaptive PD control strategy is designed, which not only enables the rehabilitation exoskeleton robot to quickly track the human gait trajectory obtained through the 3D NOKOV motion capture system, but also can well identify the structural parameters of the system and avoid excessively initial output torque for the robot. MATLAB simulation verifies that the proposed method has a better performance to realize tracking the experimental trajectory of human movement and anti-interference ability under the condition of ensuring global stability for a lower limb rehabilitation exoskeleton robot..

References Wang A, Hu N, Yu J, et al. Human-Like Robust Adaptive PD Based Human Gait Tracking for Exoskeleton Robot[J]. Journal of Robotics and Mechatronics, 2021, 33(1): 88-96.

Link https://www.jstage.jst.go.jp/article/jrobomech/33/1/33_88/article/-char/ja/

Adaptive Control of the Rehabilitation Robot with the Model Uncertainty based on Real Human Gait

Keywords

True human gait; Model uncertainty; Adaptive control; Rehabilitation robot; Structural parameters

Abstract

The lower limbs exoskeleton rehabilitation robot (LLERR) has been used widely in rehabilitation training of human lower limb injury, where the control research of LLERR is still a hot point in robotics. Therefore, in this paper, a trajectory tracking adaptive control based on human gait is proposed. Where, the real gait trajectory of human body mechanism by using NOKOV motion capture system is used in control system design. In the simulation results, the dynamics model uncertainty are also considered, and the real gait trajectory data of human body is used to the control system. According to the simulation results, when there is uncertainty in the LLERR model, the proposed trajectory tracking adaptive control method can improve the accuracy and reliability of the trajectory tracking, which makes it suitable for different patients. By adaptive control, the uncertain part of the LLERR is compensated, and the structure parameters of the robot are obtained



How NOKOV Contributes to the Research

In this paper, the NOKOV motion capture system was used to collect the gait movement of the experimenter and obtain hip and knee trajectories. The sampling frequency of this test was 120Hz, and 70 groups were collected. The resulting trajectories of the hip and knee joints will be unsmooth and need to be filtered to obtain a smooth curve. The existing filtering methods include least square method and Kalman filter, etc. In this paper, the least square method is used for filtering. The NOKOV motion capture system combines with a pressure test bench to measure plantar torque, which can be calculated to obtain hip and knee torques, as shown in Fig3 and Fig4.

References Lu J, Aihui W, Ma Z. Adaptive control of the rehabilitation robot with the model uncertainty based on real human gait[C]//2020 International Conference on Advanced Mechatronic Systems (ICAMechS). IEEE, 2020: 282-285.
Link <https://ieeexplore.ieee.org/abstract/document/9310078>

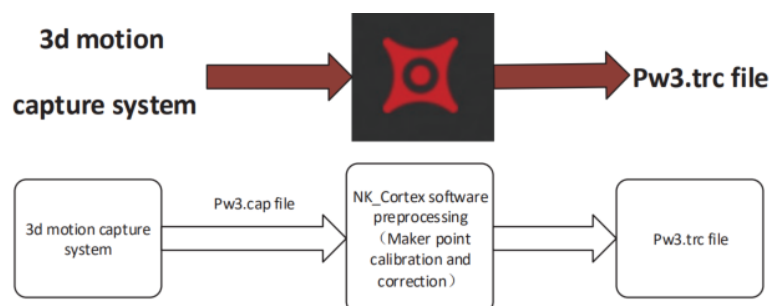
Human Gait Analysis based on OpenSim

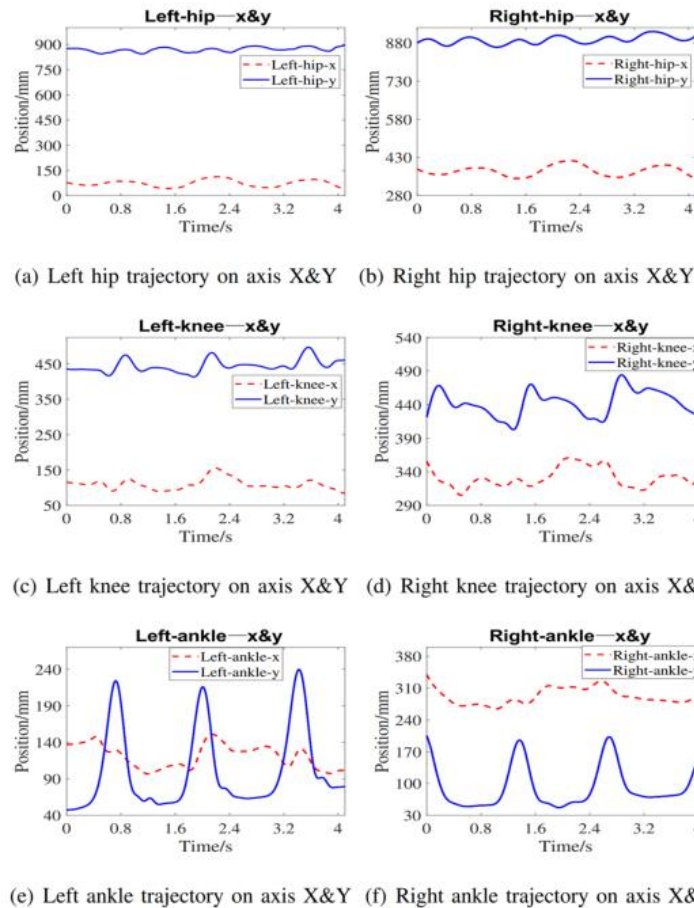
Keywords

Lower limb exoskeleton; OpenSim; Human gait; Simulation; NoKov optical 3D motion

Abstract

The gait motion is fundamental human locomotion that is associated with the human lower limb. Considering the complex structural characteristics of gait, the major challenge is the dynamics model's accuracy and certainty. This paper proposes a method for analyzing human gait in walking. This paper adopted the NoKov optical three-dimensional motion capture system to collect human gait data, built dynamics and kinematics exoskeleton model, and then designed inverse kinematics algorithms. To obtain human gait characteristics, we established a high-precision human lower limb musculoskeletal model in OpenSim, and then imported gait data into the model for kinematics and dynamics simulation. To verify the effectiveness of the proposed method, in this paper, joint angles changes obtained through inverse kinematics analysis of MATLAB are compared with joint angle changes simulated by OpenSim musculoskeletal model.





How NOKOV Contributes to the Research

The gait motion is fundamental human locomotion that is associated with the human lower limb. Considering the complex structural characteristics of gait, the major challenge is the dynamics model's accuracy and certainty. This paper proposes a method for analyzing human gait in walking. This paper adopted the NOKOV optical three-dimensional motion capture system to collect human gait data, built dynamics and kinematics exoskeleton model, and then designed inverse kinematics algorithms. To obtain human gait characteristics, we established a high-precision human lower limb musculoskeletal model in OpenSim, and then imported gait data into the model for kinematics and dynamics simulation. To verify the effectiveness of the proposed method, in this paper, joint angles changes obtained through inverse kinematics analysis of MATLAB are compared with joint angle changes simulated by OpenSim musculoskeletal model.

References Yu J, Zhang S, Wang A, et al. Human gait analysis based on OpenSim[C]//2020 International Conference on Advanced Mechatronic Systems (ICAMechS). IEEE, 2020: 278-281.

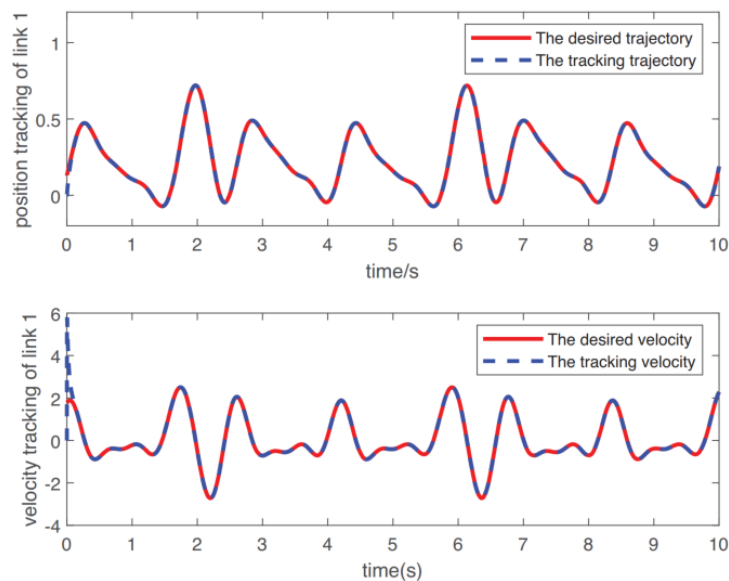
Link <https://ieeexplore.ieee.org/abstract/document/9310111>

Research on Robot Control System of Lower Limb rehabilitation Robot Based on Human Gait Comfort

Abstract

Aiming at the comfort of wearing rehabilitation training robot for patients with lower limb movement disorder, this paper proposes a control method based on real human gait data as the reference trajectory, and designs a control strategy with the comfort of human wearable robot as the objective function, and the wearing rehabilitation training robot can realize the real-time tracking of experimental trajectory of human movement. Firstly, NOKOV motion capture system is used to collect human gait data. Then, the dynamics model of the lower limb rehabilitation training robot's unilateral leg is identified by Lagrange method, and the controller with comfort level as the objective function is designed. The feasibility of the designed control method is proved by Lyapunov stability theorem. Finally, the effectiveness of the proposed method is verified by MATLAB simulation based experimental data on the lower limb rehabilitation training robot prototype.





How NOKOV Contributes to the Research

Aiming at the comfort of wearing rehabilitation training robot for patients with lower limb movement disorder, this paper proposes a control method based on real human gait data as the reference trajectory, and designs a control strategy with the comfort of human wearable robot as the objective function, and the wearing rehabilitation training robot can realize the real-time tracking of experimental trajectory of human movement. Firstly, NOKOV motion capture system is used to collect human gait data. Then, the dynamics model of the lower limb rehabilitation training robot's unilateral leg is identified by Lagrange method, and the controller with comfort level as the objective function is designed. The feasibility of the designed control method is proved by Lyapunov stability theorem. Finally, the effectiveness of the proposed method is verified by MATLAB simulation based experimental data on the lower limb rehabilitation training robot prototype.

References Wang A, Hu N, Yu J, et al. Research on robot control system of lower limb rehabilitation robot based on human gait comfort[C]//2019 International Conference on Advanced Mechatronic Systems (ICAMechS). IEEE, 2019: 34-39.

Link <https://ieeexplore.ieee.org/abstract/document/8861558>

Adaptive Control of Lower Limb Robot based on Human Comfort under Minimum Inertial Parameters

Keywords

Gait data analysis; Comfortable function; Minimum inertia parameters; Adapt tracking control

Abstract

For patients with dyskinesia caused by central nervous system diseases such as stroke, the lower limb rehabilitation robot is required to provide passive rehabilitation training during the initial rehabilitation training, and a trajectory for healthy human body is used as a reference trajectory, this paper designs an adaptive controller to help patients with rehabilitation training methods. First, the gait trajectory of a healthy human body is obtained using an optical motion capture system and used as a reference trajectory for rehabilitation training. Then, the unilateral oscillating leg of the lower limb rehabilitation robot is modeled by dynamics. Based on this, an adaptive controller is designed to enable the lower limb rehabilitation robot to quickly track the desired trajectory, and a similarity function for evaluating the follow ability of the lower limb rehabilitation robot is proposed. Finally, experiments on the prototype of MATLAB simulation and lower limb rehabilitation robot demonstrates the feasibility of the proposed method.



How NOKOV Contributes to the Research

In most human gait acquisition devices, optical motion capture is preferred because it can accurately measure small reflective markers attached to some relevant body landmarks. Therefore, in this paper, the NOKOV 3d infrared passive optical motion capture system is used to collect human movement gait data as input data of exoskeleton robot controller, and then proposes a human body comfort function based on human-computer interaction, in the case of model parameter uncertainty adaptive controller is designed for lower limb rehabilitation robot trajectory tracking control, so as to realize more comfortable natural gait trajectory, and improve the efficiency of rehabilitation, finally simulation experiment and in lower limb exoskeleton rehabilitation training robot experiments verify the feasibility and effectiveness of the proposed method.

References Hu N, Wang A, Yu J, et al. Adaptive control of lower limb robot based on human comfort under minimum inertial parameters[C]//2019 International Conference on Advanced Mechatronic Systems (ICAMechS). IEEE, 2019: 40-45.

Link <https://ieeexplore.ieee.org/abstract/document/8861655>

Adaptive Research of Lower Limb Rehabilitation Robot Based on Human Gait

Abstract

For patients with dyskinesia caused by central nervous system diseases such as stroke, the lower limb rehabilitation robot is required to provide passive rehabilitation training during the initial rehabilitation training, and a trajectory for healthy human body is used as a reference trajectory, this paper designs an adaptive controller to help patients with rehabilitation training methods. First, the gait trajectory of a healthy human body is obtained using an optical motion capture system and used as a reference trajectory for rehabilitation training. Then, the unilateral oscillating leg of the lower limb rehabilitation robot is modeled by dynamics. Based on this, an adaptive controller is designed to enable the lower limb rehabilitation robot to quickly track the desired trajectory, and a similarity function for evaluating the follow ability of the lower limb rehabilitation robot is proposed. Finally, experiments on the prototype of MATLAB simulation and lower limb rehabilitation robot demonstrates the feasibility of the proposed method.



How NOKOV Contributes to the Research

Using the NOKOV 3D infrared passive optical motion capture system can only obtain the coordinates of the outside of the lower extremities of the human body. Therefore, the center position of each joint of the human lower extremity needs to be calculated based on the anthropometric experience formula to determine the walking parameters of the human body. Take the right lower limb of the human body as an example. The key points and world coordinates are shown in Fig.3. The positive direction is the direction of the human body as the axis, the positive direction of the axis is the left-hand side, and the positive direction of the axis is the vertical direction. For the hip joint of the right lower extremity of the human body, a local coordinate is established with the marker point 8, ie, the sacrum, as the origin, and the local coordinate unit vector is defined as,

References Yu J, Cai F, Wang A, et al. Adaptive Research of Lower Limb Rehabilitation Robot Based on Human Gait[C]//2018 International Conference on Advanced Mechatronic Systems (ICAMechS). IEEE, 2018: 86-92.

Link <https://ieeexplore.ieee.org/abstract/document/8507128>

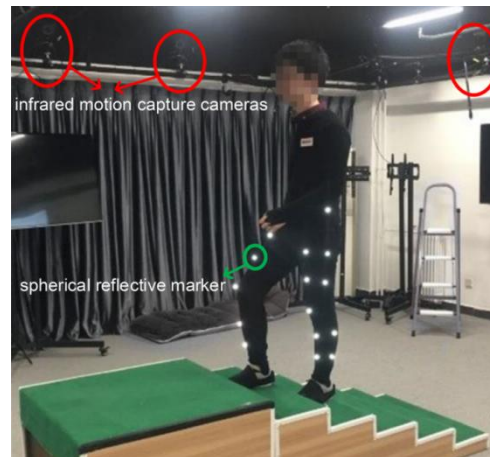
Official Website Case Link

https://www.nokov.com/en/support/case_studies_detail/Motion-planning-of-bio-inspired-robot.html

An Adaptive Stair-ascending Gait Generation Approach based on Depth Camera for Lower Limb Exoskeleton

Abstract

The mobility on stairways is a daily challenge for seniors and people with dyskinesia. Lower limb exoskeletons can be effective assistants to improve their life quality. In this paper, we present an adaptive stair-ascending gait generation algorithm based on a depth camera for lower limb exoskeletons. We first construct a linked-list-based stairway model with the point cloud captured from the depth camera. Then, an optimal foothold point is calculated based on the linked-list stair model for gait generation. Finally, the exoskeleton takes the stair-ascending gait of healthy people as a reference and generates appropriate gait for the stair. The proposed gait generation algorithm is initially validated through holistic simulation analyses. We tested the stairway modeling algorithm on varieties of indoor and outdoor stairways and evaluated the gait generation algorithm on stairs of different height. The subjects' stair walking tests with lower limb exoskeletons show the effectiveness of the proposed stairway modeling and gait generation approaches.



How NOKOV Contributes to the Research

To generate a natural stair walking gait as a reference, we built a motion capture site and collected a series of motion postures of healthy people walking on flat ground and stairs with the NOKOV motion capture system (Fig. 3). The infrared cameras capture the 3D location of each spherical reflective markers on the suit, and the motion posture analysis software associates each marker to a specific position of a human model. During the marker association process, all the incorrect markers with the wrong position are identified and removed manually. Then, the lower limbs and torso are fitted with the position of the markers. In this research, we denote the sequence of hip joint angles (α) and knee joint angles (β) as the lower limb exoskeleton walking gait. Hence, we can compute and filter the joint angles (α and β) with the fitted limbs and torso. Each gait angle data are fitted by an 8th order multiple sinusoidal functions for a smooth reference average gait curve,

References Zhao X, Chen W H, Li B, et al. An adaptive stair-ascending gait generation approach based on depth camera for lower limb exoskeleton[J]. Review of Scientific Instruments, 2019, 90(12): 125112.

Link <https://aip.scitation.org/doi/abs/10.1063/1.5109741>

Mechanical Design and Kinematic Modeling of a Cable-Driven Arm Exoskeleton Incorporating Inaccurate Human Limb Anthropomorphic Parameters

Keywords

Cable-driven exoskeleton; Rehabilitation robot; Upper limb

Abstract

Compared with conventional exoskeletons with rigid links, cable-driven upper-limb exoskeletons are light weight and have simple structures. However, cable-driven exoskeletons rely heavily on the human skeletal system for support. Kinematic modeling and control thus becomes very challenging due to inaccurate anthropomorphic parameters and flexible attachments. In this paper, the mechanical design of a cable-driven arm rehabilitation exoskeleton is proposed to accommodate human limbs of different sizes and shapes. A novel arm cuff able to adapt to the contours of human upper limbs is designed. This has given rise to an exoskeleton which reduces the uncertainties caused by instabilities between the exoskeleton and the human arm. A kinematic model of the exoskeleton is further developed by considering the inaccuracies of human-arm skeleton kinematics and attachment errors of the exoskeleton. A parameter identification method is used to improve the accuracy of the kinematic model. The developed kinematic model is finally tested with a primary experiment with an exoskeleton prototype.

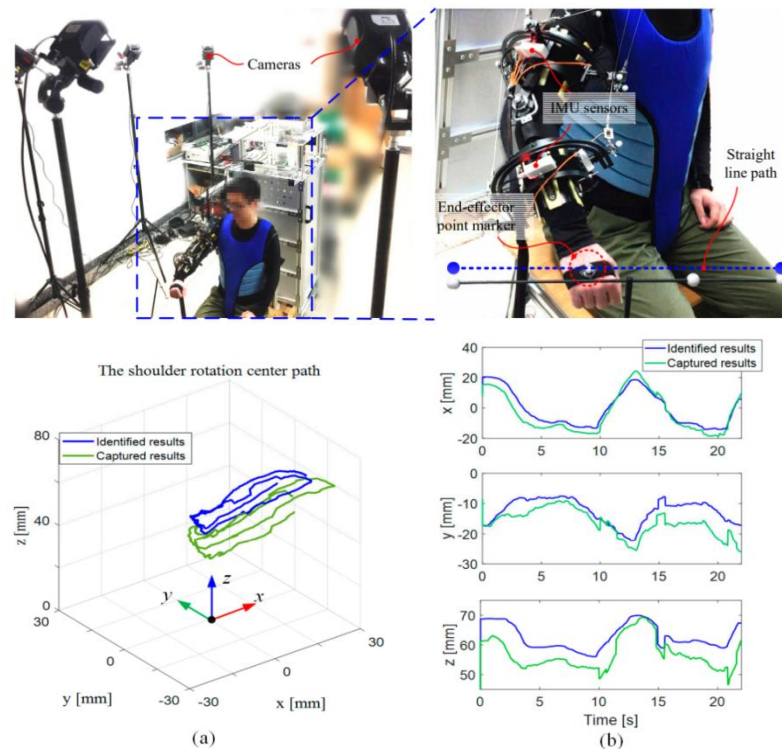


Figure 9. The movement of the shoulder joint center in (a) 3D and (b) 2D plots.

How NOKOV Contributes to the Research

To verify the accuracy of the kinematic models with identified parameters, a Nokov motion capture system was used, as shown in Figure 6. In the motion capture system, a marker was mounted onto the hand of the subjects to measure their tracking results in the experiments. Fourteen markers were mounted onto the cable-routing points of the cuffs to capture changes in the lengths of the cables in the exoskeleton in the tracking task. A marker mounted on the subject's shoulder joint recorded the movement of the joint center. Noting that as the Nokov system captures motion with sub-millimeter accuracy, the measurements can be considered as the real results in this work. The captured results can be compared with the identified results to validate the accuracy of the kinematic model.

References Chen W, Li Z, Cui X, et al. Mechanical design and kinematic modeling of a cable-driven arm exoskeleton incorporating inaccurate human limb anthropomorphic parameters[J]. Sensors, 2019, 19(20): 4461.

Link <https://www.mdpi.com/1424-8220/19/20/4461>

Robust Adaptive PD-like control of Lower Limb Rehabilitation Robot based on Human Movement Data

Keywords

Human movement data; Rehabilitation robot; Trajectory tracking; S-curve function; Similarity function

Abstract

The combination of biomedical engineering and robotics engineering brings hope of rehabilitation to patients with lower limb movement disorders caused by diseases of the central nervous system. For the comfort during passive training, anti-interference and the convergence speed of tracking the desired trajectory, this paper analyzes human body movement mechanism and proposes a robust adaptive PD-like control of the lower limb exoskeleton robot based on healthy human gait data. In the case of bounded error perturbation, MATLAB simulation verifies that the proposed method can ensure the global stability by introducing an S-curve function to make the design robust adaptive PD-like control. This control strategy allows the lower limb rehabilitation robot to track the human gait trajectory obtained through the motion capture system more quickly, and avoids excessive initial output torque. Finally, the angle similarity function is used to objectively evaluate the human body for wearing the robot comfortably.



How NOKOV Contributes to the Research

Therefore, this paper adopts an nonlinear complex exoskeleton robot as the research object, and designs a robust adaptive PD-like control (RAPLC) based on real data. The schematic diagram of the principle is shown in Fig. 1. First, NOKOV three-dimensional (3D) infrared motion capture equipment and three-dimensional force measurement platform equipment are used to collect human gait movement data and plantar force data. The collected data is analyzed for human kinematics and dynamic characteristics and the processed gait data is used as the reference expected motion trajectory of the LLRR control system. Second, the S-curve function is introduced on the basis of the traditional robust adaptive PD controller, which improves the convergence speed of the joint angular displacement and reduces the joint initial torque. Finally, the quadratic function of the error is used as the energy function of the system, and the stability of the control system of the LLRR based on the Lyapunov stability theory is guaranteed under the condition of the coupling and nonlinearity of each joint. The trajectory tracking simulations results show that the LLRR system joint angular displacement has good tracking performance. Comparing the LLRR system joint torque obtained from simulation with experimenter's lower limbs can safely and comfortably complete the rehabilitation training.

References Hu N, Wang A, Wu Y. Robust adaptive PD-like control of lower limb rehabilitation robot based on human movement data[J]. PeerJ Computer Science, 2021, 7: e394.

Link <https://peerj.com/articles/cs-394/>

Kinematics and Dynamics Analysis of Lower Limbs based on Human Motion Data

Keywords

Gait trajectory; Data processing; Kinesiology; Human biomechanics

Abstract

For patients with an aging population and patients suffering from movement disorders, the combination of robotic systems and biomedical engineering principles has brought good news to the majority of patients. The analysis of human gait laws can help better develop and research robotic systems. Therefore, this article is based on the idea of using the gait trajectory of a healthy person as the input reference trajectory of the robot, using NOKOV three-dimensional infrared motion capture equipment and three-dimensional force measurement platform equipment to collect human gait movement data and plantar force data, and analyze the collected data Kinematics and dynamics characteristics of human body data.



How NOKOV Contributes to the Research

Through the comparison of domestic motion capture systems, this paper adopts the NOKOV 3D infrared passive optical motion capture system with better effect. Its system components mainly consist of 6 Mars2H lenses (as shown in Fig.1), connectors, tripods, three-way pan/tilt, connecting wire, calibration rod (T-shaped and L-shaped calibration rods), NK- Cortex software, and reflective markers (Marker).

References Hu N, Wang A. Kinematics and dynamics analysis of lower limbs based on human motion data[C]//2020 Chinese Automation Congress (CAC). IEEE, 2020: 6727-6732.

Link <https://ieeexplore.ieee.org/abstract/document/9327180/>

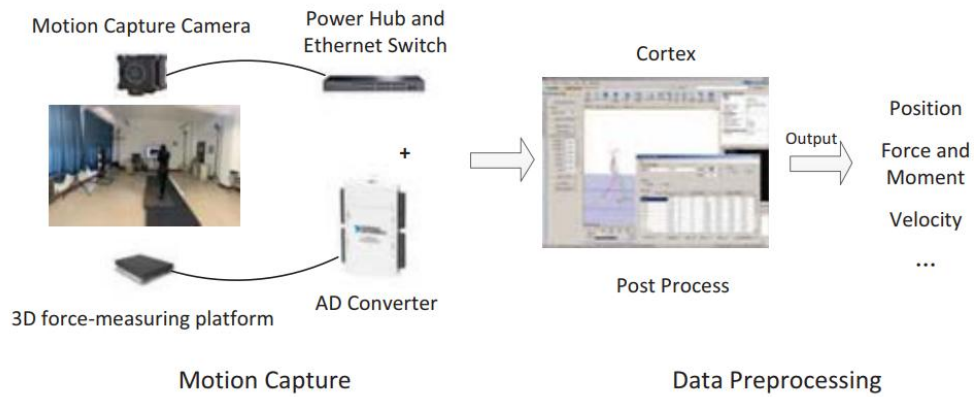
Lower Limb Rehabilitation Robot Control based on Human Gait Data and Plantar Reaction Force

Keywords

Human gait data; Plantar reaction force; Lower limb rehabilitation robot; Motion capture platform

Abstract

In this paper, to improve the safety and availability of rehabilitation training for patients, a control scheme for Lower limb rehabilitation robot (LLRR) based on human gait data and plantar reaction force is proposed. First, the real normal human gait data obtained by motion capture platform (MCP) is fed to the designed control system, and the plantar reaction force is fed to the LLRR after dynamic calculation. Then, a controller based on joint angle and angular velocity tracking error is designed to realize the trajectory tracking. Finally, for a robot simulation model, the effectiveness of the proposed method is confirmed based on the simulation results.



How NOKOV Contributes to the Research

One of the main concerns of these LLRR is to generate physiological gait trajectories with human motion characteristics, and achieve safe and effective rehabilitation training. Generally, the gait trajectories with good rehabilitation training results are generated by healthy subjects. There are a variety of motion capture techniques for acquiring gait trajectory in current, such as inertia, magnetic, and ultrasonic system [13], among them, the optical motion capture (OPC) is one of the most effective methods. The Nokov OPC equipment adopt the motion capture cameras with ten bits gray-scale depth and GrayScale image processing algorithm, can achieve the precision of millimeter [16], [17]. To realize better control effectiveness, Xu proposed a torque control strategy based on real-time gait event detection [14], which could control torque and directly drive joint motion based on real-time gait information. However, this control strategy is proposed for the control of a bionic knee exoskeleton. Wang introduced a gait trajectory measurement and planning method [15]. By measuring and planning the gait trajectory, the ideal controller input of the lower limb rehabilitation robot is obtained, which is used to realize the humanoid movement of the lower limb rehabilitation robot, and improve the safety of rehabilitation training. Wang and Hu proposed a LLRR control system based on human gait comfort [16], [17], which can well realize the tracking of human gait trajectory.

References Ge Y, Dan Y, Wang A, et al. Lower limb rehabilitation robot control based on human gait data and plantar reaction force[C]//2020 International Conference on Advanced Mechatronic Systems (ICAMechS). IEEE, 2020: 286-289.

Link <https://ieeexplore.ieee.org/abstract/document/9310077>

Simulation of Limb Rehabilitation Robot Based on OpenSim

Keywords

Lower limb rehabilitation robot; Gait trajectory; OpenSim; Inverse kinematics; Forward dynamics

Abstract

Lower limb rehabilitation robots have been widely used in medical treatment in recent years. The safety of robot in rehabilitation training should be considered. In this paper, the method to analyze complicated inverse kinematics and forward dynamics of lower limb rehabilitation robot is studied. Because of the mathematical model of lower limb is difficult to establish, the OpenSim software is used to simulate and analyze the human gait data which collected by the NOKOV optical motion capture system. According to the inverse kinematics simulation, the curve of each joint angle changing over time is obtained, and the reaction force acting on the ground is obtained through the forward dynamics simulation.



How NOKOV Contributes to the Research

In this paper, the NOKOV optical motion capture system is used to gain gait data. The cameras which can record each frame marker point movement sequence images are used to real-time motion capture. Because of the limited range of each camera lens, the 6 cameras can be crossed to produce an area. Markers which are captured by two lenses at the same time generate continuous trajectories. Besides, the motion capture system records the spatial coordinates of all markers at the rate of 60 frames per second and its capture accuracy can reach 1 mm. The collected human gait data can be simply processed and exported by the NOKOV system software. In this section, the 6 optical motion capture cameras are arranged in parallel in two columns [11]. The scene of motion capture and the marking points are shown in Fig. 1.

References Wang A., Lu J., Ge Y., Yu J., Zhang S. (2020) Simulation of Limb Rehabilitation Robot Based on OpenSim. In: Pan L., Liang J., Qu B. (eds) Bio-inspired Computing: Theories and Applications. BIC-TA 2019. Communications in Computer and Information Science, vol 1160. Springer, Singapore. https://doi.org/10.1007/978-981-15-3415-7_55
Link https://link.springer.com/chapter/10.1007%2F978-981-15-3415-7_55

Mechanical Compliance and Dynamic Load Isolation Design of Lower Limb Exoskeleton for Locomotion Assistance

Keywords

Compliant joint; constant-force suspension mechanism; exoskeleton; metabolic rate

Abstract

Improving the comfort and portability of exoskeleton structures for human locomotion assistance is challenging. Research shows that the misalignments between exoskeletons and human joints will lead to interference and undesired interaction forces in locomotion. Moreover, the weight and inertia of exoskeletons will lead to rapid changes in impact forces acting on the wearer. When the exoskeleton is tightly bound to the wearer's body, such forces are usually unavoidable. These issues not only reduce the comfort and portability of exoskeletons, but also increase the energy costs to wearers. To address these issues, we propose a novel lower limb exoskeleton with a constant-force suspension mechanism and self-adapting compliant joints. The constant-force suspension mechanism aims to isolate the impact force on the back of the wearer, while the self-adapting compliant joints aim to reduce the misalignments between the exoskeleton and the wearer. For the first time, a structure designed to isolate the impact force of exoskeletons is proposed. Moreover, compliant joints exhibiting comprehensive great performances in portability, flexibility, self-aligning, and weight-supporting ability have also been rarely reported. The experimental results demonstrate that the developed mechanisms can reduce the impact forces and misalignments in locomotion. Furthermore, the proposed exoskeleton can reduce the metabolic rate during walking at 5 km/h by $10.95 \pm 4.40\%$ and that during running at 9 km/h by $1.71 \pm 4.54\%$ compared with locomotion without the exoskeleton. These results confirm that the proposed designs can improve the performance of the exoskeleton in locomotion.

How NOKOV Contributes to the Research

The motion data under two conditions were measured and recorded by a motion capture system (NOKOV, China) with eight cameras. We defined a whole gait cycle as the motions between two consecutive right heel strikes. At these moments, the marker on the right heel reached its maximum anterior position. Thus, the motion data recorded by the motion capture system were segmented into gait cycles. Based on the data, we calculated the mean angles of the hip, knee, and ankle joints on the sagittal plane in these gait cycles ranging from 0% to 100% in the two conditions, as shown in Fig. 9(a). As the position of the markers might unavoidably move relative to the subject between the two conditions, the imperceptible differences may have arisen as a result of marker repositioning errors. Even so, these differences highlight the minimally restrictive nature of the exoskeleton.

References Li H, Sui D, Ju H, et al. Mechanical Compliance and Dynamic Load Isolation Design of Lower Limb Exoskeleton for Locomotion Assistance[J]. IEEE/ASME Transactions on Mechatronics, 2022.

Link <https://ieeexplore.ieee.org/abstract/document/9802513>

Serial–Parallel Mechanism and Controller Design of a Robotic Brace for Dynamic Trunk Support

Keywords

Dynamic trunk support; force control; robotic brace; serial–parallel manipulator;

Abstract

This article presents a robotic trunk support system that provides dynamic support and direct force/moment measurement, to assist trunk movement and train related core muscle groups. The proposed system has four degrees of freedom, three in the coronal plane and one in flexion/extension motion, which has a larger range than the existing robotic braces. The system was designed based on the serial–parallel mechanism. The serial module is responsible for the flexion/extension motion, and the parallel module is responsible for three degrees of freedom in the coronal plane. The kinematic modeling and singularity analysis of the parallel mechanism were discussed in detail. Then, the position control and force control of the serial control actuator were implemented, and the experiments were carried out to evaluate the performance of the controller. The effects on human body in force control were discussed and explained. A reduction in anterior muscles activation and an increase of dorsal muscles activation were exhibited when flexion torque was applied, while the trunk core muscle groups have opposite activation when extension torque was applied. The results indicate that the proposed robotic brace has potentials in the application to trunk core muscle groups training and corresponding rehabilitation.

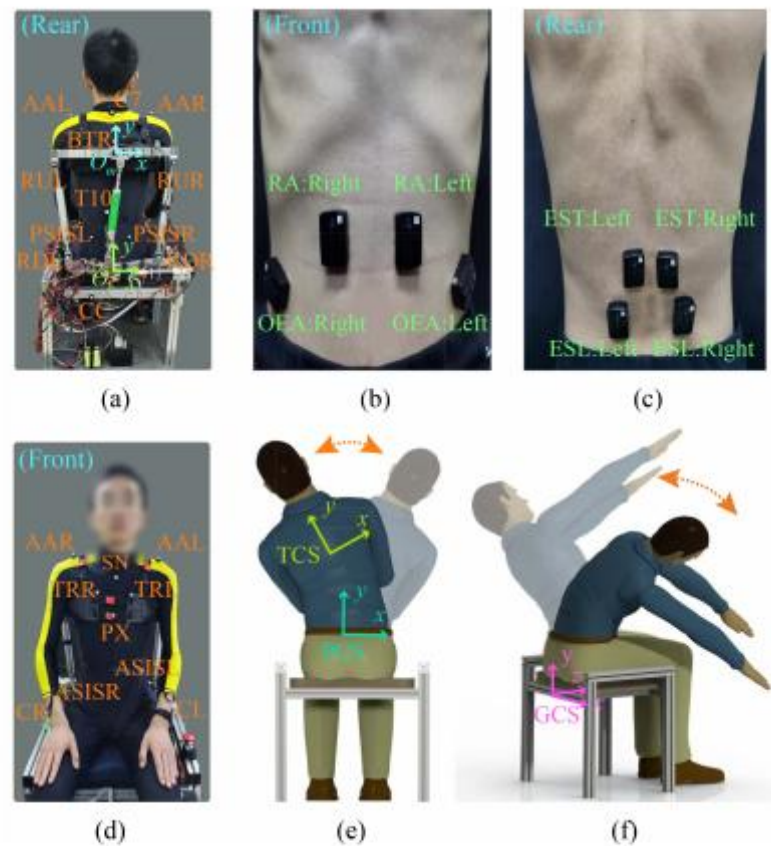


Fig. 6. (a) and (d) Placement of markers. (b) and (c) Placement of EMG electrodes. There are eight electrodes placed bilaterally on four muscle groups including EST, ESL, OEA, and RA. (e) and (f) Movement of lateral bending and the movement from right anterior flexion to left posterior extension, respectively.

How NOKOV Contributes to the Research

Kinematical Data Preprocessing: The motion capture camera system (MARS4H, NOKOV, China) was used to evaluate the performance of the RoboBDsys in terms of the accuracy of position at 100 Hz.

References Guo X, Zhou Z, Gao Y, et al. Serial-Parallel Mechanism and Controller Design of a Robotic Brace for Dynamic Trunk Support[J]. IEEE/ASME Transactions on Mechatronics, 2022.

Link <https://ieeexplore.ieee.org/abstract/document/9747946>

Artificial human balance control by calf muscle activation modelling

Keywords

Muscle stretch reflex, calf muscle activation, standing control, exoskeleton control

Abstract

The natural neuromuscular model has greatly inspired the development of control mechanisms in addressing the uncertainty challenges in robotic systems. Although the underpinning neural reaction of posture control remains unknown, recent studies suggest that muscle activation driven by the nervous system plays a key role in human postural responses to environmental disturbance. Given that the human calf is mainly formed by two muscles, this paper presents an integrated calf control model with the two comprising components representing the activations of the two calf muscles. The contributions of each component towards the artificial control of the calf are determined by their weights, which are carefully designed to simulate the natural biological calf. The proposed calf modelling has also been applied to robotic ankle exoskeleton control. The proposed work was validated and evaluated by both biological and engineering simulation approaches, and the experimental results revealed that the proposed model successfully performed over 92% of the muscle activation naturally made by human participants, and the actions led by the simulated ankle exoskeleton wearers were overall consistent with that by the natural biological response.

How NOKOV Contributes to the Research

2) EXPERIMENTAL PROTOCOL

A Nokov motion capture system (produced by Beijing Nokov Science & Technology Co., Ltd. China) with eight charge-coupled device cameras was employed for the vehicle kinematic data capturing. The camera system calibration and three-dimensional target reconstruction were performed by NK-cortex software (produced by Nokov, China). The vehicle kinematics were derived from 4 markers attached on the vehicle, and the participants kinematic were derived from 21 markers pasted on the joints of body-segments including head-arms-trunk, thigh, and shank-foot segments. The vehicle kinematic data were acquired at the frequency of 100Hz.

References Yin K, Chen J, Xiang K, et al. Artificial human balance control by calf muscle activation modelling[J]. IEEE Access, 2020, 8: 86732-86744.

Link <https://ieeexplore.ieee.org/abstract/document/9086511/>

PD Controller of a Lower Limb Exoskeleton Robot Based on Sliding Mode RBF Neural Network

Keywords

LLER, PD, SMRBF-nn, Human gait.

Abstract

The lower limb exoskeleton robot (LLER) is a human-robot interaction device that combines human functions and mechanical characteristics. Due to the complexity and strong coupling of human gait, it is difficult for LLER to be worn comfortably and safely for training. In such a scenario, the paper proposes a kind of proportional-derivative(PD) controller of LLER based on sliding mode RBF neural network(SMRBF-nn). In order to verify the effectiveness of the proposed control scheme, pertinent experiments were carried out. The gait data of the subject was collected through the motion capture system. A simulating model was established, different control methods, like conventional SMRBF-nn controller and PD controller based on SMRBF-nn, have been tested on the LLER. The experimental results show that the control strategy proposed in this paper can not only make LLER track the human body's gait trajectory, but also output appropriate torque when there is a disturbance.

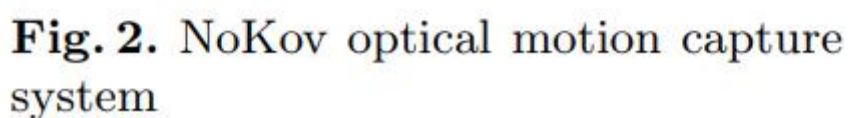


Fig. 2. NoKov optical motion capture system

In this paper, using NoKov optical three-dimensional motion capture system, its gait capture accuracy meets our experimental requirements. This paper employs a control scheme that combines SMRBF-nn with PD to approximate human gait trajectory (as shown Fig. 1). In order to verify the effectiveness of the employed control scheme, we designed a gait capture experiment and MATLAB simulation for normal human. We also compared and evaluated the corresponding simulation results of the employed control scheme.

References Wang A, Li W, yu J. PD Controller of a Lower Limb Exoskeleton Robot Based on Sliding Mode RBF Neural Network[C]//Multimedia Technology and Enhanced Learning: Third EAI International Conference, ICMTel 2021, Virtual Event, April 8–9, 2021, Proceedings, Part I 3. Springer International Publishing, 2021: 427–436. Link https://link.springer.53yu.com/chapter/10.1007/978-3-030-82562-1_40

Inverse Dynamic Model using GRU Networks Learning

Keywords

Exoskeleton robot, inverse dynamic model, GRU network, deep learning

Abstract

Considering the designing the controller of a lower limb rehabilitation robot, an accurate inverse dynamic model is usually needed to calculate the joint torque, which is the key to trajectory execution. However, the physical modeling method will have various uncertainties, resulting in inaccurate estimating and measuring physical parameters. Therefore, this paper proposes to use a Gated Recurrent Unit (GRU) to learn the inverse dynamic model. First of all, the gait data and plantar force data of healthy people are obtained synchronously using the NOKOV 3D optical motion capture system and the Bertec 3D force measurement platform. Secondly, using MATLAB, the angle, angular velocity, angular acceleration and joint torque of the hip and knee joints at different moments can be calculated when people walk normally with the data derived from the system. Finally, the calculated joint angles, angular velocities, and angular accelerations are used as inputs to the GRU network, and joint torques are used as outputs to learn the inverse dynamics model of the exoskeleton robot. The evaluation index of the training model is the Root Mean Square Error (RMSE), which eventually settles at about 0.15. The results show that the model has a good learning ability and has made some certain contribution to the development of the exoskeleton robot control system.

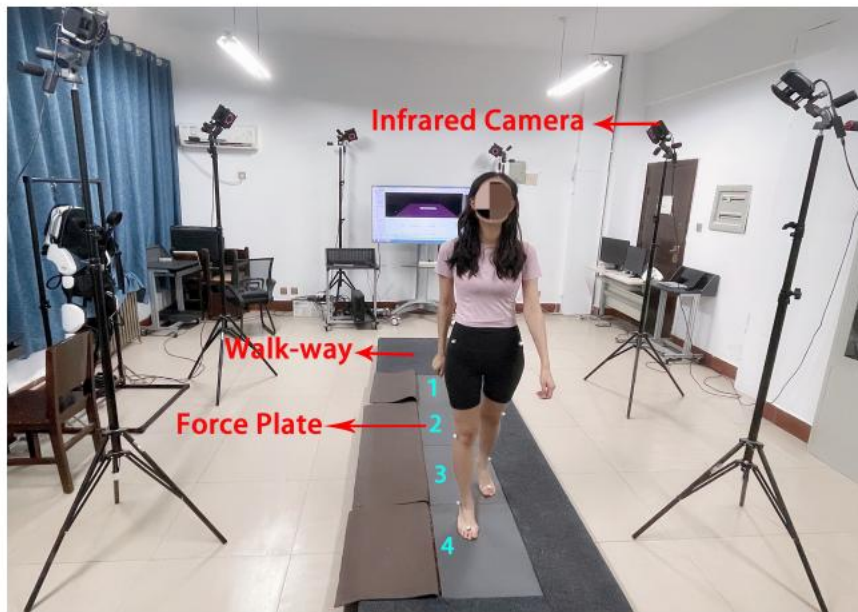


Fig. 6. Data collection scene.

How NOKOV Contributes to the Research

First of all, the gait data and plantar force data of healthy people are obtained synchronously using the NOKOV 3D optical motion capture system and the Bertec 3D force measurement platform.

References Song L, Wang A, Ren J. Inverse Dynamic Model using GRU Networks Learning[C]//2021 International Conference on Advanced Mechatronic Systems (ICAMechS). IEEE, 2021: 52-55.

Link <https://ieeexplore.ieee.org/abstract/document/9661523/>

Human Gait Prediction for Lower Limb Rehabilitation Exoskeleton Using Gated Recurrent Units

Keywords

Gait prediction, Gated recurrent units, Lower limb rehabilitation robot

Abstract

Strokes or injuries may cause lower-limb severe movement deficits. These years, lower limb rehabilitation robots show great potential in rehabilitation. In the exoskeleton control strategy, the gait prediction or gait planning becomes a fundamental problem. This paper focuses on the gait prediction using Gated Recurrent Unit Model, working on the raw data captured from an optical motion capture platform by a healthy subject. The designed model learns the history gait data and predicts the current values based on the axis values from the hip and the ankle, without any manual feature engineering. The univariate axis values are processed to a 3D shape to the supervised prediction model. Experimental results show the excellent prediction performance in terms of the error rate of 0.0 or 0.01 on all the 31 trails. The accurately predicted values can be used as the reference trajectories for later rehabilitation. Furthermore, the predicted axis values can be further transformed into the torque values or the angular velocity values used as the actuation input for an exoskeleton robot.

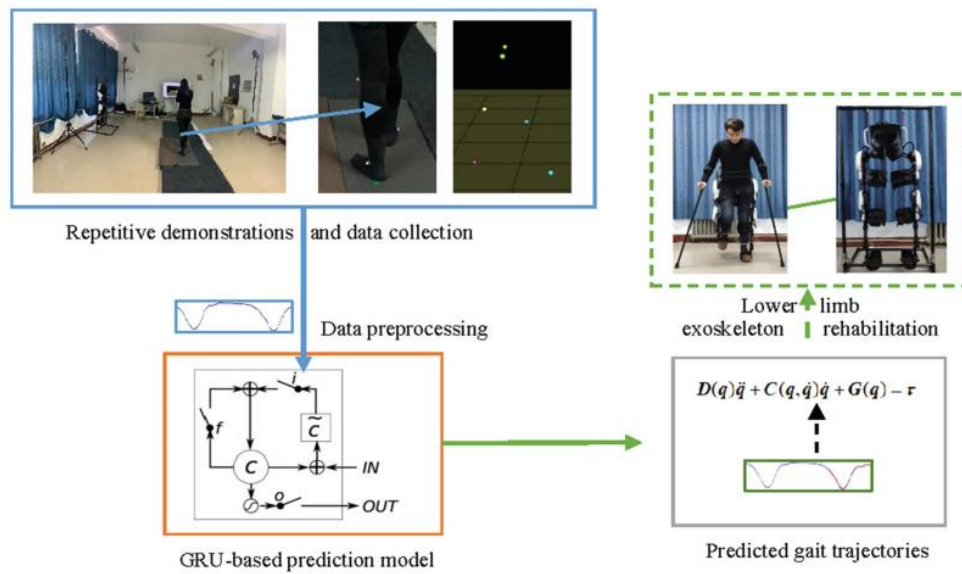


Fig. 1. Gait prediction for lower limb exoskeleton rehabilitation

How NOKOV Contributes to the Research

The study uses NOKOV passive infrared optical motion capture platform collecting the human gait data, referring to Fig. 1, and we have the same camera layout with the work in [6]. The consent had been taken from the subject before data collection. One healthy subject was instructed to walk at self-paced normal walking speeds under the environment of NOKOV. The subject is a female and aged 23, with the thigh length of 47cm, shank length of 40cm, and foot length of 22cm. We finally acquire 31 trials of the gait data from the subject. The average walking duration of all the trials is 3.45s, with the shortest of 2.95s and the longest 4.33s. The gait trajectories were captured in the format of the three-axis values from each of the six markers. The markers were in parallel attached to the hip, the knee, and the ankle of each body side during the data collection.

References Wang Y, Li Z, Chen Y, et al. Human gait prediction for lower limb rehabilitation exoskeleton using gated recurrent units[C]//RiTA 2020: Proceedings of the 8th International Conference on Robot Intelligence Technology and Applications. Springer Singapore, 2021: 128-135.

Link https://link.springer.53yu.com/chapter/10.1007/978-981-16-4803-8_15

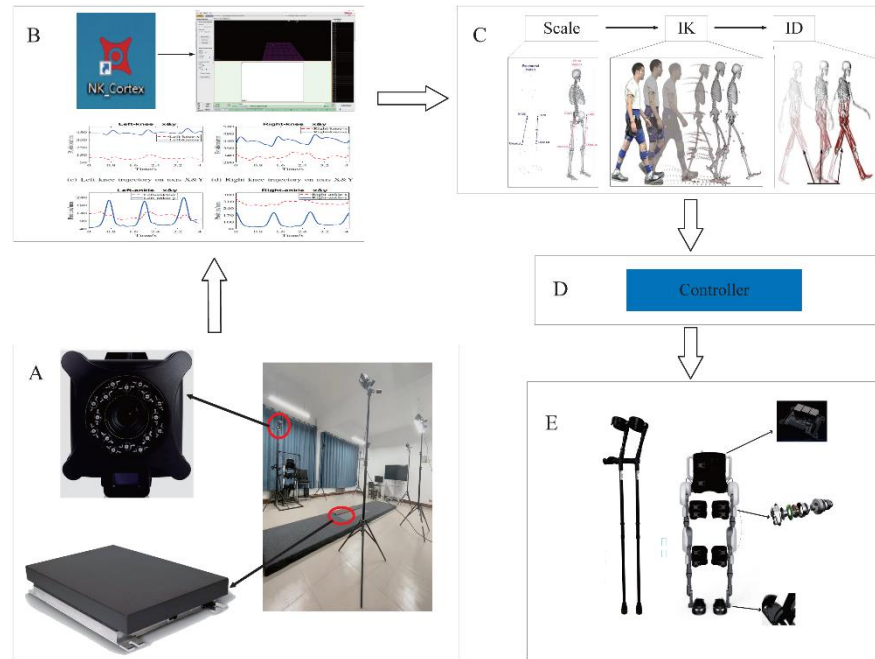
Musculoskeletal modeling and humanoid control of robots based on human gait data

Keywords

Human-gait, Musculoskeletal model, OpenSim, Humanoid control, Lower limb rehabilitation exoskeleton, Motion capture

Abstract

The emergence of exoskeleton rehabilitation training has brought good news to patients with limb dysfunction. Rehabilitation robots are used to assist patients with limb rehabilitation training and play an essential role in promoting the patient's sports function with limb disease restoring to daily life. In order to improve the rehabilitation treatment, various studies based on human dynamics and motion mechanisms are still being conducted to create more effective rehabilitation training. In this paper, considering the human biological musculoskeletal dynamics model, a humanoid control of robots based on human gait data collected from normal human gait movements with OpenSim is investigated. First, the establishment of the musculoskeletal model in OpenSim, inverse kinematics, and inverse dynamics are introduced. Second, accurate human-like motion analysis on the three-dimensional motion data obtained in these processes is discussed. Finally, a classic PD control method combined with the characteristics of the human motion mechanism is proposed. The method takes the angle values calculated by the inverse kinematics of the musculoskeletal model as a benchmark, then uses MATLAB to verify the simulation of the lower extremity exoskeleton robot. The simulation results show that the flexibility and followability of the method improves the safety and effectiveness of the lower limb rehabilitation exoskeleton robot for rehabilitation training. The value of this paper is also to provide theoretical and data support for the anthropomorphic control of the rehabilitation exoskeleton robot in the future.



How NOKOV Contributes to the Research

This paper selected Nokov 3D infrared passive optical motion capture system with high accuracy and good effect after comparing several existing motion capture systems and combining with the needs for current research topic. In the scene set up by this system, the infrared camera is used to fully cover the experimental scene, infrared light is emitted by the infrared camera array in the process of data collection, and the position information of the reflective Marker points are captured in the experimental scene. In the process of collecting gait data, the experimental subject walks in the experimental scene with affixed Marker points. In order to meet the needs of the research, the experimental platform is equipped with a three-dimensional force measuring platform, which can synchronously collect the three-dimensional ground reaction force during the movement of the experimental object.

References Yu J, Zhang S, Wang A, et al. Musculoskeletal modeling and humanoid control of robots based on human gait data[J]. PeerJ Computer Science, 2021, 7: e657. Link <https://peerj.com/articles/cs-657/>

Impedance control based on the human gait data for lower limb rehabilitation robot

Keywords

Lower limb rehabilitation robots, impedance controller, gait data, motion capture

Abstract

The lower limb rehabilitation robots (LLRR) have brought significant support to patients with movement disorders, and the research of LLRR has also become one hot topic in robotic fields. However, how to reduce the contact force between the rehabilitation robot and the human is becoming a challenge. Therefore, in order to make robots and humans form a most concerted whole, this paper designs a novel controller based on real human gait data. In detail, the Nokov motion capture system (NMCS) is used to collect human gait data, and then the collected data is fed to the designed control system as reference control signals. That is, the controller can simulate the interaction between the lower limbs of the human body and the environment to provide human-like abilities for the LLRR. To confirm the effectiveness of the proposed method, the simulation results based experimental data is given. The results show a satisfactory trajectory tracking effect, and the stability can be guaranteed.

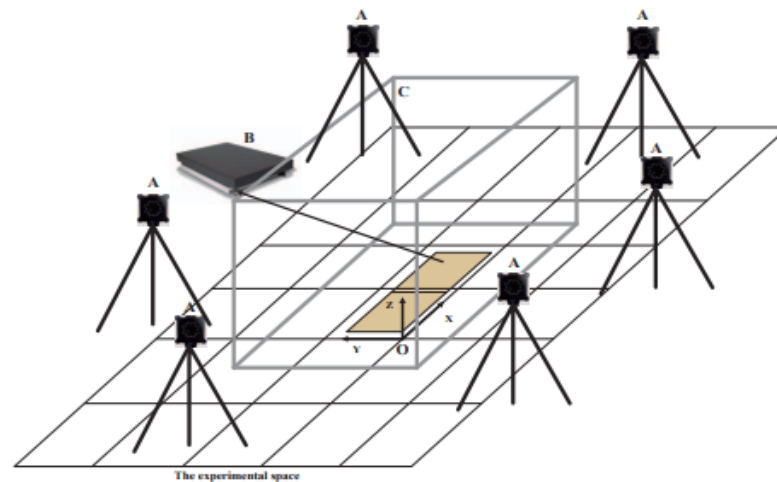


Fig. 1. The experimental space. A: Six infrared cameras. B- 3D force measurement platform. C: The collection space of gait data. O: The origin of the space coordinate after calibration.

How NOKOV Contributes to the Research

Currently, the 3D motion capture system can accurately collect human gait data in the experimental space [1]. 3D motion capture data can be used to identify the geometric parameters and gait trajectory boundaries of the kinematic human lower limb model [2]. This paper uses the Nokov motion capture system (NMCS) to collect human gait data. The system uses reflective marking points attached to the body surface, then collect the three-dimensional position of the mark on the limbs, which are used to calculate the joints kinematics of the human body.

References Wang A, Li W, Yu J, et al. Impedance control based on the human gait data for lower limb rehabilitation robot[C]//2021 China Automation Congress (CAC). IEEE, 2021: 5864-5868.

Link <https://ieeexplore.ieee.org/abstract/document/9727345>

Human Gait Process Analysis and Rehabilitation Robot Humanoid Control Based on OpenSim

Keywords

Human gait, humanoid control, rehabilitation robot, OpenSim, joint moment

Abstract

The existing exoskeleton rehabilitation equipment still has great obstacles to the compliance tracking of normal gait. Therefore, the anthropomorphic compliance control of the rehabilitation exoskeleton robot is still a hot research topic. One of the ways to improve this problem is to import the gait characteristics of human normal motion to achieve human-like tracking and control of the robot. This paper collected normal human gait data and obtained smooth gait data after preprocessing such as filtering and interpolation. Then, combine the collected data with OpenSim to build a musculoskeletal model that matches the subject and analyze the high-fidelity model's joint angle and joint moment. Finally, the joint moment signal with the characteristics of human gait is used as the controller input of the exoskeleton robot, and the exoskeleton robot is simulated, analyzed, and verified. The simulation results prove the scientificity and effectiveness of the method.

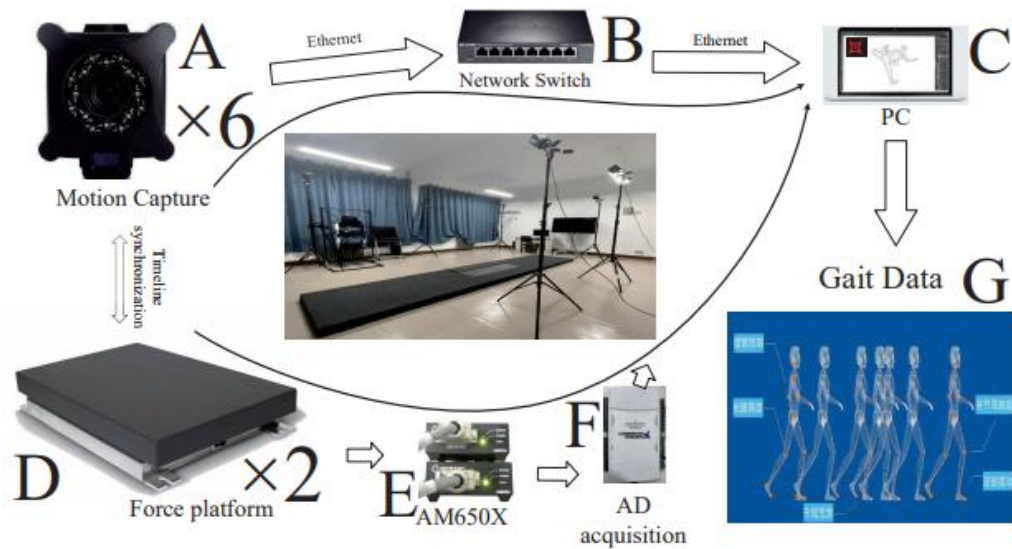


Fig. 1. Gait data acquisition system.

How NOKOV Contributes to the Research

This paper uses the NoKov optical three-dimensional motion capture system to collect human gait data and analyzes the human movement process. Then, the gait data was imported into OpenSim to establishes the kinematics and dynamics model of the joints of the lower limbs, and the gait characteristics of the humanbody are obtained by simulation. The accuracy of the OpenSim model is veriped by comparison the gait feature data. The versatility and particularity of the gait feature are obtained at the same time.

References Zhang S, Wang A, Ma Z, et al. Human Gait Process Analysis and Rehabilitation Robot Humanoid Control Based on OpenSim[C]//2021 International Conference on Advanced Mechatronic Systems (ICAMEchS). IEEE, 2021: 92-96.

Link <https://ieeexplore.ieee.org/abstract/document/9661521>

Human Gait Data Augmentation and Trajectory Prediction for Lower-Limb Rehabilitation Robot Control Using GANs and Attention Mechanism

Keywords

gait prediction; attention; lower-limb rehabilitation robot; LSTM

Abstract

To date, several alterations in the gait pattern can be treated through rehabilitative approaches and robot assisted therapy (RAT). Gait data and gait trajectories are essential in specific exoskeleton control strategies. Nevertheless, the scarcity of human gait data due to the high cost of data collection or privacy concerns can hinder the performance of controllers or models. This paper thus first creates a GANs-based (Generative Adversarial Networks) data augmentation method to generate synthetic human gait data while still retaining the dynamics of the real gait data. Then, both the real collected and the synthesized gait data are fed to our constructed two-stage attention model for gait trajectories prediction. The real human gait data are collected with the five healthy subjects recruited from an optical motion capture platform. Experimental results indicate that the created GANs-based data augmentation model can synthesize realistic-looking multi-dimensional human gait data. Also, the two-stage attention model performs better compared with the LSTM model; the attention mechanism shows a higher capacity of learning dependencies between the historical gait data to accurately predict the current values of the hip joint angles and knee joint angles in the gait trajectory. The predicted gait trajectories depending on the historical gait data can be further used for gait trajectory tracking strategies.

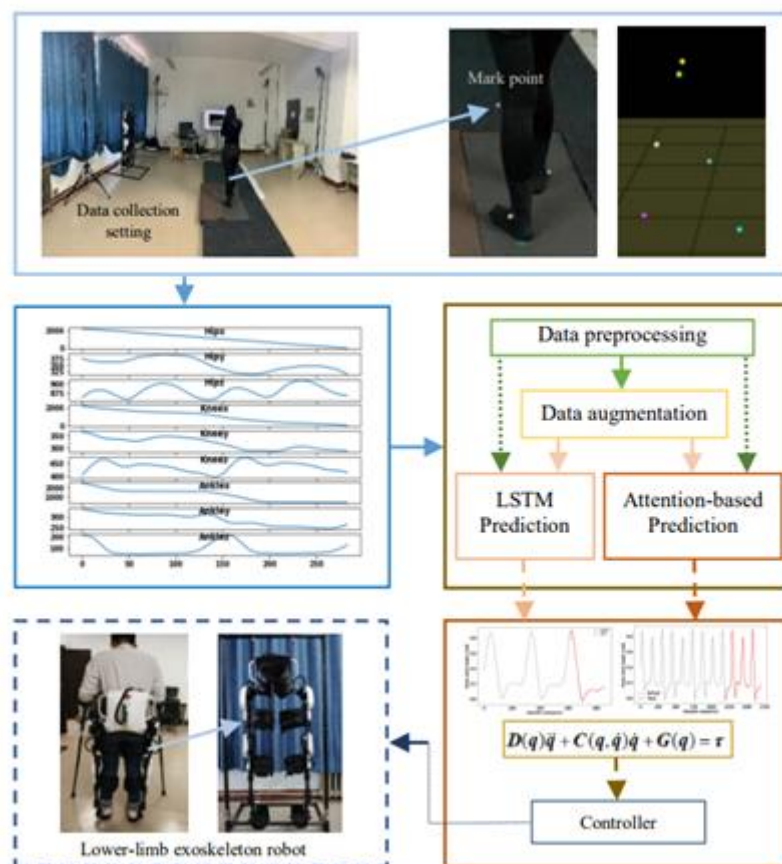


Figure 1. Gait trajectory prediction for lower-limb exoskeleton rehabilitation.

How NOKOV Contributes to the Research

The human gait data in the work are acquired from a NOKOV passive infrared optical motion capture platform [47], referring to the data collection setting in Figure 1. There are six cameras uniformly spaced in two lines along the walls in the lab environment. The data collection associated procedures were approved by Zhongyuan University of Technology Research Ethics Committee (Project identification code ZUTSEI202008-001). All subjects gave their informed consent for inclusion before they participated in the data collection. During the data collection, each participant is instructed to walk at his/her own normal way under the setting of NOKOV. The captured gait data are recorded in the format of three-axis position values from each of the six mark points placed on human body (see Figure 1).

References Wang Y, Li Z, Wang X, et al. Human gait data augmentation and trajectory prediction for lower-limb rehabilitation robot control using GANs and attention mechanism[J]. Machines, 2021, 9(12): 367.

Link <https://www.mdpi.com/2075-1702/9/12/367>

Human-Like Control System Design of the Lower Limb Rehabilitation Robot Based on Adaptive RBF Neural Network

Keywords

Human gait data, RBF neural network, Lower limb rehabilitation Robot

Abstract

Lower limb rehabilitation robot (LLRR) brings hope to the patients with lower-limb paralysis. However, the security and stability of the LLRR are still the challenging problems. To provide safer and more effective rehabilitation training for patients, a human-like control system of LLRR based on adaptive radial basis function (ARBF) neural network is proposed. The optical 3D motion capture platform (O3DMCP) is adopted to collect the human gait data (HGD) from the healthy subjects. The HGD is used as the expected trajectory for LLRR, this can make the rehabilitation training more consistent with the human kinematics characteristics. To improve the tracking accuracy, the ARBF neural network controller is adopted estimate the model uncertain parameters. Simulation results demonstrate the effectiveness of the proposed method.

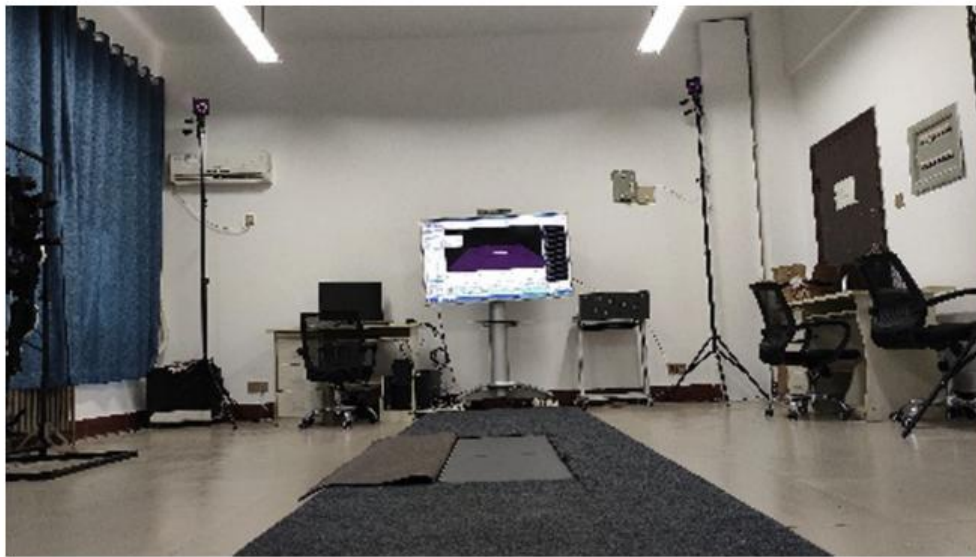


Fig. 1. *Motion capture platform.*

How NOKOV Contributes to the Research

4 Experimental Results

4.1 Data Acquisition

A male subject with 24 years, 80 kg, 180 cm participate in this experiment, his normal gait trajectory is collected through Nokov O3DMCP [6]. Because of the internal error of the O3DMCP, the waggle of mark points and the influence of some miscellaneous points in experiment, the obtained trajectory is not smooth enough with some abnormal values. In this paper, the moving average filter is used for smoothing processing, and the 8th order Fourier function is used for curve fitting.

References Dan Y, Ge Y, Wang A, et al. Human-Like Control System Design of the Lower Limb Rehabilitation Robot Based on Adaptive RBF Neural Network[C]//RITA 2020: Proceedings of the 8th International Conference on Robot Intelligence Technology and Applications. Springer Singapore, 2021: 93-100.

Link https://link.springer.53yu.com/chapter/10.1007/978-981-16-4803-8_11

Development of Robotic Ankle–Foot Orthosis With Series Elastic Actuator and Magneto-Rheological Brake

Keywords

Ankle-foot orthosis, robot-assisted therapy, energy harvesting, magneto-rheological effect, torsion spring, actuators and transmissions, mechanism design, wearable robots

Abstract

This paper illustrates the development and experimental validation of a robotic ankle-foot orthosis (AFO) with a series elastic actuator (SEA) and a magneto-rheological (MR) brake. First, the biomechanics of a human ankle joint during walking is explained. Next, the hardware design of the robotic AFO is introduced, including its mechanical structure, actuator design and configuration, and electronic system. The SEA is primarily composed of an electric motor, a planetary gearbox, a torsion spring, and a pair of bevel gears. The MR brake can modulate the viscosity of the robotic AFO and generate a large braking torque of 21.8 Nm with a low power of 8.8 W. Additionally, the modeling of the robotic AFO is presented, followed by an introduction to its control; several gait evaluation indices are proposed as well. Finally, a pilot study is conducted to verify the effectiveness of the developed robotic AFO. The experimental results demonstrate that the robotic AFO has the potential to provide dorsiflexion assistance, thus preventing foot slap and toe drag, in addition to plantarflexion assistance for the forward propulsion of the body. During a gait cycle, an average power of 0.23 W is harvested, and an 8% improvement in the system energy efficiency is achieved.

How NOKOV Contributes to the Research

In the first walking experiment, the participants walked without the robotic AFO with a normal gait pattern. The kinematic data of the participants' ankle joints were obtained using a motion capture system (NOKOV, Beijing Nokov Science & Technology Co., Ltd., China). In the second, third, and fourth walking experiments, the robotic AFO was worn on the left foot of the participants, and the sensors of the robotic AFO were used to measure the motion data of the human-AFO system. For a fair comparison, the motor energy harvesting mode was not utilized in the third walking experiment. In each walking experiment, 12 steps were extracted from each participant.

References Chen B, Zi B, Wang Z, et al. Development of robotic ankle-foot orthosis with series elastic actuator and magneto-rheological brake[J]. Journal of Mechanisms and Robotics, 2021, 13(1).

Link

<https://asmedigitalcollection.asme.org/mechanismsrobotics/article-abstract/13/1/011002/1086017/Development-of-Robotic-Ankle-Foot-Orthosis-With>

Kinematic Compatible Design and Analysis of a Back Exoskeleton via a Hyper Redundant Hybrid Mechanism

Keywords

Back exoskeleton, hyper redundant mechanism, kinematic compatibility, mechanism design.

Abstract

Back exoskeletons can reduce low-back pain and injury for workers engaged in manual handling operations. However, conventional back exoskeletons are incompatible with human trunk kinematics, arousing uncomfortable human-exoskeleton interaction and limiting natural human movement. This article proposes a novel back exoskeleton design to solve this problem. A hyper redundant hybrid mechanism is synthesized as the exoskeleton mechanism to be compatible with the multi-DOF anatomy of the human spine. We establish the kinematic model of the coupled human-exoskeleton system and analyze the kinematic incompatibility problem. The movement of the human back is characterized by a multi-joint lumbar spine combined with a rigid thoracic spine. The proposed back exoskeleton is composed sequentially of five six-DOF passive parallel mechanisms with one cable actuation, generating active assistance in the sagittal plane and passive assistance in the coronal and transverse planes. The kinetics of the coupled human-exoskeleton system is analyzed to show effective assistance. Experimental tests on kinematic compatibility are finally implemented to demonstrate that the proposed exoskeleton allows the wearer to perform natural movements in all directions with large ranges of motion.

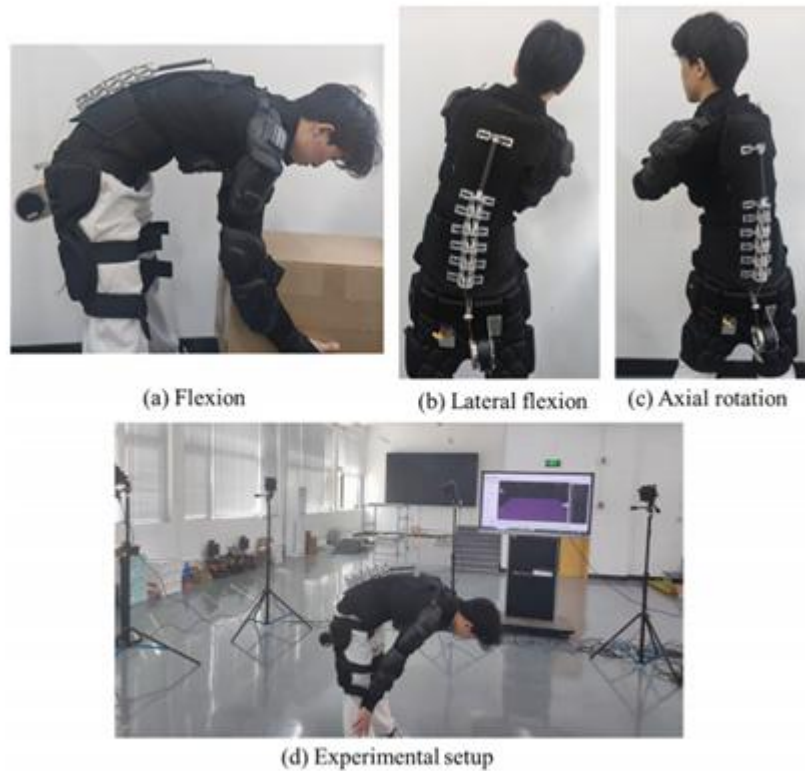


Fig. 9. The exoskeleton adapts to the shapes of the human back very well in all the three directions: (a) Flexion, (b) Lateral flexion, (c) Axial rotation. The optical motion capture system is set up to record the positions of the exoskeleton during the movement (d) Experimental Setup.

How NOKOV Contributes to the Research

We test the adaptability of the device to the different wearer postures. Three subjects participate in the tests, whose heights range from 178cm to 185cm. One can see that the exoskeleton conforms to the back shape of the wearer well in all directions (Fig. 9(a)–(c)). We then set up an optical motion capture system with eight cameras (NOKOV Mocap, CN) to record the positions of the exoskeleton on the human back during the movement (Fig. 9(d)). Accordingly, seven markers are attached to the seven disks to track the wearers' motion conveniently.

References Yang X, Zhou P, Sun Y, et al. Kinematic Compatible Design and Analysis of a Back Exoskeleton via a Hyper Redundant Hybrid Mechanism[J]. IEEE Robotics and Automation Letters, 2022, 7(4): 11322-11329.

Link <https://ieeexplore.ieee.org/abstract/document/9857997>

Estimation of the interaction force between human and passive lower limb exoskeleton device during level ground walking

Keywords

Interaction force estimation; Passive lower limb exoskeleton; Level ground walking; Bilevel optimization

Abstract

To develop sophisticated and efficient control strategies for exoskeleton devices, acquiring the information of interaction forces between the wearer and the wearable device is essential. However, obtaining the interaction force via conventional methods, such as direct measurement using force sensors, is problematic. This paper proposes a kinematic data-based estimation method to evaluate the interaction force between human lower limbs and passive exoskeleton links during level ground walking. Unlike conventional methods, the proposed method requires no force sensors and is computationally cheaper to obtain the calculation results. To obtain more accurate kinematic data, a marker refinement algorithm based on bilevel optimization framework is adopted. The interaction force is evaluated by a spring model, which is used to imitate the binding behavior between human limbs and the exoskeleton links. The deflection of the spring model is calculated based on the assumption that the phase delay between human limb and exoskeleton link can be presented by the sequence of frames of kinematic data. Experimental results of six subjects indicate that our proposed method can estimate the interaction forces during level ground walking. Moreover, a case study of bandage location optimization is conducted to demonstrate the usefulness of obtaining the interaction information.

How NOKOV Contributes to the Research

The subjects were asked to walk on a treadmill with a speed of 1.5 m/s for 1 min in one task. Each subject was asked to perform five tasks in two days. To evaluate the effectiveness of our proposed method, the markers were attached by different persons. The kinematic data were recorded by a passive optical motion capture system (sampling frequency 50 Hz, Nokov, Beijing Metrology Technology Co., Ltd., China).

References S. Bai, M.R. Islam, V. Power, L. O'Sullivan, User-centered development and performance assessment of a modular full-body exoskeleton (AXO-SUIT), Biomimetic Intell. Robot. 2 (2022) 100032, 2022/06/01/.

Link <https://www.sciencedirect.com/science/article/pii/S2667379722000213>

Gait phases recognition based on lower limb sEMG signals using LDA-PSO-LSTM algorithm

Keywords

Gait phase recognition; Surface electromyography (sEMG); Long short-term memory (LSTM); Particle swarm optimization (PSO); Linear discriminant analysis (LDA)

Abstract

Gait phases are widely used in exoskeleton movement control. Surface electromyography (sEMG) is predictive and plays an important role in gait phase recognition. The purpose of this study is to improve the stability and accuracy of gait recognition methods based on the sEMG signals of lower limbs. First, we presented a LDA-PSO-LSTM algorithm based on feature combination selection and verified its recognition accuracy through experiments. LDA-PSO-LSTM had an average recognition rate of 94.89% and a maximum accuracy of 97.02%. Second, we tested and compared the recognition accuracy of LDA-LSTM (92.17%). Experiments showed that the [PSO](#) optimization model had good recognition performance. Finally, we compared LDA-LSTM with all classifier combinations and concluded that the LDA-LSTM method has the highest recognition rate among a series of method combinations. The results indicated that LDA-PSO-LSTM as a classification model has apparent advantages in gait recognition. LDA-PSO-LSTM provides more accurate gait phase results for lower limb exoskeleton control. This method is beneficial to the development of the exoskeleton gait recognition system.

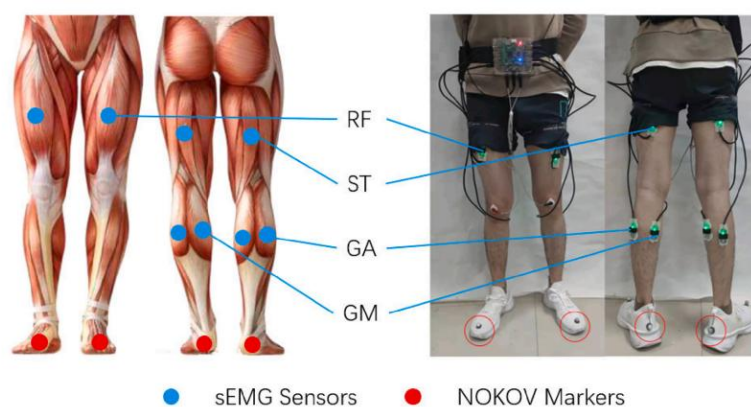


Fig. 6. The location of the sEMG sensors and the optical markers. Four optical markers were placed at the toe and heel of each foot, and eight sEMG sensors were placed in the lower limb.

How NOKOV Contributes to the Research

We designed a customized device to measure subjects' sEMG and optical signals when walking on a treadmill. The system's hardware included a treadmill, a computer, a Bluetooth transmitter, a 5 V power supply, a self-designed data acquisition board, a pair of reference electrodes, eight sEMG signal sensors, several droplet-type electrode sheets, eight NOKOV motion capture cameras Mars 12H (NOKOV motion capture system, Beijing, China), and four reflective markers.

References N. Abhayasinghe, I. Murray, Human gait phase recognition based on thigh movement computed using IMUs, 2014 IEEE Ninth International Conference on Intelligent Sensors, Sensor Networks and Information Processing (ISSNIP), IEEE, 2014, pp. 1-4.

Link <https://www.sciencedirect.com/science/article/abs/pii/S1746809422007261>

Dual-Loop Control Framework of a Self-Balancing Lower-Limb Exoskeleton for Assisted Walking

Keywords

Centroid dynamics, deformation estimator, dual-loop control, exoskeleton, linear quadratic regulator.

Abstract

This article describes a dual-loop control framework of self-balancing lower-limb exoskeletons (SBLLEs) to estimate deformations and to stabilize SBLLEs for assisted walking tasks conducted on wearers with diverse physical parameters. First, considering the nonlinear time-varying characteristic of deformation and the time dependency and periodic nature of walking motion, a bidirectional long short-term memory (Bi-LSTM) neural network is utilized to estimate multiple deformations. In particular, the force sensor and Bi-LSTM-based deformation estimator (FBDE) loop are used to estimate and compensate for the deformation based on the force and moment signals. Second, a physical parameter-independent controller (PPIC) based on centroidal dynamics is proposed to stabilize exoskeletons that can accommodate wearers with varying physical parameters. The PPIC loop is also based on force and moment signals, indicating that only 6-D force sensors are necessary to estimate deformation and stabilize SBLLEs. Finally, a series of experiments is conducted on three subjects with varying physical parameters to validate the effectiveness of the dual-loop control framework. The results of walking experiments show that the dual-loop control framework can estimate deformation and stabilize the SBLLE during walking under different loads.

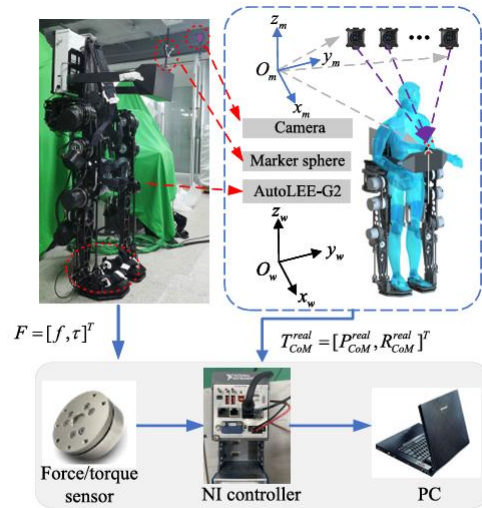


Fig. 6. Data acquisition platform for AutoLEE-G2.

How NOKOV Contributes to the Research

Fig. 6 shows the experimental platform. The SBLLE AutoLEE-G2 was placed in the NOKOV high precision motion capture environment ΣO_m .

References O. Harib et al., "Feedback control of an exoskeleton for paraplegics: Toward robustly stable, hands-free dynamic walking," IEEE Control Syst. Mag., vol. 38, no. 6, pp. 61–87, Dec. 2018.

Link <https://ieeexplore.ieee.org/document/10637283>

Human-Gait-Based Tracking Control for Lower Limb Exoskeleton Robot

Keywords

lower limb exoskeleton robot, human gait data, radial basis function network, feed-forward control, motion capture

Abstract

Research shows that it is practical for the normal human movement mechanism to assist the patients with stroke in robot-assisted gait rehabilitation. In passive training, the effect of rehabilitation training for patients can be improved by imitating normal human walking. To make the lower limb exoskeleton robot (LLER) move like a normal human, a tracking control scheme based on human gait data is proposed in this paper. The real human gait data is obtained from healthy subjects using a three-dimensional motion capture platform (3DMCP). Furthermore, the normal human motion characteristics are adopted to enhance the scientificity and effectiveness of assistant rehabilitation training using LLER. An adaptive radial basis function network (ARBFN) controller based on feed-forward control is presented to improve the trajectory tracking accuracy and tracking performance of the control system, where the ARBFN controller is deployed to predict the uncertain model parameters. The feed-forward controller based on the tracking errors is used to compensate for the input torque of LLER. The effectiveness of the presented control scheme is confirmed by simulation results based on experimental data.

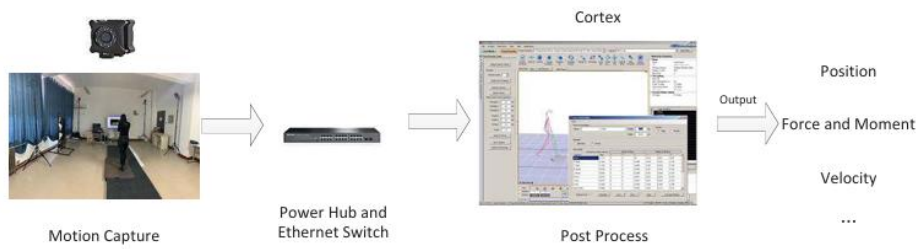


Fig. 1. Motion capture platform.

How NOKOV Contributes to the Research

Currently, a Nokov 3DMCP is utilized to obtain RHGD from healthy subjects. The 3DMCP comprises six high performance infrared lenses (HPIL), motion capture software (Cortex), reflective markers, two calibration rods, data transmission equipment, etc. The motion capture platform is shown in Fig. 1. The HPIL is adopted to capture the spatial position of passive reflective markers, and these data are uploaded to the Cortex using data transmission equipment. The accuracy of RHGD processed by Cortex is up to sub-millimeter level, which benefits from the HPIL with the highest frequency of 240 Hz and 1.3 megapixels.

References Y. Dan, Y. Ge, A. Wang, and Z. Li, "Human-Gait-Based Tracking Control for Lower Limb Exoskeleton Robot," *Journal of Robotics and Mechatronics*, vol. 34, no. 3, pp. 615-621, 2022. <https://doi.org/10.20965/jrm.2022.p0615>.

Link https://www.jstage.jst.go.jp/article/jrobomech/34/3/34_615/_article/-char/ja/

Part III Robotic Arm



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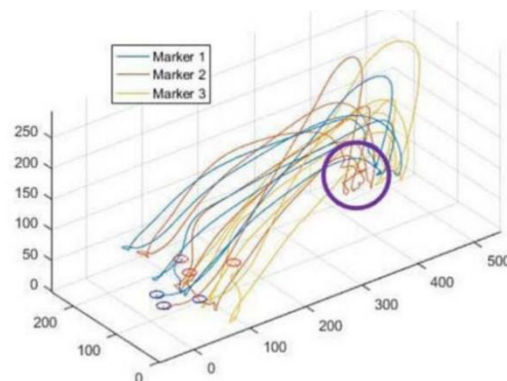
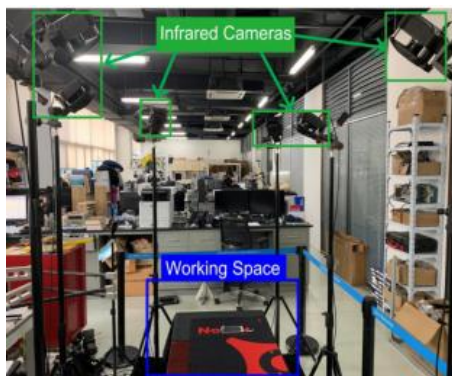


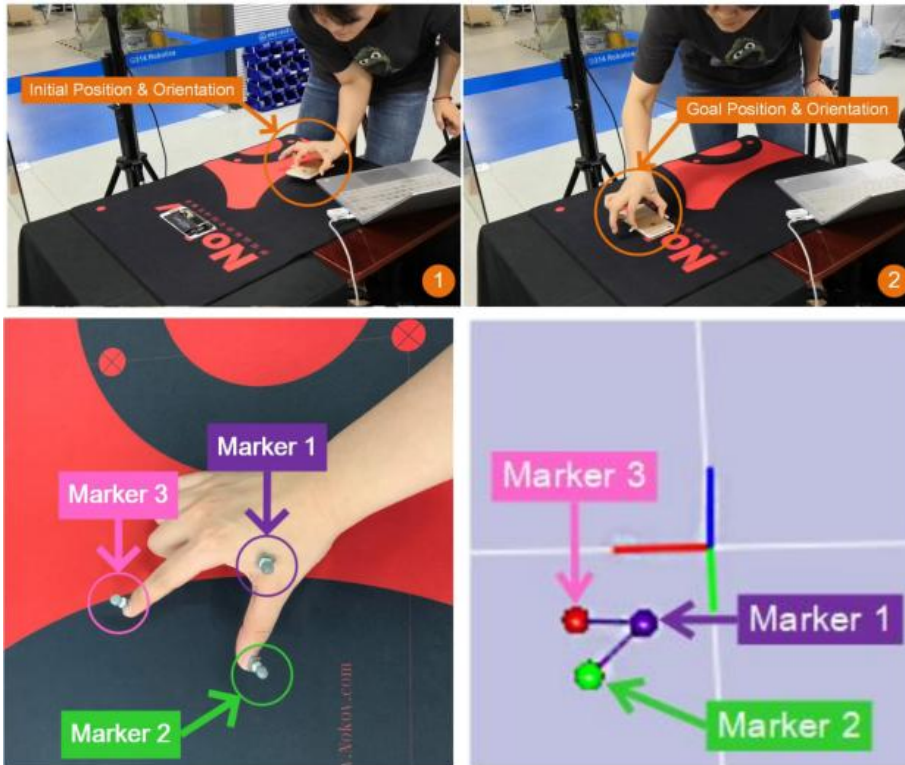
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A Robot 3C Assembly Skill Learning Method by Intuitive Human Assembly Demonstration

Abstract

With the development of the Internet and computer technology, the demand for 3C products such as mobile phones has surged. At the same time, due to the continuous improvement of labor costs, it is urgent to automate the 3C assembly lines with industrial robots. The motivation of this paper is to propose an efficient off-line programming by demonstration method to automate 3C assembly lines. This process consists of two phases. In the first phase, the optical motion capture device is used to capture the position and orientation information of human hands during assembly process. In the second phase, those information of a couple of demonstrations are learned to derive a robot control policy. To do so, the local outlier factor based anomalous point detection algorithm as well as the trajectory segmentation algorithm inspired by density-based spatial clustering is utilized in advance to pre-process the demonstration data. Then the human assembly skill represented by a probabilistic policy is learned from those data to drive the robot to accomplish the same assembly task under new environment. The Gaussian Mixture Model is utilized to pre-structure the policy. The offline programming by demonstration method efficiently transfers human experience to the robot. Experimental results demonstrate the effectiveness of the method.





How NOKOV Contributes to the Research

For the purpose of acquiring human demonstration data the optical motion capture platform from NOKOV Science and Technology Cooperation Ltd. [15], as shown in Fig.3, is selected to record the movement of reflective markers pasted on one hand of the operator at a 340fps sampling frequency. By doing so, the sub-millimeter positions of the operator's hand as well as fingers are recorded through the infrared cameras when the operator executes the assembly process. In view of the quickness and complexity characteristics of the 3C assembly process, Better still, the optical motion capture platform does not interfere the normal assembly work of the operator. In addition, the optical motion capture platform is capable of working in normal industrial environments with- out too much space usage and dense equipment distribution. Afterwards, by a series of data pre-processing techniques, several smooth assembly motion trajectories are obtained from multiple human assembly demonstrations. Finally, a policy for the robot to regenerate assembly motion trajectory that can achieve the same assembly task under strange start and goal positions and orientation is derived by the policy learning algorithm.

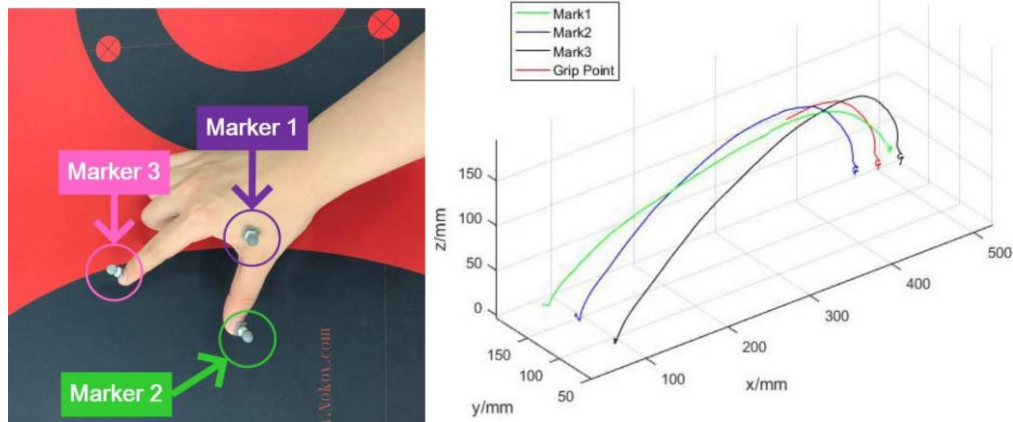
References Cao Z, Hu H, Yang X, et al. A robot 3C assembly skill learning method by intuitive human assembly demonstration[C]//2019 WRC Symposium on Advanced Robotics and Automation (WRC SARA). IEEE, 2019: 13-18.

Link <https://ieeexplore.ieee.org/document/8931930>

A Robot Programming by Demonstration Method for Precise Manipulation in 3C Assembly

Abstract

Industrial robots have been widely used in various production line but not in the 3C assembly line which is characterized by highly frequent change-over and still relies on labor heavily. To deal with this challenge, the Programming by Demonstration (PbD) method, which provides a simple and intuitive way for workers to transfer skills to the robot, will greatly reduces professional knowledge requirements for robot users. However, some practical factors in 3C assembly line, such as the mechanical diversities between the human body and the robot, make it difficult to apply PbD methods. Motivated by that, with a subtly designed demonstration data extraction platform and a posture mapping strategy, this article proposes a PbD method that transforms human assembly skill to the robot for 3C assembly tasks. Furthermore, by an iterative optimization method, the demonstration path is simplified to boost the assembly efficiency which counts for much in practice. A simulation and a 3C assembly experiment has demonstrated the effectiveness of the proposed method.



How NOKOV Contributes to the Research

A programming by demonstration (PbD) platform to transfer human 3C assembly skills to robots is proposed in this work, which consists of a demonstration data extraction platform (Fig. 3) and a robot assembly platform (Fig. 4). The demonstration data used in this work is obtained by an optical motion capture platform provided by Nokov Science & Technology Co., Ltd. [13] that tracks three reflective markers pasted on the operator's hand, as shown in Fig. 6. By doing so, intuitive human assembly actions are obtained in a natural and convenient way that sets no threshold for non-expert operators. Details of the demonstration process can be found in the supplemented video. Furthermore, the robot assembly platform is facilitated with a 3D structure light camera to estimate the initial position and orientation of the components and an Universal Robot 5kg (UR5) robot to carry out the assembly task. However, owing to the inevitable mechanical diversity between human hand and robot's end-effector, several physically different grippers coupled with quick-change mechanism are designed for a couple of 3C components. With the quick-change mechanism, the UR5 robot is capable of selecting the grippers automatically.

References Zhao Z, Hu H, Yang X, et al. A Robot Programming by Demonstration Method for Precise Manipulation in 3C Assembly[C]//2019 WRC Symposium on Advanced Robotics and Automation (WRC SARA). IEEE, 2019: 172-177.

Link <https://ieeexplore.ieee.org/document/8931947>

Official Website Case Link

https://www.nokov.com/en/support/case_studies_detail/robotic-arm-assembly-skills-learning.html

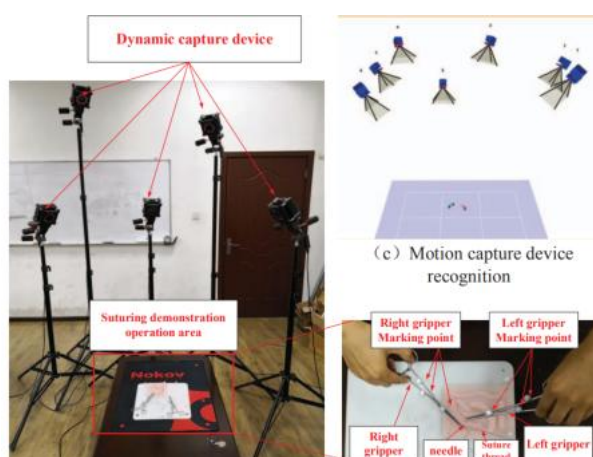
Learning from Demonstration: Dynamical Movement Primitives based Reusable Suturing Skill Modelling Method

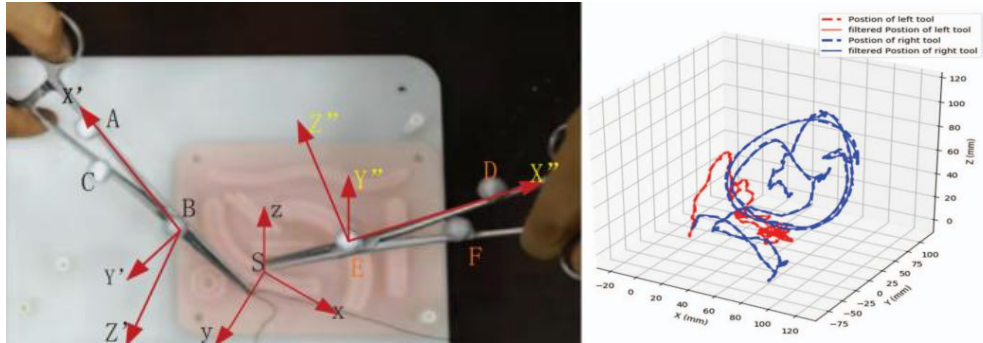
Keywords

Learn from demonstration; Suturing skill; Dynamical movement primitives; Modeling method

Abstract

Establishing a surgical skill model plays an important role for automatic robotic assisted surgery, besides that it is of great value to evaluate physicians' operational skills. In this paper, superficial tissue suture is selected as the modeling object, and suture skill learning and modeling research is conducted. In order to solve the problem of conventional models with poor migration ability for applications in new scenes, a demonstration-decomposition-modeling suture skill learning and modeling framework is proposed. The demonstration suture process is decomposed into sub-processes that repeats in various suture paradigms. The dynamic movement primitives (DMPs) method is introduced to uniformly model the suture trajectory of each sub-process. The superficial suture operation demonstration and acquisition platform is built, and the modeling ability of the proposed method and the adaptability to new scenes are preliminarily verified.





How NOKOV Contributes to the Research

The Nokov optical motion capture system is an optical three-dimensional motion capture system. In this experiment, 7 optical lenses were used to measure and track the suturing process. Six marker points were attached on the two surgical clamps. The 3D position of the marker points on the surgical forceps are collected, and the continuous motion trajectory (namely the spatial position and posture) of the surgical forceps can be calculated and recorded on the Nokov system in real time. By means of coordinate system transformation method, we can analyze the motion trajectory of the surgical forceps in the wound coordinate frame.

References Yang D, Lv Q, Liao G, et al. Learning from demonstration: dynamical movement primitives based reusable suturing skill modelling method[C]//2018 Chinese Automation Congress (CAC). IEEE, 2018: 4252-4257.

Link <https://ieeexplore.ieee.org/abstract/document/8623781>

Kinematics Optimization of a Novel Cable-Driven

7-Dof Redundant Manipulator

Keywords

Kinematics analysis; Spherical parallel mechanism; Redundant manipulator; LDWPSO;

Abstract

Redundant manipulators can accomplish complex tasks due to the redundant degree of freedom (DOF). At the same time, their kinematics calculations are complicated. In this study, a hybrid method is proposed to obtain the optimal kinematics solution for a 7-DOF redundant anthropomorphic manipulator. Kinematics analysis of the spherical parallel shoulder joint was performed by using a geometric method and coordinate transformation. Kinematics analysis of the redundant manipulator was performed by a geometric method with an optimizing arm angle. Linearly decreasing weight particle swarm optimization (LDWPSO) was used to optimize the arm angle to minimize the motion time of the manipulator. The kinematics calculation of the shoulder joint was verified by a combination of SolidWorks and MATLAB software. The kinematics calculation of the redundant manipulator was verified by MATLAB. The linear, circular and figure 8-shaped motion trajectories were used to evaluate the proposed method in the simulation. The simulation results showed that the motion time with optimization of the arm angle was only 16%–30% of that without optimization. Furthermore, the proposed method was evaluated through real manipulator experiments, and the experimental results were similar to those in the simulation.

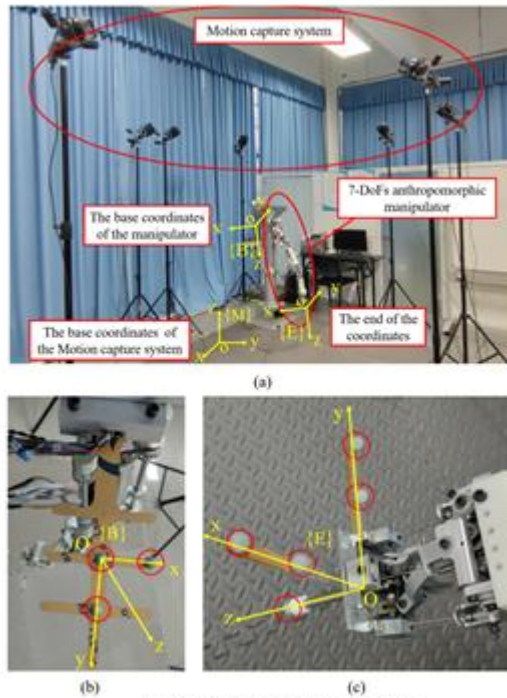


Fig. 22 Kinematics experiment on the manipulator

How NOKOV Contributes to the Research

*This section describes an experiment performed to test the kinematics of the whole arm. As shown in Fig. 22, the experimental platform is built. A motion capture system is used to evaluate the manipulator. The motion capture system is composed of eight Mars1.3H infrared high-speed cameras of the NOKOV system, with a resolution of 2048*1088 and an accuracy of 0.02 mm.*

References Chen Y, Zhang X, Yanjiang H, et al. Kinematics Optimization of a Novel Cable-Driven 7-Dof Redundant Manipulator[J]. Available at SSRN 4116420.
Link https://papers.ssrn.com/sol3/papers.cfm?abstract_id=4116420

An operational calibration approach of industrial robots through a motion capture system and an artificial neural network ELM

Keywords

Robot calibration · Kinematic identification · Artificial neural network · Motion capture system · DH parameter

Abstract

In the industry, robots' absolute accuracy is critical. This study provides a new calibration method to increase the robot's absolute accuracy. This method relies on a motion capture device as a measurement tool that can capture the robot's continuous motion state in any conceivable domain. Geometric parameters are determined by conducting overall measurements and assessing each joint individually, and the Extreme Learning Machine (ELM) neural network is in charge of compensating for non-geometric error sources that are difficult or impossible to model correctly or completely. The combination of model-based parameter identification and ELM neural network-based compensation approaches is an effective solution for the correction of all robot error sources. To verify the method's effectiveness and correctness, a six-revolute joint robot GSK RB03 is used across simulation and experiment. After calibration, the robot's position accuracy ranges from 7.440 to 0.159 mm, and its orientation accuracy ranges from 3.073 to 0.077 degrees. The method's practical applicability and correctness are determined.

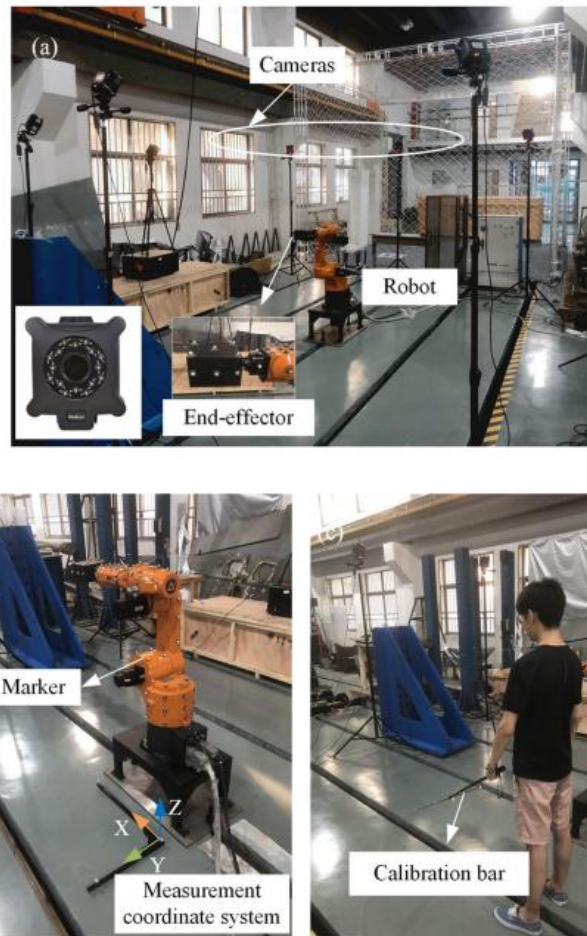


Fig. 13 (a) GSK RB03 robot and equipment, (b) the arrangement of markers and the position of the measurement coordinate system, and (c) determining the relative positions of cameras

How NOKOV Contributes to the Research

*Nokov's MARS 2H is used as the motion capture system in this paper. The pixel is 2.2 MP, the highest frame rate is 380 Hz, the image resolution is 2048*1088, and the 3D measurement accuracy is ± 0.15 mm. The robot is approximately arranged in the geometric center of 8 infrared high-speed cameras. An end effector is bolted to the end flange of the robot. The end-effector is designed as an aluminum hollow device, which not only ensures the strict and accurate geometric dimensions from the tool coordinate system to the end coordinate system but also considers the load requirements of the robot in Fig. 13(a).*

References Lu GM, Li Y, Jin S, Zheng Y, Chen WD, Zheng XX (2011) A realtime motion capture framework for synchronized neural decoding. In: 2011 IEEE international symposium on VR innovation, Singapore. <https://doi.org/10.1109/ISVRI.2011.5759656>.
Link <https://link.springer.com/article/10.1007/s00170-023-10856-w>

Ultralong Stretchable Soft Actuator (US^2A): Design, Modeling and Application

Keywords

Soft robotics, Soft actuator, Pneumatic artificial muscle, Modeling

Abstract

Actuator plays a significant role in soft robotics. This paper proposed an ultralong stretchable soft actuator (US^2A) with a variable and sizeable maximum elongation. The US^2A is composed of a silicone rubber tube and a bellows woven sleeve. The maximal extension can be conveniently regulated by just adjusting the wrinkles' initial angle of the bellows woven sleeve. The kinematics of US^2A could be obtained by geometrically analyzing the structure of the bellows woven sleeve when the silicone rubber tube is inflated. Based on the principle of virtual work, the actuating models have been established: the pressure-elongation model and the pressure-force model. These models reflect the influence of the silicone tube's shell thickness and material properties on the pneumatic muscle's performance, which facilitates the optimal design of US^2A for various working conditions. The experimental results showed that the maximum elongation of the US^2A prototype is 257%, and the effective elongation could be variably regulated in the range of 0 and 257%. The proposed models were also verified by pressure-elongation and pressure-force experiments, with an average error of 5% and 2.5%, respectively. Finally, based on the US^2A , we designed a pneumatic rehabilitation glove, soft arm robot, and rigid-soft coupling continuous robot, which further verified the feasibility of US^2A as a soft driving component.

Link <https://cjme.springeropen.com/articles/10.1186/s10033-023-00835-3>

Joint torque estimation for the human arm from sEMG using backpropagation neural networks and autoencoders

Keywords

Joint torque estimation; Surface electromyography; Backpropagation neural networks; Autoencoder

Abstract

Estimation of joint torque for human arm is important to help a robot adapt to a human partner in human-robot collaboration. In this study, we proposed an estimating model combined the back propagation neural networks (BPNN) and autoencoders to estimate the joint torque of human arm from surface electromyography (sEMG). Autoencoders were used to select the appropriate features of sEMG signals. The joint angles obtained from a vision-system were combined with the sEMG features, which were set as the input data for the estimating model. A 4-layer BPNN was used to map the sEMG features to the joint torque. The proposed estimating model was compared to other five models through experiments by estimating shoulder joint torque and elbow joint torque. The root mean square error and correlation coefficient between the estimated joint torque value and a theoretic value calculated by inverse dynamics were used to evaluate the proposed model. The root mean square error and correlation coefficient for the proposed model is 1.3842 and 0.9439, respectively. Experimental results showed that the proposed model can derive lower root mean square error and higher correlation coefficient by comparing to other models, which indicates the proposed model can estimate more accurate joint torque from sEMG than other models.



Fig. 1. Camera arrangement of the vision system.

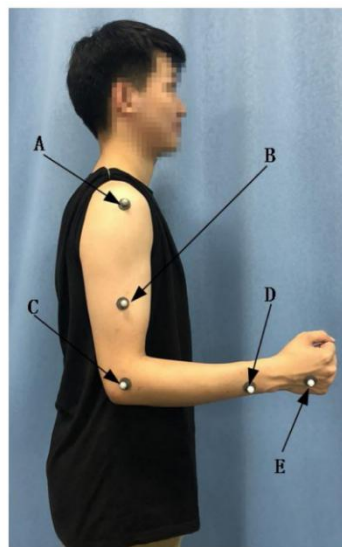


Fig. 2. The marker placement.

How NOKOV Contributes to the Research

In this paper, we collected the information from the sEMG and vision system to build a model that can estimate the joint torque of the human arm. A vision system (Nokov Mars 1.3H) was used to capture the motion of the human arm. Angles of the shoulder joint and elbow joint were calculated based on the visual information. The vision system, which consisted of eight high-speed infrared cameras, recorded the 3-D position of the markers in the worldcoordinate frame at a frequency of 100 Hz (Fig. 1). To obtain enough information, we regarded the upper arm as link L1 and combined the forearm and hand as link L2. Then, the whole arm was simplified as a two-link structure. Five markers, A, B, C, D, and E, were attached to the human arm. Marker A was on the shoulder. Marker B was on the center of mass of link L1. Marker C was on the elbow. Marker D was on the center of mass of link L2, and marker E was on the end of the hand (Fig. 2). Additionally, an L-shaped rod with four markers was placed on the ground as the reference coordinate frame for motion analysis.

References M. Ersen, E. Oztop, S. Sariel, Cognition-enabled robot manipulation in human environments: requirements, recent work, and open problems, IEEE Robot. Autom. Mag. 24 (3) (2017) 108–122.

Link <https://www.sciencedirect.com/science/article/abs/pii/S174680942030207X>

Augmented Reality-Assisted Robot Learning Framework for Minimally Invasive Surgery Task

Abstract

This paper presents an Augmented Reality (AR)assisted robot learning framework for Minimally Invasive Surgery (MIS) tasks. The proposed framework exploits an external optical tracking system to collect human demonstration. Gaussian Mixture Model (GMM) and Gaussian Mixture Regression (GMR) are utilized to encode and generate a robust desired trajectory for transferring to the real robot for the MIS task. The HoloLens 2 Head-Mounted-Display (HMD) is integrated for intuitive visualization of the robot configuration under the constraint of a small incision on the patient's abdominal cavity during the demonstration phase. Experiments are conducted to verify the feasibility and performance of the proposed framework and compared it with the kinesthetic teaching-based modality in a tumor resection MIS task. The results illustrate that the proposed AR-assisted robot learning framework requires lower workload demand, achieves higher performance and efficiency, and ensures the feasibility of the learned results for reproduction on a real robot for MIS tasks.

How NOKOV Contributes to the Research

In addition, the passive observation-based modality is also a practical solution, including optical, electromagnetic tracking, etc [17]. Passive markers were attached to the surgical gripper and tracked by the Nokov motion capture device to collect the suturing skills from the human surgeon and transfer them to the physical robot [18]. Huang et al. designed a vision-based LfD framework for multi-robot cooperation and task reproduction in a robot-assisted bimanual sewing task, infrared markers were adopted to record human demonstrated trajectories [19]. Typically, the motion scaling and mapping strategy between the input device and the real robot should be properly established to ensure the effectiveness of human demonstration and robot reproduction [14].

References E. De Momi and A. Segato, "Autonomous robotic surgery makes light work of anastomosis," *Science Robotics*, vol. 7, no. 62, p. eabn6522, 2022.
Link <https://ieeexplore.ieee.org/abstract/document/10160285>

Design and evaluation of a robotic apple harvester using optimized picking patterns

Highlights

- An integrated robotic apple harvesting prototype is presented.
- An optimized picking pattern and an anthropomorphic picking pattern were used in the field picking evaluation of the robotic apple harvester.
- The success rates using two picking motions were 80.17% and 82.93%, respectively.
- Stem-pulled or bruised apples did not appear during picking evaluation using either motion pattern.

Abstract

A robotic apple harvester consisting of a mobile platform, a manipulator, an end-effector, a stereo camera, and a host computer was constructed and evaluated using two picking motions. The field tests showed all apple picking with success rates of 80.17% and 82.93% when using anthropomorphic and “horizontal pull with bending” motions, respectively. The main reasons for picking failure were depth misalignment, detachment failure, and blocked grasp. The “horizontal pull with bending” and anthropomorphic motions took 1.14 s and 3.13 s, respectively. The full picking cycle process using “horizontal pull with bending” motion was 12.53 ± 0.53 s, 4.64 s less than the average picking time when using anthropomorphic picking motion (17.17 ± 0.36 s). The picking process using anthropomorphic motion experienced a lower dynamic payload, meaning less effort would be required by the manipulator joints; however, fruit slipping decreased the overall success rate. The “horizontal pull with bending” picking motion had a superior picking cycle time and success rate. Notably, there were no stem-pulled or bruised apples during picking process using either motion. Based on this study, both picking motions have the potential to be applied in harvesting robots.



Fig. 8. NOKOV motion capture system.

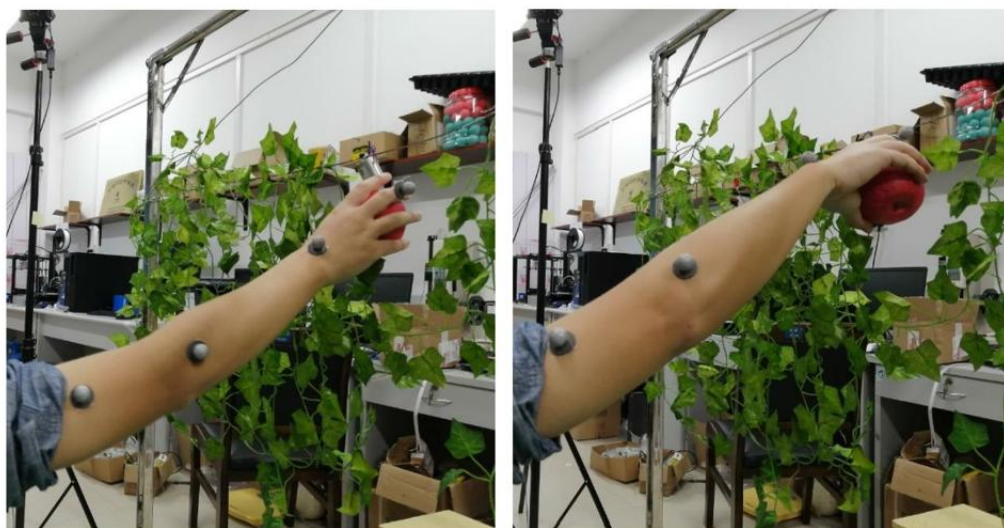


Fig. 9. Picking motion capture process.

How NOKOV Contributes to the Research

The NOKOV Motion Capture system (Beijing NOKOV Science & Technology Co., Ltd., China) was used to collect data on the movements of pickers' upper arms. As shown in Fig. 8, there were eight digital cameras with a 2048×1088 resolutions and ± 0.15 mm 3D accuracy positioned around the test area. Data were collected by tracking marked points and transferred to the host computer and processed in real-time to calculate the coordinates, velocity, and acceleration of the moving objects in space (Zhang et al., 2021b). Data acquisition started after the system was calibrated. The data acquisition frequency was set to 100 frames per second due to the small amplitude and high speed of picking motions. Meanwhile, in order to maintain high tracking accuracy, the system ensured that at least three cameras tracked each marked point. Points were marked with small balls with fluorescent surfaces; these were attached to picker's shoulder, elbow, wrist, and fingertip, as shown in Fig. 9.

References

Lingxin Bu, Chengkun Chen, Guangrui Hu, Adilet Sugirbay, Hongxia Sun, Jun Chen, Design and evaluation of a robotic apple harvester using optimized picking patterns, Computers and Electronics in Agriculture, Volume 198, 2022, 107092, ISSN 0168-1699

Link

<https://www.sciencedirect.com/science/article/abs/pii/S0168169922004094>

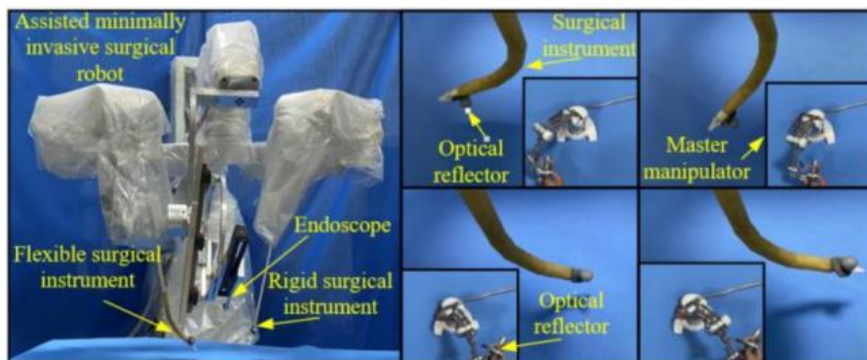
272



Kinematic Analysis of a Flexible Surgical Instrument for Robot-assisted Minimally Invasive Surgery

Abstract

Flexible surgical instruments can flexibly adjust their posture with a high degree of freedom, which makes them highly suitable for performing surgical tasks in narrow workspaces. However, redundant degrees of freedom increase their kinematic difficulty, which may cause redundant solutions, complex calculations, and low speeds. In this paper, a flexible surgical instrument is presented. The structural characteristics of this flexible instrument were explored in terms of force balance, it was concluded that the instrument had a constant curvature during bending. Based on this, the kinematics and inverse kinematics were solved via the geometric and Newton iteration methods, respectively. Our experiments showed that the proposed method for solving flexible instrument kinematics had high precision, a unique solution, and high speed; the instrument can be well controlled to perform refined operations. The proposed geometric method for solving the flexible instrument kinematics avoided the calculation of the Jacobian matrix, making it fast and capable of meeting the master-slave control requirement for real-time surgery. Furthermore, the proposed kinematics solution method is not limited by the mechanical structure, so it can be used for flexible instruments owing to its constant curvature bending.



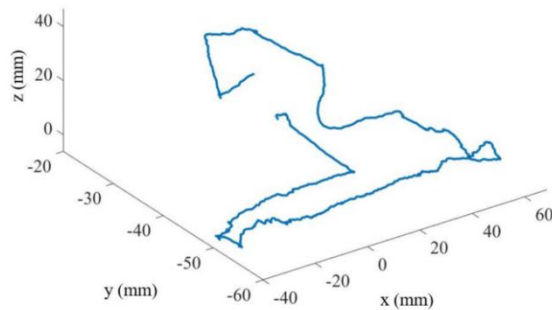
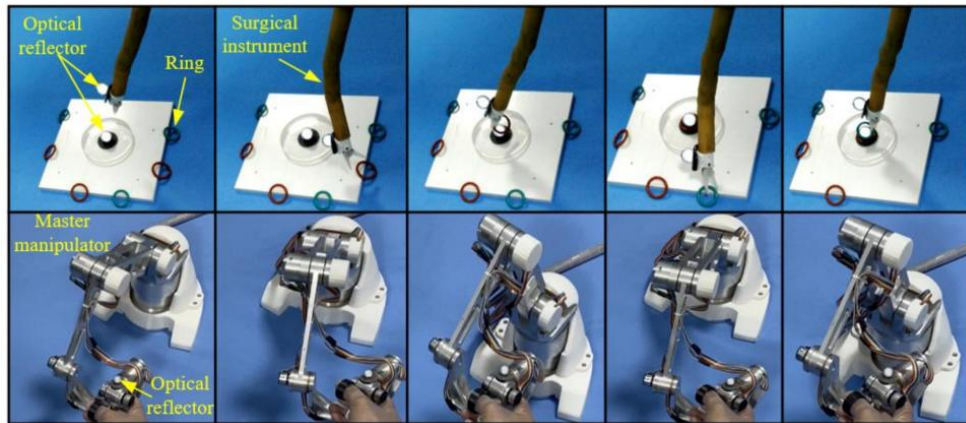


Figure 8. Trajectory of the flexible instrument mapped from the master manipulator

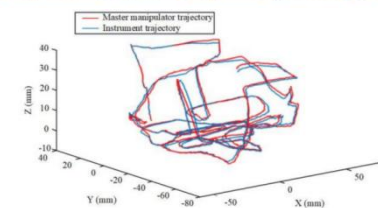


Figure 11. Trajectories of the master manipulator and instrument

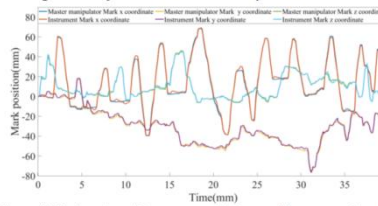


Figure 12. Trajectories of the master manipulator and instrument along the x, y, and z directions

How NOKOV Contributes to the Research

To validate the performance of the instrument under refined manipulation, it was controlled to grasp rings in target locations, as shown in Fig. 10, and a motion capture system (NOKOV Co., China) was used to follow the master and slave during operation. Optical reflectors were fixed to the end effectors of the master manipulator and instruments to collect movement information. Fig. 11 shows the movement trajectories of the master manipulator and the instrument. Fig. 12 shows the trajectories of the master manipulator and instrument along the x, y, and z directions. The RMSE of the expected and actual trajectory of the flexible surgical instrument in the x, y, and z directions were 0.498, 0.399, and 0.051, respectively. The experimental results show that the instrument can be well controlled to perform refined operations and under master-slave operation.

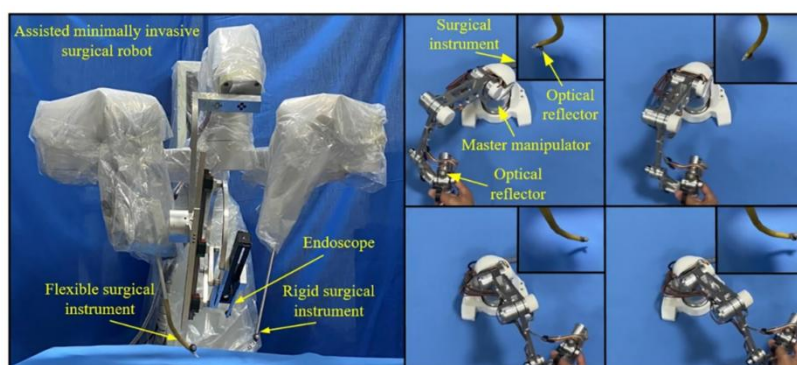
References Feng M, Ni Z, Fu Y, et al. Kinematic analysis of a flexible surgical instrument for robot-assisted minimally invasive surgery[C]//2021 IEEE International Conference on Robotics and Automation (ICRA). IEEE, 2021: 12229-12235.

Link <https://ieeexplore.ieee.org/abstract/document/9561634/>

Master Manipulator Optimisation for Robot Assisted Minimally Invasive Surgery

Abstract

In robot-assisted minimally invasive surgery, the surgeon controls a robot by operating a pair of master manipulators. Thus, the performance of a master manipulator directly affects the work of the surgeon physiologically and psychologically. In order to improve the operability and quality of operation, a structure optimisation method of master manipulator is proposed. This improves the comfort of the surgeon and can effectively avoid the problem of cutting off master-slave communication to adjust the position of the master manipulator owing to the small workspace.



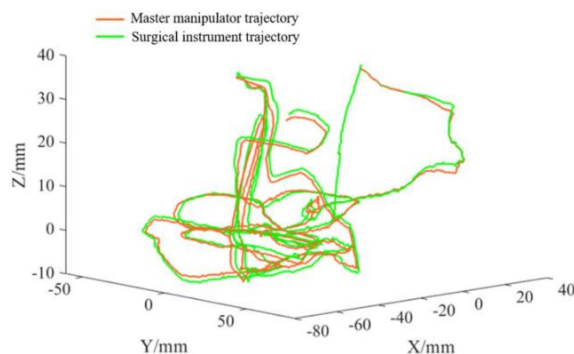


FIGURE 8 Trajectories of the end of the master manipulator and the surgical instrument

How NOKOV Contributes to the Research

The optimised structural dimensions were used to construct a prototype series-type master manipulator, as shown in Figure 4. To analyse the coverage by the master manipulator workspace of the human hand operation space, surgeons were asked to hold and move the end-effector of the manipulator in the operating posture. An optical reflector was fixed to the end-effector, and the motion capture system of China NOKOV company was used to collect the position of the optical reflector. The envelope of these positions was defined as the workspace. Figure 5 shows the master manipulator workspace and human hand operation space. The master manipulator workspace completely covered the human hand operation space, which can help avoid the problem of cutting off the master-slave communication to adjust the position of the master manipulator.

References Feng M, Ni Z X, Li A, et al. Master manipulator optimisation for robot assisted minimally invasive surgery[J]. The International Journal of Medical Robotics and Computer Assisted Surgery, 2021, 17(2): e2208.

Link <https://onlinelibrary.wiley.com/doi/abs/10.1002/rcs.2208>

Indoor Relative Positioning Method and Experiment Based on Inertial Measurement Information/Human motion model/UWB Combined System

Keywords

Indoor positioning; Inertial measurement information; Human motion model; UWB

Abstract

Navigation and positioning services are related to national security, economic development and social livelihood, they're playing a decisive and indispensable role in both military and civilian fields. Urbanization is accelerating and indoor sites are increasing. People spend more than 80% of their time living and working in the indoor environments (including large buildings, underground sites, mines, tunnels, etc.), How to achieve high-precision navigation and positioning in such a complex navigation environment? This is not only an urgent social problem but also a technical problem. Based on the introduction of indoor navigation background, an inertial measurement information/UWB combined indoor positioning method is proposed in this paper, to solve the problem of long-term positioning in a complex indoor environment, and the experimental researches have been performed. Firstly, based on the research of human motion model, the method extracts motion parameters related to navigation and establishes the mapping model between human motion parameters and navigation parameters in order to implement autonomous pedestrian navigation which is based on inertial measurement information/human motion model. Then, using the position information provided by UWB as the measurement, the Kalman filter based on inertial measurement information/ human motion model/UWB combined system is designed to integrate positioning information from the combined system of inertial measurement information/human motion model and UWB, so that it can provide long-term and high-precision relative navigation for indoor pedestrians. The paper develops a prototype of the inertial measurement information/human motion model/UWB combined system, which provides a physical platform for the experimental verification of the proposed method. What's more, it also optimizes and verifies the proposed method referring to experiments under the environment of NOKOV high-precision indoor 3D-motion

capture system by Beijing measurement technology co., LTD. There are bright application prospects that the method can be extended in single-soldier system, urban anti-terrorism operations and pedestrian positioning in the field of emergency rescue.

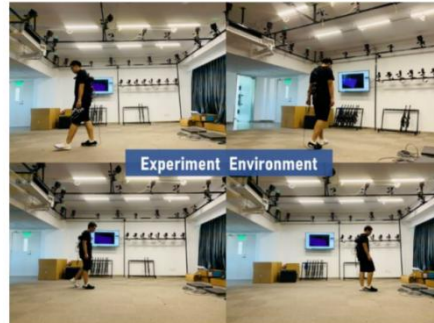


Fig. 4. NOKOV 3D optical motion capture system.

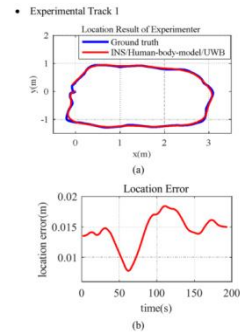


Fig. 5. (a) Location result of the indoor experimenter, (b) Location error.

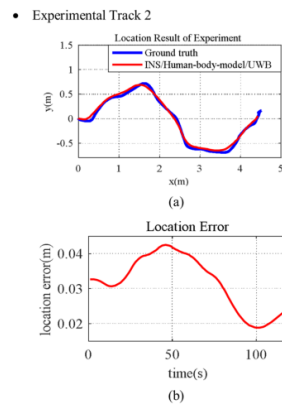


Fig. 6. (a) Location result of the indoor experimenter, (b) Location error.

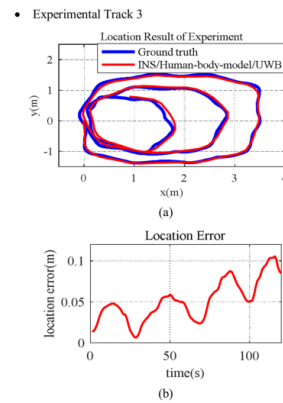


Fig. 7. (a) Location result of the indoor experimenter, (b) Location error.

How NOKOV Contributes to the Research

Nokov 3D optical motion capture system is mainly composed of infrared optical motion capture lens, operation, analysis and processing software and related accessories, which can capture and output the accurate three-dimensional data of the object to be measured. The accuracy of this motion capture system can reach sub-millimeter level, which meets the requirements of evaluating the positioning accuracy of the method proposed in this paper. The Nokov 3D optical motion capture system is shown as Fig. 4.

References Zhang Y, Wang N, Li M, et al. Indoor Relative Positioning Method and Experiment Based on Inertial Measurement Information/Human motion model/UWB Combined System[C]//2020 27th Saint Petersburg International Conference on Integrated Navigation Systems (ICINS). IEEE, 2020: 1-6.

Link <https://ieeexplore.ieee.org/abstract/document/9134055>

A General Framework of Knowledge-Based Coaching System with Application in Table Tennis Training

Keywords

Knowledge-based coaching system; Sequential decision problem; Table tennis training; Motion capture

Abstract

As the growth of people's awareness in health and fitness, high-level sports exercise coaching resources become scarce comparing to the fast increasing demands. In traditional sports training, coaches are responsible for scheduling training sessions and giving instructions based on the performance of the trainee. This paper proposes a general knowledge-based coaching system (KBCS) for the purpose of automatic and intelligent sports training. Through systematically modeling the interactive training process between the coach and the trainee as a sequential decision problem, we show that the knowledge-based training controller can be properly designed to help the trainee to make effective progress towards the preset training goal. Partial implementation of KBCS in the table tennis training has been realized to demonstrate and verify the effectiveness of the proposed method.



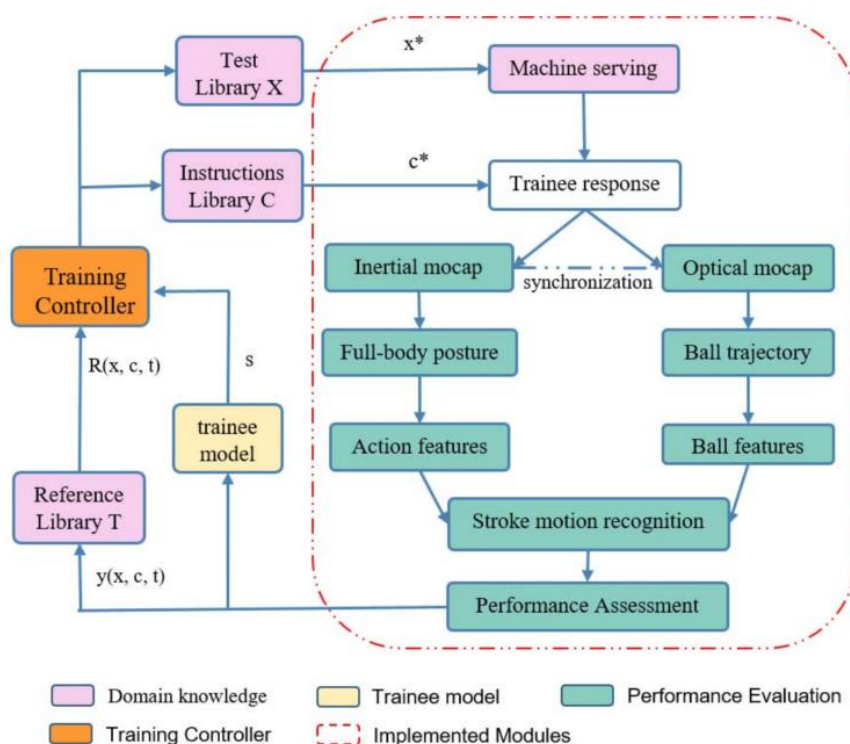


Fig. 4: KBCS prototype structure for table tennis

How NOKOV Contributes to the Research

According to the above design, an experimental platform is built as shown in Fig. 5. A NOKOV Motion Capture System was used to capture the trajectory of balls and the racket. The table tennis balls are covered with reflective material and the racket is attached reflective markers around the edge for better results. Besides, a NOITOM Perception Legacy suit is worn by the trainee for accurate posture recording. Four students are recruited and their data are analyzed.

References Yan W, Ma H, Yang Z. A General Framework of Knowledge-Based Coaching System with Application in Table Tennis Training[C]//2020 39th Chinese Control Conference (CCC). IEEE, 2020: 2902-2907.

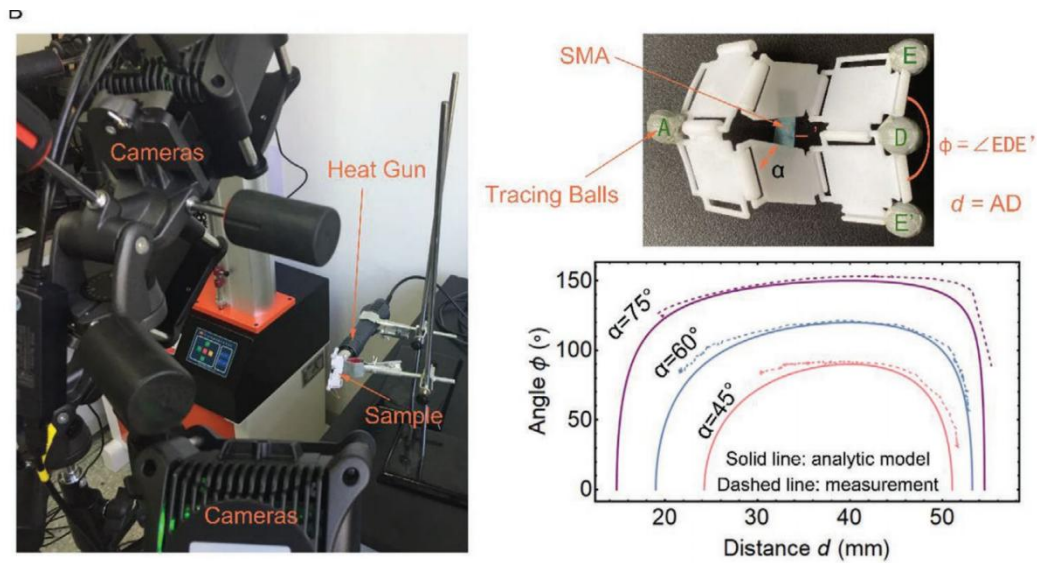
Link <https://ieeexplore.ieee.org/abstract/document/9188412>

Deployable-Structure-Based Artificial Muscles

Generating Coded Forces

Abstract

Here mechanical metamaterials with shape memory alloys (SMA) are assembled to create a new kind of artificial muscle driven by response to temperature changes that generates pulling, pushing, and structurally coded and sequentially coded forces. The deformation and force response of a muscle unit can be geometrically tuned, and different forces are generated with two units arranged in parallel and in series. SMA sheets with different orientations controlling different units of a muscle chain can generate programmable force responses as a mixture of pulling and pushing forces. Combining structural coding with sequential coding, a wide range of force responses can be generated with a single muscle chain. It is shown that the proposed muscles can be applied in an arm-like mechanism to lift weights and to enable a car to cross a barrier. The principle explored in the work can be used to transform versatile origami- and kirigami-based structures into multifunctional actuators for many applications that can also be based on a variety of materials other than SMA.



How NOKOV Contributes to the Research

With the insertion of a SMA sheet into a section of the kiri- muscle, the section exhibits the expected deformation (Figure 1A) in response to the temperature changes. To measure the structural deformation, four tracing balls were attached at points A, D, E and E' of the kiri-muscle and were tracked by a six-camera motion capture system (Figure 2B).

References Yang N, Deng Y, Niu X. Deployable-Structure-Based Artificial Muscles Generating Coded Forces[J]. Advanced Materials Technologies, 2021, 6(9): 2100493.
Link <https://onlinelibrary.wiley.com/doi/full/10.1002/admt.202100493>

Achieving Reliable Underground Positioning With Visible Light

Keywords

Location algorithm; personnel positioning; underground positioning; visible light communication (VLC); visible light positioning (VLP);

Abstract

Visible light technology shows great potential in the underground mine positioning today. Most visible light-based underground positioning technology deploy the receiver, such as photodiodes (PDs), on the mine tunnel ground and consider the miner's lamp on the ceiling as the transmitter. This incurs the unreliable reception of visible light signals and the difficult implementation of VLP systems due to the complex underground mine environment, such as the receiver being obscured or trampled by objects (e.g., Minecarts and Ore). We present a reliable underground positioning algorithm with visible light considering the actual mine environment. The key innovation is that we design a novel reverse mechanism, i.e., utilizing the existing miner's headlamp as the transmitter for broadcasting the specific light beacon representing the mine's identity information, and the PDs deployed on the ceiling of the mine tunnel as the receiver for optical signal reception. We can infer the current locations of all miners using a series of base stations constructed by PDs. If some special mine environment limits the large-scale deployment of base stations and causes the positioning blind spots, the inertial navigation is combined to achieve all miners' real-time positioning. We overcome some technical challenges for this new mechanism and the underground mine environment, such as optimal frequency selection, robust frequency identification, and accurate base station positioning. We conduct extensive experiments on our algorithm in a simulated mine tunnel, and the experimental results verify that our algorithm exhibits good performance.



Fig. 21. Experimental scene.

How NOKOV Contributes to the Research

Motion Capture System: We use the motion capture system as the reference measurement system to obtain the real location of participants. The NOKOV motion capture system that we used consists of four optical motion capture lenses and some reflective markers. The optical motion capture lens (MASR1.3H) has an FOV of $56.3^{\circ} \times 43.7^{\circ}$ and a sampling frequency of 240 Hz, and the reflective markers are reflective balls coated with special fluorescent materials. This motion capture system can achieve subcentimeter-level indoor positioning accuracy in an area of $12\text{ m} \times 12\text{ m}$. During the experiment, we deploy the reflective marker on the modulated LED light; hence, the position of the reflective marker determined by the motion capture system is taken as the real location of participants.

References Pang M, Shen G, Yang X, et al. Achieving Reliable Underground Positioning With Visible Light[J]. IEEE Transactions on Instrumentation and Measurement, 2022, 71: 1-15.

Link <https://ieeexplore.ieee.org/abstract/document/9740565>

A Dynamic Resistive Force Model for Designing Mobile Robot in Granular Media

Keywords

Contact modeling; filed robots; methods and tools for robot system design; wheeled robots;

Abstract

Locomotion failure is the challenge for wheeled robots applied in granular media, which puts a high demand on the design of wheels. Due to a lack of a mechanical analysis method that can analyze the dynamic interaction between wheel and granular media, it is a challenge to seek design guidance for the wheel geometry. To this end, we introduce a Dynamic Resistive Force Model (DRFM) method suitable for 3D dynamic intrusion. In this method, Granular Resistive Force Theory (RFT) can be extended to 3D RFT smoothly by using granular parameters, and additional velocity terms can describe the effect of intrusion velocity on intrusion force. The relationship between wheel geometry and mechanical properties can be established by performing differential processing on the wheel surface and integrating the force generated by each micro-surface. On this basis, a complete set of solutions is proposed for the design of mobile robot wheels in granular media, and the screw-propelled wheel is taken as an example to optimize it. Theories, simulations, and experiments have all proved that the optimized wheel can produce greater thrust and lift than normal screw-propelled wheel when ensuring the stability of smooth propulsion. The proposed solution and the optimized screw-propelled wheel bring insights for designing mobile robots in granular media.

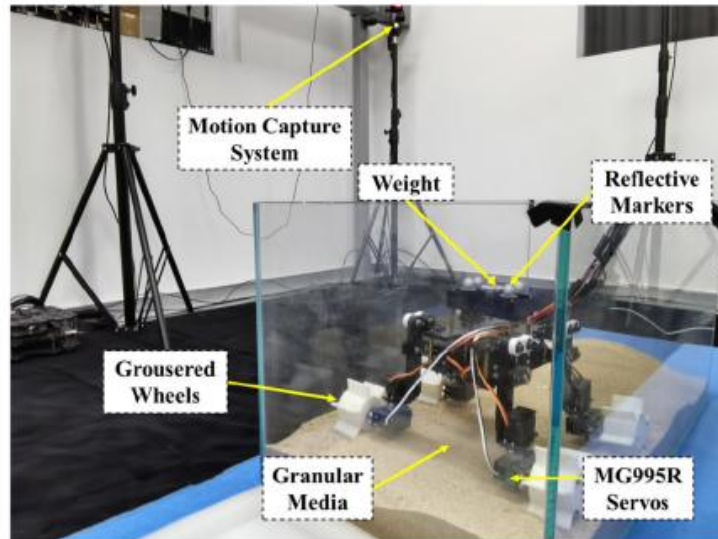


Fig. 1. An overview of the robot motion experiment. Robot equipped with grousers wheels is moving on sand media. The robot is driven by MG995R servos and its mass can be changed by weights. The motion data is captured by motion capture system through four reflective markers on the robot.

How NOKOV Contributes to the Research

ReBective markers were placed on the body of the robot, and kinematic data of the robot was captured by the NOKOV system. Eight NOKOV motion capture cameras were mounted around the bed, and the kinematic data were captured by reflective markers attached to the robot at a speed of 60 frames per second.

References Huang L, Zhu J, Yuan Y, et al. A dynamic resistive force model for designing mobile robot in granular media[J]. IEEE Robotics and Automation Letters, 2022, 7(2): 5357-5364.

Link <https://ieeexplore.ieee.org/abstract/document/9729492>

Continuous Occupancy Mapping in Dynamic Environments Using Particles

Keywords

Mapping; Aerial Systems: Perception and Autonomy; Collision Avoidance; Dynamic Environment;

Abstract

Dynamic occupancy maps were proposed in recent years to model the obstacles in dynamic environments. Among these maps, the particle-based map offers a solid theoretical basis and the ability to model complex-shaped obstacles. Current particle-based maps describe the occupancy status in discrete grid form and suffer from the grid size problem, namely: large grid size is unfavorable for path planning while small grid size lowers efficiency and causes gaps and inconsistencies. To tackle this problem, this paper generalizes the particle-based map into continuous space and builds an **efficient** 3D local map. A dual-structure subspace division paradigm, composed of a voxel subspace division and a novel pyramid-like subspace division, is proposed to propagate particles and update the map efficiently with the consideration of occlusions. The occupancy status of an arbitrary point can then be estimated with the cardinality expectation. To reduce the noise in modeling static and dynamic obstacles simultaneously, an initial velocity estimation approach and a mixture model are utilized. Experimental results show that our map can effectively and efficiently model both dynamic obstacles and static obstacles. Compared to the state-of-the-art grid-form particle-based map, our map enables continuous occupancy estimation and substantially improves the performance in different resolutions. We also deployed the map on a quadrotor to demonstrate the bright prospect of using this map in obstacle avoidance tasks of small-scale robotics systems.

How NOKOV Contributes to the Research

The velocity estimation experiments were conducted with the data collected in an indoor testing field with the Nokov motion capture system.

References Chen G, Dong W, Peng P, et al. Continuous Occupancy Mapping in Dynamic Environments Using Particles[J]. arXiv preprint arXiv:2202.06273, 2022.
Link <https://arxiv.org/abs/2202.06273>

Deep Reinforcement Learning based Multi-target Coverage with Connectivity Guaranteed

Keywords

Multi-target coverage; multi-robot system; connectivity maintenance; deep reinforcement learning;

Abstract

Deriving a distributed, time-efficient, and connectivity guaranteed coverage policy in multi-target environment poses huge challenges for a multi-robot team with limited coverage and limited communication. In particular, the robot team needs to cover multiple targets while preserving connectivity. In this paper, a novel deep reinforcement learning based approach is proposed to take both multi-target coverage and connectivity preservation into account simultaneously, which consists of four parts: a hierarchical observation attention representation, an interaction attention representation, a two-stage policy learning, and a connectivity guaranteed policy filtering. The hierarchical observation attention representation is designed for each robot to extract latent features of the relations from its neighboring robots and the targets. To promote the cooperation behavior among the robots, the interaction attention representation is designed for each robot to aggregate information from its neighboring robots. Moreover, to speed up the training process and improve the performance of the learned policy, the two-stage policy learning is presented using two reward functions based on algebraic connectivity and coverage rate. Furthermore, the learned policy is filtered to strictly guarantee connectivity based on a model of connectivity maintenance. Finally, the effectiveness of the proposed method is validated by numerous simulations. Besides, our method is further deployed to an experimental platform based on quad-rotor unmanned aerial vehicles (UAVs) and omni-directional vehicles. The experiments illustrate the practicability of the proposed method.

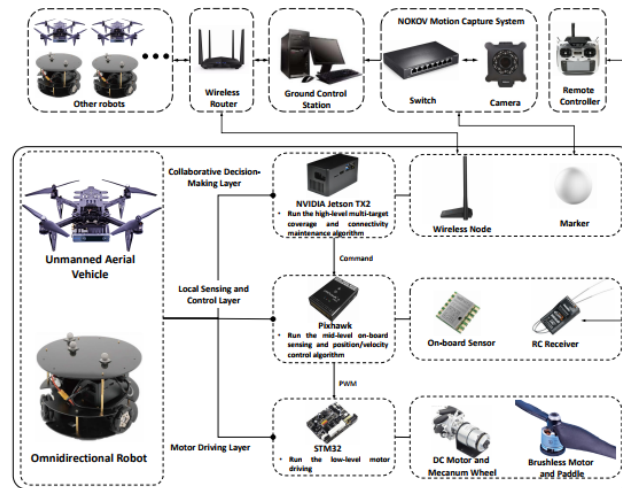


Fig. 6. Architecture of unmanned system consisting of quad-rotor UAV and omnidirectional vehicle.

How NOKOV Contributes to the Research

NOKOV motion capture system is used to calculate the positions and velocities of the mobile robots.

References Wu S, Pu Z, Qiu T, et al. Deep Reinforcement Learning based Multi-target Coverage with Connectivity Guaranteed[J]. IEEE Transactions on Industrial Informatics, 2022.

Link <https://ieeexplore.ieee.org/abstract/document/9738445>

Cost Function Determination for Human Lifting Motion via the Bilevel Optimization Technology

Keywords

inverse optimization control; bilevel optimization; direct collocation; particle swarm optimization; human lifting motion;

Abstract

Investigating the optimal control strategy involved in human lifting motion can provide meritorious insights on designing and controlling wearable robotic devices to release human low-back pain and fatigue. However, determining the latent cost function regarding this motion remains challenging due to the complexities of the human central nervous system. Recently, it has been discovered that the underlying cost function of a biological motion can be identified from an inverse optimization control (IOC) issue, which can be handled via the bilevel optimization technology. Inspired by this discovery, this work is dedicated to studying the underlying cost function of human lifting tasks through the bilevel optimization technology. To this end, a nested bilevel optimization approach is developed by integrating particle swarm optimization (PSO) with the direction collocation (DC) method. The upper level optimizer leverages particle swarm optimization to optimize weighting parameters among different predefined performance criteria in the cost function while minimizing the kinematic error between the experimental data and the result predicted by the lower level optimizer. The lower level optimizer implements the direction collocation method to predict human kinematic and dynamic information based on the human musculoskeletal model inserted into OpenSim. Following after a benchmark study, the developed method is evaluated by experimental tests on different subjects. The experimental results reveal that the proposed method is effective at finding the cost function of human lifting tasks. Thus, the proposed method could be regarded as a paramount alternative in the predictive simulation of human lifting motion.

How NOKOV Contributes to the Research

To verify the developed method on the IOC issue of human lifting motion, six healthy male subjects (age: 23 ± 2 years, mass: 62.5 ± 4.52 kg, height: 1.71 ± 0.101 m) have been recruited to carry out a pack with 10 kg in our experimental study. The experimental kinematic data of each subject is obtained by using a motion capture system (Nokov, Beijing Metrology Technology Co., Ltd.) with 39 marker points. For more details about ways of sticking markers, the reader can refer to the operation guidance from OpenSim official website (<https://simtk-confluence.stanford.edu:8443/display/OpenSim/Gait+2392+and+2354+Models>). After gaining experimental kinematic data, the musculoskeletal model is scaled by the scaling tool in OpenSim to change the anthropometry of the model in order to match the specific subject as far as possible. Note that joint kinematics and kinetics data (e.g., joint angles and torques) are gained by inverse kinematics and inverse dynamics tools in OpenSim.

References Tang B, Peng Y, Luo J, et al. Cost Function Determination for Human Lifting Motion via the Bilevel Optimization Technology[J]. Frontiers in Bioengineering and Biotechnology, 2022, 10.

Link <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC9163668/>

Design and Control of TAWL—A Wheel-Legged Rover With Terrain-Adaptive Wheel Speed Allocation Capability

Keywords

Legged robot; planetary rover; terrain adaptive capability; wheel speed allocation; wheeled robot.

Abstract

This article presents the design, control, and implementation of a novel four-legged rover for planetary exploration. A novel active-passive coupled leg mechanism is proposed for the design of the planetary rover. Then, based on the design of the actuator and leg mechanism, the novel wheel-legged robotic system is developed. Furthermore, the kinematic model of the robot is established and then a kinematics-based control strategy is presented, incorporating roll-and-pitch control, contact force control, wheel speed allocation, and steering modules. Experiments are carried out on regular indoor terrain and irregular outdoor terrain. The variations of rover attitude angles and contact forces in regular terrain decrease by more than 90% and 60%, respectively. In the outdoor environment, rover attitude angles are regulated within 1° and variations of contact forces are reduced by more than half. In addition, the proposed control strategy decreases the slippage on hard unstructured terrain by almost half. Finally, the robustness and failure responses of the robot are investigated, which are also verified by simulations and experiments.

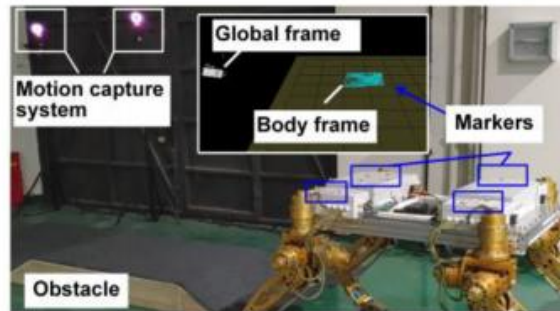


Fig. 15. Slippage experiments.

How NOKOV Contributes to the Research

Here, they are measured in real-time by the NOKOV visual motion capture system pictured in Fig. 15, which includes 8ve cameras. Serval markers bonded on the body of the rover are used to establish the body frame in the world coordinate system.

References He J, Sun Y, Yang L, et al. Design and Control of TAWL—A Wheel-Legged Rover With Terrain-Adaptive Wheel Speed Allocation Capability[J]. IEEE/ASME Transactions on Mechatronics, 2022.

Link <https://ieeexplore.ieee.org/abstract/document/9794482>

MineGPS: Battery-Free Localization Base Station for Coal Mine Environment

Keywords

Coal mine;; robot self-positioning; battery-free; base stations

Abstract

Rescue robot self-positioning is rescue. significant end, we propose a localization system with unique low-cost battery-free base stations for underground rescue robots, called MineGPS. The main idea is to design the battery-free base station with a unique arrangement sequence of reflective tags, and make robots identify each base station with computer vision-related technology to determine which base station it is near. In particular, inspired by Hamming code, we introduce a detection mechanism to correct the base station identification errors caused by dust covering in the mine. Finally, we present a robot localization algorithm by utilizing the location information of the base station. We implement MineGPS and evaluate its performance extensively. Results show that MineGPS can achieve the positioning accuracy with a median error of 15 cm. It is believed that MineGPS has a high potential to provide a low-cost and effective solution for the precise positioning of rescue robots in complex underground environments.

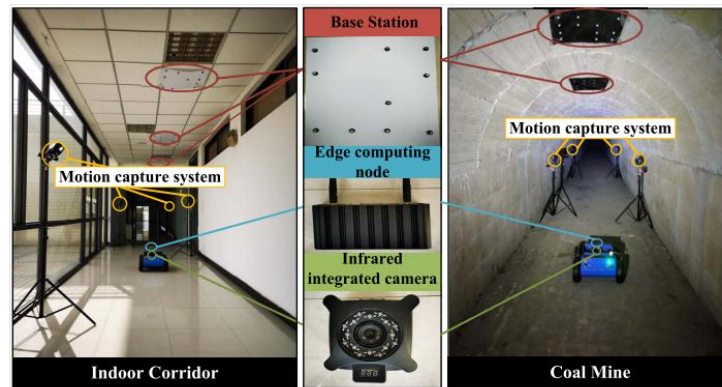


Fig. 8. Experimental scene.

References Yang X, Yu X, Zhang C, et al. MineGPS: Battery-free localization base station for coal mine environment[J]. IEEE Communications Letters, 2021, 25(8): 2579-2583.
Link <https://ieeexplore.ieee.org/abstract/document/9433550>

Model Predictive Control of a Novel Wheeled–Legged Planetary Rover for Trajectory Tracking

Keywords

mobile robot; advanced intelligent control; wheeled–legged; trajectory tracking; model predictive control;

Abstract

Amid increasing demands for planetary exploration, wide-range autonomous exploration is still a great challenge for existing planetary rovers, which calls for new planetary rovers with novel locomotive mechanisms and corresponding control strategies. This paper proposes a novel wheeled–legged mechanism for the design of planetary rovers. The leg suspension utilizes a rigid–flexible coupling mechanism with a hybrid serial–parallel topology. First, the kinematic model is derived. Then, a control strategy for the wheeled–legged rover that includes a trajectory tracking module based on the model predictive control, the steering strategy, and the wheel speed allocation algorithm is proposed. After that, three groups of cosimulations with different trajectories and speeds, and experiments are carried out. Results of both the simulations and experiments validate the proposed control method.

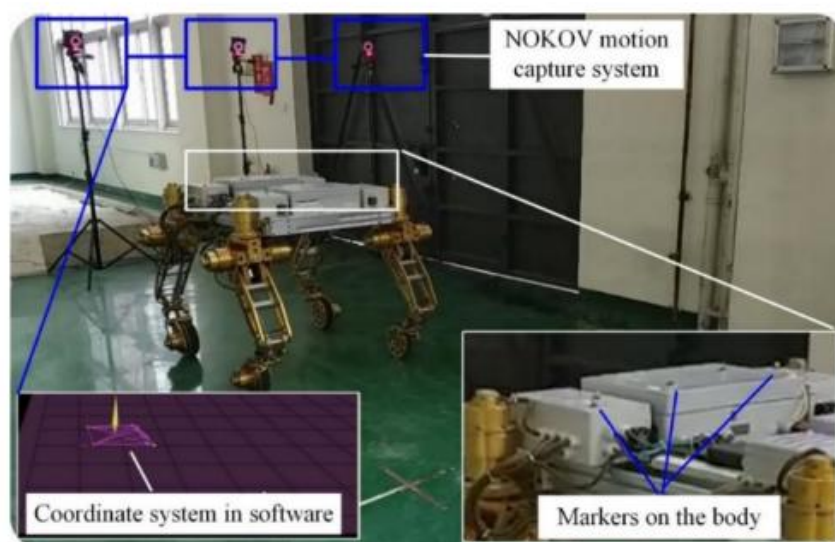


Figure 16. The experimental setup.

How NOKOV Contributes to the Research

To further verify the control strategy, an experimental setup based on the NOKOV vision motion capture system was established that consisted of six cameras, as shown in Figure 16.

References He J, Sun Y, Yang L, et al. Model Predictive Control of a Novel Wheeled–Legged Planetary Rover for Trajectory Tracking[J]. Sensors, 2022, 22(11): 4164.
Link <https://www.mdpi.com/1424-8220/22/11/4164>

Monocular-Vision-Based Relative Pose Estimation of Noncooperative Spacecraft Using Multicircular Features

Keywords

Monocular vision; multi circular features; noncooperative target spacecraft; nonlinear optimization method; pose estimation.

Abstract

Accurate relative pose (position and orientation) estimation for noncooperative target spacecraft is the basic module for on-orbit services, such as capture and repair. This article technically proposes a nonlinear optimization method based on monocular vision to estimate the relative pose for the solid-of-revolution-shaped noncooperative spacecraft. Specifically, considering the multicircular features on the target, the parameters of the ellipses derived from the circles perspective projection are first obtained by the propounded ellipse detection algorithm. Moreover, the constraint that the center of each circle falls on the center axis (normal direction) of the spacecraft is utilized to optimally solve the center position and the normal of the multi circles. In particular, the roll angle around the center axis is recovered through the geometric constraints of the solar panels, and the six-degree-of-freedom spacecraft pose estimation is then accomplished. Consequently, it is analyzed that the proposed optimization algorithm is able to provide a closed-form solution to ensure the accuracy of the relative pose and only requires limited computational resources. Finally, experimental results on the synthetic images and the physical scenes are conducted to evaluate the efficiency of the proposed approach.

How NOKOV Contributes to the Research

The noncooperative target spacecraft model is Shenzhou 12, and the Nokov motion capture system is used to obtain the ground truth pose of the noncooperative target relative to the camera.

References Long C, Hu Q. Monocular-Vision-Based Relative Pose Estimation of Noncooperative Spacecraft Using Multicircular Features[J]. IEEE/ASME Transactions on Mechatronics, 2022.

Link <https://ieeexplore.ieee.org/abstract/document/9802504>

Multi-Target Encirclement with Collision Avoidance via Deep Reinforcement Learning using Relational Graphs

Abstract

In this paper, we propose a novel decentralized method based on deep reinforcement learning using robot-level and target-level relational graphs, to solve the problem of multi-target encirclement with collision avoidance (MECA). Specifically, the robot-level relational graphs, composed of three heterogeneous relational graphs between each robot and other robots, targets and obstacles, are modeled and learned through using graph attention networks (GATs) for extracting different spatial relational representations. Moreover, for each target within the observation of each robot, a target-level relational graph is built with GAT to construct spatial relations from the robot. Furthermore, the movement of each target is modeled by the target-level relational graph and learned through supervised learning for predicting the trajectory of the target. In addition, a knowledge-embedded compound reward function is defined to solve the multi-objective problem in MECA, and guide the policy learning for deriving the behavior of MECA. An actor-critic training algorithm based on the centralized training and decentralized execution framework is adopted to train the policy network. Simulation and real-world experiment results demonstrate the effectiveness and generalization of our method.

How NOKOV Contributes to the Research

Aside from the simulations above, we also examine the trained policy in real-world experiments on an experimental platform based on omnidirectional robots. In the experimental platform, each robot is equipped with an onboard computer (Nvidia Jetson TX2) to improve the real-time performance of policy implement. Moreover, the positions and velocities of the robots are provided by the NOKOV motion capture system. The three snapshots of 6 robots encircling 2 targets in 2 obstacle environment are presented in Fig. 4. These actual results show the robots can successfully complete the MECA task, which demonstrates the effectiveness and practicability of the proposed method.

References Zhang T, Liu Z, Pu Z, et al. Multi-Target Encirclement with Collision Avoidance via Deep Reinforcement Learning using Relational Graphs[C]//2022 International Conference on Robotics and Automation (ICRA). IEEE, 2022: 8794-8800.
Link <https://ieeexplore.ieee.org/abstract/document/9812151>

Multi-Sensor State Estimation Fusion on Quadrupe Robot Locomotion

Keywords

Quadrupe robot, state estimation, preprocess data, modular filter

Abstract

In this paper, we present a effective state estimation algorithm that combined with various sensors information (Inertial measurement unit, joints encoder, camera and LIDAR) to enable balance the different frequency and accuracy for the locomotion of our quadrupe robot StarDog. Of specific interest is the approach to preprocess a series of sensors measurement which can obtain robust data information, as well as improve the system performance in terms of time and accuracy. In the IMU-centric leg odometry which fuse the inertial and kinematic data, the detection of foot contact state and lateral slip are considered to further import the accuracy of robot's base-link posiiton and velocity. Before use the NDT registration to align the point clouds, we need to feed the requirement by preprocessing these data and obtaining initial localization in map. Also a modular filtering is proposed which can fuse leg odometry based on the inertial-kinematics state estimator of the quadrupe robot and LIDAR- based localization method. This algorithm has been tested from the experimental data at indoor building and the motion capture system is used to evaluate the performance, the experimental results can demonstrate the feasibility of low error and high frequency for our quadrupe robot system.



Fig. 13. The simulation and actual experiment environment.

How NOKOV Contributes to the Research

The robot simulation platform system and complex environment are mainly built by ROS and gazebo, and the visualization can be carried out in RVIZ. The gazebo parameters of robot state are taken as the ground truth data. In order to provide the ground truth data in real time and accurately, as well as further adjust and optimize the parameters of our algorithms, a motion capture system NOKOV is selected and attached a few markers on the upper deck of the robot to record the actual pose information. The simulation and actual experiment system is detailed shown in Fig. 13.

References Yao C, Jia Z. Multi-Sensor State Estimation Fusion on Quadruped Robot Locomotion[J]. arXiv e-prints, 2020: arXiv: 2007.02679.

Link <https://ui.adsabs.harvard.edu/abs/2020arXiv200702679Y/abstract>

The Application of Digital Technology in the Protection of Intangible Cultural Heritage — Taking Beijing Palace Carpets as an Example

Keywords

Beijing palace carpet, Intangible cultural heritage, Digital technology.

Abstract

The patterns, decorations and exquisite skills of Beijing palace carpets show the meaningful national culture that is passed down through body language and oral instruction. They are not only a concrete display of ancient people's world view, outlook on life, and values, but also convey image language and textual information related to natural science and cultural knowledge. As one of the typical intangible cultural heritage, the protection and inheritance of Beijing palace carpets must be based on the principle of "taking the protection as the principal thing, rescuing firstly, using rationally, and inheriting and developing", so as to record, research and inherit the representative skills, original hand-drawn draft maps, textures and materials and even the body language of the inheritors of Beijing palace carpets in a comprehensive and brand-new way; at the same time, at the moment when digital technology is involved in social development, Beijing palace carpets, as an integral part of cultural inheritance and development, should also explore ways and methods to achieve innovative transformation and creative development.

How NOKOV Contributes to the Research

the Chinese motion capture system manufacturer Nokov is accurate and easy to use, which is more suitable for the motion capture of the Beijing palace carpet inheritors. Its collector is close to everyday clothes, light and comfortable to wear without complicated circuits, and it captures the details of the inheritor's motions with precision. The process of Beijing palace carpet is complicated. Each step of the technical motions such as slitting, spinning, twisting, underpainting, through the weft, off the machine, flattening and trimming, etc. can be meticulously recorded by the Nokov (Nokov motion compensation system). In particular, it can add positioning points to the hands, and capture the extremely fine technical motions and even the force skills. It uses computer collectors to summarize and organize to generate 3D motion data files, and to attach model bones through 3D software such as 3ds max and MAYA. It uses UE4 or U3D technology to further display and experience the exquisite and complete manufacturing skills of the Beijing palace carpet inheritors in the virtual reality space with direct screen, ring screen, arc screen, helmet and VR glasses, CAVE projection system and other equipment.

References Jiang B. The Application of Digital Technology in the Protection of Intangible Cultural Heritage—Taking Beijing Palace Carpets as an Example[C]//2021 Conference on Art and Design: Inheritance and Innovation (ADII 2021). Atlantis Press, 2022: 35-40. Link <https://www.atlantis-press.com/proceedings/adii-21/125970447>

Multi-scale Variational Autoencoder Aided Convolutional Neural Network for Pose Estimation of Tunneling Machine Using A Single Monocular Image

Keywords

Pose estimation, convolutional neural network, variational autoencoder, computer vision, tunnelling.

Abstract

With the rising demand of underground construction, intelligent tunneling techniques have been increasingly studied to improve the safety and efficiency of construction. The self-positioning technology of tunneling machines is the cornerstone of intelligent tunneling, which is particularly challenging due to the extreme environments of the underground tunnels. In this paper, a novel robust and real-time six degrees of freedom (6-DoF) pose estimation strategy is proposed for tunneling machines based on the computer vision and deep learning methods. A monocular camera is attached to the tunneling machine, and employed to capture the images of the artificial feature object that is set far behind the tunneling machine. A novel multi-scale variational autoencoder aided convolutional neural network (MSVAE-CNN) model is developed to estimate the current absolute 6-DoF pose of the tunneling machine in an end-to-end manner using a single monocular image, in which the multi-task variational learning scheme is able to enhance the generalization and robustness of the model and the multi-scale structure can improve the learning ability of the neural network. In our numerical experiments, a Motion Capture System is utilized to assist the acquisition of training dataset. The experimental results demonstrate the efficacy of the proposed MSVAE-CNN based pose estimation method.

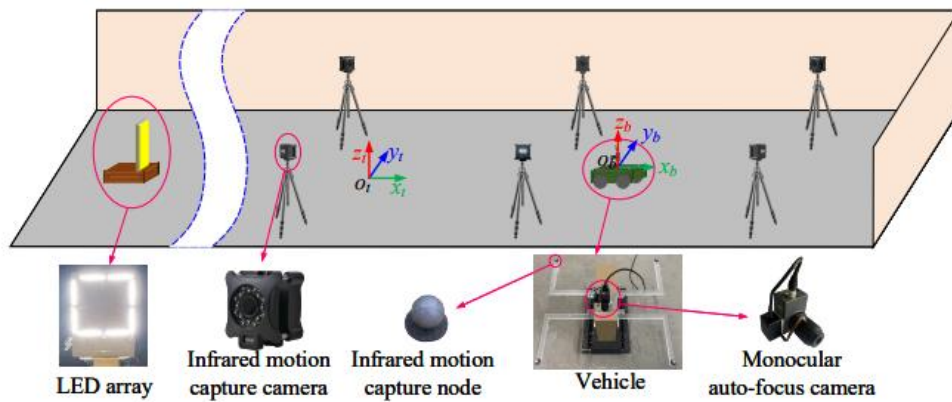


Fig. 5. Deployments of the dataset acquisition system in our experiments.

How NOKOV Contributes to the Research

In this work, the NOKOV Motion Capture System is utilized to obtain the 6-DoF pose of a vehicle (i.e., labels in the dataset), and a monocular monochrome auto-focus camera is attached to the vehicle and employed to capture the images of the artificial feature object (i.e., input images in the dataset). A visible-light LED array is used as the artificial feature object in our experiments to ensure the feasibility of the proposed pose estimation method in the underground tunnelling environment.

References Wu H, Liu S, Cheng C, et al. Multiscale Variational Autoencoder Aided Convolutional Neural Network for Pose Estimation of Tunneling Machine Using a Single Monocular Image[J]. IEEE Transactions on Industrial Informatics, 2021, 18(8): 5161-5170. Link <https://ieeexplore.ieee.org/abstract/document/9591347/>

Human Control Intent Inference Using ESNs and Input-Tracking Based Inverse Model Predictive Control

Keywords

Movement intent inference, Model predictive control, Echo state Networks

Abstract

Acquiring human motor control strategies or intents is helpful for clinical research, wearable robotic device design and human-robot cooperation control. The state-of-art method is to construct an optimal control framework which is capable to predict the target motion and take the cost function as the potential control intents. Aimed to solve this problem, an echo state networks based state space model (SSM) extraction method and input-tracking inverse MPC algorithm are proposed in this paper. By applying Taylor expansion around an operating point, it is convenient to acquire the SSM via the SSM extraction method and more detailed information about human musculoskeletal system is preserved. Setting the target of the upper level optimization as input-tracking is more rational than conventional output-tracking structure, given the consideration that human motion control is a multiple-solution problem. The effectiveness of the proposed method is verified in both simulation and real-world experiments.

How NOKOV Contributes to the Research

It is important to acquire the accurate inputs and outputs data for ESNs training. Normally, the proper inputs of a human movement control are muscle activations, muscle tendon force or joint torques. The open source program of OpenSim provides an alternative indirect approach to obtain these inputs. In order to calculate the joint torques or muscle tendon forces, kinematics data and, if available, the external forces are required. These required inputs can be easily accessed via motion capture system (Nokov, Duliang Technology Co. Ltd., CHN) and 6-axis force platform (BMS400600-1K, AMTI, USA). In this paper, ankle, knee, hip and lumbar joint torques are obtained through OpenSim inverse dynamic function and calibrated by Residual Reduction Algorithm function.

References Gong P, Pang M, Xiang K, et al. Human Control Intent Inference Using ESNs and Input-Tracking Based Inverse Model Predictive Control[C]//Intelligent Robotics and Applications: 14th International Conference, ICIRA 2021, Yantai, China, October 22–25, 2021, Proceedings, Part I 14. Springer International Publishing, 2021: 642–653.

Link https://link.springer.53yu.com/chapter/10.1007/978-3-030-89095-7_61

Predictive Simulation of Human Body Bending Motion Based on Inverse Optimization

Abstract

Low-back pain (LBP) is the leading cause of disability in the world. Low waist exoskeleton devices have formed a new strategy to reduce the possibility of the LBP. Although most of the low waist exoskeletons have been developed, the candidate design of the device must be built and the control parameters must be manually adjusted based on trial-and-error, which may be a waste of human and material resources. Recently, human-in-the-loop approaches has been used to optimize motion tasks. This method is very promising in finding the best control parameters, and it is only used in relatively few parameters. When complex parameters need to be optimized, it may be difficult for the human-in-the-loop approaches to converge in a reasonable time. This makes the development of low waist exoskeleton device still challenging, in part because the current practice of establishing and testing cycle is time-consuming and laborious. To deal with the abovementioned challenge, this paper uses predictive simulation technology and OpenSim to study the control of human bending movement. Most of the existing researches give the weight of objective function in the form of manual adjustment, which is not scientific. Herein based on the idea of inverse optimization, by constructing a bi-level optimization framework, the problem of determining the cost function is transformed into a bi-level optimization problem. The experimental results show that the predictive simulation established in this thesis can reproduce the human autonomous bend motion. Thus, the proposed motion predictive simulation framework based on inverse optimization is feasible and effective, which can provide the theoretical basis and practical reference for realizing the predictive simulation of other complex human motions.

How NOKOV Contributes to the Research

A total of six healthy male subjects (age= 23±3 years, mass = 61.25±6.25kg, height = 1.715±0.065m) voluntarily participated in and completed the bending experiment. This experiment contains a marker set of 39 Marker points, which is based on the detailed description of the marker set manual provided on the OpenSim official website. Placing markers on the corresponding key anatomical positions of the body. The overall process of the experiment is shown in Figure 2. Kinematics information is recorded through a passive optical motion capture system (Nokov, Beijing Metrology Technology Co., Ltd.). The force and moment information of the feet in three directions during bending movement was recorded by using the 6-dimensional force measuring platform (AMTI Company).

References Ba X, Zhou Y, Pang M, et al. Predictive Simulation of Human Body Bending Motion Based on Inverse Optimization[C]//2021 27th International Conference on Mechatronics and Machine Vision in Practice (M2VIP). IEEE, 2021: 109-114.

Link <https://ieeexplore.ieee.org/abstract/document/9665102/>

Modeling the Soft Continuum Robot Based on Large Deflection Theorem

Keywords

Large deflection theory, soft robot, pneumatic muscle, modeling

Abstract

How to model the dynamics and represent the behavior of soft continuum robot accurately and efficiently has always been a challenging problem. In this paper, a new soft continuum robot dynamic model modeling idea is proposed, that is, the pneumatic soft continuum robot is regarded as a variable-stiffness beam structure when bending. The stiffness of the variable-stiffness beam structure is related to the input air pressure, and a large deflection theorem is employed for force analysis of the variable-stiffness beam. Finally, the soft continuous robot pressure model is used to solve the deflection curve equation. In the experiment, the bending posture of the soft continuum robot obtained by the visual capture system was compared with the simulation results, and the average error of the model was 8.5%

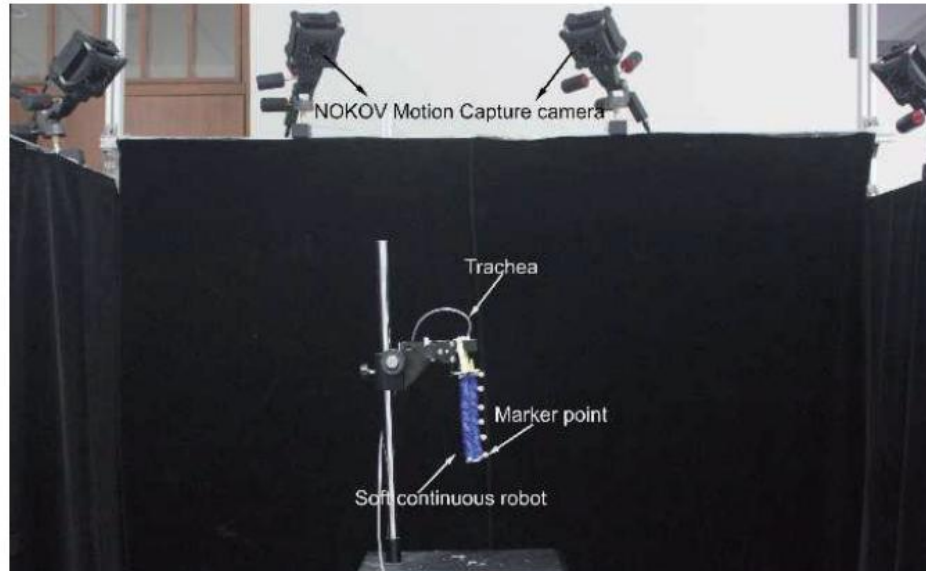


Fig. 5. Visual detection experiment platform

How NOKOV Contributes to the Research

Fig. 5 shows the soft continuum robot pneumatic platform and motion catching system. The kinematic model established in this paper describes the overall motion posture of the soft continuum robot at a given air pressure. Therefore, as shown in Fig. 5, the soft continuum robot is vertically installed on the fixture. The soft continuum robot is within the range of the visual capture system, and 6 reflective points (Marker point) are evenly attached on the side of the soft continuum robot. When the soft continuum robot is in its motion, the infrared camera (NOKOV Motion Capture System) can capture the relative position of the 6 points on the soft continuum robot in space, and later take out the 6 spatial positions and connect them with continuous smooth curves.

References Wang M, Wang W, Qiu Z, et al. Modeling the Soft Continuum Robot Based on Large Deflection Theorem[C]//2021 IEEE International Conference on Robotics and Biomimetics (ROBIO). IEEE, 2021: 209-214.

Link <https://dl.acm.org/doi/abs/10.1109/ROBIO54168.2021.9739488>

A Moving Object Detection Method for Mobile Robots Using Feature Point Sticky Tree

Keywords

moving object detection, feature points, sticky tree, moving camera

Abstract

Moving object detection is a common task for mobile robots when working in dynamic environments. Due to the nonstatic nature of the mobile robots themselves, we need to conduct the motion detection work under the moving camera assumption. Feature-point-based object detection is an important research direction to solve this problem. The distances and motion conditions of the feature points in both 2D image space and 3D physical space are critical criteria to cluster the points to objects. However, previous work never fully considered all the criteria and explored their complex relationship with object segmentation outputs. As a result, accurate segmentation can not be achieved when the differentiation condition only presents in part of the criteria space. Therefore, we proposed a novel feature point stickiness method to include all the above considerations. A fuzzy neural network is designed to assess the stickiness between any feature points, e.g., the possibility of being on the same object. The network is trained with small data set to capture the complex nonlinear relationship between the point distance and motions and the stickiness. In addition, a sticky tree growth algorithm is used to connect the feature points according to their stickiness distributions. The formation of the trees autonomously completes the object segmentation task. The method is testified in various experimental situations, including some challenging scenarios. It improved the object recall rate and precision rate to a large extent compared to previous works and showed high potential in providing precise trajectories of moving objects in the robot's workspace.

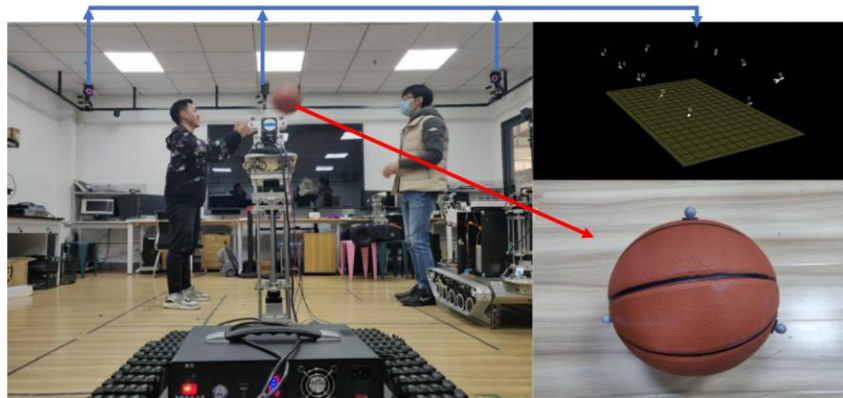


Figure 13: 3D motion tracking experiment

How NOKOV Contributes to the Research

Though it is not the focus of this paper, we present the rough estimation of the 3D trajectory measurement capability of the proposed 3D moving object detection algorithm in this subsection. The experimental system is set up in the laboratory, as shown in Fig.13. The detection object is chosen as the ball since it is the most manageable object to be measured by the motion capture system and with a relatively long trajectory in a short time. We attached markers on the ball and used a motion capture system Nokov Mars4H to measure its positions in real-time. The accuracy of the motion capture system is ± 0.1 mm, and its measurement is considered the ground truth for the ball's trajectory.

References Luo J, Zhou S, Zhu J, et al. A Moving Object Detection Method for Mobile Robots Using Feature Point Sticky Tree[J]. Available at SSRN 3994131.
Link https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3994131

The Application of Human–Computer Interaction Technology Fused With Artificial Intelligence in Sports Moving Target Detection Education for College Athlete

Keywords

artificial intelligence, framework of coaching system, motion capture technology, human-computer interaction, sports training

Abstract

The purposes are to digitalize and intellectualize current professional sports training and enrich the application scenarios of motion capture technology of moving targets based on artificial intelligence (AI) and human–computer interaction (HCI) in sports training. From an educational psychology perspective, sport techniques are a cognitive ability of sports, and a tacit knowledge. However, sports technology, language, image, and other methods play an auxiliary role in sports training. Here, a General Framework of Knowledge-Based Coaching System (KBCS) is proposed using the HCI technology and sports knowledge to accomplish autonomous and intelligent sports training. Then, the KBCS is applied to table tennis training. The athletic performance is evaluated quantitatively through the calculation of the sports features, motion recognition, and the hitting stage division in table tennis. Results demonstrate that the speed calculated by the position after mosaicking has better continuity after the initial frame of the unmarked segment is compared with the end frame of the market segment. The typical serve and return trajectories in three serving modes of slight-spin, top-spin, and back-spin, as well as the trajectories of common services and return errors, are obtained through the judgment of the serving and receiving of table tennis. Comparison results prove that the serving accuracy of slight-spin and back-spin is better than that of top-spin, and a lower serve speed has higher accuracy. Experimental results show that the level distribution of the three participants calculated by the system is consistent with the actual situation in terms of the quality of the ball returned and the standard of the motion, proving that the proposed KBCS and algorithm are useful in a small sample, thereby further improving the accuracy of pose restoration of athletes in

sports training.

How NOKOV Contributes to the Research

The running environment of the inertial capture system and optical capture system is the Window10 system. During the experiment, the computer CPU model is Intel i7-9700k, and the memory is 8G. The optical motion capture system is the Nokov, which can capture the key points of the trajectory at a higher frame rate. At the same time, the system has good stability and is suitable for the experimental requirements. The Legacy system is selected for inertial motion capture, which can output the posture of the main human bones more accurately. Meanwhile, the sensor of the Legacy system has a long life and can meet the application for a long time.

References Liu J, Wang L, Zhou H. The application of human-computer interaction technology fused with artificial intelligence in sports moving target detection education for college athlete[J]. Frontiers in Psychology, 2021, 12: 677590.

Link <https://www.frontiersin.org/articles/10.3389/fpsyg.2021.677590/full>

Simulation of Human Posture Sway Based On Reference Control

Keywords

OpenSim, Musculoskeletal model, Reference control, Reference point trajectory

Abstract

Investigating human movement mechanism is meaningful since it can provide clues to controller designs of bionic robots. However, the complicated interactions between human neural and musculoskeletal cause this issue to be challenging. Fortunately, the reference control has recently shown its potential and reliability to deal with this issue. In order to evaluate the roles of this control mechanism in the simulation of human upright swing, this paper firstly builds a simulation platform of musculoskeletal model augmented with reference controller via leveraging OpenSim. Then, the muscle-driven simulations of anti-disturbance and intentional swing in sagittal plane are conducted by using the established platform. During this process, the parameters contained in the reference controller and the reference point trajectory in swing are optimized using the `fminsearch` optimizer. Lastly, the established platform is verified by computer simulation and experimental test. The results between comparison of the simulation results and experimental data confirm that the established method can simulate the actual human upright standing motion.

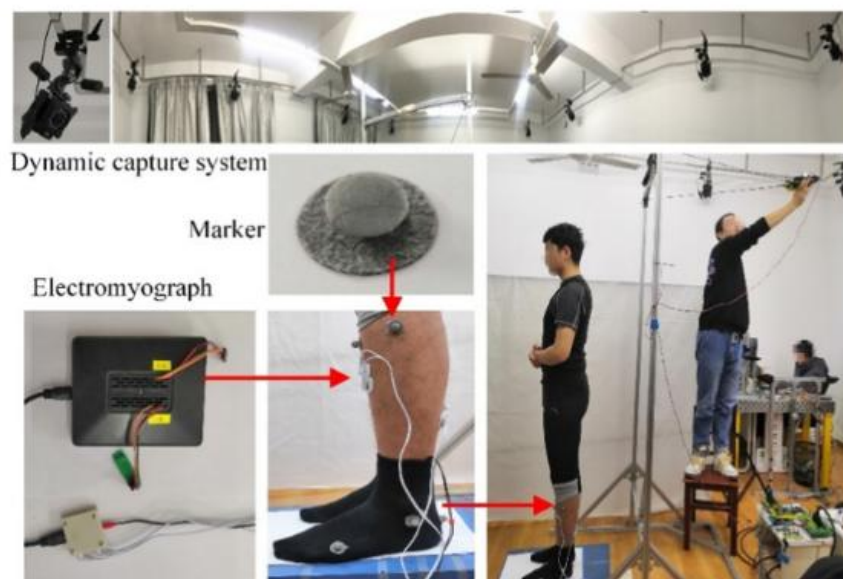


Fig. 3. Human upright anti-disturbance experiment

How NOKOV Contributes to the Research

The developed method is verified by comparing the simulation results obtained by our designed simulation platform with the experimental results. Prior to conducting experimental tests, a healthy volunteer without neuromusculoskeletal problems participate in two motor tasks in this study. The kinematics of the recruited volunteer are recorded in a Nokov optical three-dimensional motion capture system (Beijing NOKOV Science & Technology, 50 Hz) with eight high-definition cameras. Six reflective markers are placed on the medial and lateral shank and toes of the right leg to calculate the angle of the ankle. The muscle EMG signals of SOL and TA are measured by electromyography (ELONXI EMG 100-Ch-Y-RA). In the preparation of the experiment, the leg hairs of the subjects are removed and the skin is cleaned with alcohol. After that, EMG activity levels during maximal isometric voluntary contractions (MVC) are measured in standard muscle testing position for the subject before the experiment and used to normalize the EMG signals. The sampling frequencies of Nokov and EMG are 50 Hz and 1000 Hz, respectively. Moreover, we implement the real-time simulator to achieve the synchronous reception of experimental data.

References Zeng Q, Pang M, Tang B, et al. Simulation of Human Posture Sway Based on Reference Control[C]//Intelligent Robotics and Applications: 14th International Conference, ICIRA 2021, Yantai, China, October 22–25, 2021, Proceedings, Part I 14. Springer International Publishing, 2021: 631-641.

Link https://link.springer.53yu.com/chapter/10.1007/978-3-030-89095-7_60

Mechanical Metamaterials with Discontinuous and Tension/Compression-Dependent Positive/Negative Poisson's Ratio

Keywords

Additive manufacturing, auxetic materials, deformations, discontinuous Poisson's ratios, mechanical metamaterials

Abstract

Ordinary materials have positive Poisson's ratios (PRs), and the currently known mechanical metamaterials possess negative PRs both under tension and compression. Herein, two types of novel cellular structure units with tension/compression-dependent positive/negative PRs are demonstrated. For type I structure, the unit exhibits a positive PR during tension and a negative PR during compression, while for type II structure, the unit shows a negative PR during tension and a positive PR during compression. The deformation behaviors of the new metamaterials are first predicted by a simple analytical model and finite-element method calculations, and the predicted results are then confirmed experimentally. Numerical and experimental studies show that for the plane angle $\alpha < 90^\circ$, the unit of the new metamaterial switches its PR under compression and exhibits a discontinuous change in the PR. The deformation law does not depend on either the Young's modulus or the hardness of the material. The geometrically determined deformation behaviors of the proposed metamaterials are observed in 2D and 3D cellular structures and are proven to be robust through numerical and experimental studies. The proposed mechanical metamaterials can be applied as new deformation-protected materials in energy-absorbing devices, soft robotics, and tissue engineering scaffolds and as thermal deformation materials.

How NOKOV Contributes to the Research

Many half spheres used as tracing points were glued onto a sample to measure its deformation. Then, six cameras of the motion capture system emitted infrared radiation that was reflected by the half spheres and received by the cameras. Then, the positions of the sphere centers were calculated by the system. The radius of the half spheres was 2.5 mm. The centers of the half spheres were aligned with that of the connecting disks (depicted in Figure 3a). A photograph of the experimental setup is shown in Supporting Information.

References Yang N, Deng Y, Zhao S, et al. Mechanical Metamaterials with Discontinuous and Tension/Compression-Dependent Positive/Negative Poisson's Ratio[J]. Advanced Engineering Materials, 2022, 24(3): 2100787.
Link <https://onlinelibrary.wiley.com/doi/abs/10.1002/adem.202100787>

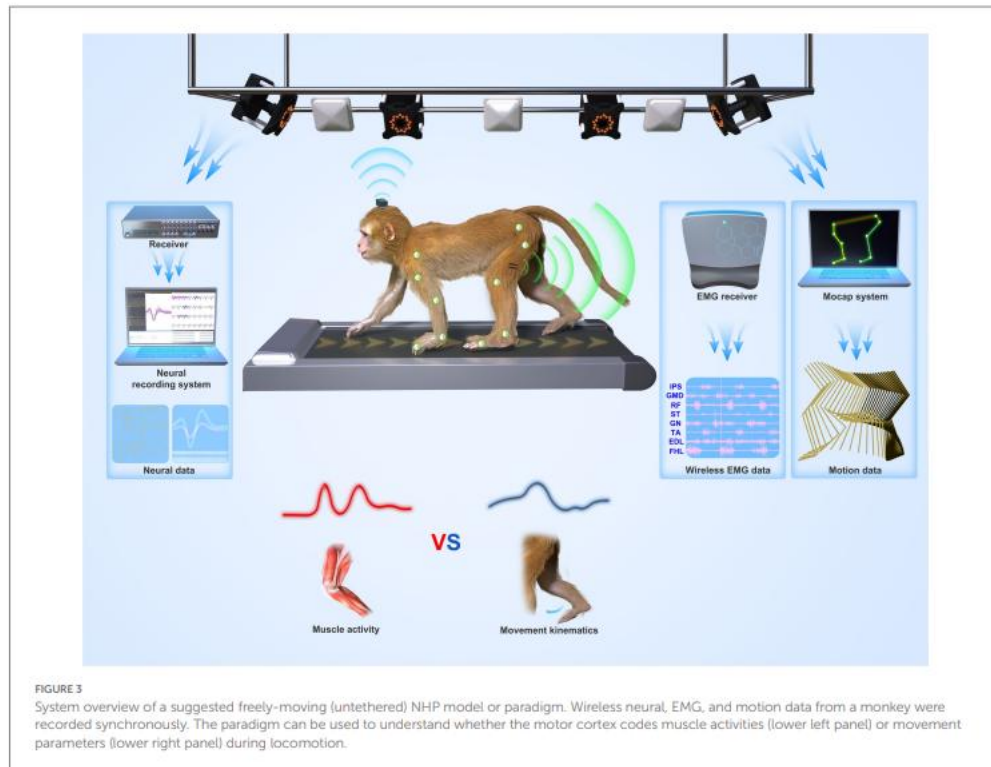
Non-human primate models and systems for gait and neurophysiological analysis

Keywords

non-human primate, gait, BCI, neurophysiological analysis, motor cortex

Abstract

Brain–computer interfaces (BCIs) have garnered extensive interest and become a groundbreaking technology to restore movement, tactile sense, and communication in patients. Prior to their use in human subjects, clinical BCIs require rigorous validation and verification (V&V). Non-human primates (NHPs) are often considered the ultimate and widely used animal model for neuroscience studies, including BCIs V&V, due to their proximity to humans. This literature review summarizes 94 NHP gait analysis studies until 1 June, 2022, including seven BCI-oriented studies. Due to technological limitations, most of these studies used wired neural recordings to access electrophysiological data. However, wireless neural recording systems for NHPs enabled neuroscience research in humans, and many on NHP locomotion, while posing numerous technical challenges, such as signal quality, data throughput, working distance, size, and power constraint, that have yet to be overcome. Besides neurological data, motion capture (MoCap) systems are usually required in BCI and gait studies to capture locomotion kinematics. However, current studies have exclusively relied on image processing-based MoCap systems, which have insufficient accuracy (error: $\geq 4^\circ$ and 9 mm). While the role of the motor cortex during locomotion is still unclear and worth further exploration, future BCI and gait studies require simultaneous, high-speed, accurate neurophysiological, and movement measures. Therefore, the infrared MoCap system which has high accuracy and speed, together with a high spatiotemporal resolution neural recording system, may expand the scope and improve the quality of the motor and neurophysiological analysis in NHPs.



How NOKOV Contributes to the Research

Recently, our group is setting up an OS-based freely-moving NHP model (Figure 3) with a high accuracy and fast speed MoCap system (Mars 4H, Nokov Science and Technology Co., Ltd., Beijing, China) along with a high spatiotemporal resolution neural recording system and wireless EMG system, to increase the accuracy of kinematics and thus expand motor and neurophysiological analysis in NHPs

References Aflalo, T., Kellis, S., Klaes, C., Lee, B., Shi, Y., Pejsa, K., et al. (2015). Decoding motor imagery from the posterior parietal cortex of a tetraplegic human. *Science* 348, 906–910. doi: 10.1126/science.aaa5417.

Link <https://www.frontiersin.org/articles/10.3389/fnins.2023.1141567/full>

Arbitrary Spatial Trajectory Reconstruction based on A Single Inertial Sensor

Abstract

Compared with visible, infrared rays, and ultrasonic positioning technologies, the signal acquisition of portable inertial sensors is not affected by the external environment, such as light and occlusion. Therefore, motion tracking based on inertial signals is a promising complement. However, accurate trajectory reconstruction based on inertial sensors is a great challenge due to intrinsic and measurement errors, especially the drift error that exacerbates the accumulative error in trajectory calculation. To address this challenge, we propose a new trajectory reconstruction method, geometric dynamic segmental reconstruction (GDSR), where we treat the movement trajectory as a combination of basic trajectories. To this end, we design a temporal and spatial interaction segmentation approach to decompose the trajectory into basic segments by combining the dynamic feature of inertial measurement unit (IMU) signals with the spatial morphological feature of motion. Accordingly, we design a geometrical model library with undetermined parameters to match these segments. For precise parameter prediction, we propose an extra-supervised learning method that integrates different prediction tasks into one framework, which can not only expand training samples, but also enable different subtasks to compete with each other, thus improving the parameter prediction accuracy of each subtask, thereby accurately approximating the trajectory segments. To quantify the trajectory reconstruction accuracy, we propose the Fréchet spline sliding error (FSSE) and length error ratio (LER) to evaluate curve similarity. The range of FSSE is $[0, 2]$, where 0 means that the two curves have the same shape. The range of LER is $[1, +\infty]$, where 1 means that the two curves have the same length. We test different IMUs in two experimental scenarios and one public available dataset. In all three tests, the FSSE of the GDSR method is less than 0.09, and the LER is less than 1.31, which is significantly better than all the comparison methods.



Fig. 11. Scene of motion collection within the optical camera facility (the left panel), and the racket attached with an inertial sensor and five optical markers (the right panel).

How NOKOV Contributes to the Research

The inertial sensor can measure the specific force, which is the vector difference between the actual body acceleration and the gravity vector and 3-D angular velocity data. The optical facility includes eight high-speed cameras, which can synchronously record the moving trajectory from their own view. When the object attached to the reflective markers moves within the view field of the cameras. The accuracy of the stereophotogrammetric system can be even higher than centimeters, up to the submillimeter, as described in [49]. It should be noted that the layout of reflective balls is asymmetric, which ensures that the distance between any two balls is different. So, the optical camera system can still record the racket movement by capturing at least any two balls. The collection of a concrete object movement is shown in Fig. 11. To keep time alignment, we uniformly adjust the sampling frequency of the optical and inertial sensors to 200 Hz. Therefore, the data collection of the two sensors can be synchronized.

References C.-S. Jao and A. M. Shkel, "A reconstruction filter for saturated accelerometer signals due to insufficient FSR in foot-mounted inertial navigation system," IEEE Sensors J., vol. 22, no. 1, pp. 695–706, Jan. 2022.

Link <https://ieeexplore.ieee.org/document/10080916>

Construction of a niche-specific spinal white matter-like tissue to promote directional axon regeneration and myelination for rat spinal cord injury repair

Keywords

Extracellular matrix; Oligodendroglial lineage cells; White matter-like tissue; Directional axon regeneration; Spinal cord injury

Abstract

Directional axon regeneration and remyelination are crucial for repair of spinal cord injury (SCI), but existing treatments do not effectively promote those processes. Here, we propose a strategy for construction of niche-specific spinal white matter-like tissue (WMLT) using decellularized optic nerve (DON) loaded with neurotrophin-3 (NT-3)-overexpressing oligodendrocyte precursor cells. A rat model with a white matter defect in the dorsal spinal cord of the T10 segment was used. The WMLT transplantation group showed significant improvement in coordinated motor functions compared with the control groups. WMLT transplants integrated well with host spinal cord white matter, effectively addressing several barriers to directional axonal regeneration and myelination during SCI repair. In WMLT, laminin was found to promote development of oligodendroglial lineage (OL) cells by binding to laminin receptors. Interestingly, laminin could also guide linear axon regeneration via interactions with specific integrins on the axon surface. The WMLT developed here utilizes the unique microstructure and bioactive matrix of DON to create a niche rich in laminin, NT-3 and OL cells to achieve significant structural repair of SCI. Our protocol can help to promote research on repair of nerve injury and construction of neural tissues and organoids that form specific cell niches.

How NOKOV Contributes to the Research

Future behavioral testing in animals with incomplete SCI would benefit from methods like the Nokov (metric) optical 3D motion capture system that provides a more detailed evaluation of paralyzed limb support and coordination of movement [39].

References J. Koffler, W. Zhu, X. Qu, O. Platoshyn, J.N. Dulin, J. Brock, L. Graham, P. Lu, J. Sakamoto, M. Marsala, S. Chen, M.H. Tuszynski, Biomimetic 3D-printed scaffolds for spinal cord injury repair, Nat. Med. 25 (2) (2019) 263–269.

Link <https://www.sciencedirect.com/science/article/pii/S2452199X21004564>

A non-contact system for intraoperative quantitative assessment of bradykinesia in deep brain stimulation surgery

Keywords

Deep brain stimulation; Intraoperative test; Quantitative assessment; Non-contact measurement; Bradykinesia

Abstract

Background and Objective: [Deep brain stimulation](#) (DBS) is an effective [treatment](#) for a number of [neurological diseases](#), especially for the advanced stage of Parkinson's disease (PD). Objective assessment of patients' motor symptoms is crucial for accurate electrode targeting and treatment. Existing approaches suffer from subjective variability or interference with voluntary motion. This work is aimed to establish an objective assessment system to quantify bradykinesia in DBS surgery. *Methods:* Based on the analysis of the requirements for intraoperative assessment, we developed a system with non-contact measurement, online movement feature extraction, and interactive data analysis and visualization. An optical sensor, Leap Motion Controller (LMC), was taken to detect hand movement in three clinical tasks. A [graphic user interface](#) was designed to process, compare and visualize the collected data and assessment results online. Quantified movement features include amplitude, frequency, velocity, their decrement and variability, etc. Technical validation of the system was performed with a motion capture system (Mocap), with respect to data-level and feature-level accuracy and reliability. Clinical validation was conducted with 20 PD patients for intraoperative assessments in DBS surgery. Treatment responses with respect to the bradykinesia movement features were analyzed. Single case analysis and group statistical analysis were performed to examine the differences between preoperative and intraoperative performance, and the correlation between the clinical ratings and the quantified assessment was analyzed. *Results:* For the movements measured by LMC and Mocap, the average Pearson's correlation coefficient was 0.986, and the mean amplitude difference was 2.11 mm. No significant difference was found for all movement features quantified by LMC and Mocap. For the clinical tests, key

movement features showed significant differences between the preoperative baseline and intraoperative performance when the brain stimulation was ON. The assessment results were significantly correlated with the MDS-UPDRS clinical ratings. *Conclusions:* The proposed non-contact system has established itself as an objective intraoperative assessment, analysis, and visualization tool for DBS treatment of Parkinson's disease.

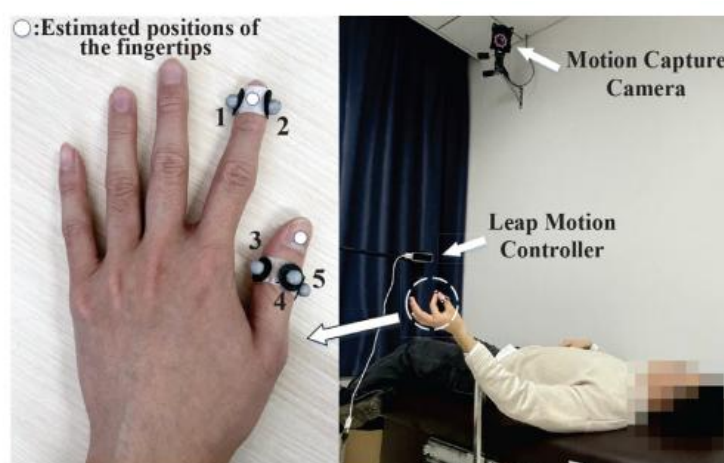


Fig. 4. Technical validation with a motion capture system.

How NOKOV Contributes to the Research

Experiments were conducted in a laboratory environment equipped with a Mocap system (NOKOV, Beijing Nokov Science and Technology Co., Ltd, China). The Mocap system has eight cameras to detect the markers from different views. Subjects were asked to lie on a ward bed to complete the task, which was consistent with the intraoperative assessment.

References E. Tolosa, A. Garrido, S.W. Scholz, W. Poewe, Challenges in the diagnosis of Parkinson's disease, *Lancet Neurol.* 20 (5) (2021) 385–397.

Link <https://www.sciencedirect.com/science/article/abs/pii/S016926072200387X>

Synchronize Dual Hands for Physics-Based Dexterous Guitar Playing

Keywords

character animation, hand animation, physics-based control, dexterous control, motion synthesis, multi-agent reinforcement learning, cooperative learning, motion capture dataset

Abstract

We present a novel approach to synthesize dexterous motions for physically simulated hands in tasks that require coordination between the control of two hands with high temporal precision. Instead of directly learning a joint policy to control two hands, our approach performs bimanual control through cooperative learning where each hand is treated as an individual agent. The individual policies for each hand are first trained separately, and then synchronized through latent space manipulation in a centralized environment to serve as a joint policy for two-hand control. By doing so, we avoid directly performing policy learning in the joint state-action space of two hands with higher dimensions, greatly improving the overall training efficiency. We demonstrate the effectiveness of our proposed approach in the challenging guitar-playing task. The virtual guitarist trained by our approach can synthesize motions from unstructured reference data of general guitar playing practice motions, and accurately play diverse rhythms with complex chord pressing and string picking patterns based on the input guitar tabs that do not exist in the references. Along with this paper, we provide the motion capture data that we collected as the reference for policy training. Code is available at: <https://pei-xu.github.io/guitar>.



How NOKOV Contributes to the Research

We took the motion capture solutions provided by NOKOV and employed 18 infrared cameras for motion capture, 7 of which have a resolution of 9MP and 11 have a resolution of 4MP. We placed 19 markers on each hand of the subject, and 4 markers on the guitar for localization. All data were captured at 120FPS. The camera deployment and marker placement are exhibited in Fig. S1.

References Xu P, Wang R. Synchronize Dual Hands for Physics-Based Dexterous Guitar Playing[J]. arXiv preprint arXiv:2409.16629, 2024.
Link <https://arxiv.org/abs/2409.16629>

GaitFormer: Leveraging dual-stream spatial–temporal Vision Transformer via a single low-cost RGB camera for clinical gait analysis

Keywords

Gait analysis; Human pose estimation; Healthcare; Vision Transformer; Single RGB camera

Abstract

Gait analysis is an essential technique in treating patients with lower limb dysfunctions. Traditional methods often rely on expensive and complex equipment, such as wearable body sensors and a multi-camera with marker tracking system. Aiming for a more cost-effective yet accurate alternative, this paper introduces GaitFormer, a novel approach that leverages Vision Transformer (ViT) for gait analysis using minimal, non invasive equipment, i.e. a single low-cost RGB camera. Initially, a unique dataset using a multi-camera system with marker tracking, comprising 6 walking patterns gathered from 80 volunteers is developed. The ViT-based GaitFormer is then proposed to automatically recognize human walking patterns through a single RGB camera. GaitFormer comprises hybrid networks for each step, including: (i) a cascaded convolutional 2D human key points estimation network; (ii) a ViT-based dual-stream spatial–temporal network extending the information of human key points into 3D; (iii) leveraging specific lower limb key joints' angle features for clinical gait analysis, capturing the geometric, kinematic, and physical attributes of human motion; (iv) employing a pure self-attention-based classification network to recognize clinical human walking patterns. The experiments are designed to comprehensively validate each step against various related baseline methods and multi-camera tracking system, with results demonstrating the promising performance of GaitFormer as an affordable, precise, and integrated solution. To the best of our knowledge, GaitFormer is the first hybrid CNN- and ViT-based end-to-end solution via low-cost device for clinically valuable gait analysis.

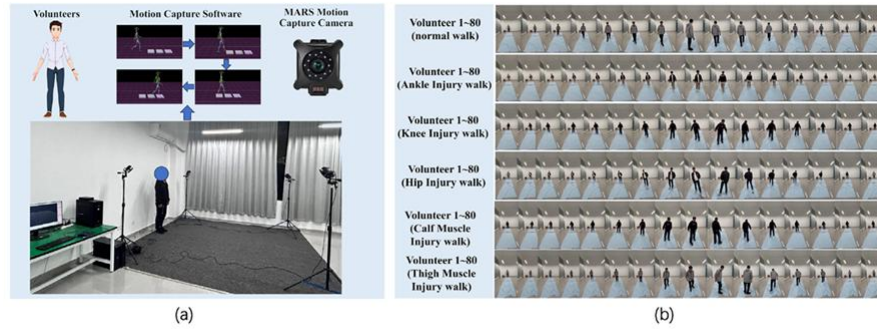


Fig. 2. The illustration of dataset development hardware and example frames of dataset with multiple volunteers and multiple walking patterns.

How NOKOV Contributes to the Research

Additionally, a NOKOV metric optical 3D motion capture system was employed to gather detailed gait data from an individual, serving as a benchmark for our analysis,¹ which is briefly illustrated in Fig. 2(a).

References Jiabao Li, Ziyang Wang, Chengjun Wang, Wenhong Su, GaitFormer: Leveraging dual-stream spatial-temporal Vision Transformer via a single low-cost RGB camera for clinical gait analysis, Knowledge-Based Systems, Volume 295, 2024,111810, ISSN 0950-7051, <https://doi.org/10.1016/j.knosys.2024.111810>.

Link <https://www.sciencedirect.com/science/article/abs/pii/S0950705124004441>

A Binocular-Camera-Assisted Sensor-to-Segment Alignment Method for Inertial Sensor-Based Human Gait Analysis

Keywords

Binocular camera, inertial measurement unit, joint angle estimation, OpenPose, sensor-to-segment alignment.

Abstract

Lower limb joint angles are powerful clinical indicators for mobility assessment and gait rehabilitation training. Inertial motion capture (IMC), which is both accurate and convenient, is becoming an ideal method to estimate joint angles. Considering that the corresponding segment and sensor frame are usually misaligned, an alignment procedure is required to determine the orientation offset between the segment and the sensor frame. This article presents a binocular-camera-assisted alignment procedure to align the frames of segment and inertial sensor. This procedure uses a binocular camera to acquire lower limb segment's initial orientations with image-based human pose algorithm and then estimates the joint angle with kinematic constraint. The alignment procedure requires only sitting posture and does not need any functional movement. Ten healthy participants were recruited for the validation experiments, which included stand-up, turn-around, and walking. The accuracy and effectiveness of our alignment procedure were validated against the optical motion capture (OMC). The mean root mean square errors (RMSEs) of hip, knee, and ankle angles were 1.6°, 2.9°, and 3.9°, respectively. The result demonstrates that our procedure performs well on joint angle estimation. The proposed alignment procedure did not require any complicated functional movement, but only sitting posture, which could enable more practical joint angle estimation for patient subjects in clinical applications.



Fig. 5. Inertial sensors and OMC markers on human lower limb.

How NOKOV Contributes to the Research

An OMC system (Mars, Nokov3DOMCP, Beijing, China) with eight infrared cameras was used as the "Gold Standard" for validation experiments.

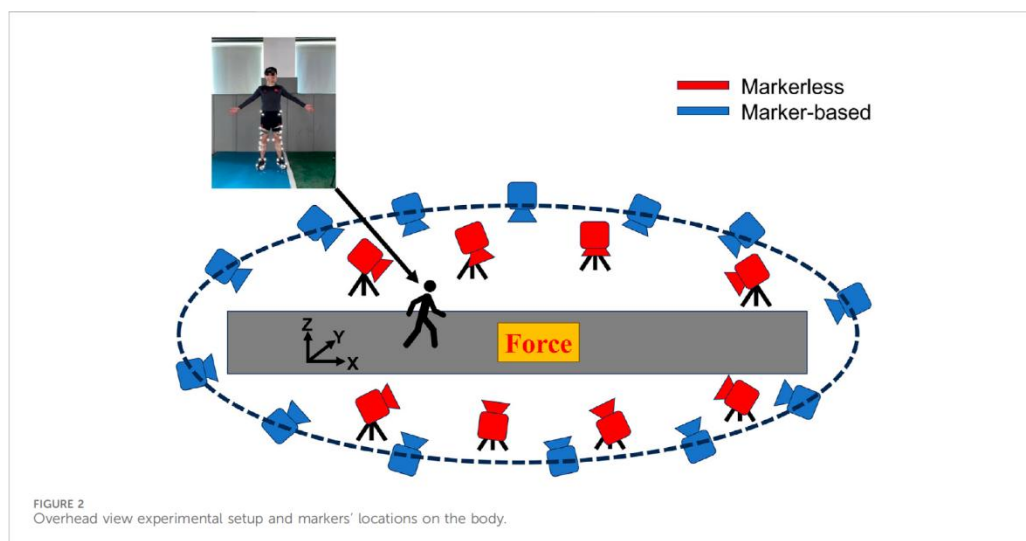
References S. Cai, M. Shao, M. Du, G. Bao and B. Fan, "A Binocular-Camera-Assisted Sensor-to-Segment Alignment Method for Inertial Sensor-Based Human Gait Analysis," in IEEE Sensors Journal, vol. 23, no. 3, pp. 2663-2671, 1 Feb.1, 2023, doi: 10.1109/JSEN.2022.3229384.

Link <https://ieeexplore.ieee.org/document/9994701>

Comparison of kinematics and joint moments calculations for lower limbs during gait using markerless and marker-based motion capture

Abstract

Lower limb joint angles are powerful clinical indicators for mobility assessment and gait rehabilitation training. Inertial motion capture (IMC), which is both accurate and convenient, is becoming an ideal method to estimate joint angles. Considering that the corresponding segment and sensor frame are usually misaligned, an alignment procedure is required to determine the orientation offset between the segment and the sensor frame. This article presents a binocular-camera-assisted alignment procedure to align the frames of segment and inertial sensor. This procedure uses a binocular camera to acquire lower limb segment's initial orientations with image-based human pose algorithm and then estimates the joint angle with kinematic constraint. The alignment procedure requires only sitting posture and does not need any functional movement. Ten healthy participants were recruited for the validation experiments, which included stand-up, turn-around, and walking. The accuracy and effectiveness of our alignment procedure were validated against the optical motion capture (OMC). The mean root mean square errors (RMSEs) of hip, knee, and ankle angles were 1.6°, 2.9°, and 3.9°, respectively. The result demonstrates that our procedure performs well on joint angle estimation. The proposed alignment procedure did not require any complicated functional movement, but only sitting posture, which could enable more practical joint angle estimation for patient subjects in clinical applications.



How NOKOV Contributes to the Research

An MB motion capture system (thirteen Mars 4H (Nokov Motion Capture System, Beijing, China), 4.1-megapixel resolution cameras) and an MMC system (eight Z-CAM-E2 (Intotech, Beijing, China), 120HZ, 8-megapixel resolution) were used to capture the movement synchronously.

References Huang T, Ruan M, Huang S, Fan L and Wu X (2024), Comparison of kinematics and joint moments calculations for lower limbs during gait using markerless and marker-based motion capture. *Front. Bioeng. Biotechnol.* 12:1280363. doi: 10.3389/fbioe.2024.1280363

Link <https://www.frontiersin.org/journals/bioengineering-and-biotechnology/articles/10.3389/fbioe.2024.1280363/full>

A HUG taxonomy of humans with potential in human–robot hugs

Abstract

Humans can easily perform various types of hugs in human contact and affection experience. With the prevalence of robots in social applications, they would be expected to possess the capability of hugs as humans do. However, it is still not an easy task for robots, considering the complex force and spatial constraints of robot hugs. In this work, we propose the HUG taxonomy, which distinguishes between different hugging patterns based on human demonstrations and prior knowledge. In this taxonomy, hugs are arranged according to (1) hugging tightness, (2) hugging style, and (3) bilateral coordination, resulting in 16 different hug types. We then further study the hug type preference of humans in different scenarios and roles. Furthermore, we propose a rule-based classification system to validate the potential of this taxonomy in human–robot hugs based on a humanoid robot with an E-skin of contact sensation. The HUG taxonomy could provide human hugging behavior information in advance, facilitating the action control of humanoid robots. We believe the results of our work can benefit future studies on human–robot hugging interactions.

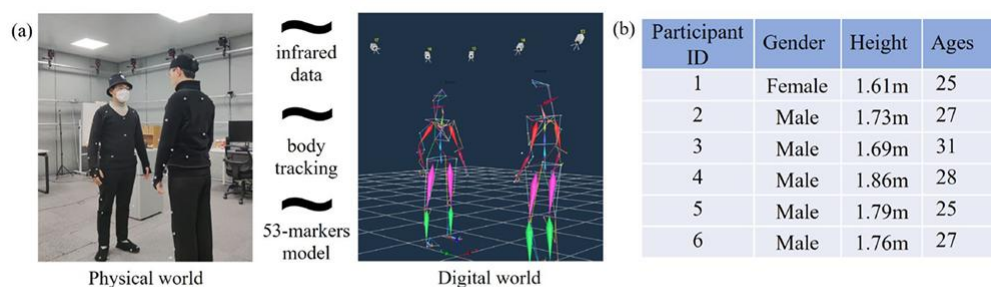


Figure 2. Motion capture experiments of the human demonstration. (a) Motion capture system. (b) Participant information.

How NOKOV Contributes to the Research

As shown in Fig. 2a, we utilize the motion capture system from NOKOV to collect data. It consists of 20 infrared cameras with 60fps, arranged in a square on the ceiling of a $6.5m \times 6.5m \times 2.5m$ quiet room, with five cameras on each side. Combined with the 53-marker human body detection and modeling algorithms, human demonstrations can be transformed into digital models, making physical properties more apparent.

References Yan, Z., Wang, Z., Ren, R. et al. A HUG taxonomy of humans with potential in human–robot hugs. *Sci Rep*14, 14212 (2024). <https://doi.org/10.1038/s41598-024-64825-8>

Link <https://www.nature.com/articles/s41598-024-64825-8>

Hybrid Nursing Robot Based on Humanoid Pick-Up Action: Safe Transfers in Living Environments for Lower Limb Disabilities

Keywords:

hybrid nursing robot; humanoid pick-up action; workspace; dynamical model; comfort analysis

Abstract

This research paper outlines the development of a modular and adjustable transfer care robot aimed at enhancing safe and comfortable transfers for individuals with lower limb disabilities. To design this robot, we utilized a 3D motion capture system to analyze the movements of a person assisting another person and to determine the necessary range of motion and workspace for the robot. Based on this analysis, we developed a 3-UPS + UPR parallel spreader to transfer the person receiving care. We also conducted kinematic and dynamic analyses of the parallel spreader to validate its operational space and to obtain the force change curve for the drive. To evaluate the robot's performance, we enlisted the help of ten volunteers with varying heights and weights. Our findings indicate that the pressure distribution during transfers remained largely consistent. Additionally, the surveys revealed that those receiving care perceived the robot as being capable of securely and comfortably transferring individuals between different assistive devices. This modular and adaptable transfer assistance robot presents a promising solution to the challenges encountered in caregiving.

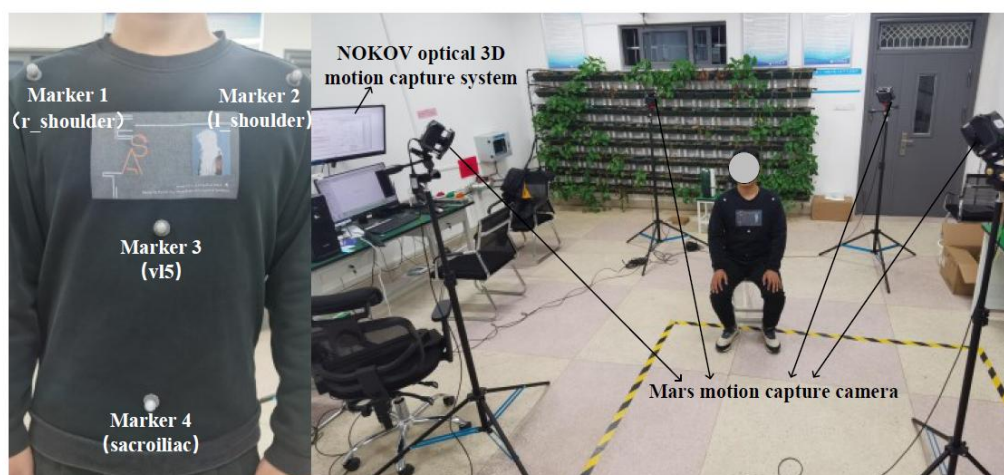


Figure 1. Test arrangement for the pick-up action.

How NOKOV Contributes to the Research

In this test, a NOKOV optical 3D motion capture system was used to capture the actions of the nursed person in the pick-up process [21,22]. According to the standard human skeleton model [23] given in the ISO/IEC19774-Human animation (H-Anim) standard, "r_shoulder", "l_shoulder", "vl5", and "sacroiliac" in the model were selected as the markers (Figure 1). As shown in the figure, a nursed person sat on a chair, nursing person picked up the nursed person and moved them

References Li, J.; Wang, C.; Deng, H. Hybrid Nursing Robot Based on Humanoid Pick-Up Action: Safe Transfers in Living Environments for Lower Limb Disabilities. *Actuators* 2023, 12, 438. <https://doi.org/10.3390/act12120438>
Link <https://www.mdpi.com/2076-0825/12/12/438>

A Modular Soft Pipe-Climbing Robot With High Maneuverability

Keywords:

Maneuverability, modular design, pipe climbing robot, soft robotics.

Abstract

Pipeline is widely deployed in infrastructures, which raises a great demand of pipe-climbing robots to provide pipe-inside inspecting and platform delivery. Soft pipe climbing robots (SPRs) are intrinsically flexible and able to operate within complex, narrow and curving pipelines more effectively than their rigid counterparts. In the state of the art, the existing SPRs have demonstrated their performances in fixed-diameter pipes that are placed in static fluid environments, which however does not show their full potential. In this article, we propose a modular design methodology for SPRs that are expected to operate inside complex pipelines. The robot's torso is composed of three PSAs in parallel, which can realize axial elongation as well as omnidirectional bending with pressurized air. The PSA-based tentacles will elongate after inflation to contact and hold the pipe wall using their silicone pads. Further more, the slim tentacles and compact torso results in quite small cross-sectional occupation ratio and subsequently low windward resistance from the liquid in the pipeline, which ensures the robot's high passivity in fluid. The proposed robot can adapt to and work in diverse pipelines with complex fluid conditions, such as horizontal and vertical straight, as well as curved pipes with varying diameters, with and without loads, in still and dynamic water flows. In addition, the robot, whose weight is 50 g, can achieve a fast speed of climbing (up to 15.0 mm/s in open air) and high load-carrying ability (up to 1500 g) due to the quick response of the tentacles and large elongation ratio of its body. The experimental results have proved the high maneuverability of the developed SPR.

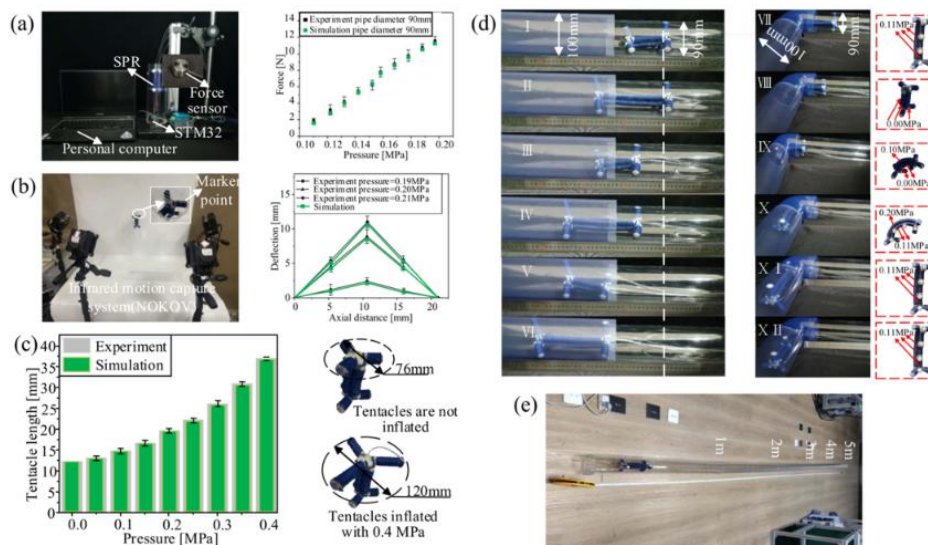


Fig. 3. Motion performance of SPR. (a) Test and simulation of tentacle force. (b) Test and simulation of tentacle bending. (c) Test and simulation of tentacle extending. (d) Crawling in pipes with different diameters and corners. (e) Crawling in 5 m long pipe.

How NOKOV Contributes to the Research

To investigate the buckling behavior of tentacles, we tested the deflection profile when the PSAs were pressurized. Three reflective markers were pasted on one of the tentacles, which were monitored by the NOKOV motion capture system.

References W. Wang et al., "A Modular Soft Pipe-Climbing Robot With High Maneuverability," in IEEE/ASME Transactions on Mechatronics, doi: 10.1109/TMECH.2024.3385506.

keywords: {Robots; Pipelines; Immune system; Force; Bellows; Mobile robots; Legged locomotion; Maneuverability; modular design; pipe-climbing robot; soft robotics},

Link <https://ieeexplore.ieee.org/document/10509550>

An adaptive lumped-mass dynamic model and its control application for continuum robots

Abstract

Dynamic modeling for continuum robots remains challenging due to their large nonlinear deformation and the variation of dynamic parameters during movement. In this paper, a lumped-mass dynamic model (LMD) for a continuum robot is constructed including elastic and viscous parameters in the robotic joints. Then the appropriate dynamic parameters (e.g. spring and damping coefficients of the LMD) with respect to the motion status (e.g. position and velocity of the robot) are estimated using a Genetic Algorithm (GA). Based on the obtained data set, a Multi-Layer Perception (MLP) is trained to establish a direct mapping from the motion status to the dynamic parameters, so the LMD can tune its parameters in real-time when moving within the workspace, resulting an adaptive lumped-mass dynamic model (ALMD). Compared to the fixed-parameter LMD, the modeling error of the ALMD is reduced by up to 60.2 %. Finally, a feedforward controller is implemented to control a continuum robotic prototype using the presented ALMD, reducing the maximum tracking error by 67.5 %.

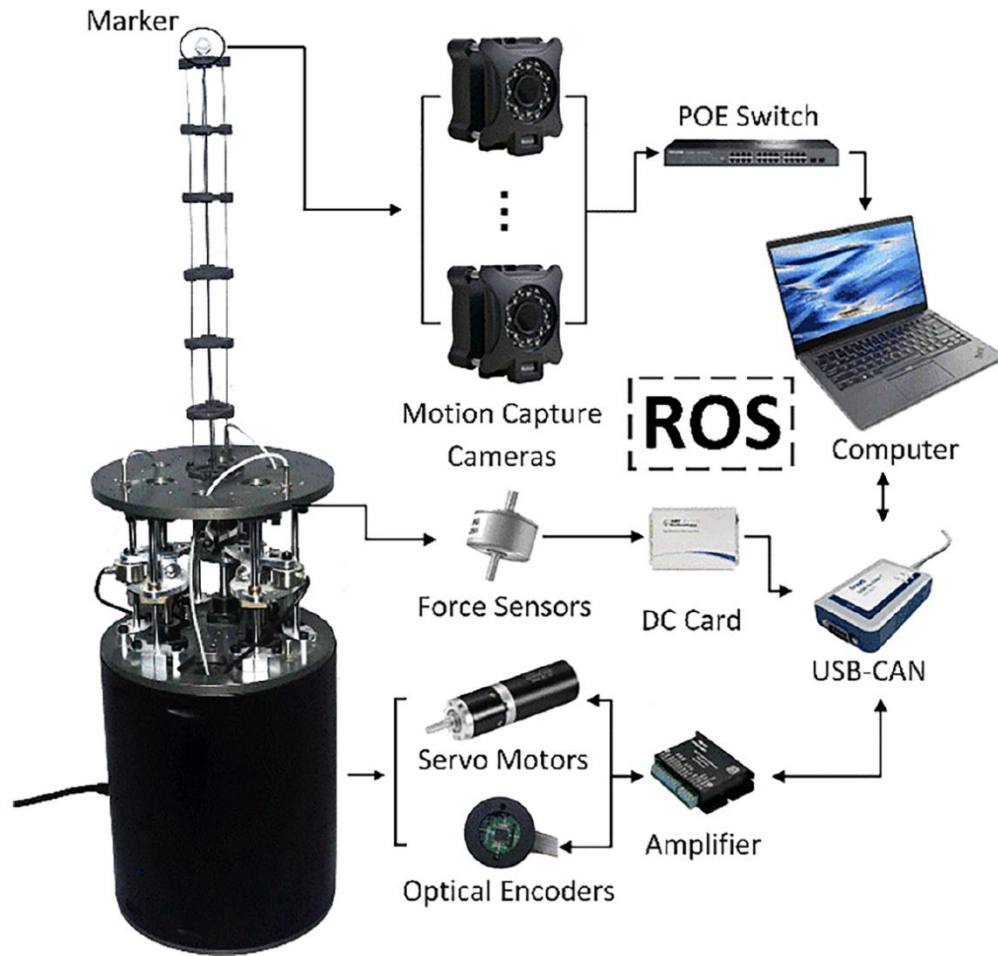


Fig. 11. Prototype of the continuum robot.

How NOKOV Contributes to the Research

The continuum arm contains a backbone, three driving rods distributed 120° apart and 7 constraint disks. The total length of the arm is 300 mm, and the sets of servo motors and ball screws to pull or push the driving rods. An optical motion capture system including four cameras and a marker (The NOKOV Inc., China) is used to measure the position of the arm tip within its operational space.

References Xu Zhang, Chenghao Yang, Zhibin Song, Mojtaba A. Khanesar, David T Branson, Jian S. Dai, Rongjie Kang, An adaptive lumped-mass dynamic model and its control application for continuum robots, Mechanism and Machine Theory, Volume 201, 2024, 105736, ISSN 0094-114X, <https://doi.org/10.1016/j.mechmachtheory.2024.105736>.
Link <https://www.sciencedirect.com/science/article/pii/S0094114X24001630>

Quantifying Uncertainties in Nonlinear Dynamics of a Modular Assembly Using the Resonance Decay Method

Keywords

uncertainty quantification; large structure assembly; resonance decay method

Abstract

Modular assembling is a promising approach to constructing large spacecraft beyond the size limitations posed by launch vehicles. However, the uncertainties and nonlinearities of the dynamics associated with the assembled structure are deeply concerned in the design stage of such spacecraft. Conventionally, this concern can be relieved by performing Ground Vibration Testing (GVT) of the structure. Nevertheless, it is challenging for a modular assembly, among which its very-low frequency behaviour and a lack of dynamic testing procedure that can incorporate with nonlinearities are two major obstacles. In this regard, the present paper firstly introduces a demonstrator for Large Structure Assembly (LSA demonstrator), which includes a soft-bungee suspension system, a 6-meter-long modular assembly, a vibration control system, and a non-contact measurement system. Secondly, a new quantification procedure for modular assembly, which utilises the resonance decay method, was proposed in this paper. Detailed test steps were illustrated through the demonstrator, in which the backbone curves were treated as key measurement targets in quantifying its nonlinear dynamics. The uncertainties in nonlinear dynamics were also evaluated by assembling and disassembling the structure multiple times. Results have shown that the proposed procedure can efficiently and accurately quantify the dynamics of a highly-flexible, large-scale modular assembly.

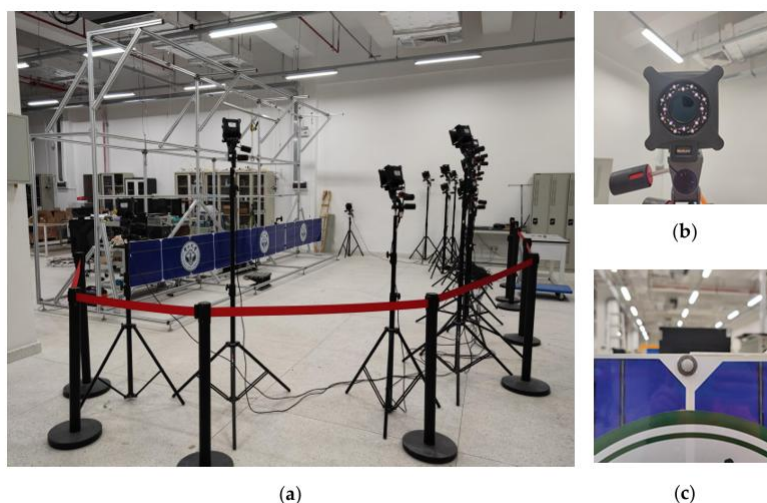


Figure 5. The measurement system: (a) overview of the non-contact motion capture system, (b) camera lens and (c) reflective marking points attached to the test structure.

How NOKOV Contributes to the Research

Considering the very-low natural frequencies of the modular assembly predicted in Table 1, a non-contact motion capture system (NOKOV) was established to perform 3D dynamic displacement measurements. As shown in Figure 1, a total of 18 passive markers were distributivity attached to the modular assembly as measurement targets and 14 cameras were installed approx. 5 meters away from the structure (see Figure 5). The NOKOV system tracked these markers simultaneously by infrared light and calculated 3D coordinates of each marker with images from at least two cameras.

References Lin, C.; Zhao, Z.; Wang, Z.; Jiang, J.; Wu, Z.; Wang, X. Quantifying Uncertainties in Nonlinear Dynamics of a Modular Assembly Using the Resonance Decay Method. *Actuators* 2022, 11, 350. <https://doi.org/10.3390/act11120350>
Link <https://www.mdpi.com/2076-0825/11/12/350>

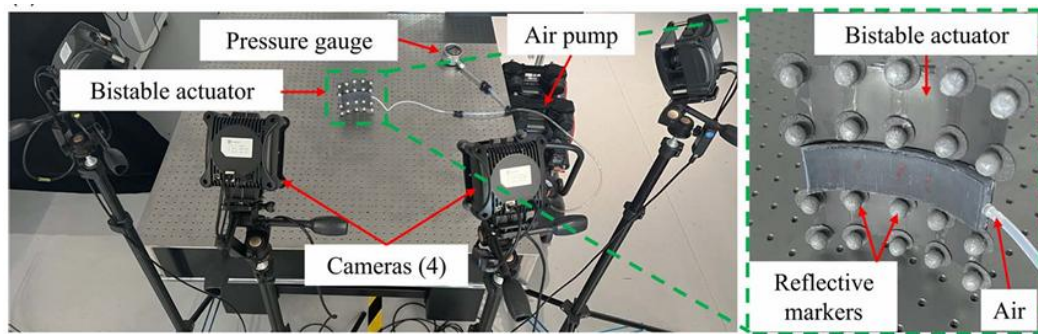
Mechanically Prestressed Pneumatically Driven Bistable Soft Actuators

Keywords

actuators and transmissions, grasping and fixturing, bistable mechanisms, soft robots

Abstract

Bistable soft robots are gaining momentum for their fast speed. This study presents a novel asymmetric mechanically prestressed, pneumatically driven, bistable laminated soft actuator. Its two orthogonal stable shapes are created by prestretching two orthogonal elastomer matrix composites before bonding them to a thin core layer. Two fluidic layers with fluid channels are bonded on either side of the core layer to actuate and trigger the snap-through process of the actuator. An analytical model is proposed as follows: the actuator net energy is calculated based on polynomials with unknown coefficients, and the stable shapes of the actuator are computed as a result of pneumatic pressure and external loads with the Rayleigh–Ritz method. Bistable actuators are fabricated with different prestrains, and motion capture and tensile loading experiments are conducted for model validation. A gripper is fabricated with two bistable actuators and demonstrated to grasp a variety of objects. Sensitivity studies are performed to identify the actuator response as a function of a variety of design parameters.



How NOKOV Contributes to the Research

In quasi static conditions, motion capture experiments are carried out to evaluate the shape of the actuator as it varies with pneumatic pressure (Fig. 6(a)). The image of the actuator surface is acquired using a NOKOV motion capture system including four Mars2H cameras (resolution: 2.2 megapixels) and multiple reflective markers glued on the actuator surface.

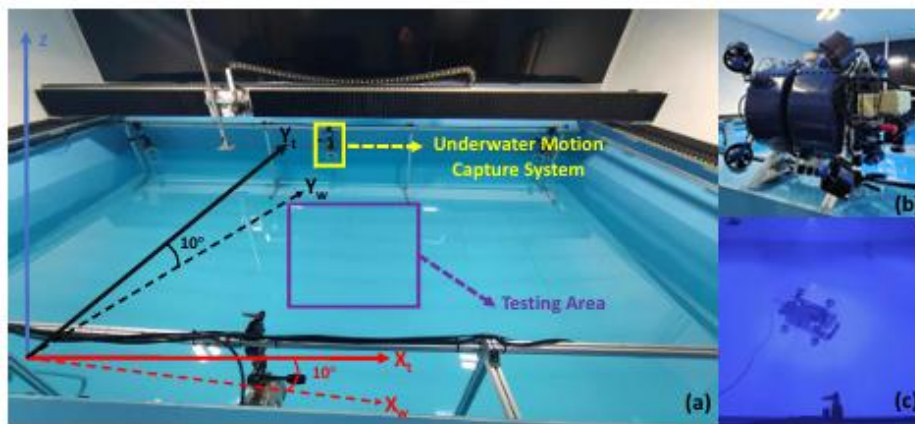
References Zhou, Y., and Xu, Z. (July 20, 2023). "Mechanically Prestressed Pneumatically Driven Bistable Soft Actuators." ASME. J. Mechanisms Robotics. May 2024; 16(5): 051006. <https://doi.org/10.1115/1.4062949>

Link <https://asmedigitalcollection.asme.org/mechanismsrobotics/article-abstract/16/5/051006/1164338/Mechanically-Prestressed-Pneumatically-Driven?redirectedFrom=fulltext>

Model Predictive Control for an Autonomous Underwater Robot with Fully Vectored Propulsion

Abstract

Due to the low motion efficiency and maneuver-ability of underwater robots with six degrees of freedom, it is challenging for them to respond quickly to the attitude requirements during underwater autonomous manipulation. This paper presents a novel autonomous underwater robot with fully vectored propulsion and a model predictive control method to achieve more agile and efficient movements autonomously. In detail, we first design a robot with eight vector-distributed thruster layouts for fully vectored propulsion and construct the software architecture based on the robot operating system (ROS). Then, we establish the hydrodynamic model by adopting the Fossen approach and construct a 13-dimensional system state-space equation, which is discretized using the explicit fourth-order Runge-Kutta method. To achieve autonomous manipulation, model predictive control is employed along with physical constraints of the custom-built robot to enable real-time prediction and optimization of the robot's states for control purposes. Finally, numerical simulations and experiments of the Point-to-Point Motion are conducted to test the robot's performance. Experimental results reveal that the average error of each direction is 0.0027 m, 0.0031 m, and 0.0368 m in the x-axis, y-axis, and z-axis, respectively, and 0.8502°, 2.1941°, 0.2408° corresponding to three attitude angles, which verify the performance of employing MPC to control an autonomous underwater robot with fully vectored propulsion.



How NOKOV Contributes to the Research

The experiment was conducted in a $5\text{ m} \times 3\text{ m} \times 1.5\text{ m}$ indoor rectangular tank, surrounded by a set of NOKOV Motion Capture System with six underwater cameras, and the sampling rate of the system is 100 Hz.

References T. Gao et al., "Model Predictive Control for an Autonomous Underwater Robot with Fully Vectored Propulsion," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 10482-10488, doi: 10.1109/ICRA57147.2024.10611025.

Link <https://ieeexplore.ieee.org/abstract/document/10611025>

ROV6D: 6D Pose Estimation Benchmark Dataset for Underwater Remotely Operated Vehicles

Abstract

Accurately localization between multi-robots is crucial for many underwater applications, such as tracking, convoying and subsea intervention tasks. 6D pose estimation is a fundamental task that enables precise object localization in 3D space with full six degrees of freedom. However, one critical challenge is the lack of available large-scale datasets due to the unbearable cost of labelled data collection. To overcome this difficulty, we propose a benchmark dataset, ROV6D, for 6D pose estimation of remotely operated vehicles (ROVs). The training subset consists of a large number of synthetic images with 6D pose ground truth for ROVs. These synthetic images are generated using BlenderProc and further rendered with the underwater neural rendering (UWNR) strategy to enhance their realism. The testing subsets cover different real-world scenarios, including the Pool subset and Maoming subset, focusing on challenging cases that involve partial occlusion and low visibility. Diverse recent methods are evaluated on the constructed dataset. The results show that methods based on dense coordinates currently perform best, outperforming both the keypoint-based method and the refinement-based method. Our dataset will be made publicly available soon.

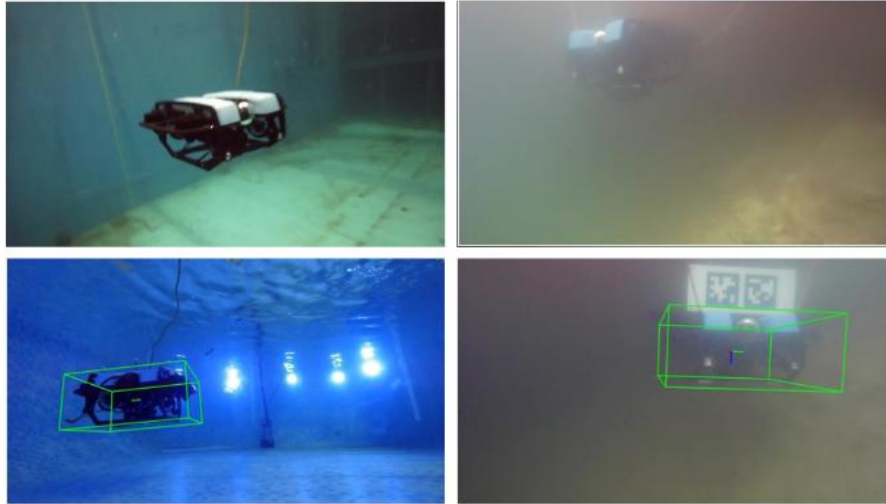


Fig. 1. The BlueROV Heavy is commonly utilized underwater, such as conducting experiments in engineering pools (left) and exploring the open lakes (right). We can provide the 6D pose annotations obtained through a motion capture system or a marker-based system (bottom). The green wireframe represents the projection of the 3D bounding box of the ground-truth pose.

How NOKOV Contributes to the Research

The experimental pool with a dimension of about $6.0 \times 3.5 \times 1.5\text{m}$ is equipped with a NOKOV motion capture system, which supports the capture area of about $4.0 \times 2.5 \times 1.0\text{m}$ and tracks the reflective markers on the ROV with millimetre precision, enabling accurate annotations collection.

References

J. Tang et al., "ROV6D: 6D Pose Estimation Benchmark Dataset for Underwater Remotely Operated Vehicles," in IEEE Robotics and Automation Letters, vol. 9, no. 1, pp. 65-72, Jan. 2024, doi: 10.1109/LRA.2023.3331624.

Link

<https://ieeexplore.ieee.org/document/10313927>

Design and locomotion characteristic analysis of two kinds of tensegrity hopping robots

Highlights

- Two kinds of tensegrity hopping with new actuation methods robots were proposed
- The self-righting characteristics were analyzed and verified by simulation
- Both robots have better locomotion performance over other tensegrity robots
- The proposed robots can be applied in the fields of surveillance and detection

Summary

This study proposed two kinds of tensegrity hopping robots, which were actuated by push-pull electromagnets and servo motors, respectively. Both tensegrity robots are able to conduct stable and consecutive hopping actions. This paper covers the robots' structural designs, theoretical modeling of the hopping actuators, and detailed analysis of the robot's self-righting properties, all of which are validated by corresponding experimental and simulational results. The first hopping robot could hop forward at an average speed of 0.641 body length/s. Although the second robot has a lower moving speed of 0.237 body length/s, its average jumping height of 0.301 m is nearly 2.5 times higher than that of the first robot. Then compared with other tensegrity rolling robots, the proposed two robots show obvious advantages in locomotion performance over their counterparts. Therefore, the proposed robots can have large potential in many fields such as space exploration, urban search, military surveillance, etc.

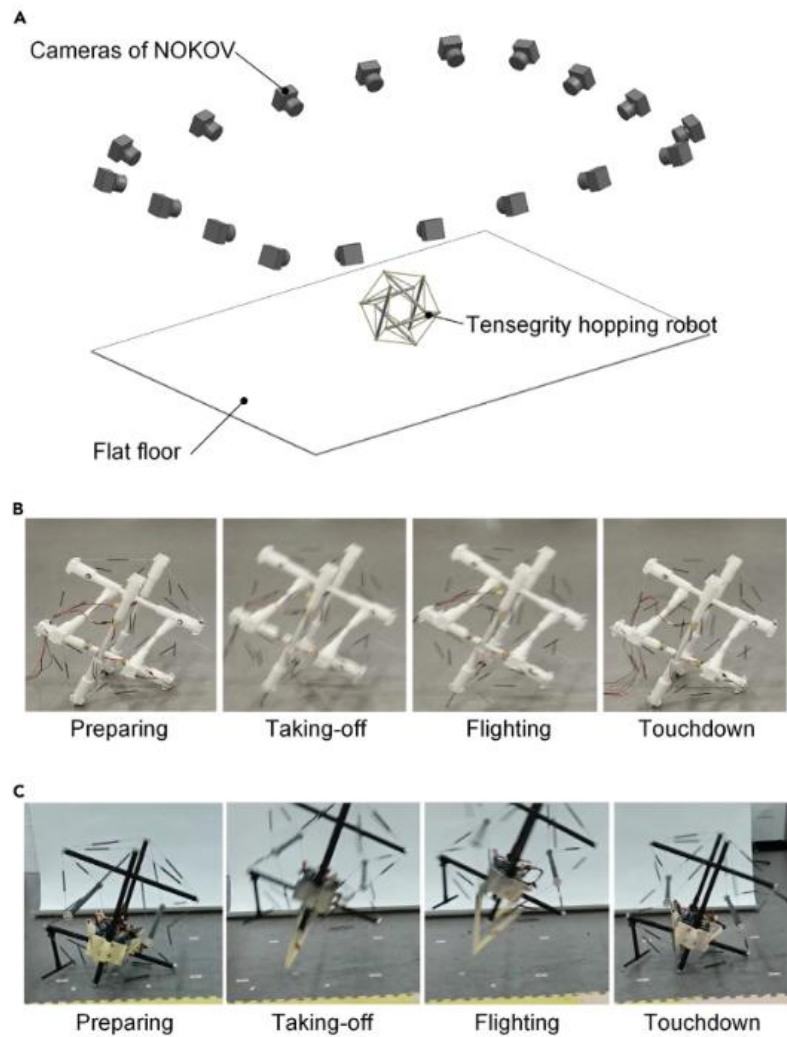


Figure 12. Experimental scene of the proposed robot's hopping experiment

(A) Experimental setup.

(B) Single hopping process of the first robot.

(C) Single hopping process of the second robot.

How NOKOV Contributes to the Research

The armature's movement, the 5-bar mechanism's movement and both robots' movement processes on flat ground were recorded by the NOKOV motion capture system with a sampling frequency of 200Hz.

References

Mo J, Fang H, Yang Q. Design and locomotion characteristic analysis of two kinds of tensegrity hopping robots[J]. Iscience, 2024, 27(3).

Link

[https://www.cell.com/iscience/fulltext/S2589-0042\(24\)00447-4](https://www.cell.com/iscience/fulltext/S2589-0042(24)00447-4)

Continuous adaptive gaits manipulation for three-fingered robotic hands via bioinspired fingertip contact events

Abstract

The remarkable skill of changing its grasp status and relocating its fingers to perform continuous in-hand manipulation is essential for a multifingered anthropomorphic hand. A commonly utilized method of manipulation involves a series of basic movements executed by a high-level controller. However, it remains unclear how these primitives evolve into sophisticated finger gaits during manipulation. Here, we propose an adaptive finger gait-based manipulation method that offers real-time regulation by dynamically changing the primitive interval to ensure the force/moment balance of the object. Successful manipulation relies on contact events that act as triggers for real-time online replanning of multifinger manipulation. We identify four basic motion primitives of finger gaits and create a heuristic finger gait that enables the continuous object rotation of a round cup. Our experimental results verify the effectiveness of the proposed method. Despite the constant breaking and reengaging of contact between the fingers and the object during manipulation, the robotic hand can reliably manipulate the object without failure. Even when the object is subjected to interfering forces, the proposed method demonstrates robustness in managing interference. This work has great potential for application to the dexterous operation of anthropomorphic multifingered hands.

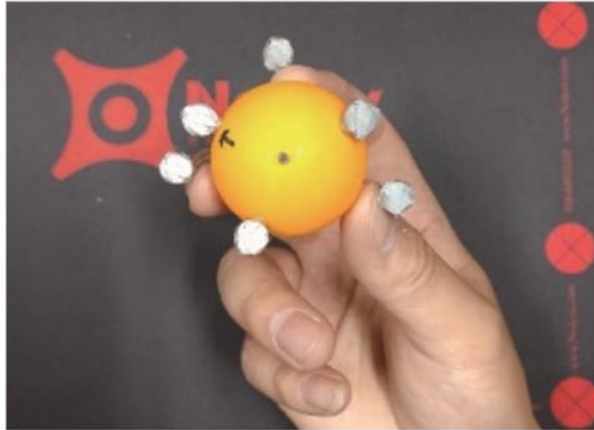


Fig. 1. Location of markers.

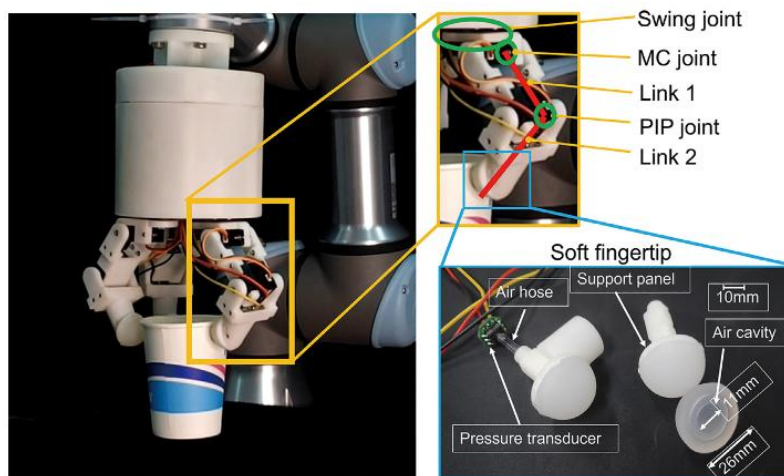


Fig. 4. Three-fingered robotic hand prototype and soft fingertip with embedded sensor.

How NOKOV Contributes to the Research

The posture parameters of the object are sampled and calculated using the NOKOV optical 3D motion capture system.. The NOKOV optical 3D motion capture system estimates the object's angular velocity during manipulation.

References

Xiaolong Ma, Jianhua Zhang, Binrui Wang, Jincheng Huang, Guanjin Bao, Continuous adaptive gaits manipulation for three-fingered robotic hands via bioinspired fingertip contact events, Biomimetic Intelligence and Robotics, Volume 4, Issue 1, 2024, 100144, ISSN 2667-3797

Link

<https://www.sciencedirect.com/science/article/pii/S2667379724000020>

Locomotion control of a rigid-soft coupled snake robot in multiple environments

Abstract

The versatile motion capability of snake robots offers themselves robust adaptability in varieties of challenging environments where traditional robots may be incapacitated. This study reports a novel flexible snake robot featuring a rigid-flexible coupling structure and multiple motion gaits. To better understand the robot's behavior, a bending model for the soft actuator is established. Furthermore, a dynamic model is developed to map the relationship between the input air pressure and joint torque, which is the model base for controlling the robot effectively. Based on the wave motion generated by the joint coupling direction function in different planes, multiple motion gait planning methods of the snake-like robot are proposed. In order to evaluate the adaptability and maneuverability of the developed snake robot, extensive experiments were conducted in complex environments. The results demonstrate the robot's effectiveness in navigating through intricate settings, underscoring its potential for applications in various fields.

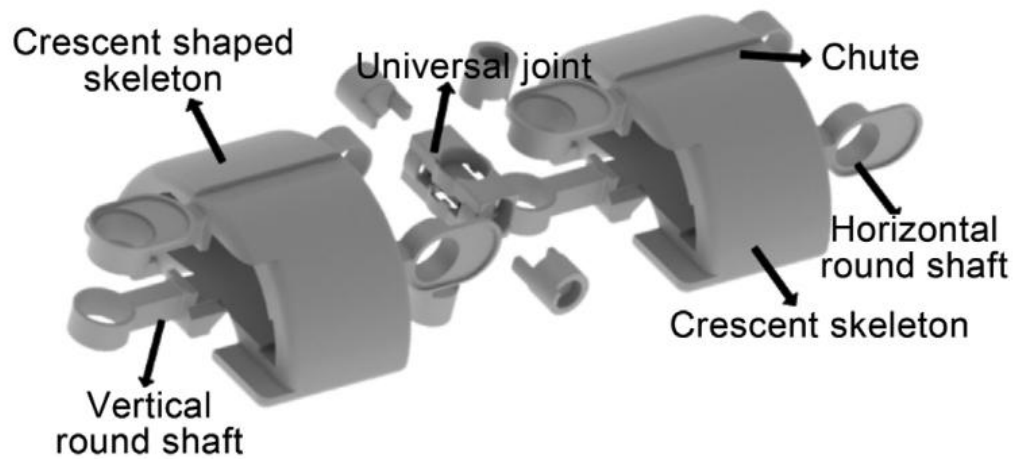


Fig. 1. Exploded view of rigid skeleton.

How NOKOV Contributes to the Research

During the experiments, markers were affixed to the snake robot, and an external vision system, NOKOV, was employed to record the data of motion.

References

Xuanyi Zhou, Yuqiu Zhang, Zhiwei Qiu, Zhecheng Shan, Shibo Cai, Guanjun Bao, Locomotion control of a rigid-soft coupled snake robot in multiple environments, Biomimetic Intelligence and Robotics, Volume 4, Issue 2, 2024, 100148, ISSN 2667-3797
Link

<https://www.sciencedirect.com/science/article/pii/S2667379724000068>

Enhancing Generalizable 6D Pose Tracking of an In-Hand Object With Tactile Sensing

Abstract

When manipulating an object to accomplish complex tasks, humans rely on both vision and touch to keep track of the object's 6D pose. However, most existing object pose tracking systems in robotics rely exclusively on visual signals, which hinder a robot's ability to manipulate objects effectively. To address this limitation, we introduce TEG-Track, a tactile-enhanced 6D pose tracking system that can track previously unseen objects held in hand. From consecutive tactile signals, TEG-Track optimizes object velocities from marker flows when slippage does not occur, or regresses velocities using a slippage estimation network when slippage is detected. The estimated object velocities are integrated into a geometric-kinematic optimization scheme to enhance existing visual pose trackers. To evaluate our method and to facilitate future research, we construct a real-world dataset for visual-tactile in-hand object pose tracking. Experimental results demonstrate that TEG-Track consistently enhances state-of-the-art generalizable 6D pose trackers in synthetic and real-world scenarios.

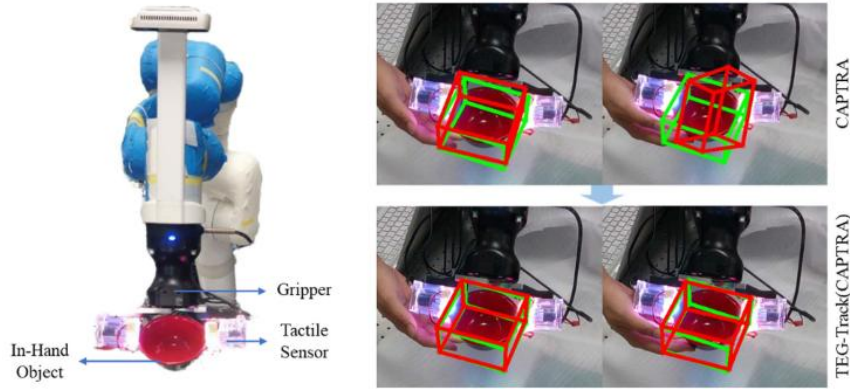


Fig. 1. We propose a general in-hand object pose tracking framework TEG-Track, and evaluate it on our synthetic and real datasets. Our approach enhances generalizable visual trackers such as BundleTrack [7] with tactile sensing. Here we visualize the tracking task as tracking the object's 3D bounding box: green boxes denote ground truth poses whereas red boxes denote estimated poses.



Fig. 5. (a) Data capturing system. Red, green, blue and yellow boxes indicate the Xarm robot arm with Geisight tactile sensors, the Realsense D415 RGB-D sensor, the NOKOV motion capture suite and the in-hand tracked object, respectively. (b) Object categories and instances.

How NOKOV Contributes to the Research

As illustrated in Fig. 5, in our real-world scenario, we additionally set up a NOKOV motion capture system to obtain the ground-truth object poses.

References

Y. Liu et al., "Enhancing Generalizable 6D Pose Tracking of an In-Hand Object With Tactile Sensing," in *IEEE Robotics and Automation Letters*, vol. 9, no. 2, pp. 1106-1113, Feb. 2024, doi: 10.1109/LRA.2023.3337690.

Link

<https://ieeexplore.ieee.org/document/10333330>

Efficient RRT*-based Safety-Constrained Motion Planning for Continuum Robots in Dynamic Environments

Abstract

Continuum robots, characterized by their high flexibility and infinite degrees of freedom (DoFs), have gained prominence in applications such as minimally invasive surgery and hazardous environment exploration. However, the intrinsic complexity of continuum robots requires a significant amount of time for their motion planning, posing a hurdle to their practical implementation. To tackle these challenges, efficient motion planning methods such as Rapidly Exploring Random Trees (RRT) and its variant, RRT*, have been employed. This paper introduces a unique RRT*-based motion control method tailored for continuum robots. Our approach embeds safety constraints derived from the robots' posture states, facilitating autonomous navigation and obstacle avoidance in rapidly changing environments. Simulation results show efficient trajectory planning amidst multiple dynamic obstacles and provide a robust performance evaluation based on the generated postures. Finally, preliminary tests were conducted on a two-segment cable-driven continuum robot prototype, confirming the effectiveness of the proposed planning approach. This method is versatile and can be adapted and deployed for various types of continuum robots through parameter adjustments.

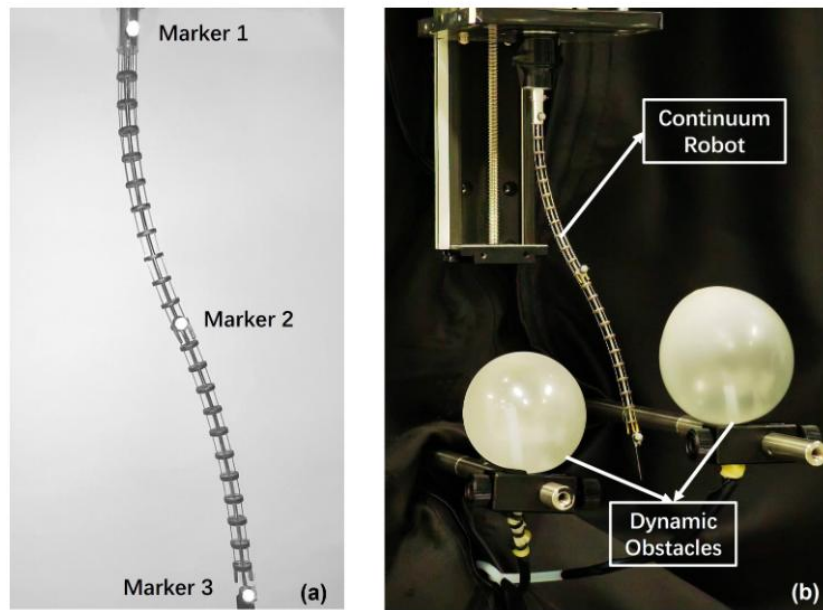


Fig. 1. The robot configuration. (a). Two-section continuum robot. (b). Real-world dynamic environment implementation.

How NOKOV Contributes to the Research

Employing five NOKOV® motion capture cameras with 0.3mm accuracy, we instrumented markers to track the robot's base, the proximal segment's tip, and the distal segment's end (see Fig. 1).

References

P. Luo, S. Yao, Y. Yue, J. Wang, H. Yan and M. Q. . -H. Meng, "Efficient RRT*-based Safety-Constrained Motion Planning for Continuum Robots in Dynamic Environments," 2024 IEEE International Conference on Robotics and Automation (ICRA), Yokohama, Japan, 2024, pp. 9328-9334, doi: 10.1109/ICRA57147.2024.10610309.

Link

<https://ieeexplore.ieee.org/document/10610309>

A markless 3D human motion data acquisition method based on the binocular stereo vision and lightweight open pose algorithm

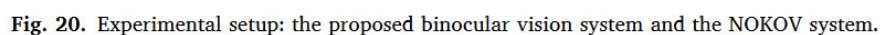
Highlights

- A lightweight Open Pose model was developed for community settings' applications.
- The proposed binocular vision system showed good feasibility in static experiments.
- The proposed system proved good effective compared to the NOKOV system.
- The proposed system exhibited better stability than the Kinect series devices.

Abstract

This study aimed to achieve precise 3D human motion data acquisition without the need for markers. Specifically, the proposed system utilized two affordable cameras to capture human motion from diverse angles. It developed a lightweight Open Pose algorithm to extract 2D human body joints from recorded videos. Subsequently, a stereo-matching method was applied to derive accurate 3D human body joint data. Comprehensive experiments with 10 healthy participants showcased the system's exceptional performance, demonstrating superior measurement accuracy compared to established standards like the NOKOV system (RMSE values consistently below 9°) and commonly used Microsoft Kinect devices (V2 and Azure). This innovative binocular vision system presents a new approach to motion capture, potentially offering a cost-effective and precise tool for practical applications across various scenarios (e.g., motor functional assessment, virtual avatar streaming), contributing significantly to both scientific and academic communities.

Abstract



How NOKOV Contributes to the Research

In the experiments, this research compared the human motion data obtained by the proposed binocular vision system at different distances with data from the gold standard NOKOV and commercially available devices, Kinect and Azure Kinect, both of which do not require markers. This comparison aimed to validate the reliability, effectiveness, and consistency of the proposed binocular vision system

References

Bo Sheng, Linfeng Chen, Jian Cheng, Yanxin Zhang, Zikai Hua, Jing Tao, A markless 3D human motion data acquisition method based on the binocular stereo vision and lightweight open pose algorithm, Measurement, Volume 225, 2024, 113908, ISSN 0263-2241

Link

<https://www.sciencedirect.com/science/article/abs/pii/S0263224123014720>

Research on Enhanced Gait Phase Segmentation Based on Multi-Modal Spatiotemporal Information Fusion

Keywords

Sensors, Motion segmentation, Accuracy, Wearable sensors, Tracking, Skeleton, Legged locomotion, Optical sensors, Estimation, Biomedical optical imaging

Abstract

Gait phase segmentation, pivotal for understanding lower limb motion, finds applications in diverse fields like medicine and sports. While existing method often struggle with accuracy and adaptability in real-world settings, this study presents a novel methodology employing particle filters for precise lower limb motion capture (MoCap) utilizing inertial sensors, which can be used in more everyday environments and in a wider range of applications over a longer period of time. The innovative approach adeptly tracks walking movements, labeling six gait phases via skeleton reconstruction facilitated by the MoCap algorithm. Subsequently, we propose a neural network architecture amalgamating Temporal Convolutional Network (TCN), Graph Convolutional Network (GCN), and Long Short-Term Memory (LSTM). This architecture integrates raw data from inertial sensors with joint angles derived from reconstructed motion, achieving accurate segmentation of the six gait phases. Experimental validation compares the MoCap algorithm against an optical motion capture system, and the neural network's performance against state-of-the-art methods. Results demonstrate our method's superior accuracy of 96.94%, highlighting its efficacy in addressing gait phase segmentation challenges and propelling advancements in gait analysis.



Fig. 10: Overview of IMU-MoCap validation experiment with comparison to Opt-MoCap.

How NOKOV Contributes to the Research

To validate our proposed IMU-based MoCap algorithm, an Opt-MoCap system, NOKOV, shown in Fig. 10, served as the golden standard to measure the accuracy of motion capture. The NOKOV system employed the Mars 1.3H camera, with specific parameters detailed in Table III.

References Zhang H, Liu X, Li J, et al. Research on Enhanced Gait Phase Segmentation Based on Multi-Modal Spatiotemporal Information Fusion[J]. IEEE Internet of Things Journal, 2024.

Link <https://ieeexplore.ieee.org/document/10758819>

Part V VR





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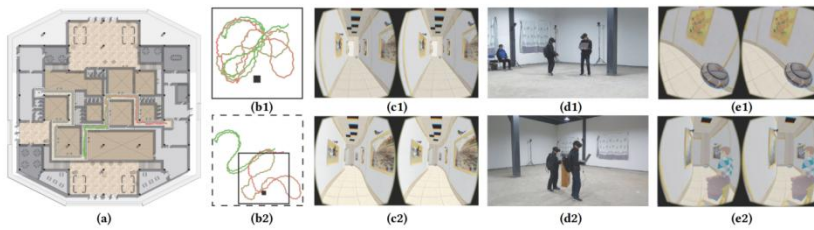
Redirected Smooth Mappings for Multi-user Real Walking in VR

Keywords

Virtual reality; Multi-user real walking; Multi-user physical interaction; Redirected smoothing mapping; Automatic collision avoidance

Abstract

We propose a novel technique to provide multiuser real walking experiences with physical interactions in virtual reality (VR) applications. In our system, multiple users walk freely while navigating a large virtual environment within a smaller physical workspace. These users can interact with other real users or physical props in the same physical locations. The key of our method is a redirected smooth mapping that incorporates the redirected walking technique to warp the input virtual scene with small bends and low distance distortion. Users possess a wide field of view to explore the mapped virtual environment while being redirected in the real workspace. To keep multiple users away from the overlaps of the mapped virtual scenes, we present an automatic collision avoidance technique based on dynamic virtual avatars. These avatars naturally appear, move, and disappear, producing as little influence as possible on users' walking experiences. We evaluate our multiuser real walking system through formative user studies, and demonstrate the capability and practicability of our technique in two multiuser applications.



How NOKOV Contributes to the Research

Tracking System. As shown in Figures 1 and 8, our multiuser real walking and interaction experiments run in a $10\text{m} \times 10\text{m}$ physical room with 12 Nokov motion capture cameras (<http://www.nokov.com/>). An iron pillar (Figure 8(b)) in the room is treated as an interior obstacle (the black box in Figure 8(a)) that is avoided by $f(V)$. Since users are redirected in $f(V)$, they can walk freely without hitting the iron pillar. The average position of three markers, which are placed on top of a user's head, determines the user's position.

References Dong Z C, Fu X M, Yang Z, et al. Redirected smooth mappings for multiuser real walking in virtual reality[J]. ACM Transactions on Graphics (TOG), 2019, 38(5): 1-17.
Link <https://dl.acm.org/doi/abs/10.1145/3345554>

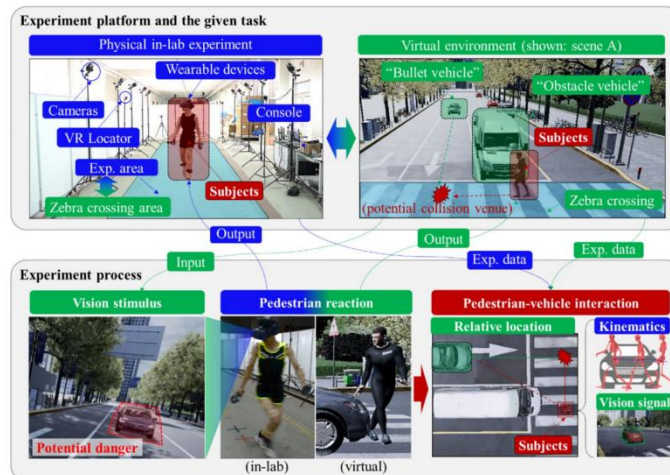
Safety Envelope of Pedestrians upon Motor Vehicle Conflicts Identified via Active Avoidance Behaviour

Keywords

Virtual reality; Multi-user real walking; Multi-user physical interaction; Redirected smoothing mapping; Automatic collision avoidance

Abstract

Human reaction plays a key role in improved protection upon emergent traffic situations with motor vehicles. Understanding the underlying behaviour mechanisms can combine active sensing system on feature caption and passive devices on injury mitigation for automated vehicles. The study aims to identify the distance-based safety boundary ("safety envelope") of vehicle–pedestrian conflicts via pedestrian active avoidance behaviour recorded in well-controlled, immersive virtual reality-based emergent traffic scenarios. Via physiological signal measurement and kinematics reconstruction of the complete sequence, we discovered the general perception–decision–action mechanisms under given external stimulus, and the resultant certain level of natural harm-avoidance action. Using vision as the main information source, 70% pedestrians managed to avoid the collision by adapting walking speeds and directions, consuming overall less "decision" time (0.17–0.24 s vs. 0.41 s) than the collision cases, after that, pedestrians need enough "execution" time (1.52–1.84 s) to take avoidance action. Safety envelopes were generated by combining the simultaneous interactions between the pedestrian and the vehicle. The present investigation on emergent reaction dynamics clears a way for realistic modelling of biomechanical behaviour, and preliminarily demonstrates the feasibility of incorporating in vivo pedestrian behaviour into engineering design which can facilitate improved, interactive on-board devices towards global optimal safety.



How NOKOV Contributes to the Research

Kinematics capture system. The kinematic capture system (No. Mars 2H; Beijing Nokov Science & Technology Co., Ltd) recorded the key motion information of subjects during the experiments. Setup of the kinetic capture system consists of 12 cameras, 54 on-body markers and a software module. The cameras (sampling frequency: 100 Hz) were fixed on the edge of the subjects' movement range by tripods. Motion of the subject throughout the whole event was recorded using 54 markers adhered externally to the skin on body. Algorithm processing of the kinematic images yielded signals of the quantified whole-body kinematics. Feature velocity of the pedestrian was extracted at the centre of the pelvis.

References Nie, B., Li, Q., Gan, S. et al. Safety envelope of pedestrians upon motor vehicle conflicts identified via active avoidance behaviour. Sci Rep 11, 3996 (2021). <https://doi.org/10.1038/s41598-021-82331-z>
Link <https://www.nature.com/articles/s41598-021-82331-z>

Kinetic and Kinematic Features of Pedestrian Avoidance Behavior in Motor Vehicle Conflicts

Keywords

Pedestrian safety; Active behavior; Kinematics; Biomechanics; Integrated safety; Volunteer testing

Abstract

The active behaviors of pedestrians, such as avoidance motions, affect the resultant injury risk in vehicle–pedestrian collisions. However, the biomechanical features of these behaviors remain unquantified, leading to a gap in the development of biofidelic research tools and tailored protection for pedestrians in real-world traffic scenarios. In this study, we prompted subjects (“pedestrians”) to exhibit natural avoidance behaviors in well-controlled near-real traffic conflict scenarios using a previously developed virtual reality (VR)-based experimental platform. We quantified the pedestrian–vehicle interaction processes in the pre-crash phase and extracted the pedestrian postures immediately before collision with the vehicle; these were termed the “pre-crash postures.” We recorded the kinetic and kinematic features of the pedestrian avoidance responses—including the relative locations of the vehicle and pedestrian, pedestrian movement velocity and acceleration, pedestrian posture parameters (joint positions and angles), and pedestrian muscle activation levels—using a motion capture system and physiological signal system. The velocities in the avoidance behaviors were significantly different from those in a normal gait ($p < 0.01$). Based on the extracted natural reaction features of the pedestrians, this study provides data to support the analysis of pedestrian injury risk, development of biofidelic human body models (HBM), and design of advanced on-vehicle active safety systems.

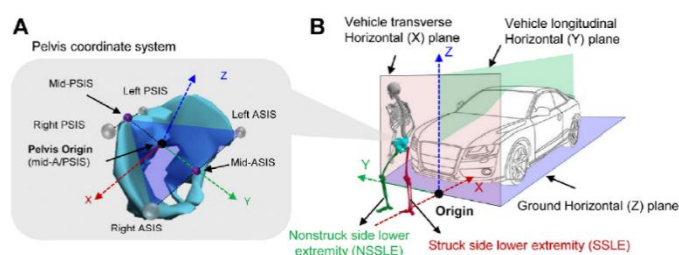
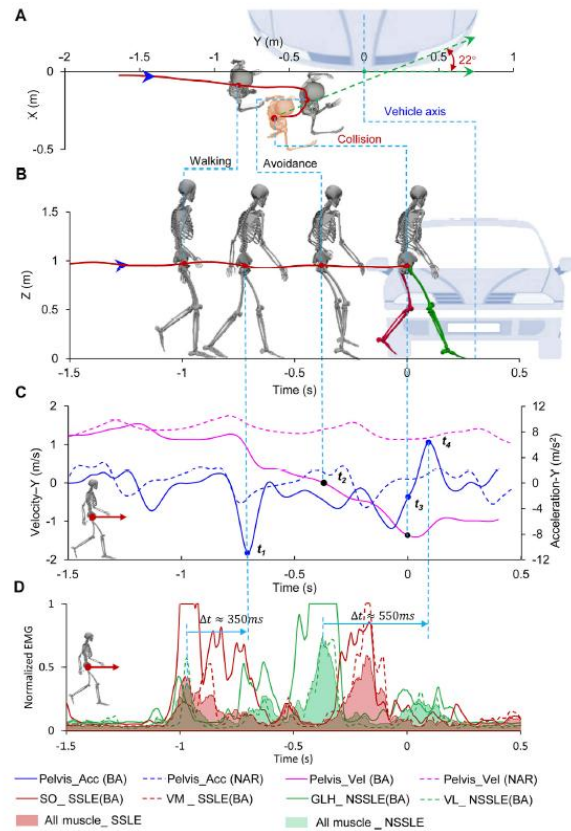


FIGURE 1 | Illustration of the locations of the coordinate system: (A) pedestrian local coordinate system, (B) vehicle coordinate system.



How NOKOV Contributes to the Research

The pedestrian kinetic and kinematic features include the relative locations of the vehicle and pedestrian, the pedestrian's movement velocity and acceleration, their posture (joint positions and angles), and their muscle activation levels. The data were recorded and extracted using the 12 cameras (100 Hz) of the motion capture system (No. Mars 2H; Beijing Nokov Science and Technology Co., Ltd.) by tracking 54 markers. In addition, 11 joints were used to describe the pedestrian posture (Arnold et al., 2010) (Supplementary Figure S1). The pedestrian kinematics were inversed using OpenSim and the "Full Body Model" (Rajagopal et al., 2016).

References Li Q, Shang S, Pei X, et al. Kinetic and Kinematic Features of Pedestrian Avoidance Behavior in Motor Vehicle Conflicts[J]. *Frontiers in Bioengineering and Biotechnology*, 2021, 9.

Link <https://www.frontiersin.org/articles/10.3389/fbioe.2021.783003/full>

Iterative Pose Estimation for a Planar Object Using Virtual Sphere

Keywords

Planar object; pose ambiguity; optimization Algorithm; local minimum; singular value decomposition;

Abstract

This paper proposed an iterative pose estimation for the planar object to deal with the pose ambiguity in the Perspective n Points (PnP) problem. Specifically, by utilizing the unit virtual sphere, the PnP pose estimation problem can be performed as a minimization of the error function with three independent transitional parameters, referred to as the coupling position. Then, Levenberg-Marquardt (LM) optimization algorithm is applied to acquire the coupling position of the first local minimum. Furthermore, the coupling position is represented as the two Euler angles and the vector length, which are combined with the planar points to compute the second local minimum approximation as the initialization of the second LM algorithm. Consequently, once the global coupling position in the two local minimum is decided by the lower error, the orientation and position are directly decoupled by using the singular value decomposition (SVD). It is shown that the designed pose estimation is able to achieve prescribed performance for locating and distinguishing two local minimum, and meanwhile guarantee the superior computation behavior under the pose ambiguity for the planar object. Finally, numerical simulation and physical experiment are conducted to validate the effectiveness of the proposed method, compared to the state-of-the-art PnP methods.

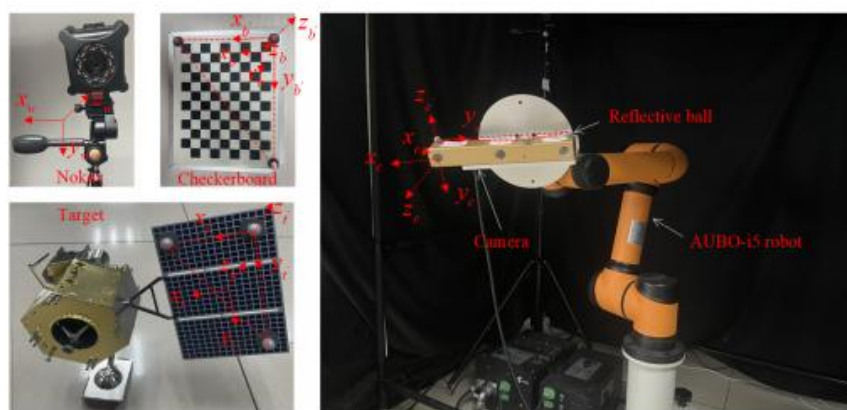


Fig. 10. The composition of the ground experimental system.

How NOKOV Contributes to the Research

To verify the effectiveness of the proposed approach, the ground experimental system is designed, shown in Fig. 10. The experimental system is mainly composed of the Bumblebee camera, the AUBO-i5 robot, the FY 4 target model, the checkerboard, and the Nokov tracker system, wherein seven reference frames are all introduced in Table I. The camera model that is mounted on the end of the robotic arm is applied to acquire the target pictures, while the Nokov tracker system is regarded as the ground truth data.

References Jiang C, Hu Q. Iterative Pose Estimation for a Planar Object Using Virtual Sphere[J]. IEEE Transactions on Aerospace and Electronic Systems, 2022.

Link <https://ieeexplore.ieee.org/abstract/document/9684975>

In vivo Pedestrian Behaviour upon Motor Vehicle Conflicts: Experimental Framework and Initial Results

Abstract

Pedestrians remain a significant portion, i.e., 26%, of all road traffic deaths given their extreme vulnerability in motor vehicle collisions (MVCs) [1]. The instinctive reactions of pedestrians upon traffic conflicts largely determine the occurrence and severity of accidents [2-3]. Yet, the absence of human-like behaviour in existing research surrogates, e.g., crash dummies, human body models, has limited their potential and application in advanced vehicle safety systems on roads. Up to now, no systematic investigation on such in vivo behaviour is available from existing epidemiological, computational, or dummy-based experimental studies [4-5]. This study proposes an experimental framework for extracting typical in vivo pedestrian behaviour in motor vehicle conflicts. When the volunteers, i.e., pedestrians, entered near-real immersive virtual reality (VR) traffic scenarios. A well-controlled dangerous traffic scene was generated as a visual stimulus to pedestrians. Real-time biomechanical signals of the subjects were recorded during the subsequent natural reactions.

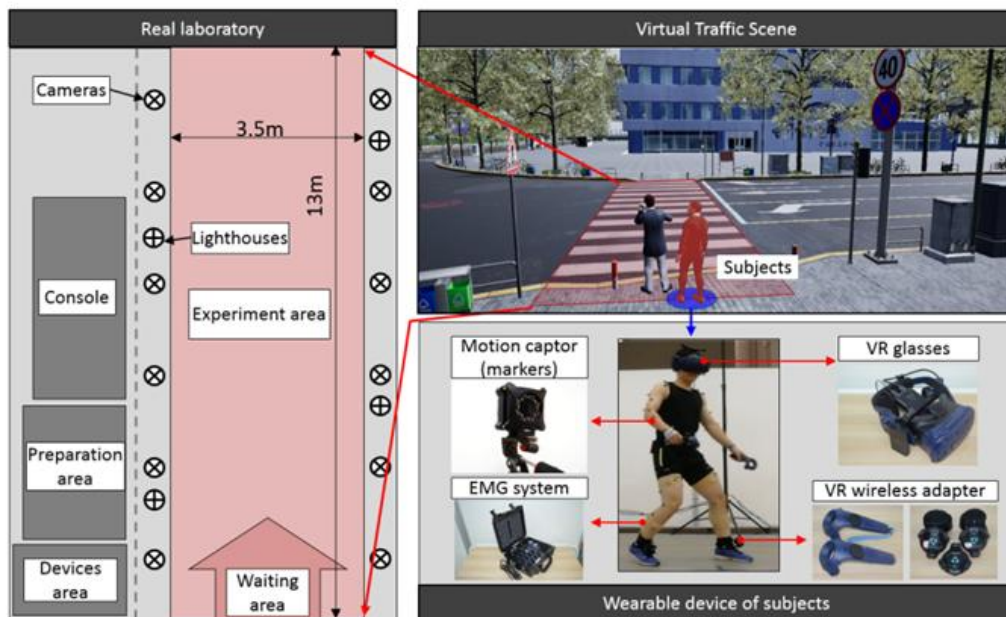


Fig. 1. The overall experiment design

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Module II: Kinetic capture system. Recording the motion information of subjects during the experiment (No. Mars 2H; Beijing Nokov Science & Technology Co., Ltd). This system consists of 12 motion capture cameras (100 Hz), 54 markers and the relevant software module. 12 cameras were fixed on the edge of the subjects' movement range by tripods (Fig. 1). Fifty-four markers were attached to the surface of subjects' body through double-sided tape [6].

References Li Q, Zhou Q, Chen W, et al. In Vivo Pedestrian Behaviour upon Motor Vehicle Conflicts: Experimental Framework and Initial Results[C]//Proceedings of IRCOBI Asia Conference. 2021: 1-2.

Link

https://www.researchgate.net/publication/343306944_In_vivo_Pedestrian_Behaviour_u pon_Motor_Vehicle_Conflicts_Experimental_Framework_and_Initial_Results

Frogman Self-Navigation Method Based on Virtual Transponder Array and Dead Reckoning

Keywords

Frogman self-navigation, integrated navigation, single transponder, underwater acoustic positioning, virtual transponder array.

Abstract

Using single transponder ranging (STR) information to aid positioning satisfies the needs of frogman autonomous positioning in waters emergency rescue, where miniaturized and low-cost devices are preferred. On this basis, a frogman self-navigation method based on virtual transponder array (VTA) and dead reckoning (DR) is proposed in this article, to solve the problem that calculation results of DR still accumulate with time when STR information is taken as the external measurement. In the proposed method, firstly, by constructing the VTA with depth information constraint, frogman autonomous positioning under the condition of single transponder configuration is realized. Then, to combine the characteristics that DR positioning data is smooth and short-term stable, and that VTA positioning error is not cumulative, the DR navigation system is used to describe frogman's motion law, the frogman positioning coordinates that calculated by VTA are taken as external measurement and the Kalman filter is designed, which solves the problem of DR accumulated errors. Compared to traditional STR-DR method, the proposed VTA-DR method further improves the accuracy and stability of frogman navigation and positioning. Finally, based on the high-precision three-dimensional motion capture system, the semi-physical simulation experimental environment is built to verify the proposed method. The experimental results under different tracks indicate that the average total location error of VTA-DR method is 0.149m, which is reduced by 53.5% compared with STR-DR method. The proposed VTA-DR method can better suppress the accumulation of positioning errors and has better positioning accuracy and robustness.



FIGURE 6. Nokov 3D Optical motion capture system.

How NOKOV Contributes to the Research

A. EXPERIMENTAL EQUIPMENT AND SCHEME

1) EXPERIMENTAL EQUIPMENT

2) Nokov 3D Optical motion capture system Nokov 3D optical motion capture system can capture and output the accurate three-dimensional data of the object to be measured, which is mainly composed of infrared optical motion capture lens, operation, analysis and processing software and related accessories. The accuracy of system can reach sub-millimeter level, which meets the requirements of evaluating the positioning accuracy of the method we have proposed in this article. The Nokov 3D optical motion capture system is shown in Fig. 6. It is used as the recorder of experimenter's movement parameter, and the high-precision position, velocity and attitude information output by motion capture system will be the evaluation benchmark of the navigation results. In addition, part of the data will be processed to simulate the truth value of the output signals of underwater navigation subsystems such as DVL and single transponder acoustic ranging system. The speciBc simulation scheme will be described in detail in the next part of this section.

References Zhang Y, Wang N, Li M, et al. Frogman Self-Navigation Method Based on Virtual Transponder Array and Dead Reckoning[J]. IEEE Access, 2020, 8: 177329-177340.

Link <https://ieeexplore.ieee.org/abstract/document/9207928>

Part VI Psychology



The influence of Prior Intention on Joint Action: an fNIRS-based Hyperscanning Study

Keywords

Prior intention; Joint action, Premotor and supplementary motor cortex, Temporo-parietal junction; Hyperscanning

Abstract

Motor performances of the same action are affected by prior intentions to move unintentionally, cooperatively or competitively. Here, a back-and-forth movement task combined with a motion capture system and functional near-infrared spectroscopy (fNIRS)-based hyperscanning technology was utilized to record both the behavioral and neural data of 18 dyads of participants acting in pairs [joint conditions: no-intention, cooperative (Coop) and competitive (Comp)] or alone (single conditions: self-paced and fast-speed). The results revealed that Coop or Comp intentions in the joint conditions significantly sped up motor performance compared with similar single conditions, e.g. shorter movement times (MTs) in the Coop/Comp condition than the self-paced/fast-speed condition. Hemodynamic response analysis demonstrated that stronger activities for all joint conditions than the single conditions in the premotor and the supplementary motor cortex (Brodmann area 6) were independent of variations of MTs, indicating that they might reflect more complex aspects of action planning rather than simple execution-based processes. The comparisons of joint conditions across distinct prior intentions before acting yielded significant results for both behavioral and neural measures, with the highest activation of the temporo-parietal junction (TPJ) and the shortest MTs in the Comp condition considered to be implications for the top-down influence of prior intentions on joint performance.

How NOKOV Contributes to the Research

A Nokov Optical 3D Motion Capture System (<http://www.nokov.com/en>) was used to obtain movement data of two motion capture markers attached to the top of the rod at a sampling rate of 100 Hz, the coordination system was established at the calibration phase by putting a calibration tool at the experimental table. Movement data were preprocessed using Cortex 6.0 to acquire spatial coordinates for each marker, which includes the displacements of the markers in the x-y-z axes over time. And the x-y-z axes represent the vertical axis (x), the longitudinal axis (y) and the horizontal axis (z), respectively

References Chen Y, Zhang Q, Yuan S, et al. The influence of prior intention on joint action: an fNIRS-based hyperscanning study[J]. Social Cognitive and Affective Neuroscience, 2020, 15(12): 1340-1349.

Link <https://academic.oup.com/scan/article/15/12/1340/5995789?login=true>

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